

National Technical University of Athens
School of Electrical & Computer Engineering
Division of Electric Power

Control and Current Limitation of Grid-Forming Converters under Asymmetrical Grid Faults

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DIPLOMA THESIS

Ioannis O. Chatzigeorgiadis

Supervisor: Antonios Antonopoulos
Associate Professor, National Technical University of Athens

Athens, October 2026



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Οι απόψεις και τα συμπεράσματα που περιέχονται σε αυτό το έγγραφο εκφράζουν τον συγγραφέα και δεν πρέπει να ερμηνευθεί ότι αντιπροσωπεύουν τις επίσημες θέσεις του Εθνικού Μετσόβιου Πολυτεχνείου.

The ideas and conclusions presented in this thesis are the author's and do not necessarily reflect the official views of the National Technical University of Athens.

Περίληψη

Η αυξανόμενη διείσδυση των μετατροπέων ηλεκτρονικών ισχύος στα σύγχρονα συστήματα ηλεκτρικής ενέργειας έχει δημιουργήσει την ανάγκη ανάπτυξης στρατηγικών ελέγχου διαμόρφωσης δικτύου (grid forming, GFM), οι οποίες συνεισφέρουν στην ευσταθή λειτουργία του συστήματος κατά τη διάρκεια διαταραχών στο δίκτυο, τόσο συμμετρικών αλλά και ασύμμετρων. Μία από τις προκλήσεις των μετατροπέων ηλεκτρονικών ισχύος είναι η αντοχή των υλικών τους σε υψηλά ρεύματα, που προκαλούνται κατά τη διάρκεια σφαλμάτων, που ως επί το πλείστον είναι ασύμμετρα.

Η παρούσα διπλωματική εργασία εξετάζει διαφορετικές τεχνικές περιορισμού ρεύματος και στρατηγικές κατανομής του για έναν μετατροπέα διαμόρφωσης δικτύου, ο οποίος λειτουργεί κυρίως υπό ασύμμετρες αλλά και υπό συμμετρικές διαταραχές του δικτύου. Ιδιαίτερη έμφαση δίνεται στην απόκριση του ρεύματος του μετατροπέα και στην τάση στο σημείο κοινής σύνδεσης (PCC) μεταξύ του μετατροπέα και του δικτύου. Στο λογισμικό PLECS αναπτύσσεται ένα μοντέλο μέσης τιμής του μετατροπέα, το οποίο περιλαμβάνει ξεχωριστό έλεγχο για τη θετική και την αρνητική ακολουθία, εξωτερικούς βρόχους ελέγχου διαμόρφωσης δικτύου και εσωτερικούς ελεγκτές ρεύματος. Υλοποιούνται και συγκρίνονται διάφοροι μέθοδοι περιορισμού ρεύματος, συμπεριλαμβανομένου του ελλειπτικού περιοριστή ρεύματος (αναλογικός περιορισμός), μιας μεθόδου που βασίζεται στην εικονική αγωγιμότητα (virtual admittance), καθώς και στρατηγικών κατανομής ρεύματος με προτεραιότητα σε συγκεκριμένη ακολουθία (θετική ή αρνητική).

Οι υλοποιημένες μέθοδοι αξιολογούνται μέσω προσομοιώσεων στο πεδίο του χρόνου. Οι ασύμμετρες περιπτώσεις περιλαμβάνουν μονοφασικές και διφασικές βυθίσεις τάσης, καθώς και μονοφασική μεταβολή της φάσης της τάσης του δικτύου, ενώ ως συμμετρική διαταραχή εξετάζεται μία τριφασική συμμετρική βύθιση τάσης. Η απόδοση των μεθόδων αξιολογείται κυρίως ως προς την αποτελεσματικότητα του περιορισμού του ρεύματος του μετατροπέα στο όριο που έχει οριστεί, τη στήριξη της τάσης στο κοινό σημείο σύνδεσης, τη συμπεριφορά των συνιστωσών θετικής και αρνητικής ακολουθίας και την επαναφορά του συστήματος ύστερα από το σφάλμα.

Τα αποτελέσματα δείχνουν ότι όλες οι μέθοδοι μπορούν να περιορίσουν αποτελεσματικά το ρεύμα του μετατροπέα κατά τη διάρκεια των εξεταζόμενων διαταραχών. Ωστόσο, ο τρόπος με τον οποίο κατανέμεται το διαθέσιμο ρεύμα επηρεάζει σημαντικά την τελική απόκριση του συστήματος. Ο ελλειπτικός περιοριστής και η μέθοδος που βασίζεται στην εικονική αγωγιμότητα παρουσιάζουν πολύ παρόμοια αποτελέσματα στον περιορισμό του ρεύματος και την στήριξη της τάσης για τις συνθήκες λειτουργίας που εξετάζονται, παρόλο που επιτυγχάνουν τον περιορισμό του ρεύματος μέσω διαφορετικών μηχανισμών. Αντίθετα, οι στρατηγικές προτεραιότητας ακολουθίας ανακατανέμουν σκόπιμα τη διαθέσιμη ικανότητα ρεύματος μεταξύ της θετικής και της αρνητικής ακολουθίας, οδηγώντας σε διαφορετικούς συμβιβασμούς μεταξύ της στήριξης της τάσης της θετικής και της αρνητικής ακολουθίας. Η εργασία αναδεικνύει επομένως τη σημασία τόσο του περιορισμού του ρεύματος όσο και της στρατηγικής κατανομής μεταξύ των ακολουθιών κατά τον σχεδιασμό ελέγχων μετατροπέων διαμόρφωσης δικτύου για λειτουργία υπό ασύμμετρες συνθήκες δικτύου.

Λέξεις κλειδιά: αντιστροφείς διαμόρφωσης δικτύου, περιορισμός ρεύματος, ασύμμετρα σφάλματα, θετική ακολουθία, αρνητική ακολουθία, προτεραιότητα ακολουθίας, εικονική αγωγιμότητα, ελλειπτικός περιοριστής

Abstract

The increasing penetration of power-electronic converters in modern power systems has created the need for grid forming control strategies that can enhance stability during grid disturbances, both symmetrical and asymmetrical. One of the main challenges associated with power-electronic converters is their limited overcurrent capability under faults, the majority of which are asymmetrical, where both positive- and negative-sequence current components may be present, resulting in unbalanced phase currents and as a result, a more complex problem.

This diploma thesis investigates different current-limiting techniques and current-allocation strategies for a grid-forming converter operating under mainly unbalanced but also balanced grid disturbances. Particular attention is given to the converter current response and to the voltage at the point of common coupling of the converter. An averaged converter model is developed in PLECS, including positive- and negative-sequence control paths, grid forming outer control loops, and inner current controllers. Several current limiting approaches are implemented and compared, including an elliptical current limiter, a virtual-admittance-based method, and sequence priority current allocation strategies.

The implemented methods are evaluated through time-domain simulations. The asymmetrical cases include single and two-phase voltage sags and a single-phase phase-angle jump, while a three-phase symmetrical voltage sag is considered as the symmetrical disturbance. Their performance is assessed mainly in terms of accurate current limitation, voltage support, positive- and negative-sequence behavior, waveform quality, and post-fault recovery.

The results show that all examined approaches can effectively restrict the converter current during severe disturbances, although the way in which the available current capability is distributed has an important influence on the resulting response. The elliptical and virtual-admittance based methods produce very similar external converter behavior for the operating conditions considered, despite achieving current limitation through completely different mechanisms. In contrast, the sequence priority strategies redistribute the available converter current capability between the positive and negative sequences, leading to different compromises between positive-sequence and negative-sequence support. Therefore, this study highlights the importance of both the current limiting mechanism and the sequence allocation strategy when designing grid-forming converter controls for operation under asymmetrical grid conditions.

Keywords: Grid-forming inverter, current limitation, asymmetrical faults, positive-sequence, negative-sequence, sequence priority, virtual admittance, elliptical limitation.

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Εκτεταμένη Ελληνική Περίληψη

1. Εισαγωγή

Η παραδοσιακή παραγωγή ενέργειας, απαρτιζόμενη από μεγάλες εγκαταστάσεις παραγωγής ενέργειας αλλάζει, καθώς η ανάγκη για την αποφυγή μίας ενεργειακής κρίσης γίνεται όλο και πιο επιτακτική, όσο μία κρίση πλησιάζει στο να γίνει πραγματικότητα.

Η αυξανόμενη διείσδυση ανανεώσιμων πηγών ενέργειας (ΑΠΕ) στα σύγχρονα συστήματα ηλεκτρικής ενέργειας (ΣΣΗΕ) έχει οδηγήσει σε σημαντική αύξηση της χρήσης μετατροπέων, καθώς, αντίθετα από τις σύγχρονες γεννήτριες, οι ΑΠΕ δεν μπορούν να συνδεθούν κατευθείαν με το ηλεκτρικό δίκτυο. Οι μετατροπείς αυτοί, αντίθετα απο τις συμβατικές σύγχρονες γεννήτριες, δεν διαθέτουν εγγενώς τις ηλεκτρομηχανικές ιδιότητες που χαρακτήριζαν παραδοσιακά τη λειτουργία των συστημάτων ηλεκτρικής ενέργειας. Για αυτόν τον λόγο, η συμπεριφορά τους καθορίζεται σε μεγάλο βαθμό από τη στρατηγική ελέγχου που εφαρμόζεται.

Οι περισσότεροι μετατροπείς που έχουν χρησιμοποιούνται μέχρι και σήμερα λειτουργούν σύμφωνα με την αρχή της ακολουθίας δικτύου, κατά την οποία ο μετατροπέας συγχρονίζεται με μία ήδη υπάρχουσα τάση δικτύου και εγχέει την επιθυμητή ενεργό και άεργο ισχύ. Ωστόσο, όσο αυξάνεται το ποσοστό της παραγωγής που χρησιμοποιεί ηλεκτρονικά ισχύος, η ύπαρξη ενός ισχυρού ηλεκτρικού δικτύου το οποίο παρέχει από μόνο του σταθερή τάση και συχνότητα δεν μπορεί πάντοτε να θεωρείται δεδομένη. Έτσι, εδιαφέρον παρουσιάζουν οι μετατροπείς διαμόρφωσης δικτύου (grid forming converters), οι οποίοι συμπεριφέρονται ως ελεγχόμενες πηγές τάσης και συμβάλλουν ενεργά στη διαμόρφωση της τάσης και της συχνότητας του συστήματος.

Παρά τα πλεονεκτήματα των μετατροπέων διαμόρφωσης δικτύου, η λειτουργία τους περιορίζεται από τη σχετικά μικρή ικανότητά τους να αντέχουν υψηλά ρεύματα ενώ ταυτόχρονα να διατηρούν τον χαρακτήρα διαμόρφωσης δικτύου, υποστηρίζοντας την τάση και την συχνότητα του δικτύου. Τα ημιαγωγικά στοιχεία του μετατροπέα μπορούν να λειτουργήσουν μόνο για περιορισμένο χρονικό διάστημα πάνω από το ονομαστικό τους ρεύμα, με αποτέλεσμα η προστασία από υψηλά ρεύματα να αποτελεί βασική απαίτηση του συστήματος ελέγχου. Το ζήτημα αυτό γίνεται ιδιαίτερα σημαντικό κατά τη λειτουργία διαμόρφωσης δικτύου, καθώς μεγάλες μεταβολές του πλάτους ή της γωνίας της τάσης δικτύου μπορούν να προκαλέσουν σημαντική διαφορά μεταξύ της εσωτερικής τάσης του μετατροπέα και της τάσης στο σημείο κοινής σύνδεσης (PCC), οδηγώντας σε απότομη αύξηση του ρεύματος.

Η αντιμετώπιση του προβλήματος γίνεται πιο σύνθετη κατά τη διάρκεια ασύμμετρων διαταραχών. Σε αντίθεση με μία συμμετρική τριφασική διαταραχή, όπου η απόκριση μπορεί να περιγραφεί κυρίως μέσω της θετικής ακολουθίας, ένα ασύμμετρο σφάλμα δημιουργεί ταυτόχρονα συνιστώσες θετικής και αρνητικής ακολουθίας. Οι δύο αυτές συνιστώσες περιστρέφονται σε αντίθετες κατευθύνσεις και συνδυάζονται διαφορετικά στις τρεις φάσεις. Ως αποτέλεσμα, τα ρεύματα των φάσεων δεν έχουν απαραίτητα το ίδιο μέτρο και ο περιορισμός της συνολικής καταπόνησης του μετατροπέα δεν μπορεί να αντιμετωπιστεί πάντοτε με έναν απλό περιορισμό.

Η δυνατότητα ξεχωριστού ελέγχου της θετικής και της αρνητικής ακολουθίας εισάγει επιπλέον βαθμούς ελευθερίας, αλλά ταυτόχρονα δημιουργεί το ζήτημα του τρόπου κατανομής της περιορισμένης ικανότητας ρεύματος του μετατροπέα. Κατά τη διάρκεια ενός σοβαρού ασύμμετρου σφάλματος, ο μετατροπέας ενδέχεται να μην μπορεί να ικανοποιήσει ταυτόχρονα τις απαιτήσεις και της θετικής και της αρνητικής ακολουθίας χωρίς να παραβιάσει το μέγιστο επιτρεπόμενο ρεύμα κάποιας φάσης. Επομένως, πέρα από την ίδια τη μέθοδο περιορισμού, σημαντικό ρόλο αποκτά και η στρατηγική με την οποία κατανέμεται το διαθέσιμο ρεύμα μεταξύ των δύο ακολουθιών.

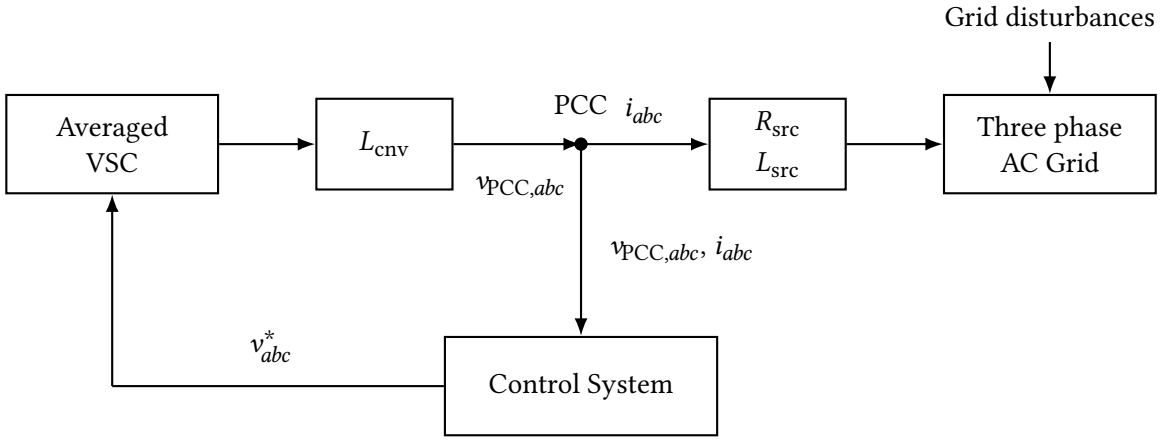
Η παρούσα διπλωματική εργασία επικεντρώνεται στη μελέτη του περιορισμού του ρεύματος ενός μετατροπέα διαμόρφωσης δικτύου υπό συμμετρικές και, κυρίως, ασύμμετρες διαταραχές του δικτύου. Αναπτύσσεται στο λογισμικό PLECS ένα μοντέλο μεσοποιημένης λειτουργίας του μετατροπέα (average model), στο οποίο υλοποιούνται ξεχωριστές στρατηγικές ελέγχου για τη θετική και την αρνητική ακολουθία. Η θετική ακολουθία συνδέεται με τις βασικές λειτουργίες διαμόρφωσης δικτύου του μετατροπέα, ενώ η αρνητική ακολουθία ελέγχεται ανεξάρτητα με στόχο τη διαμόρφωση της απόκρισης κατά τη διάρκεια ασύμμετρων συνθηκών.

Στο πλαίσιο της εργασίας υλοποιούνται και συγκρίνονται διαφορετικές μέθοδοι περιορισμού ρεύματος. Εξετάζεται αρχικά ένας ελλειπτικός περιοριστής, ο οποίος περιορίζει άμεσα τις αναφορές ρεύματος διατηρώντας τη σχετική συνεισφορά των δύο ακολουθιών. Παράλληλα μελετάται μία μέθοδος βασισμένη στη μεταβολή της εικονικής αγωγιμότητας (virtual admittance), στην οποία ο περιορισμός επιτυγχάνεται έμμεσα μέσω της μεταβολής της ηλεκτρικής σχέσης μεταξύ της εσωτερικής τάσης του μετατροπέα και στο σημείο κοινής σύνδεσης. Επιπλέον, υλοποιούνται στρατηγικές κατανομής με προτεραιότητα ακολουθίας, στις οποίες το διαθέσιμο ρεύμα κατανέμεται κατά προτεραιότητα είτε στη θετική είτε στην αρνητική ακολουθία.

Η αξιολόγηση των διαφορετικών προσεγγίσεων πραγματοποιείται μέσω προσομοιώσεων στο πεδίο του χρόνου για μονοφασικές και διφασικές βυθίσεις τάσης, για τριφασική συμμετρική βύθιση τάσης και για μονοφασική μεταβολή της γωνίας φάσης του δικτύου. Η σύγκριση επικεντρώνεται κυρίως στην αποτελεσματικότητα του περιορισμού του ρεύματος και την προστασία του μετατροπέα, στη στήριξη της τάσης στο σημείο κοινής σύνδεσης, στη συμπεριφορά των συνιστωσών θετικής και αρνητικής ακολουθίας, στην ποιότητα των κυματομορφών και στην επαναφορά του συστήματος μετά την εκκαθάριση της διαταραχής. Με τον τρόπο αυτό διερευνάται όχι μόνο κατά πόσο κάθε μέθοδος μπορεί να διατηρήσει το ρεύμα εντός των επιτρεπόμενων ορίων, αλλά και πώς ο μηχανισμός περιορισμού και η επιλογή της προτεραιότητας επηρεάζουν τη συνολική απόκριση του μετατροπέα διαμόρφωσης δικτύου.

2. Σύστημα και Μοντελοποίηση

Το σύστημα που εξετάζεται αποτελείται από έναν τριφασικό μετατροπέα, ο οποίος συνδέεται με το ηλεκτρικό δίκτυο μέσω μίας σύνθετης αντίστασης, με ωμικό και επαγωγικό μέρος. Το δίκτυο αναπαρίσταται ως ιδανική τριφασική πηγή τάσης πίσω από ισοδύναμη αντίσταση και επαγωγή. Το σημείο σύνδεσης μεταξύ του μετατροπέα και του δικτύου ορίζεται ως σημείο κοινής σύνδεσης (Point of Common Coupling – PCC) και αποτελεί το βασικό σημείο στο οποίο αξιολογείται η επίδραση των διαφορετικών στρατηγικών ελέγχου και περιορισμού ρεύματος.



Σχήμα 1.: Απλοποιημένη αναπαράσταση του συστήματος μετατροπέα-δικτύου.

Η επαγωγή L_{cnv} αντιπροσωπεύει την ισοδύναμη επαγωγή μεταξύ των ακροδεκτών του μετατροπέα και του σημείου σύνδεσης του με το δίκτυο, ενώ οι παράμετροι R_{src} και L_{src} περιγράφουν την ισοδύναμη εμπέδηση του δικτύου.

Η τάση και το ρεύμα στο σημείο κοινής σύνδεσης χρησιμοποιούνται τόσο από τους βρόχους ελέγχου όσο και για την αξιολόγηση της απόκρισης του συστήματος κατά τη διάρκεια των διαταραχών. Ιδιαίτερα στις ασύμμετρες συνθήκες, οι τριφασικές ποσότητες αναλύονται σε συνιστώσες θετικής και αρνητικής ακολουθίας, ώστε οι δύο ακολουθίες να μπορούν να ελέγχονται ξεχωριστά.

Για τις προσομοιώσεις χρησιμοποιείται μοντέλο μεσοποιημένης λειτουργίας του μετατροπέα (averaged model). Στην προσέγγιση αυτή δεν αναπαρίσταται ρητά η διακοπτική λειτουργία των ημιαγωγικών στοιχείων. Η συγκεκριμένη μοντελοποίηση επιλέγεται επειδή η παρούσα εργασία επικεντρώνεται κυρίως στη δυναμική απόκριση των βρόχων ελέγχου, στον περιορισμό του ρεύματος και στη συμπεριφορά της τάσης. Με αυτόν τον τρόπο αποφεύγεται η επιβάρυνση της προσομοίωσης από τη διακοπτική συχνότητα και τις αντίστοιχες υψηλής συχνότητας αρμονικές, χωρίς να επηρεάζεται η μελέτη των δυναμικών φαινομένων που αποτελούν το κύριο αντικείμενο της εργασίας.

Η παραγόμενη τριφασική τάση του μετατροπέα προκύπτει από τις αναφορές τάσης που δίνονται από τον εσωτερικό βρόγχο ελέγχου του ρεύματος σε *dq*. Οι θετικές και αρνητικές συνιστώσες ελέγχονται σε διαφορετικά περιστρεφόμενα συστήματα αναφοράς, με γωνιακές ταχύτητες αντίστοιχα

$$\omega^+ = \omega_{\text{GFM}}, \qquad \omega^- = -\omega_{\text{GFM}}. \tag{1}$$

Μετά τον υπολογισμό των αντίστοιχων αναφορών τάσης, οι ποσότητες μετασχηματίζονται εκ νέου στο τριφασικό σύστημα *abc*, ώστε να σχηματιστεί η συνολική τάση αναφοράς του μετατροπέα.

Η χρήση ξεχωριστών συστημάτων αναφοράς για τη θετική και την αρνητική ακολουθία είναι ιδιαίτερα σημαντική κατά τις ασύμμετρες διαταραχές. Στο αντίστοιχο σύγχρονο πλαίσιο κάθε ακολουθίας, οι θεμελιώδεις συνιστώσες μπορούν να αντιμετωπιστούν ως σχεδόν σταθερές ποσότητες, γεγονός που διευκολύνει την ανεξάρτητη επεξεργασία και τον έλεγχό τους.

2.1. Βασικές Ηλεκτρικές Παράμετροι

Ο μετατροπέας του μελετώμενου συστήματος έχει βασική φαινόμενη ισχύ και βασική πολική τάση γραμμής

$$S_b = 60\ \text{MVA}, \qquad V_b = 130\ \text{kV}, \tag{2}$$

και ονομαστική συχνότητα

$$f_b = 50\ \text{Hz}. \tag{3}$$

Η αντίστοιχη βασική αντίσταση και βασική επαγωγή υπολογίζονται από

$$Z_b = \frac{V_b^2}{S_b}, \tag{4}$$

και

$$L_b = \frac{Z_b}{\omega_b}, \qquad \omega_b = 2\pi f_b. \tag{5}$$

Η βασική τιμή του ρεύματος είναι

$$I_b = \frac{S_b}{\sqrt{3} V_b}, \tag{6}$$

ενώ το μέγιστο επιτρεπόμενο ρεύμα του μετατροπέα ορίζεται ως

$$I_{\max} = 1.2\, I_b. \tag{7}$$

Οι βασικές ηλεκτρικές παράμετροι που χρησιμοποιούνται στις προσομοιώσεις συνοψίζονται στον Πίνακα 1.

Πίνακας 1.: Βασικές παράμετροι του μελετώμενου συστήματος.

Παράμετρος	Περιγραφή	Τιμή
S_b	Βασική φαινόμενη ισχύς	60 MVA
V_b	Βασική τάση γραμμής	130 kV
f_b	Ονομαστική συχνότητα	50 Hz
R_{src}	Ισοδύναμη αντίσταση δικτύου	16.8 Ω
L_{src}	Ισοδύναμη επαγωγή δικτύου	0.179 H
L_{cnv}	Επαγωγή σύνδεσης μετατροπέα	0.134 H
$I_{\max}$	Μέγιστο επιτρεπόμενο ρεύμα	1.2 p.u.

Οι παραπάνω παράμετροι παραμένουν σταθερές στις προσομοιώσεις που ακολουθούν, ώστε οι διαφορές στην απόκριση του συστήματος να μπορούν να αποδοθούν στις διαφορετικές τεχνικές περιορισμού και κατανομής του ρεύματος και όχι σε μεταβολές του ηλεκτρικού συστήματος.

3. Δομή Ελέγχου του Μετατροπέα Διαμόρφωσης Δικτύου

Η δομή ελέγχου του μετατροπέα διαχωρίζεται σε συνιστώσες θετικής και αρνητικής ακολουθίας. Η μετρούμενη τριφασική τάση και το μετρούμενο ρεύμα μετασχηματίζονται σε δύο σύγχρονα συστήματα αναφοράς dq, τα οποία περιστρέφονται με γωνιακές ταχύτητες $+\omega_{\text{GFM}}$ και $-\omega_{\text{GFM}}$, αντίστοιχα. Με αυτόν τον τρόπο γίνεται ο διαχωρισμός θετικής και αρνητικής ακολουθίας.

Κατά την παρουσία μιας ασύμμετρης διαταραχής, η αρνητική ακολουθία εμφανίζεται σε κάθε σύγχρονο σύστημα αναφοράς ως κυμάτωση με συχνότητα διπλάσια της θεμελιώδους. Για τον περιορισμό αυτής της αλληλεπίδρασης χρησιμοποιείται κατάλληλο φιλτράρισμα των μετρούμενων σημάτων, πριν αυτά εισαχθούν στους αντίστοιχους βρόχους ελέγχου. Συγκεκριμένα χρησιμοποιείται ένα φίλτρο notch στα 100 Hz. Η τελική δομή περιλαμβάνει τους εξωτερικούς βρόχους διαμόρφωσης δικτύου, το στάδιο παραγωγής των αναφορών ρεύματος και τους εσωτερικούς ελεγκτές ρεύματος.

3.1. Έλεγχος Θετικής Ακολουθίας

Η θετική ακολουθία αντιπροσωπεύει τη συμμετρική συνιστώσα του τριφασικού συστήματος και αποτελεί τη βασική συνιστώσα κατά την κανονική λειτουργία του μετατροπέα. Οι αντίστοιχες τάσεις και τα ρεύματα εκφράζονται σε ένα σύγχρονο σύστημα αναφοράς dq, το οποίο περιστρέφεται με τη γωνιακή ταχύτητα του μετατροπέα διαμόρφωσης δικτύου,

$$\omega^{+} = \omega_{\text{GFM}}. \tag{8}$$

Η θετική ακολουθία αποτελεί τη βασική συνιστώσα μέσω της οποίας πραγματοποιείται η λειτουργία διαμόρφωσης δικτύου του μετατροπέα. Αυτό διότι οι εξωτερικοί βρόχοι ενεργού και αέργου ισχύος βασίζονται στις μετρούμενες ποσότητες θετικής ακολουθίας P^{+} και Q^{+}.

Ο βρόχος ενεργού ισχύος χρησιμοποιείται για τον καθορισμό της γωνίας του μετατροπέα. Η χαρακτηριστική εξίσωση στατισμού μπορεί να γραφτεί ως

$$\omega_{\text{GFM}} = \omega_0 + k_p \left(P^{*} - P^{+}\right), \tag{9}$$

όπου ω_0 είναι η ονομαστική γωνιακή συχνότητα, P^{*} η αναφορά ενεργού ισχύος και k_p ο αντίστοιχος συντελεστής στατισμού. Η γωνία του μετατροπέα προκύπτει από την ολοκλήρωση της συχνότητας,

$$\theta_{\text{GFM}} = \int \omega_{\text{GFM}}\, dt. \tag{10}$$

Η γωνία αυτή χρησιμοποιείται στη συνέχεια στους μετασχηματισμούς των διάφορων ηλεκτρικών μεγεθών και καθορίζει τον προσανατολισμό της εσωτερικής τάσης του μετατροπέα διαμόρφωσης δικτύου.

Παράλληλα, ο βρόχος αέργου ισχύος καθορίζει το επιθυμητό μέτρο της τάσης του μετατροπέα. Η αντίστοιχη χαρακτηριστική εκφράζεται ως

$$E^{+*} = E_0 + k_q \left(Q^* - Q^+\right), \tag{11}$$

όπου E_0 είναι η ονομαστική τάση, Q^* η αναφορά άεργου ισχύος και k_q ο συντελεστής στατισμού αέργου ισχύος. Με τον προσανατολισμό του θετικού dq συστήματος αναφοράς ως προς την γωνία του μετατροπέα, η αναφορά τάσης ορίζεται ως

$$E_d^{+*} = E^{+*}, \qquad E_q^{+*} = 0. \tag{12}$$

Η χρήση μόνο των μεγεθών θετικής ακολουθίας στους κύριους βρόχους διαμόρφωσης δικτύου επιτρέπει στη διαμόρφωση της συχνότητας, της γωνίας και του μέτρου της τάσης να βασίζεται στη βασική σύγχρονη συνιστώσα του συστήματος. Η αρνητική ακολουθία αντιμετωπίζεται ανεξάρτητα από ξεχωριστό βρόχο ελέγχου, όπως περιγράφεται στη συνέχεια.

3.2. Έλεγχος Αρνητικής Ακολουθίας

Κατά τη διάρκεια συμμετρικής λειτουργίας, η αρνητική ακολουθία είναι πρακτικά μηδενική. Αντίθετα, κατά την εμφάνιση μιας ασύμμετρης διαταραχής δημιουργείται αρνητική ακολουθία τάσης και ρεύματος, η οποία απαιτεί ξεχωριστή επεξεργασία από το σύστημα ελέγχου. Για τον λόγο αυτό χρησιμοποιείται δεύτερο σύγχρονο σύστημα αναφοράς, το οποίο περιστρέφεται προς την αντίθετη κατεύθυνση,

$$\omega^- = -\omega_{\mathrm{GFM}}. \tag{13}$$

Ο βασικός στόχος του αρνητικού βρόχου είναι ο έλεγχος της αρνητικής ακολουθίας της τάσης στο σημείο κοινής σύνδεσης. Η επιθυμητή τιμή της αρνητικής ακολουθίας της τάσης τίθεται ίση με μηδέν, έτσι ώστε

$$V_d^{-*} = 0, \qquad V_q^{-*} = 0. \tag{14}$$

Η ύπαρξη ξεχωριστών αναφορών ρεύματος θετικής και αρνητικής ακολουθίας αποκτά ιδιαίτερη σημασία κατά τον περιορισμό ρεύματος. Σε σοβαρές ασύμμετρες διαταραχές οι δύο απαιτήσεις ενδέχεται να μην μπορούν να ικανοποιηθούν ταυτόχρονα χωρίς να ξεπεραστεί το μέγιστο επιτρεπόμενο ρεύμα του μετατροπέα. Ο τρόπος με τον οποίο κατανέμεται η διαθέσιμη ικανότητα ρεύματος μεταξύ των δύο ακολουθιών εξετάζεται αναλυτικά στην Ενότητα 4.

3.3. Εικονική Αγωγιμότητα

Μετά τον υπολογισμό της επιθυμητής εσωτερικής τάσης θετικής ακολουθίας, απαιτείται ένα ενδιάμεσο στάδιο για τη δημιουργία των αντίστοιχων αναφορών ρεύματος. Στην παρούσα εργασία χρησιμοποιείται εικονική αγωγιμότητα (virtual admittance), η οποία εισάγει στον έλεγχο μία τεχνητή ηλεκτρική σχέση μεταξύ της εσωτερικής τάσης του μετατροπέα και της μετρούμενης τάσης στο σημείο κοινής σύνδεσης.

Στο σύγχρονο σύστημα αναφοράς, η σχέση εκφράζεται ως

$$\begin{bmatrix} I^{+}_{d,\mathrm{ref}} \\ I^{+}_{q,\mathrm{ref}} \end{bmatrix} = \frac{1}{R_v^2 + X_v^2} \begin{bmatrix} R_v & X_v \\ -X_v & R_v \end{bmatrix} \begin{bmatrix} E^{+*}_{d} - V^{+}_{d} \\ E^{+*}_{q} - V^{+}_{q} \end{bmatrix}. \tag{15}$$

όπου R_v και X_v είναι η εικονική αντίσταση και αντίδραση, αντίστοιχα. Ακριβώς το ίδιο εφαρμόζεται και για την αρνητική ακολουθία, αλλά για σφάλματα τάσης, έχουμε τη διαφορά μεταξύ της μετρούμενης τάσης αρνητικής ακολουθίας και την αναφοράς της.

Με αυτόν τον τρόπο, μία διαφορά μεταξύ της εσωτερικής τάσης του μετατροπέα και της τάσης στο σημείο κοινής σύνδεσης δημιουργεί την αντίστοιχη αναφορά ρεύματος, όπως θα συνέβαινε μέσω μιας πραγματικής ηλεκτρικής εμπέδησης.

Η ίδια αρχή αξιοποιείται αργότερα και ως μηχανισμός περιορισμού ρεύματος. Μεταβάλλονται οι παράμετροι της εικονικής αγωγιμότητας κατά τη διάρκεια μιας υπερέντασης, ώστε να μειωθεί η προκύπτουσα απαίτηση ρεύματος. Η συγκεκριμένη μέθοδος παρουσιάζεται αναλυτικότερα στην Ενότητα 4.2.

3.4. Εσωτερικός Ελεγκτής Ρεύματος

Οι αναφορές ρεύματος θετικής και αρνητικής ακολουθίας οδηγούνται στους αντίστοιχους εσωτερικούς ελεγκτές ρεύματος. Οι ελεγκτές αυτοί έχουν ως στόχο την παραγωγή των τελικών αναφορών τάσης που θα χρησιμοποιήσει ο μετατροπέας.

Για κάθε ακολουθία ορίζονται τα σφάλματα

$$e^{\pm}_{i,d} = I^{\pm}_{d,\mathrm{ref}} - I^{\pm}_{d}, \qquad e^{\pm}_{i,q} = I^{\pm}_{q,\mathrm{ref}} - I^{\pm}_{q}. \tag{16}$$

Τα σφάλματα εισάγονται σε PI ελεγκτές,

$$u^{\pm}_{d} = k_{pc} e^{\pm}_{i,d} + k_{ic} \int e^{\pm}_{i,d} dt, \tag{17}$$

$$u^{\pm}_{q} = k_{pc} e^{\pm}_{i,q} + k_{ic} \int e^{\pm}_{i,q} dt. \tag{18}$$

Για την αντιστάθμιση της σύζευξης μεταξύ των αξόνων d και q προστίθενται όροι αποσύζευξης που εξαρτώνται από την εκτιμώμενη επαγωγή σύνδεσης L_{est}. Οι τελικές αναφορές τάσης δίνονται από

$$V_{d,\text{ref}}^{\pm} = V_d^{\pm} + u_d^{\pm} - \omega^{\pm} L_{\text{est}} I_{q,\text{ref}}^{\pm}, \tag{19}$$

$$V_{q,\text{ref}}^{\pm} = V_q^{\pm} + u_q^{\pm} + \omega^{\pm} L_{\text{est}} I_{d,\text{ref}}^{\pm}, \tag{20}$$

με

$$\omega^{+} = \omega_{\text{GFM}}, \qquad \omega^{-} = -\omega_{\text{GFM}}. \tag{21}$$

Η αλλαγή του προσήμου της γωνιακής ταχύτητας στην αρνητική ακολουθία οδηγεί αντίστοιχα σε αντιστροφή των όρων αποσύζευξης. Οι τελικές αναφορές τάσης θετικής και αρνητικής ακολουθίας μετασχηματίζονται στο τριφασικό σύστημα αναφοράς και προστίθενται για τη δημιουργία της συνολικής τάσης αναφοράς του μετατροπέα.

Κατά τη διάρκεια υπερέντασης, οι αναφορές που λαμβάνει ο εσωτερικός ελεγκτής δεν είναι απαραίτητα οι αρχικές αναφορές που παράγουν οι εξωτερικοί βρόχοι. Μεταξύ των δύο σταδίων παρεμβάλλονται οι μέθοδοι περιορισμού και κατανομής ρεύματος, οι οποίες τροποποιούν τις αναφορές ώστε να ικανοποιείται το επιτρεπόμενο όριο του μετατροπέα. Οι μέθοδοι αυτές αποτελούν το αντικείμενο της επόμενης ενότητας.

4. Μέθοδοι Περιορισμού Ρεύματος

Το χαμηλό όριο αντοχής των μετατροπέων ηλεκτρονικών ισχύος σε υψηλά ρεύματα καθιστά απαραίτητη την εφαρμογή κατάλληλου μηχανισμού περιορισμού κατά τη διάρκεια των διαταραχών του δικτύου. Στην παρούσα εργασία το μέγιστο επιτρεπόμενο ρεύμα του μετατροπέα ορίζεται ως

$$I_{\text{max}} = 1.2\, I_b. \tag{22}$$

Η διαδικασία περιορισμού γίνεται πιο σύνθετη υπό ασύμμετρες συνθήκες, καθώς το συνολικό τριφασικό ρεύμα του μετατροπέα προκύπτει από τον συνδυασμό των συνιστωσών θετικής και αρνητικής ακολουθίας. Επειδή οι δύο ακολουθίες περιστρέφονται σε αντίθετες κατευθύνσεις, η συνεισφορά τους δεν είναι ίδια την ίδια στιγμή σε κάθε φάση. Επομένως, ο περιορισμός του ρεύματος των δύο ακολουθιών ανεξάρτητα, δεν εγγυάται από μόνος του ότι το ρεύμα κάθε φάσης θα παραμένει εντός του επιτρεπόμενου ορίου, εκτός αν τα όρια των δύο ακολουθιών είναι αθροιστικά κάτω απο το μέγιστο όριο του μετατροπέα, που σε αυτή την περίπτωση ο περιορισμός θα είναι συντηριτικός και θα υπάρχει ικανότητα για περισσότερο ρεύμα που δεν θα αξιοποιείται.

Στο πλαίσιο της εργασίας εξετάζονται τέσσερις διαφορετικές προσεγγίσεις. Οι δύο πρώτες, ο ελλειπτικός περιοριστής και ο περιορισμός μέσω μεταβολής της εικονικής αγωγιμότητας, δεν επιβάλλουν προτεραιότητα μεταξύ της θετικής και της αρνητικής ακολουθίας. Αντίθετα, οι δύο επόμενες μέθοδοι κατανέ-

μουν σκόπιμα το διαθέσιμο ρεύμα, δίνοντας προτεραιότητα είτε στη θετική είτε στην αρνητική ακολουθία.

4.1. Ελλειπτικός Περιοριστής Ρεύματος

Ο ελλειπτικός περιοριστής αποτελεί άμεση μέθοδο περιορισμού, καθώς τροποποιεί τις αναφορές ρεύματος πριν αυτές οδηγηθούν στους εσωτερικούς ελεγκτές. Βασική αρχή της μεθόδου είναι ότι οι αναφορές θετικής και αρνητικής ακολουθίας περιορίζονται ταυτόχρονα με τον ίδιο συντελεστή, ώστε να διατηρείται η σχετική συνεισφορά των δύο ακολουθιών.

Εάν οι αρχικές αναφορές είναι

$$\mathbf{i}_{dq}^{+*} = \begin{bmatrix} I_{d,\mathrm{ref}}^{+} \\ I_{q,\mathrm{ref}}^{+} \end{bmatrix}, \qquad \mathbf{i}_{dq}^{-*} = \begin{bmatrix} I_{d,\mathrm{ref}}^{-} \\ I_{q,\mathrm{ref}}^{-} \end{bmatrix}, \tag{23}$$

οι περιορισμένες αναφορές υπολογίζονται ως

$$\mathbf{i}_{dq,\mathrm{lim}}^{+} = \kappa\, \mathbf{i}_{dq}^{+*}, \qquad \mathbf{i}_{dq,\mathrm{lim}}^{-} = \kappa\, \mathbf{i}_{dq}^{-*}, \tag{24}$$

όπου

$$0 < \kappa \leq 1. \tag{25}$$

Όταν οι απαιτούμενες αναφορές δεν οδηγούν σε υπέρβαση του ορίου, ο συντελεστής $\kappa = 1$ και ο περιοριστής δεν επηρεάζει τη λειτουργία του συστήματος. Όταν όμως κάποια από τις τρεις φάσεις πρόκειται να υπερβεί το όριο $I_{\max}$, ο συντελεστής μειώνεται μέχρι η φάση αυτή να φτάσει στο όριο, και εφαρμόζεται ταυτόχρονα και στις δύο ακολουθίες.

Με αυτόν τον τρόπο διατηρείται η σχετική αναλογία των δύο ακολουθιών. Η μέθοδος δηλαδή κατανέμει αναλογικά την απαραίτητη μείωση μεταξύ θετικής και αρνητικής ακολουθίας.

4.2. Περιορισμός Ρεύματος μέσω Εικονικής Αγωγιμότητας (Virtual Admittance)

Σε αντίθεση με τον ελλειπτικό περιοριστή, η μέθοδος αυτή δεν περιορίζει άμεσα τις αναφορές ρεύματος. Ο περιορισμός επιτυγχάνεται έμμεσα, μέσω της μεταβολής της εικονικής εμπέδησης που συνδέει την εσωτερική τάση αναφοράς του μετατροπέα διαμόρφωσης δικτύου με την τάση του σημείου κοινής σύνδεσης.

Υπό κανονικές συνθήκες, η αναφορά ρεύματος θετικής ακολουθίας δίνεται από

$$\underline{I}_{\mathrm{ref}}^{+} = \frac{\underline{E}^{+*} - \underline{V}_{\mathrm{PCC}}^{+}}{Z_v}, \tag{26}$$

όπου

$$Z_v = R_v + jX_v \tag{27}$$

είναι η επιλεγμένη εικονική εμπέδηση. Κατά την εμφάνιση έντασης ρεύματος υψηλότερης από το όριο, η τιμή της εικονικής εμπέδησης αυξάνεται, ώστε να μειωθεί η αντίστοιχη απαίτηση ρεύματος. Η μεταβολή μπορεί να εκφραστεί γενικά ως

$$Z_{v,\mathrm{lim}} = k_v Z_v, \qquad k_v \geq 1, \tag{28}$$

οπότε η περιορισμένη αναφορά γίνεται

$$\underline{I}^{+}_{\mathrm{lim}} = \frac{\underline{E}^{+*} - \underline{V}^{+}_{\mathrm{PCC}}}{k_v Z_v}. \tag{29}$$

Η αύξηση λοιπόν της εικονικής εμπέδησης μειώνει το ρεύμα που προκύπτει για τη δεδομένη διαφορά μεταξύ της τάσης του μετατροπέα και της τάσης του κοινού σημείου σύνδεσης, μέχρι το σημείο που το ρεύμα της φάσης ή των φάσεων που ξεπέρασαν το όριο, πέσει κάτω από αυτό.

4.3. Περιορισμός με Προτεραιότητα στη Θετική Ακολουθία

Οι προηγούμενες μέθοδοι δεν επιβάλλουν συγκεκριμένη προτεραιότητα μεταξύ των δύο ακολουθιών. Ωστόσο, κατά τη διάρκεια μιας σοβαρής ασύμμετρης διαταραχής, το διαθέσιμο ρεύμα του μετατροπέα δεν θα είναι αρκετό για την πλήρη ικανοποίηση τόσο της θετικής όσο και της αρνητικής αναφοράς. Για τον λόγο αυτό υλοποιείται στρατηγική στην οποία η θετική ακολουθία λαμβάνει προτεραιότητα στο διαθέσιμο ρεύμα.

Αρχικά υπολογίζεται το μέτρο της απαιτούμενης θετικής ακολουθίας,

$$I^{+}_{\mathrm{ref}} = \sqrt{\left(I^{+}_{d,\mathrm{ref}}\right)^2 + \left(I^{+}_{q,\mathrm{ref}}\right)^2}. \tag{30}$$

Εάν η απαίτηση αυτή υπερβαίνει από μόνη της το μέγιστο επιτρεπόμενο ρεύμα, περιορίζεται αναλογικά μέσω του συντελεστή

$$k_+ = \min\left(1, \frac{I_{\max}}{\max(I^{+}_{\mathrm{ref}}, C)}\right), \tag{31}$$

όπου C είναι μία μικρή θετική σταθερά για την αποφυγή διαίρεσης με το 0. Οι περιορισμένες αναφορές θετικής ακολουθίας είναι

$$I^{+}_{d,\mathrm{lim}} = k_+ I^{+}_{d,\mathrm{ref}}, \qquad I^{+}_{q,\mathrm{lim}} = k_+ I^{+}_{q,\mathrm{ref}}. \tag{32}$$

Αφού καθοριστεί η θετική ακολουθία, η διαθέσιμη δυνατότητα για την αρνητική δεν υπολογίζεται απλώς από τη διαφορά $I_{\max} - I^{+}$. Μία τέτοια προσέγγιση θα ήταν υπερβολικά συντηρητική, επειδή οι συνιστώσες θετικής και αρνητικής ακολουθίας περιστρέφονται σε αντίθετες κατευθύνσεις και συνδυάζονται διαφορετικά σε κάθε φάση.

Για τον λόγο αυτό, η μέθοδος που υλοποιείται φροντίζει η ποσότητα του ρεύματος που θα δοθεί στην αρνητική ακολουθία να είναι τέτοια ώστε να αξιοποιείται ολόκληρο το διαθέσιμο ρεύμα. Ορίζεται

$$P^2 = \left(I^{+}_{d,\mathrm{lim}}\right)^2 + \left(I^{+}_{q,\mathrm{lim}}\right)^2, \tag{33}$$

και το διαθέσιμο τετραγωνικό περιθώριο

$$H_{-} = \max\left(I^2_{\max} - P^2, 0\right). \tag{34}$$

Η κατεύθυνση της ζητούμενης αρνητικής ακολουθίας διατηρείται μέσω του μοναδιαίου διανύσματος

$$u^{-}_{d} = \frac{I^{-}_{d,\mathrm{ref}}}{\max(I^{-}_{\mathrm{ref}}, C)}, \qquad u^{-}_{q} = \frac{I^{-}_{q,\mathrm{ref}}}{\max(I^{-}_{\mathrm{ref}}, C)}. \tag{35}$$

Η αλληλεπίδραση μεταξύ της ήδη κατανεμημένης θετικής ακολουθίας και της κατεύθυνσης της αρνητικής περιγράφεται από

$$A_{-} = I^{+}_{d,\mathrm{lim}} u^{-}_{d} - I^{+}_{q,\mathrm{lim}} u^{-}_{q}, \tag{36}$$

$$B_{-} = I^{+}_{d,\mathrm{lim}} u^{-}_{q} + I^{+}_{q,\mathrm{lim}} u^{-}_{d}. \tag{37}$$

Λαμβάνοντας υπόψη τη μετατόπιση 120° μεταξύ των τριών φάσεων, προκύπτουν οι συντελεστές

$$L_{a,-} = 2A_{-}, \tag{38}$$

$$L_{b,-} = -A_{-} + \sqrt{3}B_{-}, \qquad L_{c,-} = -A_{-} - \sqrt{3}B_{-}. \tag{39}$$

Για κάθε φάση $\phi \in \{a, b, c\}$, το μέγιστο επιτρεπόμενο μέτρο αρνητικής ακολουθίας προκύπτει από

$$I^{-}_{\phi,\max} = \frac{-L_{\phi,-} + \sqrt{L^2_{\phi,-} + 4H_{-}}}{2}. \tag{40}$$

Τελικά, η αρνητική ακολουθία περιορίζεται σύμφωνα με την πιο απαιτητική από τις τρεις φάσεις,

$$I^-_{\text{alloc}} = \min\left(I^-_{a,\max}, I^-_{b,\max}, I^-_{c,\max}, I_{\max}\right). \tag{41}$$

Οι τελικές αναφορές αρνητικής ακολουθίας είναι επομένως

$$I^-_{d,\lim} = I^-_{\text{alloc}} u^-_d, \qquad I^-_{q,\lim} = I^-_{\text{alloc}} u^-_q. \tag{42}$$

Η συγκεκριμένη προσέγγιση αξιοποιεί καλύτερα την πραγματική διαθέσιμη ικανότητα του μετατροπέα σε σχέση με έναν συντηρητικό περιορισμό της μορφής

$$I^+ + I^- \leq I_{\max}, \tag{43}$$

καθώς εξετάζει άμεσα το αποτέλεσμα του συνδυασμού των δύο ακολουθιών σε κάθε φάση. Παράλληλα, διαφέρει ουσιαστικά από τον ελλειπτικό περιοριστή, καθώς δεν διατηρεί σταθερή τη σχετική αναλογία των δύο ακολουθιών, αφού στη θετική ακολουθία δίνεται προτεραιότητα.

4.4. Περιορισμός με Προτεραιότητα στην Αρνητική Ακολουθία

Η στρατηγική προτεραιότητας στην αρνητική ακολουθία εφαρμόζει παρόμοια λογική με αυτή της θετικής προτεραιότητας. Κατά τη διάρκεια ασύμμετρης διαταραχής, προτεραιότητα δίνεται στην αναφορά της αρνητικής ακολουθίας, ενώ το υπολειπόμενο ρεύμα χρησιμοποιείται για την κάλυψη της θετικής ακολουθίας.

Η επιλογή αυτή έχει ως κύριο στόχο την ενίσχυση της ικανότητας του μετατροπέα να αντισταθμίζει την ασυμμετρία της τάσης του δικτύου. Μεγαλύτερη διαθέσιμη αρνητική ακολουθία επιτρέπει ισχυρότερη επίδραση στην αρνητική ακολουθία της τάσης του σημείου κοινής σύνδεσης και επομένως μεγαλύτερη αντιστάθμιση της ασυμμετρίας. Η προτεραιότητα αυτή όμως περιορίζει την ικανότητα που παραμένει διαθέσιμη για τη θετική ακολουθία.

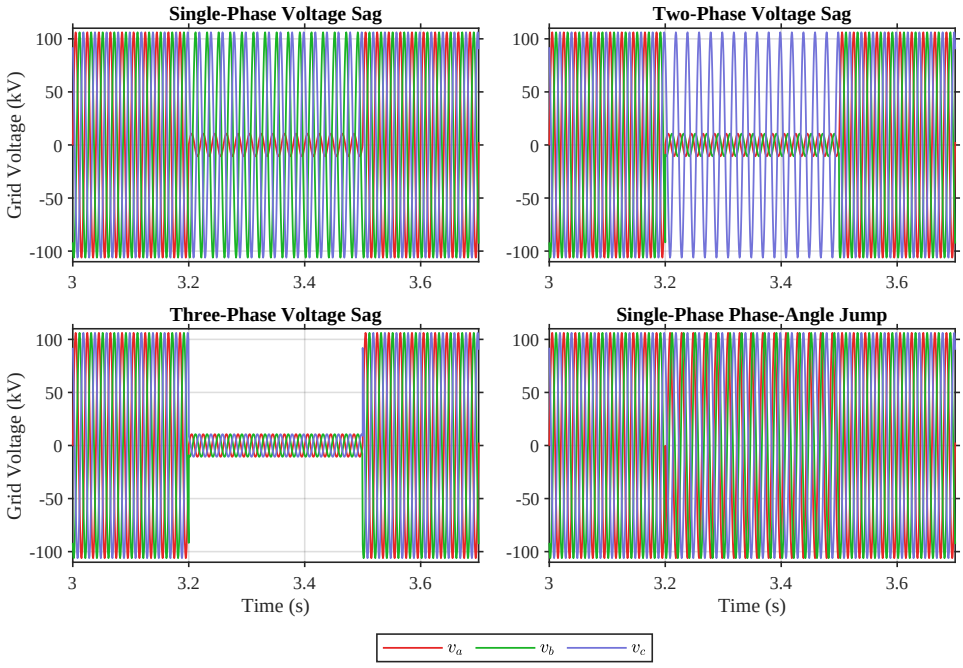
Η εφαρμογή ακολουθεί την ίδια βασική αρχή με την προηγούμενη στρατηγική, αλλά με αντίστροφη σειρά κατανομής. Αρχικά προσδιορίζεται το ρεύμα αρνητικής ακολουθίας που πρέπει να διατηρηθεί και στη συνέχεια υπολογίζεται η επιτρεπόμενη θετική συνεισφορά με βάση το διαθέσιμο περιθώριο των τριών φάσεων.

Στην υλοποίηση διατηρείται μία ελάχιστη θετική συνιστώσα ρεύματος, ώστε η θετική ακολουθία να μην μηδενίζεται πλήρως ακόμη και όταν δεν αρκεί το ρεύμα για να ολοκληρωθεί πλήρως η αρνητική ακολουθία. Με τον τρόπο αυτό διατηρείται μία βασική θετική συνιστώσα που χρειάζεται για την ασφαλή λειτουργία του ελέγχου διαμόρφωσης δικτύου.

Οι δύο στρατηγικές προτεραιότητας αποτελούν επομένως δύο διαφορετικές επιλογές κατανομής της ίδιας περιορισμένης ικανότητας του μετατροπέα. Η προτεραιότητα στη θετική ακολουθία ευνοεί κυρίως τη διατήρηση της θετικής συνιστώσας και τη στήριξη της αντίστοιχης τάσης, ενώ η προτεραιότητα στην αρνητική ακολουθία διαθέτει μεγαλύτερο μέρος του ρεύματος για την αντιμετώπιση της ασυμμετρίας. Η επίδραση των δύο επιλογών στη συνολική απόκριση του μετατροπέα εξετάζεται στα αποτελέσματα των προσομοιώσεων.

4.5. Εξεταζόμενες Διαταραχές

Για την αξιολόγηση των μεθόδων περιορισμού του ρεύματος εξετάστηκαν τέσσερις χαρακτηριστικές διαταραχές δικτύου, οι οποίες καλύπτουν τόσο ασύμμετρες όσο και συμμετρικές συνθήκες λειτουργίας. Οι ασύμμετρες διαταραχές περιλαμβάνουν μία μονοφασική βύθιση τάσης, μία διφασική βύθιση τάσης και μία μονοφασική μεταβολή της γωνίας φάσης, ενώ ως συμμετρική διαταραχή εξετάζεται μία τριφασική βύθιση τάσης.



Σχήμα 2.: Εξεταζόμενες διαταραχές δικτύου

Ειδικότερα, στη μονοφασική βύθιση τάσης το πλάτος της τάσης της φάσης *A* μειώνεται από 1 p.u. σε 0.1 p.u. για διάρκεια 0.3 s, ενώ οι υπόλοιπες δύο φάσεις διατηρούνται στην ονομαστική τους τιμή. Στη διφασική βύθιση τάσης, οι φάσεις *A* και *B* υφίστανται την ίδια μείωση από 1 p.u. σε 0.1 p.u., επίσης για διάρκεια 0.3 s. Η μονοφασική μεταβολή γωνίας φάσης εφαρμόζεται στη φάση *A*, με άλμα 60° για διάρκεια 0.3 s, ενώ οι υπόλοιπες φάσεις παραμένουν αμετάβλητες. Τέλος, στη συμμετρική διαταραχή εξετάζεται τριφασική βύθιση τάσης, κατά την οποία και οι τρεις φάσεις μειώνονται ταυτόχρονα από 1 p.u. σε 0.1 p.u. επίσης για 0.3 s.

Η επιλογή των παραπάνω σεναρίων επιτρέπει τη διερεύνηση της συμπεριφοράς του μετατροπέα διαμόρφωσης δικτύου τόσο σε συνθήκες ασυμμετρίας, όπου συνυπάρχουν συνιστώσες θετικής και αρνητικής ακολουθίας, όσο και συμμετρικά σφάλματα, όπου κυριαρχεί η θετική ακολουθία. Με τον τρόπο αυτό καθίσταται δυνατή η συστηματική σύγκριση των διαφορετικών μεθόδων περιορισμού ρεύματος ως προς την προστασία του μετατροπέα, τη στήριξη της τάσης στο σημείο κοινής σύνδεσης και την ποιότητα της μεταβατικής απόκρισης τους.

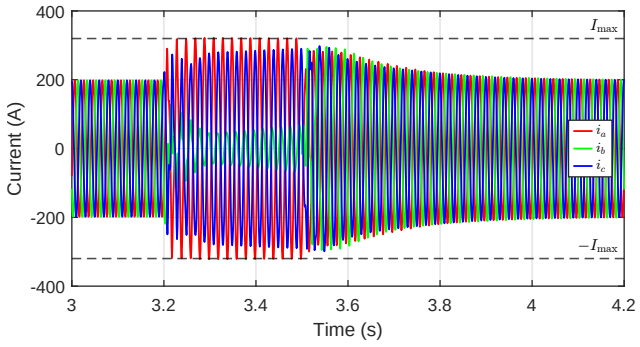
5. Αποτελέσματα Προσομοιώσεων

Η παρουσίαση των αποτελεσμάτων στην εκτεταμένη περίληψη επικεντρώνεται στο τριφασικό ρεύμα του μετατροπέα, καθώς ο περιορισμός του αποτελεί το αντικείμενο της παρούσας εργασίας, μαζί με την αξιολόγηση της αποτελεσματικότητας των διαφορετικών μεθόδων υπό συμμετρικές και ασύμμετρες διαταραχές του δικτύου. Για τον λόγο αυτό, δίνεται έμφαση στις κυματομορφές των ρευμάτων φάσης, οι οποίες αποτυπώνουν την ικανότητα κάθε μεθόδου να διατηρεί το ρεύμα εντός των επιτρεπόμενων ορίων αλλά και τη μεταβατική απόκριση του μετατροπέα κατά τη διάρκεια και μετά του σφάλμα.

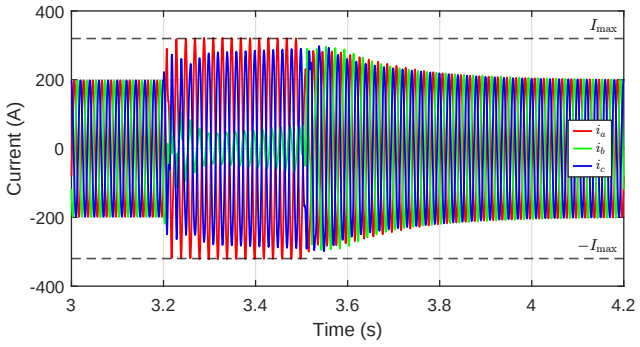
Για κάθε εξεταζόμενη διαταραχή παρουσιάζεται συγκριτικά η απόκριση των τεσσάρων μεθόδων περιορισμού. Με τον τρόπο αυτό καθίσταται δυνατή η άμεση οπτική σύγκριση της συμπεριφοράς τους ως προς το μέγιστο ρεύμα, τον βαθμό ασυμμετρίας των φασικών ρευμάτων, την ταχύτητα επέμβασης του περιορισμού και την επαναφορά του συστήματος μετά το πέρας της διαταραχής.

5.1. Μονοφασική Βύθιση Τάσης

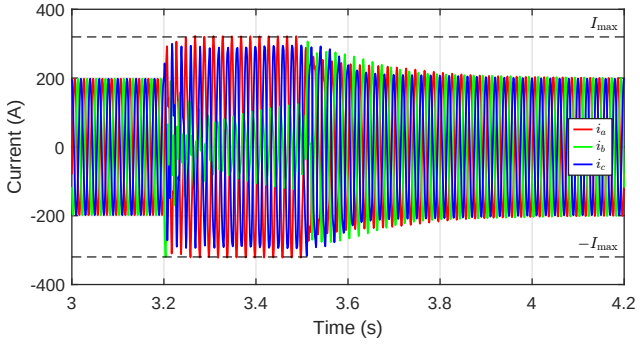
Το Σχ. 3 παρουσιάζει συγκριτικά το τριφασικό ρεύμα του μετατροπέα για τις τέσσερις εξεταζόμενες μεθόδους κατά τη διάρκεια της μονοφασικής βύθισης τάσης. Παρατηρείται ότι όλες οι μέθοδοι επιτυγχάνουν τον περιορισμό του ρεύματος εντός του επιτρεπόμενου ορίου, παρουσιάζουν όμως διαφοροποιήσεις ως προς την κατανομή των ρευμάτων φάσης και τη μεταβατική τους συμπεριφορά.



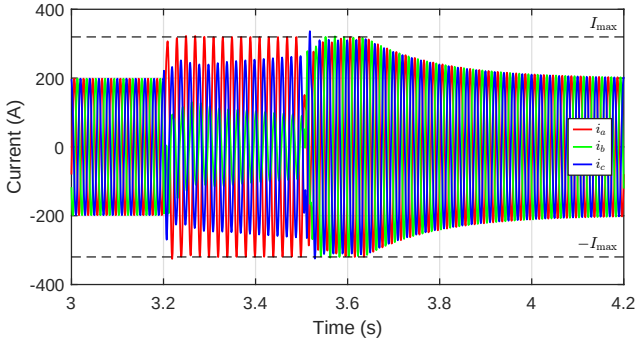
(α′) Elliptical limiter.



(β′) Virtual admittance limiter.



(γ′) Positive-sequence priority.



(δ′) Negative-sequence priority.

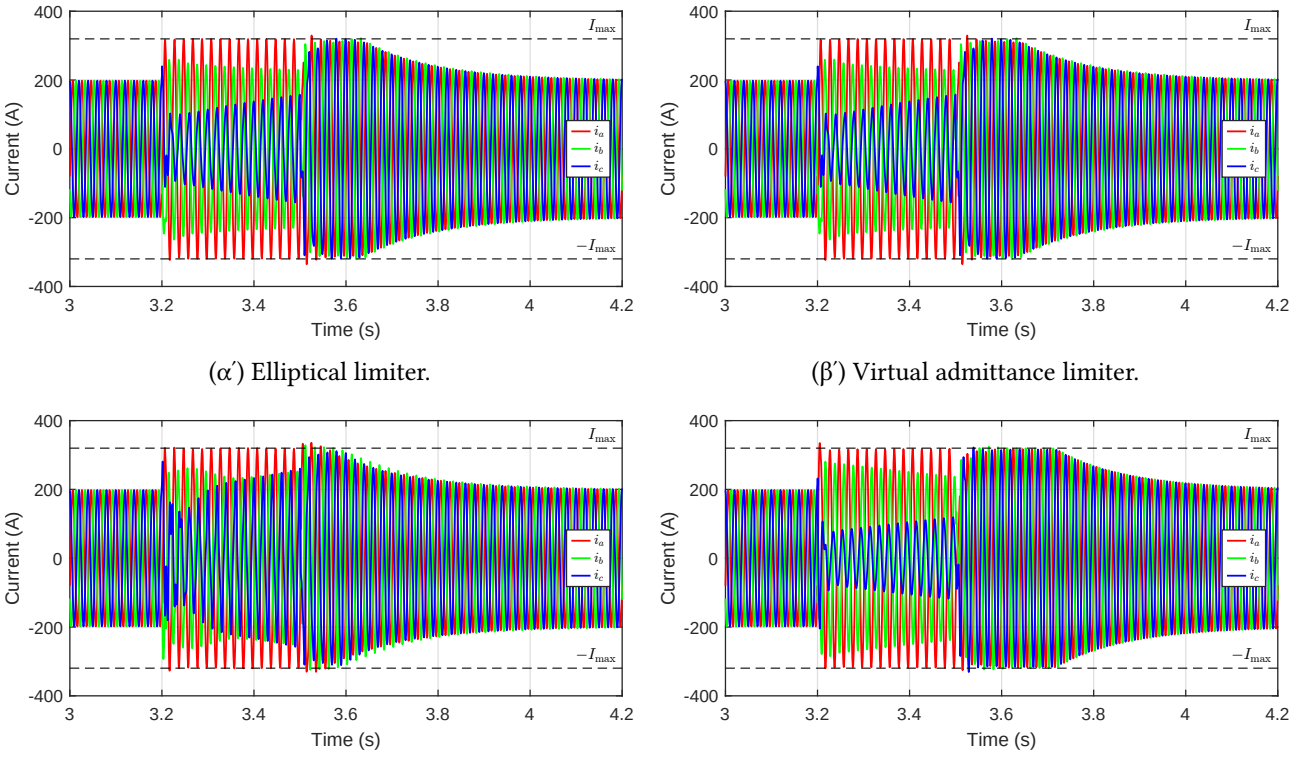
Σχήμα 3.: Σύγκριση ρευμάτων κάθε στρατηγικής περιορισμού κατά τη μονοφασική βύθιση τάσης.

Οι μέθοδοι του ελλειπτικού περιοριστή και της εικονικής αγωγιμότητας παρουσιάζουν πολύ παρόμοια απόκρι-

ση, γεγονός που δείχνει ότι για τη συγκεκριμένη διαταραχή οδηγούν ουσιαστικά σε πανομοιότυπη εξωτερική συμπεριφορά του μετατροπέα. Αντίθετα, οι στρατηγικές προτεραιότητας θετικής και αρνητικής ακολουθίας, οδηγούν σε διαφορετική κατανομή του διαθέσιμου ρεύματος, ανάλογα με το αν δίνεται έμφαση στη θετική ή στην αρνητική ακολουθία. Μετά την εκκαθάριση του σφάλματος, όλες οι μέθοδοι επαναφέρουν το ρεύμα στη μόνιμη κατάσταση, με προφανείς διαφορές ως προς το μεταβατικό φαινόμενο.

5.2. Διφασική Βύθιση Τάσης

Στη δεύτερη ασύμμετρη περίπτωση εξετάζεται διφασική βύθιση τάσης. Η διαταραχή δημιουργεί σημαντική ασυμμετρία στο σύστημα. Έτσι, έχουμε πάλι αρνητική ακολουθία και τα ρεύματα των τριών φάσεων δεν παρουσιάζουν το ίδιο μέτρο κατά τη διάρκειά της.



(α΄) Elliptical limiter.

(β΄) Virtual admittance limiter.

(γ΄) Positive-sequence priority.

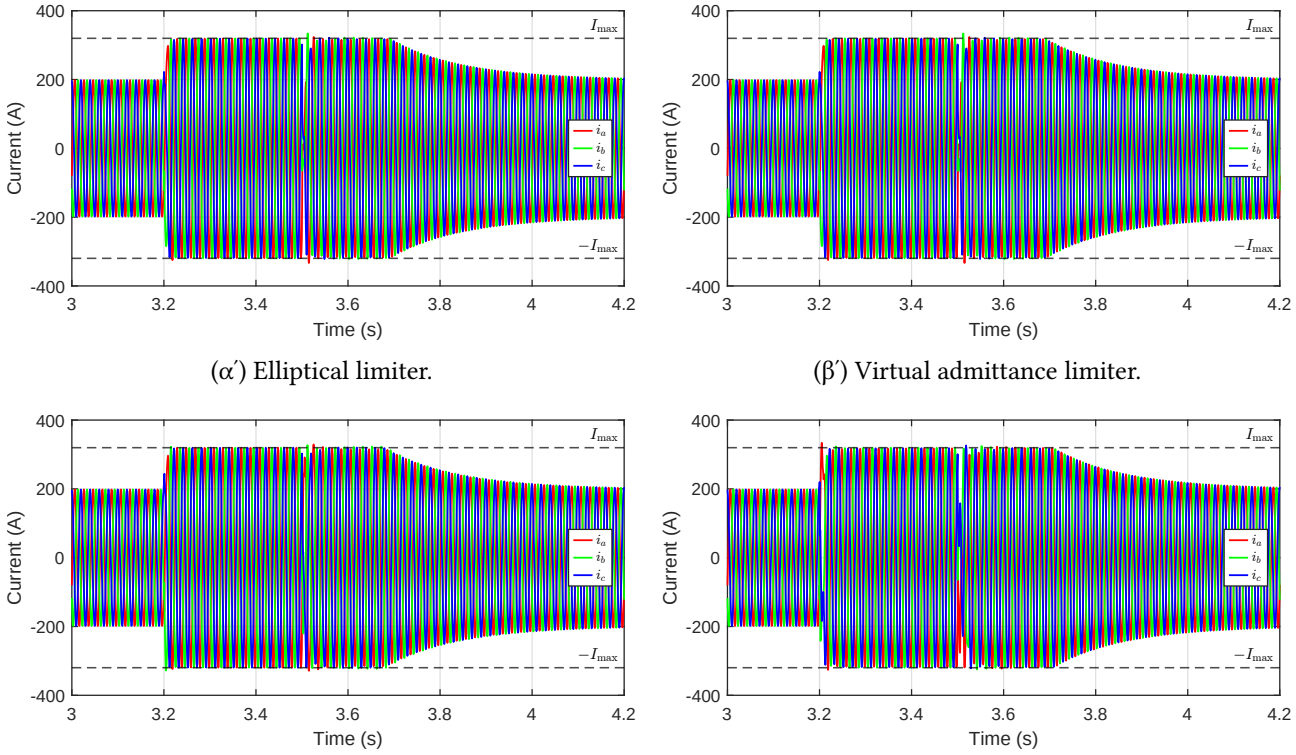
(δ΄) Negative-sequence priority.

Σχήμα 4.: Σύγκριση ρευμάτων κάθε στρατηγικής περιορισμού κατά τη διφασική βύθιση τάσης.

Όπως φαίνεται στο Σχ. 4, και οι τέσσερις μέθοδοι διατηρούν τα ρεύματα των φάσεων εντός του καθορισμένου ορίου, παρά την ασυμμετρία που προκαλεί η διφασική βύθιση. Ο ελλειπτικός περιοριστής και η μέθοδος μεταβολής της εικονικής αγωγιμότητας παρουσιάζουν πολύ παρόμοια απόκριση, τόσο ως προς τα μέγιστα ρεύματα όσο και ως προς τη μορφή των κυματομορφών κατά τη διάρκεια και το πέρας της διαταραχής. Αντίθετα, οι δύο στρατηγικές προτεραιότητας θετικής και αρνητικής ακολουθίας διαφέρουν αρκετά και μεταξύ τους και με τους προηγούμενους περιορισμούς. Η διαφορά αυτή οφείλεται στη διαφορετική κατανομή του διαθέσιμου ρεύματος μεταξύ της θετικής και της αρνητικής ακολουθίας, ανάλογα με την επιλεγμένη προτεραιότητα. Μετά την εκκαθάριση της διαταραχής, το ρεύμα επανέρχεται στην πριν της βύθισης ισορροπημένης κατάστασης για όλες τις εξεταζόμενες μεθόδους.

5.3. Τριφασική Συμμετρική Βύθιση Τάσης

Στη συμμετρική περίπτωση εξετάζεται τριφασική βύθιση τάσης. Σε αντίθεση με τις ασύμμετρες διαταραχές, κατά το συγκεκριμένο σφάλμα δεν προκύπτει σημαντική αρνητική ακολουθία και επομένως η απόκριση του μετατροπέα παραμένει συμμετρική ως προς τις τρεις φάσεις.



(α′) Elliptical limiter.

(β′) Virtual admittance limiter.

(γ′) Positive-sequence priority.

(δ′) Negative-sequence priority.

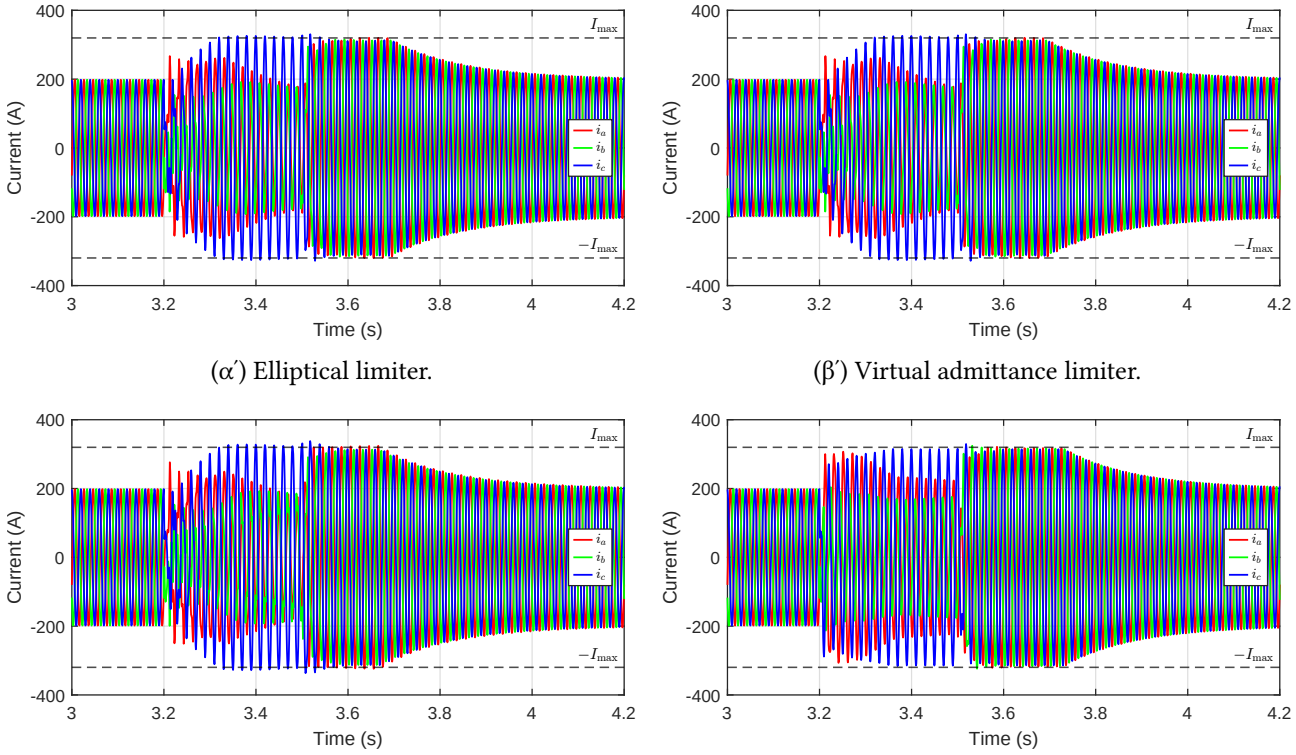
Σχήμα 5.: Σύγκριση ρευμάτων κάθε στρατηγικής περιορισμού κατά τη τριφασική βύθιση τάσης.

Το Σχ. 5 παρουσιάζει την απόκριση του ρεύματος του μετατροπέα για τις τέσσερις μεθόδους κατά την τριφασική συμμετρική βύθιση τάσης. Παρατηρείται ότι όλες οι μέθοδοι διατηρούν το ρεύμα εντός του επιτρεπόμενου ορίου και παρουσιάζουν συμμετρικές κυματομορφές στις τρεις φάσεις, όπως αναμένεται για μία ισορροπημένη διαταραχή. Ο ελλειπτικός περιοριστής και η μέθοδος εικονικής αγωγιμότητας εμφανίζουν και εδώ πολύ παρόμοια συμπεριφορά, ενώ οι στρατηγικές προτεραιότητας θετικής και αρνητικής ακολουθίας, δεν έχουν πολύ διαφορετική απόκριση, καθώς η απουσία αρνητικής ακολουθίας περιορίζει τον ρόλο της κατανομής ρεύματος μεταξύ των δύο ακολουθιών. Μετά την εκκαθάριση του σφάλματος, όλες οι μέθοδοι επαναφέρουν ομαλά το ρεύμα στη μόνιμη κατάσταση.

5.4. Μονοφασική Μεταβολή Γωνίας Φάσης

Η τέταρτη εξεταζόμενη διαταραχή είναι μία μονοφασική μεταβολή της γωνίας φάσης ενώ τα πλάτη των τάσεων διατηρούνται αμετάβλητα. Παρόλο που δεν πρόκειται για βύθιση τάσης, η διαταραχή αυτή προκαλεί έντονη ανισορροπία, καθώς δημιουργεί στιγμιαία σημαντική διαφορά μεταξύ της τάσης του μετατροπέα και της τάσης του δικτύου.

Στο Σχ. 6 παρουσιάζεται η απόκριση του ρεύματος του μετατροπέα για τις τέσσερις μεθόδους κατά τη διάρκεια της μονοφασικής μεταβολής γωνίας φάσης. Όλες οι μέθοδοι περιορίζουν αποτελεσματικά το



(α΄) Elliptical limiter.

(β΄) Virtual admittance limiter.

(γ΄) Positive-sequence priority.

(δ΄) Negative-sequence priority.

Σχήμα 6.: Σύγκριση ρευμάτων κάθε στρατηγικής περιορισμού κατά τη μονοφασική μεταβολή γωνίας φάσης.

ρεύμα εντός του επιτρεπόμενου ορίου, παρότι η απότομη μεταβολή της γωνίας προκαλεί έντονη μεταβατική διέγερση και ανισορροπία στο σύστημα. Ο ελλειπτικός περιοριστής και η μέθοδος εικονικής αγωγιμότητας παρουσιάζουν και σε αυτή την περίπτωση πολύ παρόμοια συμπεριφορά, ενώ οι στρατηγικές προτεραιότητας θετικής και αρνητικής ακολουθίας, διαφοροποιούν περισσότερο την κατανομή των φασικών ρευμάτων κατά τη διάρκεια της διαταραχής. Μετά την αποκατάσταση της αρχικής γωνίας, όλες οι μέθοδοι οδηγούν το ρεύμα πίσω στη μόνιμη κατάσταση, με μικρές διαφοροποιήσεις ως προς τον χρόνο και την μεταβατική συμπεριφορά.

6. Συμπεράσματα

Η συνολική αξιολόγηση των εξεταζόμενων μεθόδων δεν βασίζεται αποκλειστικά στο μέγιστο ρεύμα του μετατροπέα. Στο πλήρες κείμενο της εργασίας εξετάζονται επιπλέον η τάση στο σημείο κοινής σύνδεσης, καθώς και η κατανομή των συνιστωσών θετικής και αρνητικής ακολουθίας του ρεύματος. Τα μεγέθη αυτά επιτρέπουν την πληρέστερη και πιο κατανοητή ερμηνεία του τρόπου με τον οποίο κάθε μέθοδος αξιοποιεί τη διαθέσιμη ικανότητα ρεύματος και επηρεάζει τη συμπεριφορά του μετατροπέα κατά τη διάρκεια ασύμμετρων διαταραχών. Στην παρούσα εκτεταμένη περίληψη, ωστόσο, η παρουσίαση περιορίστηκε κυρίως στα φασικά ρεύματα, ώστε να δοθεί έμφαση στον βασικό στόχο του περιορισμού υψηλών ρευμάτων χωρίς να επαναληφθεί το σύνολο της αναλυτικής σύγκρισης.

Ως προς τον περιορισμό του ρεύματος, όλες οι εξεταζόμενες στρατηγικές επιτυγχάνουν τον βασικό στόχο της διατήρησης του φασικού ρευματος εντός του καθορισμένου ορίου. Παρά τις διαφοροποιήσεις κατά τη διάρκεια των ασύμμετρων διαταραχών, η απόκριση μετά τα σφάλματα είναι συνολικά παρόμοια και το σύστημα επανέρχεται στην αρχική κατάσταση λειτουργίας. Επομένως, η ουσιαστικότερη διαφοροποίηση

μεταξύ των μεθόδων δεν αφορά τόσο το αν μπορούν να περιορίσουν το ρεύμα, αλλά τον τρόπο με τον οποίο κατανέμουν τη διαθέσιμη ικανότητα του μετατροπέα και την επίδραση αυτής της κατανομής στην τάση στο κοινό σημείο σύνδεσης.

Η στρατηγική προτεραιότητας στη θετική ακολουθία διαθέτει μεγαλύτερο μέρος του διαθέσιμου ρεύματος στη θετική συνιστώσα και αυτό έχει ως αποτέλεσμα να ευνοεί τη διατήρηση της θετικής ακολουθίας της τάσης σε υψηλότερα επίπεδα κατά τη διάρκεια της διαταραχής. Αντίθετα, η προτεραιότητα στην αρνητική ακολουθία διαθέτει μεγαλύτερο μέρος του ρεύματος στην αντιμετώπιση της ασυμμετρίας και επιτυγχάνει ισχυρότερη καταστολή της αρνητικής ακολουθίας της τάσης. Ο ελλειπτικός περιοριστής και ο περιορισμός μέσω εικονικής αγωγιμότητας δεν επιβάλλουν αντίστοιχη προτεραιότητα μεταξύ των δύο ακολουθιών και παρουσιάζουν μία περισσότερο ισορροπημένη ενδιάμεση συμπεριφορά, παρέχοντας ταυτόχρονα στήριξη της θετικής ακολουθίας και περιορισμό της ασυμμετρίας.

Κατά συνέπεια, δεν υπάρχει μία στρατηγική που να υπερέχει ανεξάρτητα από τον επιθυμητό στόχο λειτουργίας. Ο ελλειπτικός περιοριστής και η μέθοδος εικονικής αγωγιμόττας θεωρούνται οι καταλληλότερες επιλογές όταν στόχος είναι ο περιορισμός του ρεύματος με τη μικρότερη δυνατή αλλοίωση της κατεύθυνσης και της σχετικής σύνθεσης των διανυσμάτων ρεύματος που ζητούνται από το σύστημα ελέγχου. Οι στρατηγικές προτεραιότητας παραμένουν χρήσιμες σε εφαρμογές όπου απαιτείται συγκεκριμένη ενίσχυση της θετικής ακολουθίας ή μεγαλύτερη καταστολή της αρνητικής ακολουθίας, ωστόσο η επιλογή τους εξαρτάται από το τι επιθυμούμε να επιτύχουμε.

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Declaration on the Use of Generative AI

During the preparation and compilation of this thesis, ChatGPT (OpenAI) was used as a supporting tool for grammar and vocabulary assistance, language editing, and discussion of technical implementation issues. All decisions made about simulation models, results, interpretations, and conclusions were reviewed and validated by the author, who retains full responsibility for the content of this thesis.

Chapter 1.

Introduction

1.1. Transformation of modern power systems

Traditional power systems were developed around massive, centrally located power plants where electrical energy was generated primarily by synchronous generators. While supplying active power, these machines played a fundamental role in regulating the grid and supporting its dynamic behavior and performance. The mechanical inertia provided by these generators helped the power system resist sudden frequency changes, while their electromagnetic characteristics contributed to voltage regulation and helped damp oscillations created in the power system. Consequently, most established practices for grid operation, stability assessment, and protection were built on the assumption that synchronous generation would remain the dominant source of electrical energy.

The imperative need to prevent an energy crisis and the continuing and growing development of renewable energy technologies (decarbonisation, renewable integration and electrification) are progressively changing this conventional structure. Wind turbines, photovoltaic plants, battery energy storage systems and other types of distributed energy resources, are being integrated into power networks on a large scale.

In contrast to conventional synchronous machines, most of these resources are not able to connect directly to an AC grid. Their generated or stored energy is interfaced with the power system through power-electronic converters. For this reason, they are often collectibely referred to as *inverter-based resources* (IBRs) [1], [2].

The transition from synchronous machine dominated generation to the converter-dominated power systems is not merely a replacement of one energy source by another. It changes the mechanisms through which the power system responds to disturbances. The behavior of a synchronous generator is strongly influenced by its inertia, electromagnetic design, and excitation system. On the contrary, the response of an IBR is mainly determined by its measurements, control algorithms, current limits, and modulation strategy. As a result, converter-dominated systems may exhibit faster dynamics, lower natural inertia, lower short circuit current, and interactions between control systems. These characteristics become profoundly important in weak grids and during disturbances, such as voltage sags, phase jumps, frequency change events, and short circuit faults [3].

A high penetration of conventionally controlled IBRs can therefore reduce some of the stabilizing properties that were naturally provided by synchronous machines. Grid-following converters, which represent the majority of currently installed grid-connected inverters, regulate their injected current after synchronizing with an already established grid voltage. Although this approach performs effectively in strong networks, its operation becomes more challenging as the number of synchronous generators decreases and the grid voltage is less stiff.

In such conditions, relying exclusively on grid-following control may pose challenges to grid synchronization, voltage regulation, frequency support, and system strength [1].

Nevertheless, the growing use of power electronics brings new possibilities. The dynamic response of an inverter is determined by its control system, since its behavior can be determined through the implemented control algorithms. Moreover, power-electronic converters operate at high switching frequencies, allowing them to react to changes in the grid fast. With a suitable control structure, grid-forming inverters can contribute to voltage and frequency regulation, emulate inertial behavior, support weak grids, and contribute to a black start. All in all, grid-forming control has emerged as a promising solution for enabling IBR to provide a larger share of the services traditionally supplied by synchronous generators [3], [4]. The distinction between grid-following and grid-forming operation is introduced and analyzed in the following section.

1.2. Grid-Following Inverters

Grid-following (GFL) control is currently the most widely used approach for connecting IBRs to the power system. A GFL inverter operates under the assumption that the voltage magnitude, frequency, and phase angle at the point of common coupling (PCC) are established by the external grid. Its main objective is therefore to synchronize with this voltage and regulate the current injected into the network [2], [3].

Synchronization is commonly achieved through a Phase Locked Loop (PLL), which estimates the angle and frequency of the PCC voltage. The estimated angle is used to transform the measured three phase voltages and currents into a synchronous rotating reference frame. When the d-axis is aligned with the PCC voltage vector, the active and reactive power can be regulated through the corresponding current components. In this case, the d-axis current is mainly associated with active power exchange, while the q-axis current is mainly associated with reactive power exchange [5].

The current references are normally generated by outer power, DC-link voltage, or grid support control loops. An inner current controller then regulates the converter output current and produces the voltage reference applied to the modulation stage. Since the injected current is directly controlled, the external behavior of a GFL inverter can be approximated as a controlled current source with a high parallel impedance [1].

Additional control functions can be included to support the power system during voltage or frequency disturbances. For example, the active and reactive current references can be modified to provide frequency support, or reactive current exchange during voltage deviations. However, these responses depend on voltage measurement, synchronization, and the dynamics of the additional control loops. Therefore, this support is provided through the converter control system rather than arising directly from its basic electrical characteristics [1], [3].

The dependence on an external voltage reference represents an important limitation of grid-following control. In weak grids, where the short circuit ratio is low, the interaction between the PLL, the current controller, and the grid impedance may reduce the converter's stability margin. Moreover, a conventional GFL inverter cannot independently establish the voltage and frequency of an islanded network. Consequently, a system composed exclusively of GFL inverters requires another voltage forming unit to provide the reference needed for synchronization [1], [2], [4].

1.3. Grid-forming inverters

Although grid-forming control has received much attention in recent years, a single and widely accepted definition has not yet been established. Various control approaches and technical specifications emphasize different capabilities. Nevertheless, the most universally accepted characteristic of a grid-forming (GFM) inverter is its ability to behave as a controlled voltage source behind an impedance [1], [4].

In this representation, the impedance describes the opposition encountered by the AC current flowing between the converter and the grid. In sinusoidal steady state operation, it is generally expressed as

$$Z = R + jX, \tag{1.1}$$

where R is the resistance and X is the reactance. The physical impedance between the internal converter voltage and the PCC may include the converter's filter, the transformer's leakage impedance and the impedance of the connecting line. This total impedance determines how much current flows when there is a difference in voltage magnitude or phase angle between the converter and the grid [1], [5].

Together with the physical impedance, a grid-forming controller can implement a *virtual impedance*. Virtual impedance does not correspond to an additional physical resistor or inductor. Instead, it is created through the control system by calculating a virtual voltage drop from the measured converter current and subtracting it from the original voltage reference. A general time domain expression is

$$v_{\text{virt}} = R_v i + L_v \frac{\mathrm{d}i}{\mathrm{d}t}, \tag{1.2}$$

where R_v and L_v are the selected virtual resistance and inductance, respectively. The modified converter voltage reference is therefore given by

$$v_{\text{ref}} = v^{\star}_{\text{GFM}} - v_{\text{virt}}. \tag{1.3}$$

Through this control action, the effective output impedance of the converter can be shaped without installing a correspondingly large physical component. Virtual impedance can improve power sharing and damping, reduce the sensitivity of the output current to grid voltage variations, and contribute to current limitation during faults. However, excessively large virtual impedance values may produce a considerable reduction in the converter terminal voltage and weaken its voltage support capability [3], [6].

Unlike a grid-following inverter, which regulates the current injected into an already established grid voltage, a GFM inverter generates its own internal voltage magnitude and phase angle. The power exchanged with the network is then determined by the difference between this internal voltage and the voltage at the point of common coupling, together with the impedance between them. As a result, a GFM inverter can provide a voltage reference to nearby loads and converters, even when the external grid is weak or absent [1].

In most GFM control approaches, synchronization is achieved through the active power exchange with the network rather than a PLL being used as the primary synchronization mechanism. The internal frequency and voltage magnitude are generated by outer control loops, such as droop control, virtual synchronous machine

control, or other power synchronization methods. These control structures can reproduce selected dynamic characteristics of synchronous generators, including frequency response, damping, and synthetic inertial behavior [1], [3].

The voltage-source behavior of GFM inverters makes them particularly suitable for operation in weak grids and islanded systems. Depending on the implemented control and available energy source, they can contribute to voltage and frequency regulation, support system strength, enable the operation of networks with a high share of inverter-based resources, and provide black start capability [4].

However, the same voltage-source behavior also creates an important challenge during grid disturbances. Since the output current is not imposed directly, but is determined by the difference between the converter and grid voltages, sudden voltage dips or phase angle changes may produce a rapid increase in current. Therefore, grid-forming inverters require dedicated current-limiting and fault ride through strategies to protect their semiconductor devices, while maintaining stable operation and supporting the power system [1], [3].

1.4. Asymmetrical Faults and Negative-Sequence Quantities

Power system faults can be broadly classified as symmetrical or asymmetrical. A symmetrical three phase fault affects all three phases in the same manner and therefore preserves the balance of the three phase system. In contrast, asymmetrical faults affect the phases unequally and result in unbalanced voltage and current waveforms. Common examples include single line to ground, line to line, and double line to ground faults. During such disturbances, the phase voltages may experience different magnitudes and phase angle deviations, while the converter currents can become significantly unbalanced.

The method of symmetrical components provides a convenient representation of unbalanced three phase quantities. According to this method, an arbitrary three phase voltage or current can be expressed as the superposition of positive-, negative-, and zero-sequence components:

$$x_{abc} = x^{+}_{abc} + x^{-}_{abc} + x^{0}_{abc}, \tag{1.4}$$

where the superscripts +, −, and 0 denote the positive-, negative-, and zero-sequence components, respectively. The positive-sequence component consists of three balanced phasors separated by 120° and following the normal phase order, meaning that the phases appear in the sequence *a-b-c*, with a phase displacement of 120° between consecutive phases. The negative-sequence component is also balanced, but it rotates in the opposite direction and therefore follows the reverse phase order. The zero-sequence components are equal in magnitude and phase in all three phases [5].

In a three-phase, three-wire system without a neutral or another zero-sequence current path, zero-sequence current cannot flow. Consequently, the behavior of the converter during asymmetrical faults is primarily determined by the positive- and negative-sequence voltage and current components. Under balanced operating conditions, only the positive-sequence component is ideally present. When an asymmetrical fault occurs, the positive-sequence voltage generally decreases, while a negative-sequence voltage component appears at the point of common coupling.

The presence of negative-sequence quantities introduces additional challenges for inverter control. In a synchronous reference frame rotating in the same direction as the positive-sequence voltage, the positive-sequence

quantities appear as DC components. However, the negative-sequence components rotate in the opposite direction and therefore appear as oscillations at twice the fundamental frequency. Conventional positive-sequence controllers based on proportional-integral regulators are not designed to independently regulate these oscillating components. Therefore, separate sequence extraction and control methods are required when independent positive- and negative-sequence regulation is desired [3].

Negative-sequence voltage can adversely affect rotating electrical machines, increase voltage unbalance, and alter the operation of protection systems. IBRs may contribute negative-sequence current during unbalanced conditions, but its magnitude and phase depend strongly on the adopted control structure, the inner controller parameters, and the employed current-limiting method. Consequently, the negative-sequence response of an inverter cannot generally be assumed to be identical to that of a synchronous generator [7], [8].

The interaction between asymmetrical faults, negative-sequence control, and current limitation constitutes the main focus of this thesis. The following sections first introduce the need for current limitation in grid-forming inverters and subsequently present the control methods developed to regulate and prioritize the positive- and negative-sequence quantities.

1.5. Overcurrent Limiting for Grid-Forming Inverters

1.5.1. Need for Current-Limiting Methods for Grid-Forming Inverters

The different control principles of grid-following and grid-forming inverters lead to substantially different responses during grid disturbances. A grid-following inverter primarily regulates its injected current and therefore behaves approximately as a controlled current source. In contrast, a grid-forming inverter establishes an internal voltage phasor and behaves as a controlled voltage source behind an impedance. Consequently, immediately after a sudden change in the grid voltage, the internal voltage of the GFM inverter tends to remain approximately unchanged, while its output current varies according to the new network conditions [1].

Considering a simplified equivalent circuit, the inverter output current can be expressed as

$$i = \frac{e - v_g}{Z_{eq}}, \tag{1.5}$$

where e denotes the internal voltage source of the GFMI, v_g is the grid voltage, and Z_{eq} represents the equivalent impedance between the two voltage sources. Equation (1.5) indicates that the output current is highly sensitive to variations in the magnitude and phase angle of the grid voltage. Therefore, severe voltage sags, short circuits, phase jumps, and other large disturbances can produce a rapid increase in the converter current [1], [6].

The resulting overcurrent is particularly critical because power-electronic converters have a considerably lower short term overcurrent capability than synchronous generators. Conventional synchronous machines can supply fault currents several times greater than their rated current for a limited period, whereas the allowable current of an inverter is typically restricted to a relatively small margin above its rated value. The exact limit depends on the semiconductor technology, thermal design, switching frequency, and duration

of the overcurrent, but values in the approximate range of 1.1–2 p.u. are commonly considered in previous research [2], [3], [6].

If this limit is exceeded, the semiconductor switches, filter components and other elements of the converter may experience excessive electrical and thermal stress. Hardware oversizing could increase the available fault current capability; however, it would also increase the cost, physical size, and cooling requirements of the converter. As a result, software-based current limiting methods are generally employed to protect the converter without requiring substantial hardware oversizing [3].

At the same time, the converter may be required to remain connected and support the grid during a disturbance. Current limitation must therefore be coordinated with fault ride through and grid support objectives. For instance, when a voltage sag occurs, the available current capacity may need to be used for reactive current injection, active power transfer, voltage support, or a combination of these services. Hence, the objective of a current limiter is not simply to restrict the output current, but to determine the converter behavior while the available current capability is constrained [2], [6].

This issue becomes more complex under asymmetrical faults. In this case, the positive-sequence and negative-sequence current components simultaneously contribute to the instantaneous phase currents. Therefore, current-limiting methods for unbalanced conditions must consider not only the total current magnitude, but also the allocation of the available converter capacity between the positive and negative sequences. In particular, the resulting currents of all three phases should be evaluated, since the positive and negative sequence components combine differently in each phase. Relying only on the individual sequence magnitudes or on a simple addition of their maximum values does not fully represent the actual phase current distribution and may lead to an unnecessarily conservative allocation of the available current capability.

1.5.2. Challenges of Current-Limiting Methods for Grid-Forming Inverters

When the current limit is reached, the GFM inverter can no longer maintain its unrestricted normal operation behavior. The current-limiting mechanism can include modifying the voltage reference, the current reference, the virtual impedance, the power references, or even the switching signals of the converter. Consequently, once activated, the limiter may become the dominant element determining the dynamic behavior of the inverter during the disturbance and the subsequent recovery process [3].

A wide range of current-limiting approaches has been proposed for grid-forming inverters. Direct current-limiting methods constrain the current reference or act directly on the converter switching signals and can therefore provide fast and accurate current limitation. Nevertheless, current reference saturation may alter the external behavior of the inverter from that of a voltage source towards that of a controlled current source. It may also cause integrator windup and complicate the transition back to normal operation.

In contrast, virtual impedance methods limit the current indirectly. Instead of directly saturating the current reference, they generate an additional virtual voltage drop that increases the effective output impedance of the converter. As a result, the sensitivity of the output current to disturbances in the grid voltage is reduced. This approach can preserve the voltage source behavior of the grid-forming inverter to a certain extent, although its current-limiting accuracy and transient response depend on the selected virtual impedance parameters, the controller bandwidth, and the network conditions [3], [6].

There is not a single current-limiting method providing optimum performance while respecting all relevant criteria. The current-limiting design involves a compromise between device protection, grid support,

synchronization stability, power quality, and recovery performance [2], [3].

Asymmetrical disturbances introduce an additional allocation problem. The limited converter capacity must be distributed between positive-sequence and negative-sequence currents and, within each sequence, between the corresponding d- and q-axis components. Prioritizing the positive sequence current can support the positive-sequence voltage, whereas prioritizing the negative-sequence current can reduce voltage unbalance. Since these objectives may conflict, the selected priority strategy directly affects the grid support performance and the distribution of the converter current among the three phases, while the maximum phase current remains constrained by the same predefined limit [8].

For all of these reasons, the current-limiting strategies are evaluated not only for their ability to maintain the phase currents below their maximum permissible value, but also according to the positive- and negative-sequence components, PCC voltage support, waveform quality, and post-fault recovery.

1.6. Objective and Structure of the Thesis

1.6.1. Objective and Scope of the Thesis

The primary objective of this thesis is to investigate the operation and current limitation of a grid-forming inverter under asymmetrical grid disturbances.

To achieve this objective, a grid-forming inverter control system is implemented in the PLECS simulation environment. The control structure includes separate extraction and regulation of the positive- and negative-sequence voltage and current components. Several current-limiting strategies are then implemented and evaluated under different asymmetrical fault conditions.

The investigated methods include direct limitation of the current references and virtual-admittance-based current limitation. Different priority rules are also examined for distributing the available current capacity between the positive and negative sequences. Particular attention is given to maintaining the phase currents below the maximum permissible converter current, since the superposition of the two sequence components may result in unequal phase current magnitudes during unbalanced operation.

The performance of the implemented control methods is assessed through time-domain simulations using an averaged converter model. This choice allows the control response and the interaction between the positive- and negative-sequence to be investigated without the additional high-frequency effects caused by the switching process.

The scope of this thesis is limited to converter-level control and simulation studies of a three phase voltage-source converter. This work does not cover detailed thermal modeling of semiconductor devices, coordination with power system protection, hardware implementation, or hardware-in-the-loop testing.

1.6.2. Structure of the Thesis

The remainder of this thesis is organized as follows.

Chapter 2 presents the theoretical background required for the development of the studied grid-forming converter. It includes the principles of voltage-source converters, grid-forming operation and the control structure. Also the positive- and negative-sequence control loops and the different control stages leading to the generation of the converter voltage references applied to the averaged converter model are explained.

Chapter 3 describes the implementation of the studied system and its control structure in PLECS. The converter, grid, measurement and sequence-extraction systems and the positive- and negative-sequence controllers are presented from a practical implementation perspective.

Chapter 4 presents the different current-limiting methods implemented, while also mentioning the disturbances used for their evaluation.

Chapter 5 presents the simulation results for each of the considered grid disturbances, while the behavior of the converter is examined.

Chapter 6 discusses the main observations obtained from the simulation results and compares the performance in terms of current limiting and voltage support of the investigated current-limiting methods.

Finally, Chapter 7 summarizes the main conclusions of the thesis and outlines possible directions for future work based on the findings of the present study.

Chapter 2.

Theoretical Background

This chapter presents the theoretical background required to describe the operation and control of the grid-forming converter implemented in this thesis. Initially, the basic structure and operating principles of three phase voltage source converters are introduced and then the average converter model used is described, together with its AC side interface and connection to the grid.

The GFM structure is presented in detail. The calculation of active and reactive power is first introduced, followed by the outer control loops responsible for creating the magnitude and phase angle of the internal voltage source of the GFMI. The way the inner control structure is implemented is then explained, including the control stages used to regulate the converter response. Eventually, the complete control process leading to the generation of the converter voltage reference applied to the averaged model is presented.

These concepts establish the theoretical basis for the practical PLECS implementation and the current-limiting strategies presented in the following chapters.

2.1. Voltage Source Converters in Power Systems

2.1.1. Basic Structure and Operating Principle

A voltage source converter (VSC) is a power electronic converter that interfaces a DC subsystem with an AC system. Its main function is to generate a manageable AC side voltage from a DC link by controlling power semiconductor switches. Depending on the direction of the active power flow, the converter may operate either as an inverter, transferring power from the DC side to the AC system, or as a rectifier, transferring power from the AC system to the DC side [5].

In this thesis The converter is represented using an averaged converter model. This representation is suitable for the development and evaluation of the grid-forming control system, the positive- and negative-sequence controllers, and the implemented current-limiting methods.

Also, the converter is connected to the AC system through an interface impedance, which may represent the filter impedance, transmission line impedance,transformer leakage impedance, or a combination of these. Its current is determined by the voltage difference across the interface impedance. Consequently, sudden changes in the magnitude or phase angle of the grid voltage may produce a rapid variation in the converter current. The current rate of change is also inversely proportional to the interface inductance, which directly affects the transient current response during grid disturbances.

Lastly, the GFM is controlled to behave as a voltage source behind an impedance. During a grid fault, the internally generated converter voltage cannot always follow the sudden change in the PCC voltage. The resulting voltage difference may therefore cause the converter current to exceed its permissible value, making an appropriate current-limiting mechanism necessary. This will be more clear as we move on to the following sections.

2.1.2. Averaged Converter Model

In this thesis, as already mentioned, the voltage source converter is represented by an averaged model. In this modeling approach, the individual semiconductor switches and the high-frequency switching process are not represented explicitly. Instead, the converter is modeled as a controllable three phase voltage source that reproduces the fundamental frequency component of the converter output voltage [5], [9].

By neglecting the switching frequency components, the averaged model significantly reduces the computational requirements of the simulations while preserving the electrical and control dynamics needed for the objectives of this study. It is therefore suitable for the development and evaluation of the grid-forming control system, the positive- and negative-sequence controllers, and the implemented current-limiting strategies.

The converter voltage is directly determined by the voltage reference generated by the control system, subject to the voltage limits of the adopted converter representation. Consequently, the analysis focuses on the fundamental frequency response of the converter rather than on switching harmonics, semiconductor losses, or PWM-induced current ripple.

2.1.3. AC Side Interface and Grid Equivalent

The averaged converter model is connected to an equivalent three phase grid through a series impedance in each phase. The converter and the grid are represented by controlled voltage sources. The grid voltage magnitude and phase angle can be modified independently to reproduce balanced and asymmetrical disturbances.

The external grid is represented by three controlled sinusoidal voltage sources. The magnitude and phase angle of each phase can be modified independently, allowing both balanced and unbalanced grid disturbances to be imposed. In particular, asymmetrical voltage sags and phase-angle disturbances are introduced by changing the corresponding grid voltage references, rather than by explicitly modelling a physical short circuit connection.

The converter and the equivalent grid voltage source are connected through a series impedance in each phase. The resistive and inductive elements shown in the model are represented by the total equivalent resistance R_Σ and inductance L_Σ of the electrical path between the two voltage sources. Assuming identical parameters in the three phases, the current dynamics can be expressed as

$$L_\Sigma \frac{\mathrm{d}i_{abc}}{\mathrm{d}t} = v_{c,abc} - v_{g,abc} - R_\Sigma i_{abc}, \tag{2.1}$$

where $v_{c,abc}$ is the three phase voltage generated by the averaged converter model, $v_{g,abc}$ is the equivalent grid voltage, and i_{abc} is the converter output current [5].

Equivalently, the rate of change of the current is given by

$$\frac{\mathrm{d}i_{abc}}{\mathrm{d}t} = \frac{1}{L_\Sigma}\left(v_{c,abc} - v_{g,abc} - R_\Sigma i_{abc}\right). \tag{2.2}$$

These equations show that the converter current is determined by the voltage difference between the internally generated converter voltage and the grid voltage, together with the impedance connecting them. This behavior is consistent with the representation of a grid-forming converter as a controlled voltage source behind an impedance [10].

The value of the equivalent inductance has a direct influence on the transient current response. For the same voltage difference, a smaller inductance produces a larger rate of change of the current and therefore allows the current to rise more rapidly after a disturbance. A larger inductance slows the initial current response. However, the resulting current magnitude also depends on the resistance, the duration and severity of the disturbance, and the action of the converter control and current-limiting methods.

2.2. Grid-Forming Control Model

Most grid-forming control methods can be described by a common structure consisting of an outer control layer and an inner control layer. The outer layer determines the magnitude, angular frequency, and phase angle of the internal converter voltage, whereas the inner layer regulates the electrical quantities required to reproduce this voltage source behavior at the converter terminals [1].

The main inputs of the implemented control system are the measured three phase voltages and currents, together with the active and reactive power references.

Under asymmetrical grid conditions, the positive- and negative-sequence quantities are processed in separate rotating reference frames. The voltage references generated by the positive- and negative-sequence controllers in their respective rotating reference frames, are transformed back into three phase quantities and combined to produce the final converter voltage reference. The sequence separation and negative-sequence control methods are presented in detail in the following sections.

2.2.1. Power Calculation

The outer control loops require measurements of the active and reactive power exchanged between the converter and the network. In the implemented model, these quantities are evaluated at the PCC and calculated from the positive-sequence voltage and current components expressed in the synchronous dq reference frame.

Using the adopted amplitude invariant transformation, the positive-sequence active power is expressed as

$$P^{+} = \frac{3}{2}\left(v_d^{+} i_d^{+} + v_q^{+} i_q^{+}\right), \tag{2.3}$$

while the positive-sequence reactive power is given by

$$Q^{+} = \frac{3}{2}\left(v_q^{+} i_d^{+} - v_d^{+} i_q^{+}\right), \tag{2.4}$$

where v_d^{+}, v_q^{+}, i_d^{+}, and i_q^{+} denote the positive-sequence voltage and current components in the synchronous reference frame [5].

The positive-sequence active and reactive powers are used as feedback signals for the outer grid-forming control loops. This choice is not only intended to remove the double frequency power oscillations that appear under unbalanced conditions. Although a notch filter tuned at $2\omega_0$ could attenuate these oscillations, the filtered total power would generally still contain contributions associated with both the positive- and negative-sequence systems.

Under asymmetrical conditions, the instantaneous active and reactive powers can be expressed in a general form as

$$P(t) = P^{+} + P^{-} + \widetilde{P}_{2\omega}(t), \tag{2.5}$$

$$Q(t) = Q^{+} + Q^{-} + \widetilde{Q}_{2\omega}(t), \tag{2.6}$$

where P^{+} and Q^{+} are associated with the positive-sequence voltage and current, P^{-} and Q^{-} are associated with the negative-sequence quantities, and the oscillatory terms result from the interaction between the two sequences.

As previously mentioned, a notch filter could suppress the oscillatory terms, but it would not separate the remaining positive- and negative-sequence power components. Consequently, the negative-sequence response could still modify the frequency, angle, and voltage magnitude generated by the main grid-forming droop loops.

In the implemented control structure, the grid-forming synchronization mechanism is associated with the balanced positive-sequence internal voltage. Therefore, P^{+} is used by the active power frequency droop loop to generate the positive-sequence angle, while Q^{+} is used by the reactive power voltage droop loop to determine the positive-sequence voltage magnitude. The negative-sequence response is regulated separately by a dedicated controller, since it has a different control objective and rotates in the opposite direction [7].

This separation prevents the negative-sequence voltage support action from directly disturbing the positive-sequence synchronization and voltage forming mechanisms. It also reduces the coupling between the two sequence controllers and avoids the additional delay and frequency dependent behavior that would be introduced if the droop feedback relied only on filtered total power measurements.

2.2.2. Outer Grid-Forming Control

The outer grid-forming control determines the internal positive-sequence voltage reference using the positive-sequence active and reactive powers calculated in the previous subsection. In the implemented model, it consists of two control branches. The active power branch determines the internal angular frequency and phase angle required for synchronization, whereas the reactive power branch determines the magnitude of the internal positive-sequence voltage [1].

The outputs of the two branches define the internal voltage phasor as

$$\underline{E}^{+\star} = E^{+\star} e^{j\theta_{\mathrm{GFM}}}, \tag{2.7}$$

where $E^{+\star}$ is the positive-sequence voltage magnitude reference and θ_{GFM} is the internally generated grid-forming angle.

The conventional active-power angular-frequency and reactive-power voltage droop relationships are adopted. For a predominantly inductive connection, active power is mainly influenced by the phase angle difference between the converter and the grid, while reactive power is mainly influenced by their voltage magnitude difference. This provides the basis for using the active power error to modify the internal frequency and the reactive power error to modify the voltage magnitude [1], [11]. Nevertheless, this separation is approximate, since the resistance of the connecting impedance, the operating point, and large disturbances may introduce coupling between active and reactive power.

The implementation of the two outer control branches is described in the following subsections.

Active Power Synchronization Loop

The active power synchronization loop generates the angular frequency and phase angle of the internal positive-sequence voltage, as already mentioned. It is therefore responsible for synchronizing the converter with the connected grid. In the implemented model, this functionality is provided by a conventional active-power angular-frequency droop controller. [1], [12], [13].

The active power error is defined as

$$e_P = P^{\star} - P^{+}, \tag{2.8}$$

where $P^{\star}$ is the active power reference and P^{+} is the measured positive-sequence active power. The resulting frequency deviation is calculated as

$$\Delta\omega = k_p e_P = k_p \left(P^{\star} - P^{+}\right), \tag{2.9}$$

where k_p is the active power droop coefficient. The internal angular frequency is then given by

$$\omega_{\mathrm{GFM}} = \omega_0 + \Delta\omega = \omega_0 + k_p \left(P^\star - P^+\right), \tag{2.10}$$

where ω_0 denotes the nominal angular frequency.

The synchronization mechanism can be understood from the active power transfer across the predominantly inductive connection between the converter and the grid. In simplified form, the transferred active power is approximately related to the angle difference δ between the internal converter voltage and the grid voltage according to

$$P \approx \frac{EV_g}{X_\Sigma} \sin\delta, \tag{2.11}$$

where E and V_g are the converter and grid voltage magnitudes, respectively, and X_Σ is the equivalent reactance between them.

When $P^+ < P^\star$, the droop controller produces a positive frequency deviation. The internal converter angle therefore advances relative to the grid angle, increasing the power-angle and, within the stable region of the power angle characteristic, increasing the transferred active power. Conversely, when $P^+ > P^\star$, the generated frequency decreases, causing the power angle and the active power output to decrease. Through this feedback mechanism, the converter can synchronize with the grid without requiring a conventional PLL for the generation of its main internal angle during normal operation [12], [13].

The internal angle is obtained by integrating the generated angular frequency:

$$\theta_{\mathrm{GFM}} = \int \omega_{\mathrm{GFM}} \, \mathrm{d}t. \tag{2.12}$$

Substituting Equation (2.10) into Equation (2.12) gives

$$\theta_{\mathrm{GFM}} = \omega_0 t + k_p \int \left(P^\star - P^+\right) \mathrm{d}t. \tag{2.13}$$

Equation (2.13) has the same mathematical form as the power synchronization control proposed for grid-connected voltage-source converters. Thus, the implemented active power droop controller can also be viewed as a power synchronization controller, even though droop control and power synchronization control were originally developed for different applications. [1], [12].

The structure of the implemented active power synchronization loop is shown in Fig. 2.1.

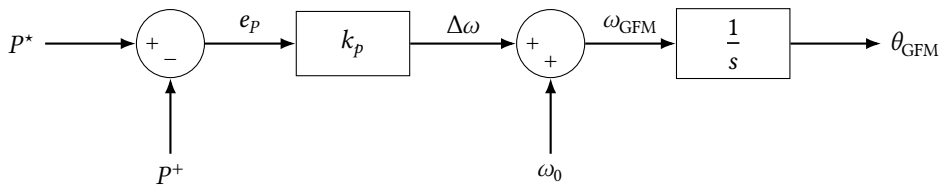


Figure 2.1.: Block diagram of the implemented active power synchronization loop.

The generated angle is used to construct the internal positive-sequence voltage reference and to define the corresponding synchronous reference frame.

No low-pass filter, swing equation model, or additional inertia emulating state is included in the implemented active power loop. Consequently, the controller belongs to the non-inertial class of grid-forming controls. According to Equation (2.10), a change in the active power error is transferred directly to the frequency command, whereas the phase angle remains continuous because it is obtained through integration. By contrast, introducing a low-pass filter into the active power path would add a dynamic state, smooth the frequency response, and produce an inertia-like behavior [1].

Reactive Power-Voltage Loop

The reactive power-voltage loop determines the magnitude of the internal positive-sequence voltage. In the implemented model, this function is realised through a conventional reactive power-voltage droop controller. The controller modifies the voltage magnitude according to the difference between the positive sequence reactive power and its reference value [11], [13], [14].

The reactive power error is defined as

$$e_Q = Q^\star - Q^+, \tag{2.14}$$

where $Q^\star$ is the reactive power reference and Q^+ is the measured positive-sequence reactive power. The corresponding change in the internal voltage magnitude is

$$\Delta E = k_q e_Q = k_q \left(Q^\star - Q^+\right), \tag{2.15}$$

where k_q denotes the reactive power droop coefficient. The positive-sequence voltage magnitude reference is therefore calculated as

$$E^{+\star} = E_0 + k_q \left(Q^\star - Q^+\right), \tag{2.16}$$

where E_0 is the nominal voltage magnitude.

The operation of the controller is related to the reactive power transfer through the predominantly inductive connection between the converter and the grid. For a simplified inductive connection, the reactive power supplied by the converter can be approximated by

$$Q_{\mathrm{PCC}} \approx \frac{E V_{\mathrm{PCC}} \cos\delta - V_{\mathrm{PCC}}^2}{X_\Sigma}, \tag{2.17}$$

where E and V_{PCC} are the internal converter voltage magnitude and the PCC voltage magnitude, respectively, X_Σ is the equivalent reactance between them, and δ is their phase angle difference. Consequently, for a given

PCC voltage and a relatively small phase angle difference, increasing the internal voltage magnitude generally increases the reactive power supplied by the converter at the PCC.

When $Q^+ < Q^\star$, the reactive power error is positive and the controller increases the internal voltage magnitude reference. This tends to increase the reactive power injection. Conversely, when $Q^+ > Q^\star$, the voltage reference is reduced, thereby decreasing the reactive power output.

The positive-sequence synchronous reference frame is defined using the internally generated angle θ_{GFM}, so that its d-axis is aligned with the internal positive-sequence voltage reference. With this choice, the voltage magnitude is assigned entirely to the d-axis, while the corresponding q-axis reference is set to zero:

$$E_d^{+\star} = E^{+\star}, \qquad E_q^{+\star} = 0. \tag{2.18}$$

Therefore, the internal positive-sequence voltage reference is expressed as

$$E_{dq}^{+\star} = \begin{bmatrix} E^{+\star} \\ 0 \end{bmatrix}. \tag{2.19}$$

This alignment is a reference frame convention and does not imply that the measured q-axis voltage or the final q-axis converter voltage command remains zero during transient operation.

The implemented reactive power droop controller does not contain an integrator or any other additional dynamic state. It establishes a direct proportional relationship between the reactive power error and the internal voltage magnitude. Therefore, conventional droop control does not generally force the reactive power to follow its reference exactly, and a steady state deviation from $Q^\star$ may remain.

In the present implementation, a relatively low value of k_q is selected. This reduces the sensitivity of the internal voltage reference to reactive power variations and avoids aggressive voltage corrections, resulting in a more damped and robust response. The corresponding disadvantage is weaker reactive power regulation, since the generated voltage adjustment may not be sufficient to bring Q^+ close to its reference. The selected value therefore represents a compromise between reactive-power tracking and the stability and robustness of the overall converter control system.

It should be noted that increasing the droop coefficient does not guarantee exact steady state tracking, since the controller does not include integral action. If zero steady state reactive power error were required, a PI-based reactive power controller could be employed, although this would introduce an additional dynamic state and require further stability-oriented tuning.

2.2.3. Inner Control Structure

The outer grid-forming control determines the magnitude and angle of the internal positive-sequence voltage reference. In the implemented multi-loop structure, this reference is compared with the measured positive-sequence voltage at the point of common coupling. The resulting voltage error is then converted into a positive-sequence current reference.

Two alternative methods are considered for this purpose: a conventional PI voltage controller and a virtual admittance controller. The PI controller regulates the voltage error through proportional and integral action, whereas the virtual admittance approach establishes a direct relationship between the voltage difference and the generated current reference. The two voltage control approaches are described in the following subsections.

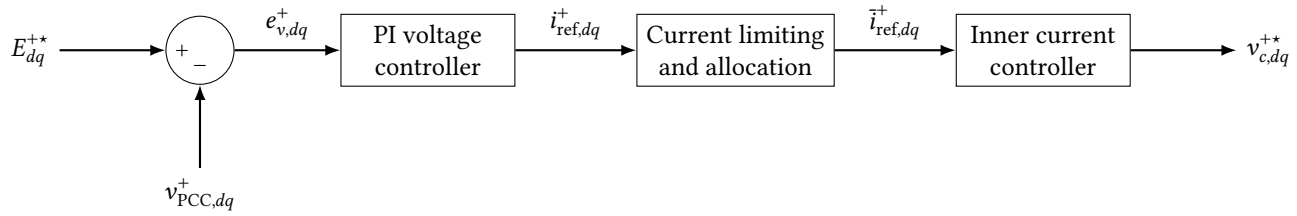


Figure 2.2.: Positive-sequence control structure based on a PI voltage controller.

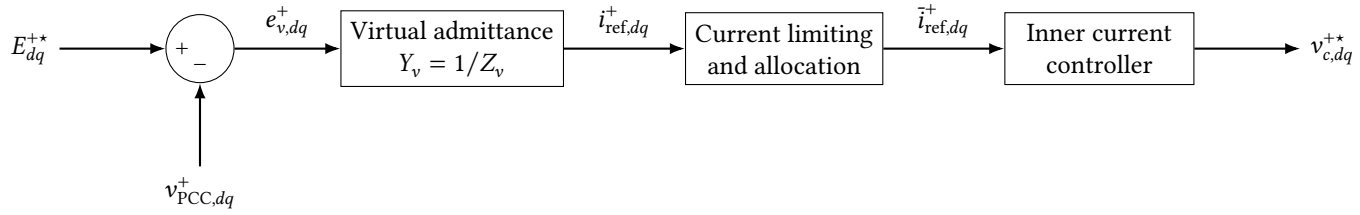


Figure 2.3.: Positive-sequence control structure based on a virtual admittance loop.

Figures 2.2 and 2.3 illustrate the two alternative methods considered for converting the positive-sequence voltage error into a current reference. In both cases, the generated current reference is subsequently processed by the current limiting and current allocation stage before being applied to the inner current controller.

PI Voltage Control

The positive-sequence voltage error can be processed by a PI controller. The controller converts the difference between the internal voltage reference and the measured PCC voltage into positive-sequence current references [1], [5].

The current references are calculated as

$$i^{+}_{d,\mathrm{ref}} = k_{pv} e^{+}_{v,d} + k_{iv} \int e^{+}_{v,d}\, \mathrm{d}t, \tag{2.20}$$

$$i^{+}_{q,\mathrm{ref}} = k_{pv} e^{+}_{v,q} + k_{iv} \int e^{+}_{v,q}\, \mathrm{d}t, \tag{2.21}$$

where k_{pv} and k_{iv} are the proportional and integral gains of the voltage controller. The proportional term provides a rapid response to voltage deviations, while the integral term improves steady state voltage tracking.

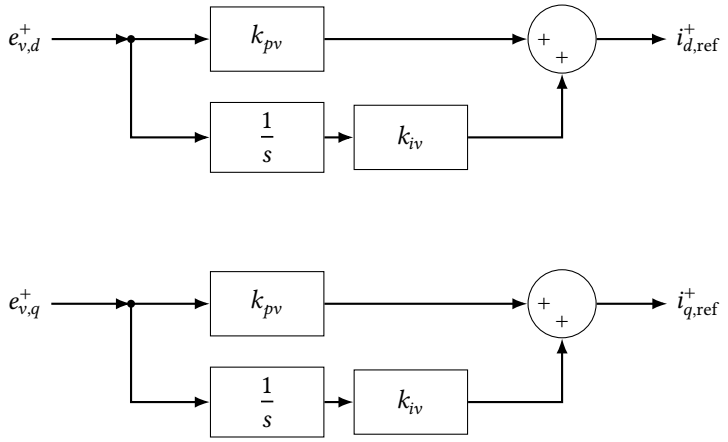


Figure 2.4.: Implementation of the positive-sequence PI voltage controller.

When the current references generated by the voltage controller exceed the available converter capability, the current-limiting stage prevents the inner controller from following the original references. Under these conditions, the voltage error may remain non zero and the integral terms of the PI controllers may continue to increase. This phenomenon, known as integrator windup, can cause overshoot and delay the post-fault recovery. Anti-windup or conditional integration mechanisms are therefore required to prevent excessive accumulation of the controller states during current limitation.

Virtual Admittance Control

The virtual admittance control emulates a series impedance between the internal voltage source of the grid-forming converter and the measured PCC voltage. It converts the voltage difference across this virtual impedance into a current reference, thereby reproducing the behavior of a controlled voltage source behind an impedance [1], [15].

The positive-sequence voltage error vector is defined as

$$e^+_{v,dq} = E^{+\star}_{dq} - v^+_{\mathrm{PCC},dq}, \tag{2.22}$$

or, in terms of its components,

$$e^+_{v,d} = E^{+\star}_{d} - v^+_{\mathrm{PCC},d}, \qquad e^+_{v,q} = E^{+\star}_{q} - v^+_{\mathrm{PCC},q}. \tag{2.23}$$

The emulated virtual impedance is expressed as

$$Z_v = R_v + jX_v, \tag{2.24}$$

where R_v is the virtual resistance and X_v is the virtual reactance. When the reactance represents a virtual inductance L_v, it is defined at the nominal angular frequency as

$$X_v = \omega_0 L_v. \tag{2.25}$$

Using complex vector notation, the positive-sequence current reference is calculated as

$$\underline{i}^{+}_{\text{ref}} = \frac{\underline{E}^{+\star} - \underline{v}^{+}_{\text{PCC}}}{R_v + jX_v} = \frac{\underline{e}^{+}_{v}}{R_v + jX_v}. \tag{2.26}$$

In the positive-sequence synchronous dq reference frame, Equation (2.26) becomes

$$\begin{bmatrix} i^{+}_{d,\text{ref}} \\ i^{+}_{q,\text{ref}} \end{bmatrix} = \frac{1}{R_v^2 + X_v^2} \begin{bmatrix} R_v & X_v \\ -X_v & R_v \end{bmatrix} \begin{bmatrix} e^{+}_{v,d} \\ e^{+}_{v,q} \end{bmatrix}. \tag{2.27}$$

Equivalently, the individual current references are given by

$$i^{+}_{d,\text{ref}} = \frac{R_v e^{+}_{v,d} + X_v e^{+}_{v,q}}{R_v^2 + X_v^2}, \tag{2.28}$$

$$i^{+}_{q,\text{ref}} = \frac{-X_v e^{+}_{v,d} + R_v e^{+}_{v,q}}{R_v^2 + X_v^2}. \tag{2.29}$$

Unlike a PI voltage controller, the virtual admittance loop does not attempt to force the voltage error to zero. A non zero difference between the internal voltage reference and the PCC voltage represents the voltage drop across the emulated impedance and is therefore required to produce the converter current. The current reference is determined by both the magnitude and the angle of this voltage difference.

The virtual resistance provides a resistive component and contributes to the damping of the converter response, whereas the virtual reactance introduces the desired predominantly inductive characteristic. The magnitude of Z_v determines the current reference produced for a given voltage error, while its X/R ratio influences the direction of the current vector and the coupling between active and reactive power.

A larger virtual impedance magnitude produces a smaller current reference for the same voltage difference. Nevertheless, a fixed virtual admittance does not impose a strict upper limit on the converter current. A sufficiently severe voltage disturbance may still generate a current reference above the permissible value. Therefore, the current references generated by Equation (2.27) are subsequently processed by the current-limiting and current allocation methods implemented in this thesis [6].

The limited current references are finally supplied to the inner current controller. Consequently, the emulated virtual impedance is accurately reproduced only within the frequency range in which the current controller can track the generated references.

The same control principles are also applied to the negative-sequence system. However, the corresponding controllers are formulated in the negative-sequence synchronous reference frame and therefore require the appropriate modifications. In the virtual admittance implementation, the negative-sequence voltage errors are formed from the corresponding reference and measured quantities, while the signs of the cross-coupling

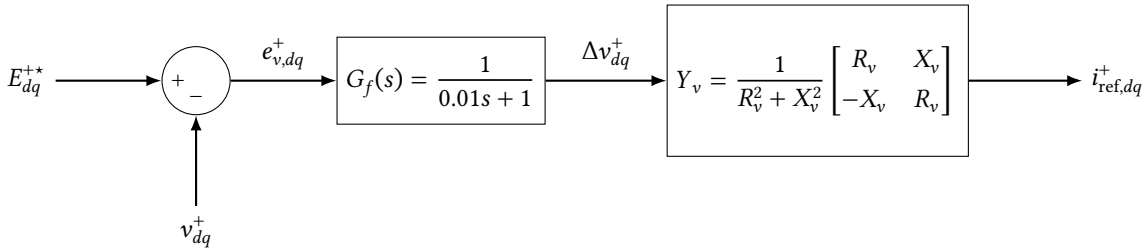


Figure 2.5.: Implementation of the positive-sequence virtual admittance controller.

terms are adjusted to account for the opposite direction of rotation. In the PI-based implementation, the same controller structure is retained, although the proportional and integral gains may be selected independently from those of the positive-sequence controller. This allows the positive- and negative-sequence components to be regulated separately within a common control architecture.

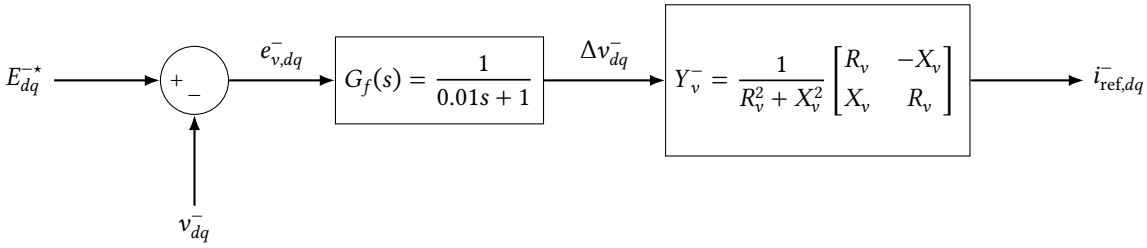


Figure 2.6.: Implementation of the negative-sequence virtual admittance controller.

Measurement Processing and Sequence Extraction

The measured three phase voltages and currents are transformed into synchronous dq reference frames before being supplied to the control system. For a general three phase quantity $x_{abc} = [x_a\ x_b\ x_c]^{\mathrm{T}}$, where x may represent either voltage or current, the transformation implemented in PLECS is expressed as

$$\begin{bmatrix} x_d \\ x_q \end{bmatrix} = \frac{2}{3}\begin{bmatrix} \cos\theta & \cos\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) \\ -\sin\theta & -\sin\left(\theta - \frac{2\pi}{3}\right) & -\sin\left(\theta + \frac{2\pi}{3}\right) \end{bmatrix}\begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix}. \tag{2.30}$$

The transformation angle determines the direction of rotation of the reference frame. The positive-sequence quantities are obtained using the internally generated angle θ_{GFM}, whereas the negative-sequence quantities are obtained using $-\theta_{\mathrm{GFM}}$, where $T_{dq}(\theta)$ denotes the abc-to-dq transformation matrix defined in Equation 2.30:

$$x^{+}_{dq} = T_{dq}\left(\theta_{\mathrm{GFM}}\right) x_{abc}, \qquad x^{-}_{dq} = T_{dq}\left(-\theta_{\mathrm{GFM}}\right) x_{abc}. \tag{2.31}$$

In the positive-sequence synchronous reference frame, a balanced positive-sequence component appears as a constant dq quantity, whereas the negative-sequence component appears as an oscillation at twice the fundamental frequency. The opposite occurs in the negative-sequence synchronous reference frame: the negative-sequence component becomes constant, while the positive-sequence component produces a double frequency oscillation [3].

To separate the two sequence components, a notch filter tuned at $2\omega_0$ is applied to the transformed voltage and current measurements. The transfer function of the implemented filter is

$$G_{\mathrm{N}}(s) = \frac{s^2 + \omega_{\mathrm{N}}^2}{s^2 + 2\zeta_{\mathrm{N}}\omega_{\mathrm{N}}s + \omega_{\mathrm{N}}^2}, \tag{2.32}$$

where

$$\omega_{\mathrm{N}} = 2\omega_0. \tag{2.33}$$

For the 50 Hz system considered in this thesis, $\omega_0 = 2\pi 50\,\mathrm{rad/s}$, and the notch frequency is therefore equal to 100 Hz. In the implemented model, the filter damping parameter is selected as

$$\zeta_{\mathrm{N}} = 0.7. \tag{2.34}$$

The notch filter suppresses the double frequency component produced by the sequence rotating in the opposite direction. Consequently, the filtered positive-sequence frame quantities mainly contain the positive sequence, while the filtered negative-sequence frame quantities mainly contain the negative sequence. The filtered signals are denoted by

$$\tilde{v}_{dq}^{+}, \quad \tilde{i}_{dq}^{+}, \quad \tilde{v}_{dq}^{-}, \quad \tilde{i}_{dq}^{-}, \tag{2.35}$$

and are used as feedback and feedforward signals in the corresponding positive- and negative-sequence controllers.

The filtering stage enables separate sequence control under asymmetrical conditions. However, it also introduces additional dynamics and delay into the measurement path. Its parameters must therefore provide sufficient rejection of the double frequency component without unnecessarily reducing the bandwidth and stability margin of the inner control loops [3].

Inner Current Controller and Reference Processing

The PI voltage controller and the virtual admittance controller described in the previous subsections generate the positive- and negative-sequence current references. These references represent the currents required to regulate the corresponding voltage components. However, they are not necessarily applied directly to the inner current controllers. Before this stage, they are processed by the selected current-limiting and current allocation method, as illustrated by

$$\left(i_{d,\mathrm{ref}}^{\pm}, i_{q,\mathrm{ref}}^{\pm}\right) \longrightarrow \left(\bar{i}_{d,\mathrm{ref}}^{\pm}, \bar{i}_{q,\mathrm{ref}}^{\pm}\right), \tag{2.36}$$

where $i^{\pm}_{d,\mathrm{ref}}$ and $i^{\pm}_{q,\mathrm{ref}}$ are the original references generated by the voltage-control stage, while $\bar{i}^{\pm}_{d,\mathrm{ref}}$ and $\bar{i}^{\pm}_{q,\mathrm{ref}}$ denote the references supplied to the inner current controllers after current limitation and allocation. The corresponding methods are presented in detail in the following sections.

For the positive-sequence system, the current errors are defined as

$$e^{+}_{i,d} = \bar{i}^{+}_{d,\mathrm{ref}} - i^{+}_{d}, \qquad e^{+}_{i,q} = \bar{i}^{+}_{q,\mathrm{ref}} - i^{+}_{q}, \tag{2.37}$$

where i^{+}_{d} and i^{+}_{q} are the measured positive-sequence currents. The implemented PI current controller generates the positive-sequence converter voltage references according to

$$v^{+\star}_{c,d} = v^{+}_{d} + k_{pi} e^{+}_{i,d} + k_{ii} \int e^{+}_{i,d}\,\mathrm{d}t - \omega_{\mathrm{GFM}} L_{\mathrm{est}} \bar{i}^{+}_{q,\mathrm{ref}}, \tag{2.38}$$

$$v^{+\star}_{c,q} = v^{+}_{q} + k_{pi} e^{+}_{i,q} + k_{ii} \int e^{+}_{i,q}\,\mathrm{d}t + \omega_{\mathrm{GFM}} L_{\mathrm{est}} \bar{i}^{+}_{d,\mathrm{ref}}, \tag{2.39}$$

where k_{pi} and k_{ii} are the proportional and integral gains of the current controller, respectively, and L_{est} is the estimated inductance of the converter grid interface [5].

The measured voltage components v^{+}_{d} and v^{+}_{q} are included as feedforward terms. This reduces the voltage disturbance that must be compensated by the PI controller and improves the current response. The terms proportional to $\omega_{\mathrm{GFM}} L_{\mathrm{est}}$ compensate for the coupling between the d- and q-axis current dynamics in the rotating reference frame.

The controller gains are selected using the estimated interface parameters and the desired current control bandwidth. In the implemented model, they are calculated as

$$k_{pi} = \alpha_c L_{\mathrm{est}}, \qquad k_{ii} = \alpha_c R_{\mathrm{est}}, \tag{2.40}$$

where α_c is the selected current control bandwidth and R_{est} is the estimated series resistance.

A separate current controller is applied to the negative-sequence quantities in the negative-sequence synchronous reference frame. Its PI structure, voltage feedforward, and gain selection method are equivalent to those of the positive-sequence controller. However, because the negative-sequence reference frame rotates in the opposite direction, the signs of the cross coupling compensation terms are reversed. The negative-sequence voltage references are therefore expressed as

$$v^{-\star}_{c,d} = v^{-}_{d} + k^{-}_{pi} e^{-}_{i,d} + k^{-}_{ii} \int e^{-}_{i,d}\,\mathrm{d}t + \omega_{\mathrm{GFM}} L_{\mathrm{est}} \bar{i}^{-}_{q,\mathrm{ref}}, \tag{2.41}$$

$$v^{-\star}_{c,q} = v^{-}_{q} + k^{-}_{pi} e^{-}_{i,q} + k^{-}_{ii} \int e^{-}_{i,q}\,\mathrm{d}t - \omega_{\mathrm{GFM}} L_{\mathrm{est}} \bar{i}^{-}_{d,\mathrm{ref}}. \tag{2.42}$$

2.2.4. Generation of the Final Converter Voltage Reference

The positive- and negative-sequence current controllers generate separate converter voltage references in their corresponding synchronous reference frames. The positive-sequence voltage reference is transformed back into three phase quantities using the internally generated grid-forming angle:

$$v_{c,abc}^{+\star} = T_{dq}^{-1}\left(\theta_{\mathrm{GFM}}\right) v_{c,dq}^{+\star}. \tag{2.43}$$

Similarly, the negative-sequence voltage reference is transformed using a reference frame rotating in the opposite direction:

$$v_{c,abc}^{-\star} = T_{dq}^{-1}\left(-\theta_{\mathrm{GFM}}\right) v_{c,dq}^{-\star}. \tag{2.44}$$

The final three phase converter voltage reference is obtained by combining the two sequence contributions:

$$v_{c,abc}^{\star} = v_{c,abc}^{+\star} + v_{c,abc}^{-\star}. \tag{2.45}$$

The resulting reference is first passed through a saturation block before being applied to the three controlled voltage sources representing the average model of the converter. The saturation limits the voltage reference to the range $[-V_d, V_d]$, where

$$V_d = \frac{\sqrt{2}\,1.1\,V_b}{\sqrt{3}}. \tag{2.46}$$

This limitation represents the maximum voltage that can be produced by the converter. The saturated reference is then applied to the averaged converter model. In this way, the converter output voltage contains the positive-sequence component required for grid-forming operation together with the negative-sequence component generated by the dedicated controller, while respecting the available converter voltage limits.

Chapter 3.

PLECS Implementation

3.1. Studied System

The system considered throughout this work is illustrated in Fig. 3.1. It consists of a three-phase grid-forming voltage-source converter connected to an equivalent AC grid through the converter-side inductance L_{cnv} and the grid impedance, represented by R_{src} and L_{src}. The converter is represented in PLECS by an averaged model, as previously mentioned, since the purpose of the simulations was to examine the dynamic response of the implemented control system and current-limiting methods during grid disturbances.

The three phase converter current and the voltage at the point of common coupling (PCC) are measured and supplied to the control system, which generates the voltage reference applied to the averaged converter. The external grid is represented by a three phase voltage source behind its equivalent impedance. Grid disturbances are introduced by modifying the individual phase voltages of this source, allowing both symmetrical and asymmetrical voltage sags and phase-angle disturbances to be reproduced.

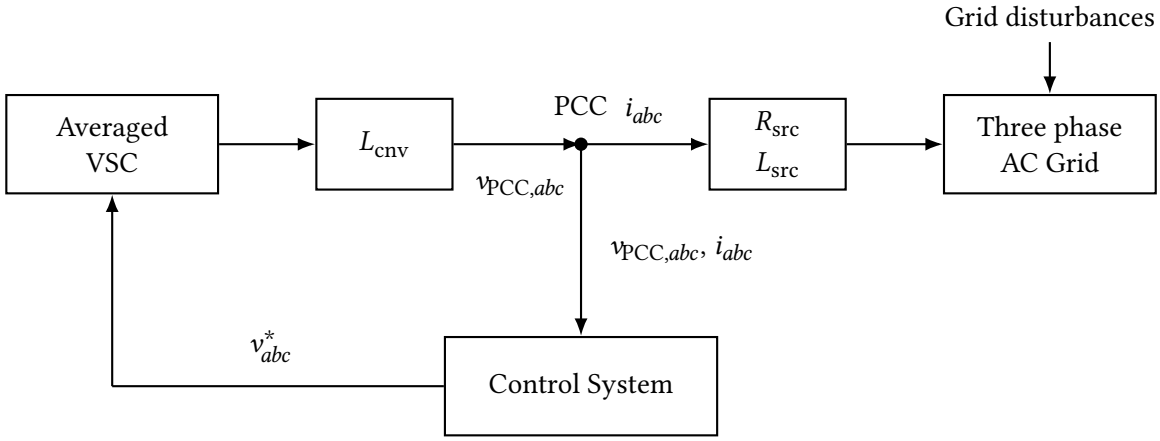


Figure 3.1.: Simplified representation of the studied grid-forming converter system.

The nominal apparent power and line-to-line voltage of the studied system are 60 MVA and 130 kV, respectively, while the nominal frequency is 50 Hz. These quantities define the base values used for the per unit representation of the electrical system. The main electrical and simulation parameters are summarized in Table 3.1. Parameters related to the individual control loops and current-limiting methods are provided later together with their corresponding implementations.

Table 3.1.: Main electrical and simulation parameters of the studied system.

Parameter	Symbol	Value
Base apparent power	S_b	60 MVA
Base line-to-line voltage	V_b	130 kV
Nominal frequency	f_b	50 Hz
Base impedance	Z_b	281.67 Ω
Base inductance	L_b	0.8966 H
Grid resistance	R_{src}	16.8 Ω
Grid inductance	L_{src}	0.1793 H (0.2 p.u.)
Converter-side inductance	L_{cnv}	0.1345 H (0.15 p.u.)
DC-link voltage	V_d	233.52 kV
Simulation time step	T_s	10 μs
Maximum converter current	I_{max}	319.76 A (1.2 p.u.)

3.2. Control System Implementation

The control system described theoretically in Chapter 2 was implemented in PLECS using the same hierarchical structure adopted in the analytical development. The implementation includes the calculation of the electrical quantities required by the controller, the outer grid-forming control loops, and the inner voltage and current control stages. Positive- and negative-sequence quantities are processed through dedicated control paths, while the corresponding measurement and sequence extraction blocks provide the signals required by each controller. The following subsections present the implementation of each part of the control system in the same order as introduced in Chapter 2, allowing a direct connection between the theoretical formulation and its realization in the simulation model.

Voltage and Current Measurement Processing

Before being used by the control system, the measured three phase voltages and currents are transformed into synchronous dq reference frames. The measured grid voltage V_{abc} and converter current $I_{\text{cnv},abc}$ are processed separately for the positive- and negative-sequence components, as shown in Fig. 3.2.

The filtering stage is implemented using a second-order notch filter, as already mentioned in subsection 2.2.3

These filtered dq quantities are subsequently used by the positive- and negative-sequence control loops. In this way, the controllers operate on constant sequence quantities while the influence of the opposite sequence on the measured signals is significantly reduced.

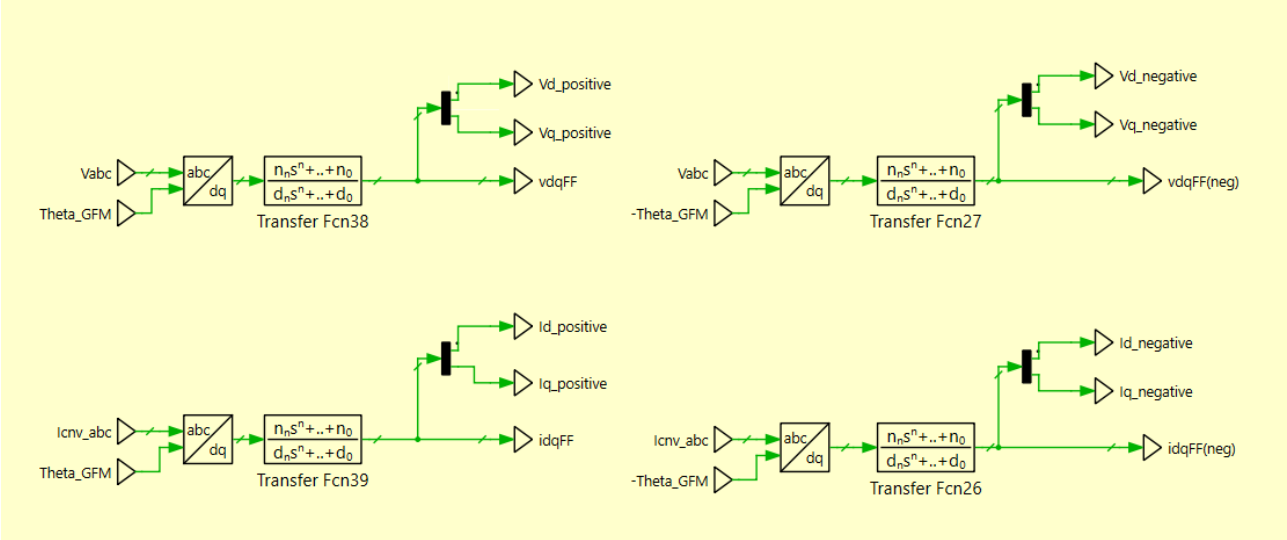


Figure 3.2.: PLECS implementation of the voltage and current measurement processing for the positive- and negative-sequence components.

3.2.1. Power Calculation Implementation

The active and reactive powers used by the grid-forming control system are calculated from the filtered positive-sequence voltage and current components obtained in the synchronous *dq* reference frame. The corresponding implementation in PLECS is shown in Fig. 3.3.

Using the adopted amplitude invariant convention, the positive-sequence active and reactive powers are calculated as

$$P^{+} = \frac{3}{2}\left(V_d^{+} I_d^{+} + V_q^{+} I_q^{+}\right), \tag{3.1}$$

and

$$Q^{+} = \frac{3}{2}\left(V_q^{+} I_d^{+} - V_d^{+} I_q^{+}\right). \tag{3.2}$$

The filtered signals V_d^{+}, V_q^{+}, I_d^{+}, and I_q^{+} are multiplied according to these expressions, and the resulting terms are combined and scaled by the factor $3/2$. The calculated quantities P^{+} and Q^{+} are then supplied to the active power synchronization loop and the reactive power-voltage control loop, respectively.

Only the positive-sequence components are used for the power calculation employed by the outer grid-forming loops. This prevents the oscillatory power contributions associated with the negative sequence during unbalanced conditions from directly affecting the synchronization and voltage regulation loops. The negative-sequence quantities are instead handled separately by the corresponding negative-sequence control structure.

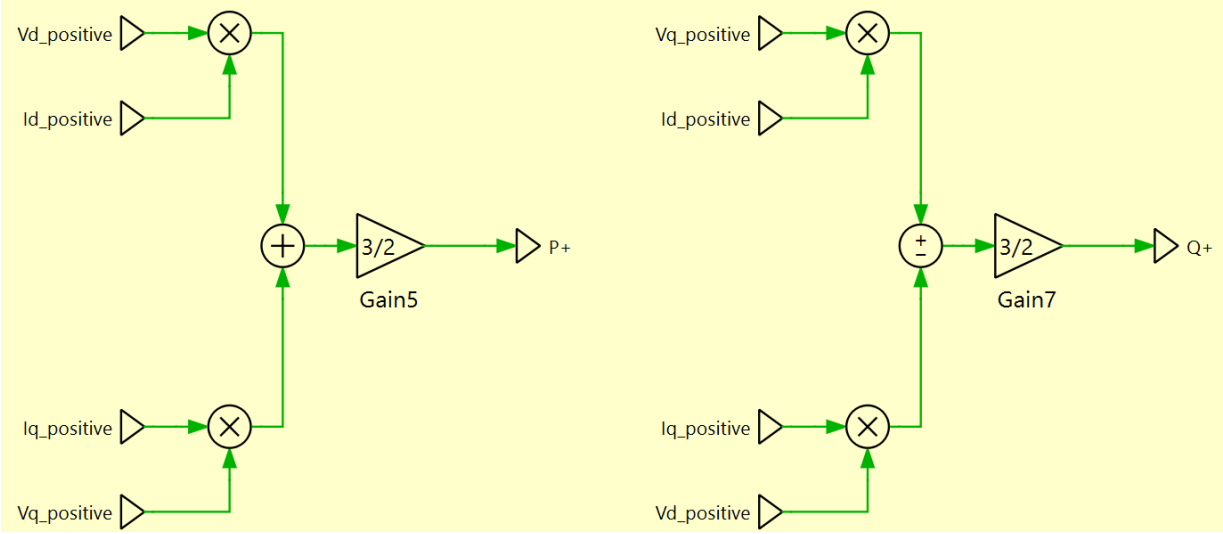


Figure 3.3.: PLECS implementation of the positive-sequence active and reactive power calculation.

3.2.2. Outer Grid-Forming Control Implementation

The outer grid-forming control loops described in Chapter 2 were implemented in PLECS using the positive-sequence active and reactive powers. The complete implementation is shown in Fig. 3.4. The upper part of the control structure generates the converter angular frequency and phase angle from the active power error, while the lower part determines the voltage magnitude reference from the reactive power error.

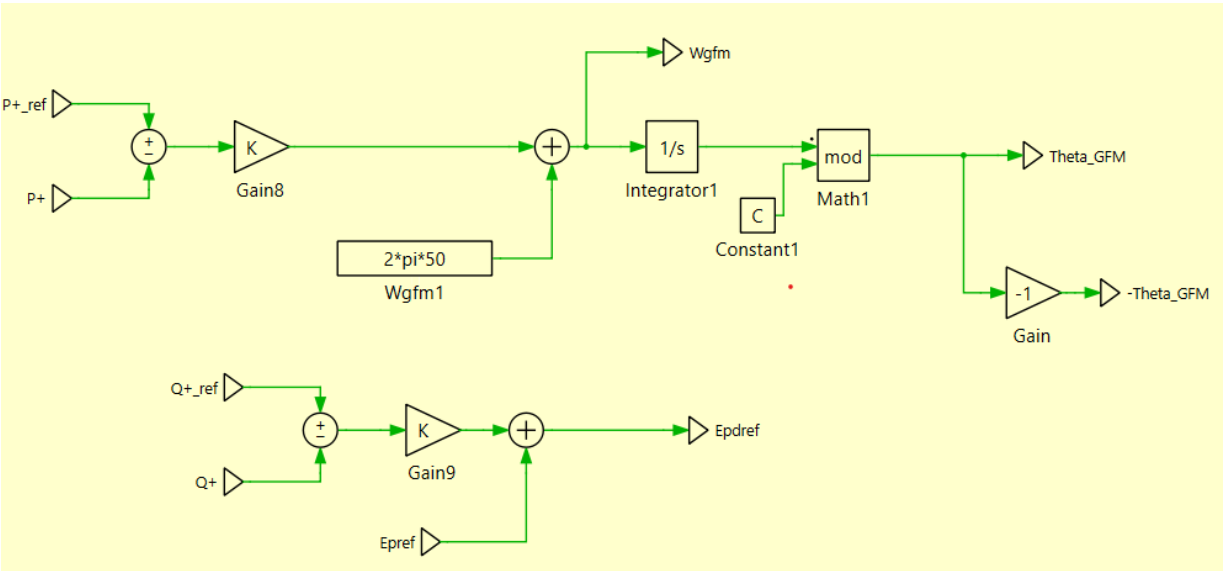


Figure 3.4.: PLECS implementation of the active power synchronization and reactive power-voltage control loops.

Active Power Synchronization Loop

The active power synchronization loop is implemented in the upper part of Fig. 3.4. The positive-sequence active power P^+ is compared with its reference P^+_{ref}, and the resulting error is multiplied by the selected active power droop gain. The resulting frequency correction is added to the nominal angular frequency $\omega_0 = 2\pi 50$ rad/s, producing the internal grid-forming angular frequency ω_{GFM}.

The generated angular frequency is then integrated to obtain the internal converter angle. A modulo operation is applied to keep the angle within the required angular range before generating the signal θ_{GFM}. In addition, the opposite angle $-\theta_{\text{GFM}}$ is generated and is used in the negative-sequence reference frame transformations. In this way, the same internally generated grid-forming angle is used consistently by both the positive- and negative-sequence control paths.

Reactive Power-Voltage Loop

The reactive powervoltage control loop is implemented in the lower part of Fig. 3.4. The measured positive-sequence reactive power Q^+ is compared with the corresponding reference Q^+_{ref}. The resulting error is multiplied by the reactive power droop gain and combined with the nominal voltage magnitude reference E_{pref}. The output of this loop, denoted by E_{pdref}, provides the positive-sequence voltage magnitude reference required by the subsequent inner control stages.

3.2.3. Inner Control Structure Implementation

The inner control structure is responsible for generating the converter current references from the corresponding voltage references and measured voltage components. Two alternative implementations are considered in this work: a PI-based voltage controller and a virtual admittance controller. Both structures are implemented independently for the positive- and negative-sequence components.

PI Voltage Controller

The first implementation uses PI voltage controllers to convert the voltage tracking errors into current references. The positive-sequence implementation is shown in Fig. 3.5.

The internal implementation of the PI voltage controller is shown in Fig. 3.6. The voltage error is applied in parallel to a proportional and an integral branch. The proportional branch multiplies the input error by the proportional gain k_{pv}, while the integral branch applies the integral gain k_{iv} before integration. The two contributions are then added to generate the corresponding current reference.

An anti-windup mechanism is also included in the integral branch. During normal operation, the voltage error is integrated normally. However, when the current-limiting algorithm becomes active, the signal `saturation_boolean` changes the state of the switch and sets the input of the integrator to zero. Consequently, integration is temporarily disabled while the converter current is constrained.

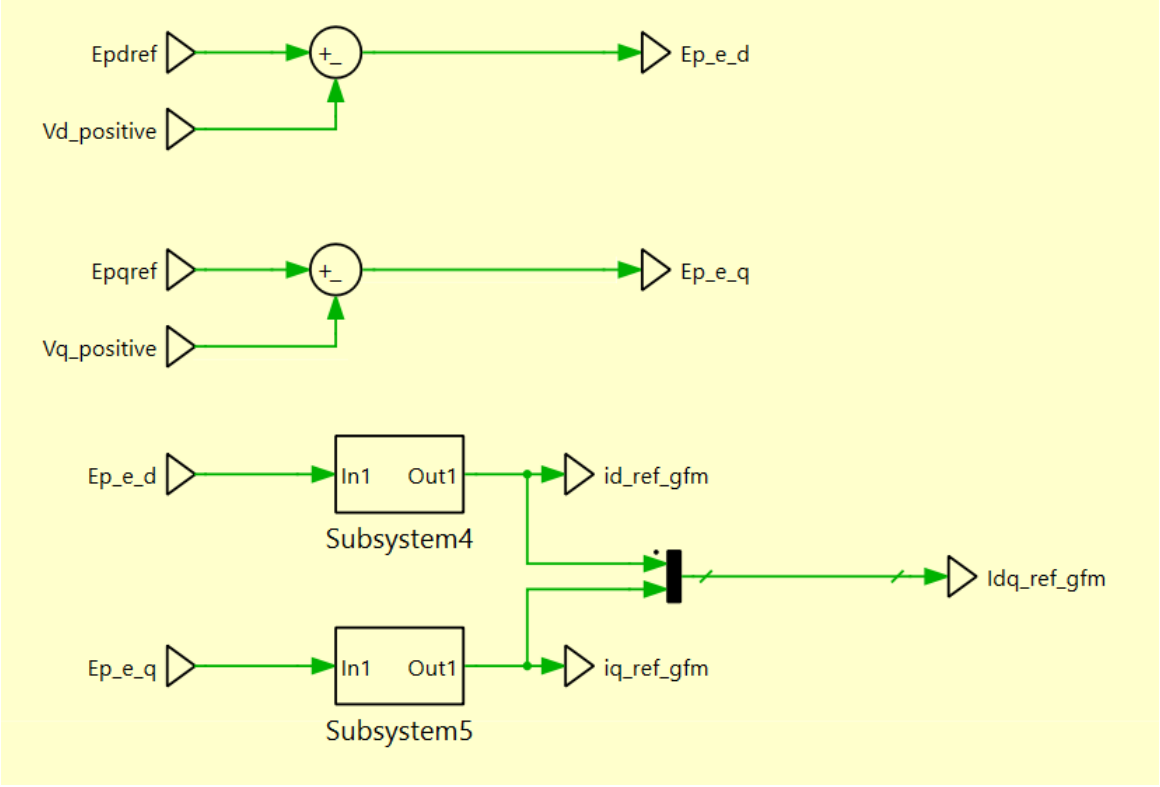


Figure 3.5.: PLECS implementation of the positive-sequence voltage error generation and PI voltage controller.

This prevents the integral term from continuing to increase when the requested current reference cannot be applied because of the imposed current limit. Without this mechanism, a large accumulated integral term could remain after the limiting condition is removed, leading to overshoot and a slower post-fault recovery. Once the current limitation is deactivated, the normal integration process is restored.

The input of the integrator can therefore be expressed as

$$u_{\text{int}} = \begin{cases} k_{iv} e_v, & \text{when current limitation is inactive,} \\ 0, & \text{when current limitation is active.} \end{cases} \tag{3.3}$$

The resulting PI controller output is

$$I_{\text{ref}} = k_{pv} e_v + x_{\text{int}}, \tag{3.4}$$

where x_{int} represents the integrator state. During current limitation, this is the state that is held constant rather than continuously increasing. The same anti-windup structure is applied to the PI voltage controllers of both the positive- and negative-sequence components.

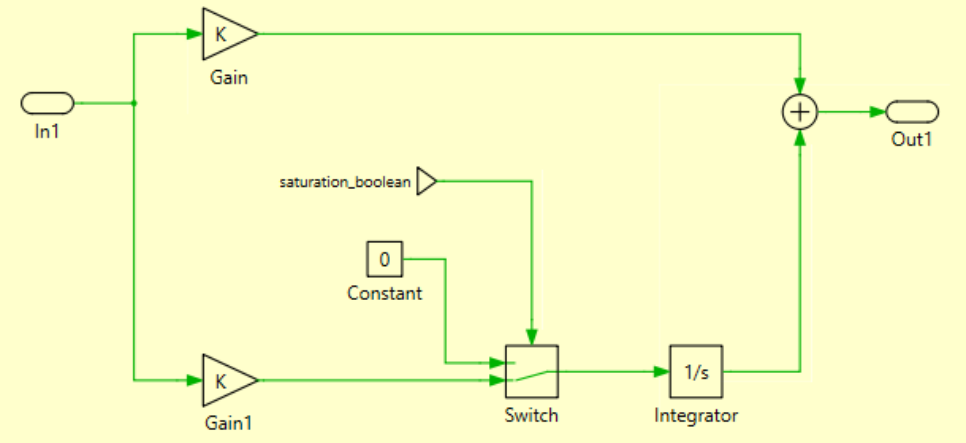


Figure 3.6.: Internal PLECS implementation of the PI voltage controller with conditional integration for anti-windup during current limitation.

The negative-sequence voltage controller follows the same implementation principle. The negative-sequence voltage errors are defined as

$$e_{v,d}^{-} = E_{d,\mathrm{ref}}^{-} - V_{d}^{-}, \tag{3.5}$$

and

$$e_{v,q}^{-} = E_{q,\mathrm{ref}}^{-} - V_{q}^{-}. \tag{3.6}$$

These errors are processed by separate PI controllers to generate the negative-sequence current references:

$$I_{d,\mathrm{ref}}^{-} = k_{pv}^{-} e_{v,d}^{-} + k_{iv}^{-} \int e_{v,d}^{-}\, dt, \tag{3.7}$$

$$I_{q,\mathrm{ref}}^{-} = k_{pv}^{-} e_{v,q}^{-} + k_{iv}^{-} \int e_{v,q}^{-}\, dt. \tag{3.8}$$

Although the positive- and negative-sequence controllers have the same basic structure, their gains can be selected independently. This allows the response of the negative-sequence controller during unbalanced operating conditions to be adjusted without modifying the positive-sequence voltage control dynamics. The generated positive- and negative-sequence current references are then processed by the current-limiting strategy before being applied to the inner current control stage.

Virtual Admittance Controller

As an alternative to the PI voltage controller, a virtual admittance implementation is employed to generate the converter current references directly from the voltage errors, as was already explained in subsection 2.2.3

accompanied with the equations. The positive- and negative-sequence components are treated independently, allowing the converter response to be adjusted separately, for each sequence.

The denominator $R_v^2 + X_v^2$ is calculated once and used by both axes. The PLECS implementation of the positive-sequence virtual admittance controller is shown in Fig. 3.7.

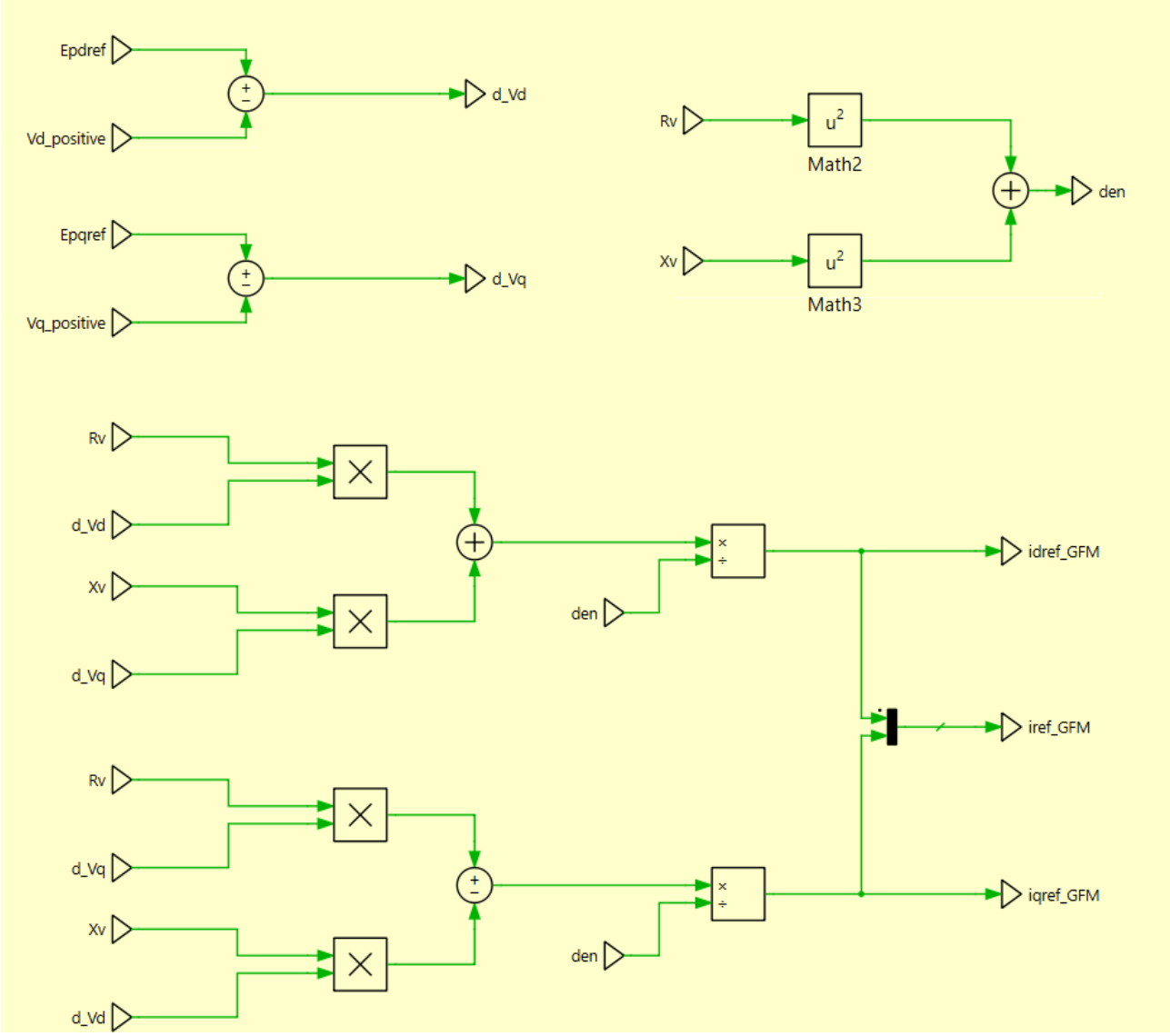


Figure 3.7.: PLECS implementation of the positive-sequence virtual admittance controller.

The same principle is applied to the negative-sequence components. However, because the negative-sequence dq reference frame rotates in the opposite direction, the signs of the cross-coupling terms are reversed. The negative-sequence current references are therefore calculated as

$$I_{d,\mathrm{ref}}^{-} = \frac{R_v \Delta V_d^{-} - X_v \Delta V_q^{-}}{R_v^2 + X_v^2}, \tag{3.9}$$

and

$$I^{-}_{q,\mathrm{ref}} = \frac{R_v \Delta V_q^{-} + X_v \Delta V_d^{-}}{R_v^2 + X_v^2}. \tag{3.10}$$

This sign reversal preserves the correct virtual impedance behavior in the negative-sequence synchronous reference frame. In this way, independent positive- and negative-sequence current references are generated from their respective voltage errors while maintaining the same virtual resistance and reactance concept.

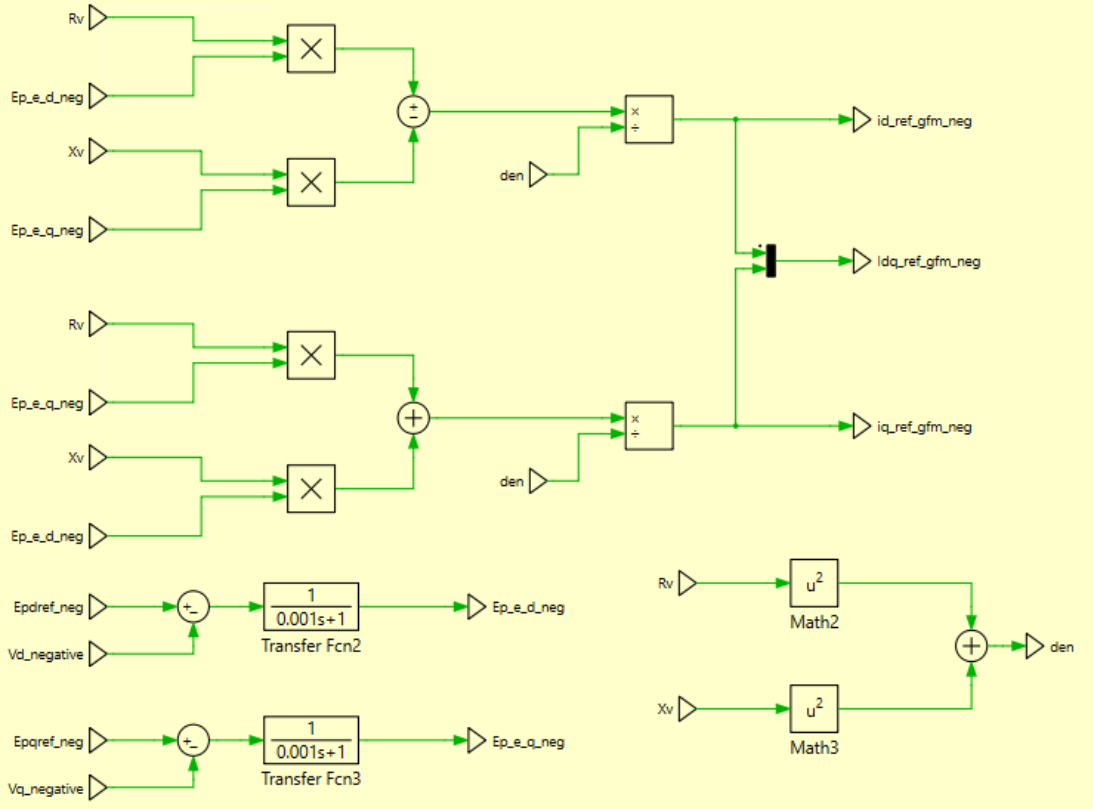


Figure 3.8.: PLECS implementation of the negative-sequence virtual admittance controller.

Inner Current Controller

The final stage of the cascaded control structure is the inner current controller. Its purpose is to force the converter current to follow the positive- and negative-sequence current references generated by the preceding voltage control and current-limiting stages. The PLECS implementation of the positive-sequence current control loop is shown in Fig. 3.9.

The positive-sequence current controller is implemented through a C-Script block and operates in the corresponding synchronous *dq* reference frame. The current reference vector, the filtered voltage feedforward terms, and the internally generated GFM angular frequency are provided as inputs to the controller. The C-Script evaluates the current control law and generates the corresponding *dq* converter voltage reference. This voltage reference is subsequently transformed back to the three phase *abc* frame using the grid-forming angle θ_{GFM}. The next equations are already mentioned in the section Section 2.2.3.

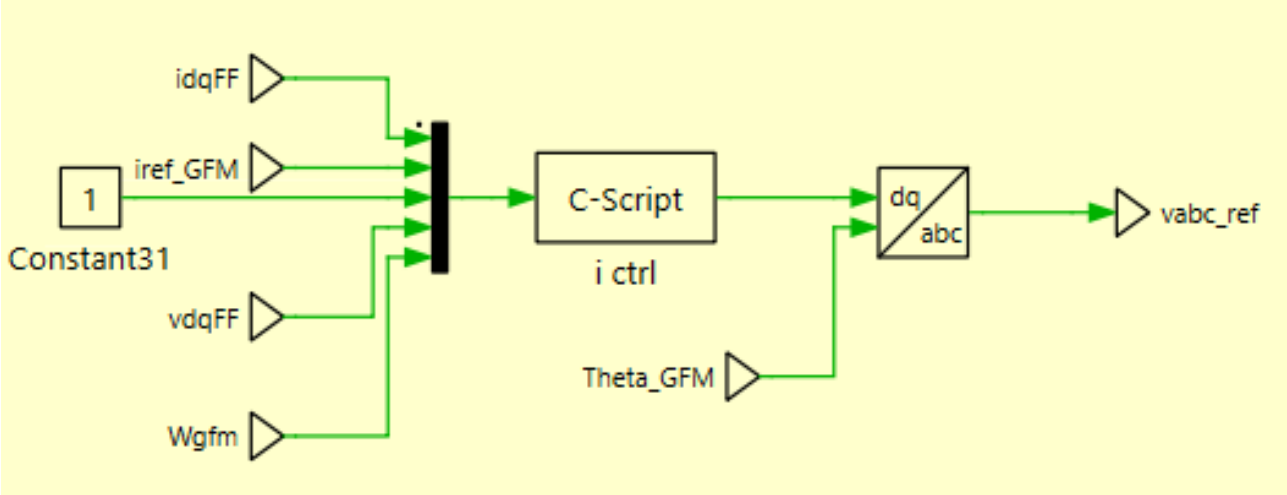


Figure 3.9.: PLECS implementation of the positive-sequence inner current control loop.

The control algorithm implemented inside the C-Script consists of a PI current controller combined with voltage feedforward and dq-axis decoupling. The positive-sequence current errors are defined as

$$e_{i,d}^{+} = I_{d,\mathrm{ref}}^{+} - I_{d}^{+}, \tag{3.11}$$

and

$$e_{i,q}^{+} = I_{q,\mathrm{ref}}^{+} - I_{q}^{+}. \tag{3.12}$$

The integral states of the current controller are updated at each simulation step according to

$$\xi_{d}^{+}[k+1] = \xi_{d}^{+}[k] + k_{ic} e_{i,d}^{+}[k] T_{s}, \tag{3.13}$$

$$\xi_{q}^{+}[k+1] = \xi_{q}^{+}[k] + k_{ic} e_{i,q}^{+}[k] T_{s}, \tag{3.14}$$

where T_s denotes the controller sampling period. The PI controller outputs are therefore

$$u_{d}^{+} = k_{pc} e_{i,d}^{+} + \xi_{d}^{+}, \tag{3.15}$$

and

$$u_{q}^{+} = k_{pc} e_{i,q}^{+} + \xi_{q}^{+}. \tag{3.16}$$

The voltage commands are obtained by adding the measured voltage feedforward terms and the cross-coupling compensation associated with the converter side inductance:

$$V_{d,\mathrm{ref}}^{+} = V_{d}^{+} + u_{d}^{+} - \omega_{\mathrm{GFM}} L_{\mathrm{est}} I_{q,\mathrm{ref}}^{+}, \tag{3.17}$$

$$V_{q,\mathrm{ref}}^{+} = V_{q}^{+} + u_{q}^{+} + \omega_{\mathrm{GFM}} L_{\mathrm{est}} I_{d,\mathrm{ref}}^{+}, \tag{3.18}$$

where L_{est} is the estimated converter-side inductance. The voltage feedforward terms improve the dynamic response of the current loop, while the cross-coupling terms compensate for the interaction between the d- and q-axis current dynamics.

The negative-sequence current controller is implemented using almost the same C-Script control structure. The main difference arises from the opposite direction of rotation of the negative-sequence synchronous reference frame. Accordingly, the angular frequency used in the decoupling terms is

$$\omega^{-} = -\omega_{\mathrm{GFM}}. \tag{3.19}$$

Therefore, the signs of the dq cross-coupling terms are automatically reversed for the negative-sequence controller. Apart from this change, the same PI control, voltage feedforward, and discrete integration principles are applied to the negative-sequence current components.

In general, the current control voltage references for either sequence can be expressed as

$$V_{d,\mathrm{ref}}^{\pm} = V_{d}^{\pm} + u_{d}^{\pm} - \omega^{\pm} L_{\mathrm{est}} I_{q,\mathrm{ref}}^{\pm}, \tag{3.20}$$

$$V_{q,\mathrm{ref}}^{\pm} = V_{q}^{\pm} + u_{q}^{\pm} + \omega^{\pm} L_{\mathrm{est}} I_{d,\mathrm{ref}}^{\pm}, \tag{3.21}$$

with

$$\omega^{+} = \omega_{\mathrm{GFM}}, \qquad \omega^{-} = -\omega_{\mathrm{GFM}}. \tag{3.22}$$

The complete C-Script implementation used for the inner current controller is provided in Appendix B. The code is presented separately for the variable declarations, the update function, and the output function, following the structure used in the PLECS C-Script block.

Chapter 4.

Current-Limiting implementation

4.1. Current-Limiting strategies

This chapter presents the current-limiting methods used in this work and describes their implementation in PLECS. The proposed schemes are placed between the current reference generation stage and the inner current controller, where they modify the current references before these are tracked by the converter.

4.1.1. Elliptical Current Limiter

The elliptical current limiter operates on the positive- and negative-sequence current references generated by the preceding control stages. The maximum current magnitude that would result in each individual phase is first calculated from the positive- and negative-sequence dq current references.

The PLECS implementation of this calculation is shown in Fig. 4.1. For phase a, the corresponding current magnitude is calculated as

$$I_{a,\mathrm{ref}} = \sqrt{\left(I^{+}_{d,\mathrm{ref}} + I^{-}_{d,\mathrm{ref}}\right)^2 + \left(I^{+}_{q,\mathrm{ref}} - I^{-}_{q,\mathrm{ref}}\right)^2}. \tag{4.1}$$

The current reference magnitudes of phases b and c are calculated in the same manner, taking into account the corresponding 120° phase displacement between the three phases. In the PLECS model, the required coordinate relationships are implemented using the factors $1/2$ and $\sqrt{3}/2$, followed by squaring and square root operations. In this way, the contribution of both sequence components to each phase is taken into account before the current-limiting action is applied.

The largest of the three calculated phase current references is then selected:

$$I_{\mathrm{ref,max}} = \max\left(I_{a,\mathrm{ref}}, I_{b,\mathrm{ref}}, I_{c,\mathrm{ref}}\right). \tag{4.2}$$

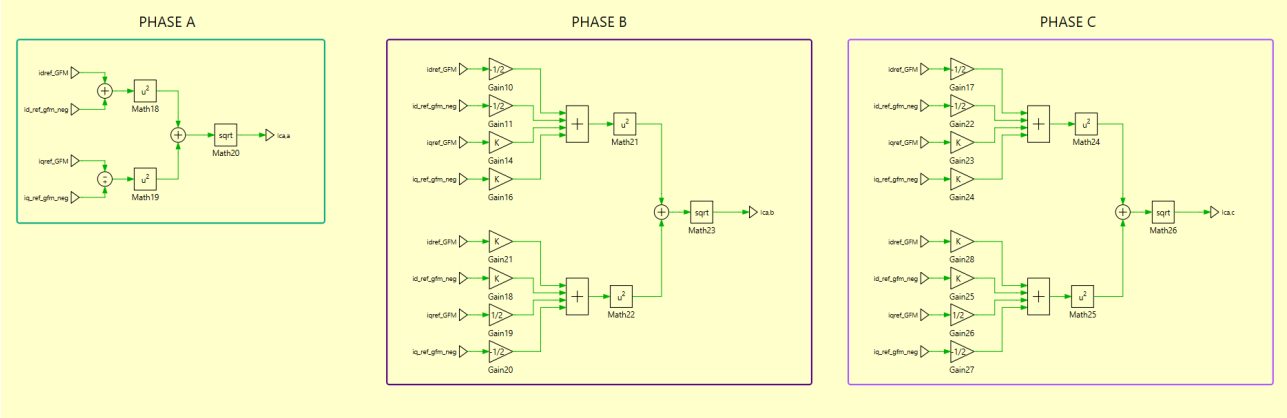


Figure 4.1.: PLECS implementation of the phase current reference magnitude calculation from the positive- and negative-sequence dq components.

Based on this value, a common scaling factor is calculated as

$$k_{\mathrm{ph}} = \min\left(1, \frac{I_{\max}}{\max\left(I_{\mathrm{ref,max}}, C\right)}\right), \tag{4.3}$$

where $I_{\max}$ is the imposed allowable maximum phase current magnitude and C is a small positive constant used to avoid numerical problems (division by zero) when the calculated current reference is close to zero.

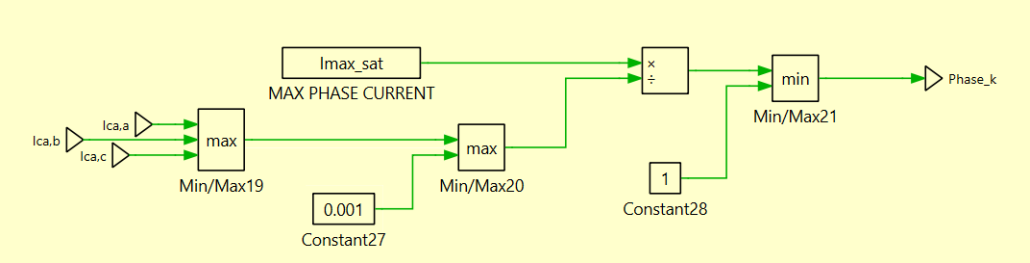


Figure 4.2.: PLECS implementation of the common scaling factor calculation for the elliptical current limiter.

When all three phase current references remain within the specified limit, $k_{\mathrm{ph}} = 1$, and no limiting action is applied. If at least one phase exceeds the allowed current magnitude, then $k_{\mathrm{ph}} < 1$, and the same scaling factor is applied to all positive- and negative-sequence dq current-reference components:

$$I^{+}_{d,\mathrm{lim}} = k_{\mathrm{ph}} I^{+}_{d,\mathrm{ref}}, \qquad I^{+}_{q,\mathrm{lim}} = k_{\mathrm{ph}} I^{+}_{q,\mathrm{ref}}, \tag{4.4}$$

$$I^{-}_{d,\mathrm{lim}} = k_{\mathrm{ph}} I^{-}_{d,\mathrm{ref}}, \qquad I^{-}_{q,\mathrm{lim}} = k_{\mathrm{ph}} I^{-}_{q,\mathrm{ref}}. \tag{4.5}$$

Consequently, the current reference associated with the most heavily loaded phase determines the degree of limitation. Since all four sequence components are multiplied by the same factor, their relative proportions are preserved while the resulting three phase current references are kept within the specified current capability.

The location at which the scaling factor is applied within the control structure is illustrated in Fig. 4.3. The calculated factor k_{ph} multiplies the current reference signal before it enters the inner current control loop. In this way, the limited current reference is directly supplied to the current controller, which then generates the corresponding converter voltage reference. The same procedure is applied to both the positive- and negative-sequence current references. Therefore, the limiter acts as an intermediate stage between the current reference generation block and the inner current controller.

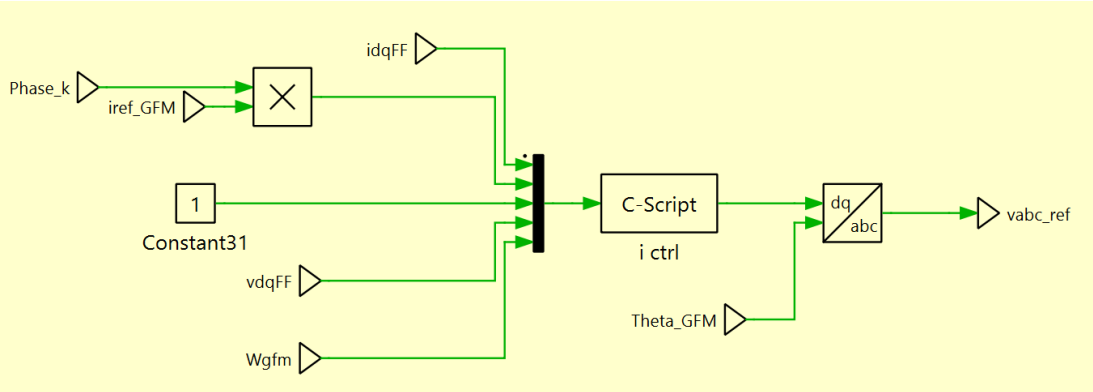


Figure 4.3.: Insertion of the scaling factor k_{ph} in the positive-sequence current control path. The same implementation is used for the negative-sequence path.

4.1.2. Virtual-Admittance-Based Current Limitation

The second implemented current-limiting method modifies the parameters of the virtual admittance introduced in Section 2.2.3. In contrast to the elliptical limiter, the generated current references are not directly scaled. Instead, the virtual resistance and reactance are increased when the voltage error becomes sufficiently large to produce a current reference above the specified converter limit.

The corresponding PLECS implementation is shown in Fig. 4.4. First, the magnitudes of the filtered positive- and negative-sequence voltage error vectors are calculated. Their sum is used to form the voltage difference quantity employed by the limiting algorithm,

$$V_{\mathrm{diff}} = \left|\Delta \mathbf{v}^{+}\right| + \left|\Delta \mathbf{v}^{-}\right|. \tag{4.6}$$

Based on the maximum permissible current I_{max}, the required virtual impedance magnitude is then calculated as

$$Z_{\mathrm{req}} = \frac{V_{\mathrm{diff}}}{I_{\mathrm{max}}}. \tag{4.7}$$

In the implemented controller, the X/R ratio of the virtual impedance is kept constant. With $X_v/R_v = 3$, the corresponding limiting values are calculated as

$$R_{\mathrm{lim}} = \frac{Z_{\mathrm{req}}}{\sqrt{1+3^2}}, \qquad X_{\mathrm{lim}} = 3R_{\mathrm{lim}}. \tag{4.8}$$

The final virtual admittance parameters are selected as the maximum between their nominal values and the values required by the limiting algorithm,

$$R_v = \max(R_{v,0}, R_{\text{lim}}), \qquad X_v = \max(X_{v,0}, X_{\text{lim}}). \tag{4.9}$$

Consequently, the nominal virtual admittance remains unchanged during normal operation. During a severe voltage disturbance, the virtual impedance magnitude increases only when required, thereby reducing the current reference generated by the virtual admittance controller, while preserving its selected X/R ratio.

An important implementation requirement is that this current-limiting method can only be used when the voltage control stage operates through the virtual admittance controller. In this case, the current references depend directly on R_v and X_v, and their adaptive modification produces the desired limiting action. When the PI voltage controller is selected instead, these parameters are not involved in the current reference generation and the present limiting mechanism therefore has no effect.

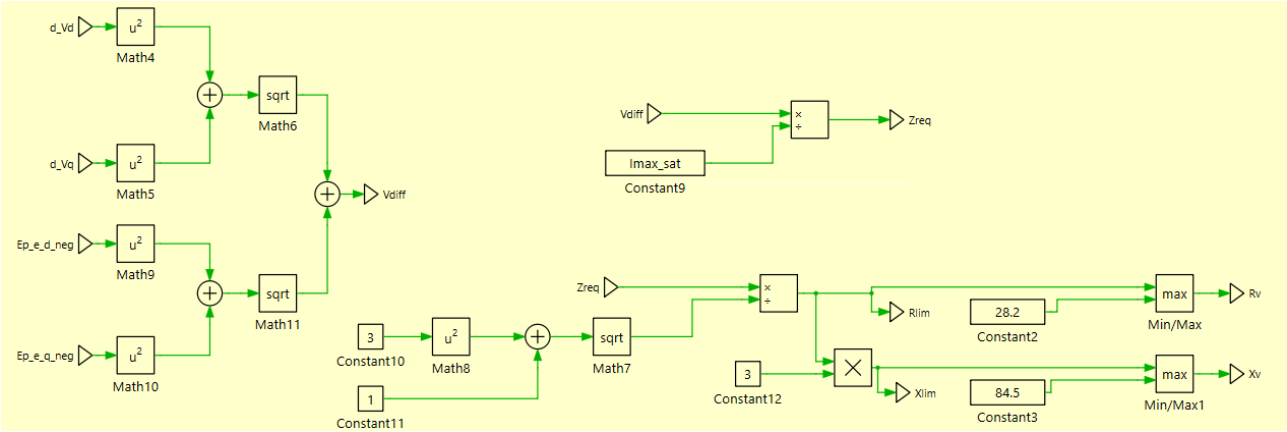


Figure 4.4.: PLECS implementation of the adaptive virtual admittance parameters used for current limitation.

4.1.3. Positive-Sequence Priority Current Limiter

The positive-sequence priority current limiter allocates the available converter current capability by giving preferential access to the positive-sequence current reference. Unlike the elliptical limiter, which applies the same scaling factor to the positive- and negative-sequence components, the present strategy deliberately modifies their relative contribution during asymmetrical disturbances.

The positive-sequence current demand is satisfied first, up to the specified converter current limit. Once this contribution has been established, the remaining phase current capability is allocated to the negative sequence. The allocation order can therefore be summarized as

$$I_{\text{ref}}^{+} \quad \rightarrow \quad I_{\text{ref}}^{-}, \tag{4.10}$$

where the arrow indicates the order in which the available converter current capability is assigned.

Giving priority to the positive sequence is particularly relevant for grid-forming operation, since the positive-sequence control path is associated with the main voltage forming and synchronization functions of the converter. The strategy therefore favours positive-sequence voltage support, while the negative-sequence controller uses the remaining converter capability to improve the response to voltage unbalance [2], [8].

Positive-Sequence Current Allocation

The magnitude of the unrestricted positive-sequence current reference is first calculated as

$$I_{\mathrm{ref}}^{+} = \sqrt{\left(I_{d,\mathrm{ref}}^{+}\right)^2 + \left(I_{q,\mathrm{ref}}^{+}\right)^2}. \tag{4.11}$$

Since the positive sequence has the highest priority, its current reference is retained whenever its magnitude remains below the converter current limit. If the positive-sequence demand alone exceeds the available current capability, it is proportionally reduced according to

$$k_{+} = \min\left(1, \frac{I_{\max}}{\max\left(I_{\mathrm{ref}}^{+}, C\right)}\right), \tag{4.12}$$

where $I_{\max}$ is the maximum permissible phase current magnitude and C is a small positive constant introduced to avoid numerical problems (division by zero) when the positive-sequence current reference approaches zero.

The positive-sequence current references supplied by the allocation stage are therefore

$$I_{d,\mathrm{lim}}^{+} = k_{+} I_{d,\mathrm{ref}}^{+}, \tag{4.13}$$

$$I_{q,\mathrm{lim}}^{+} = k_{+} I_{q,\mathrm{ref}}^{+}. \tag{4.14}$$

The squared magnitude of the resulting positive-sequence current vector is

$$P^2 = \left(I_{d,\mathrm{lim}}^{+}\right)^2 + \left(I_{q,\mathrm{lim}}^{+}\right)^2. \tag{4.15}$$

The remaining squared current margin available for the negative-sequence contribution is then calculated as

$$H_{-} = \max\left(I_{\max}^2 - P^2, 0\right). \tag{4.16}$$

The implemented calculation of the positive-sequence current magnitude, its corresponding limitation, and the remaining current margin is shown in Fig. 4.5.

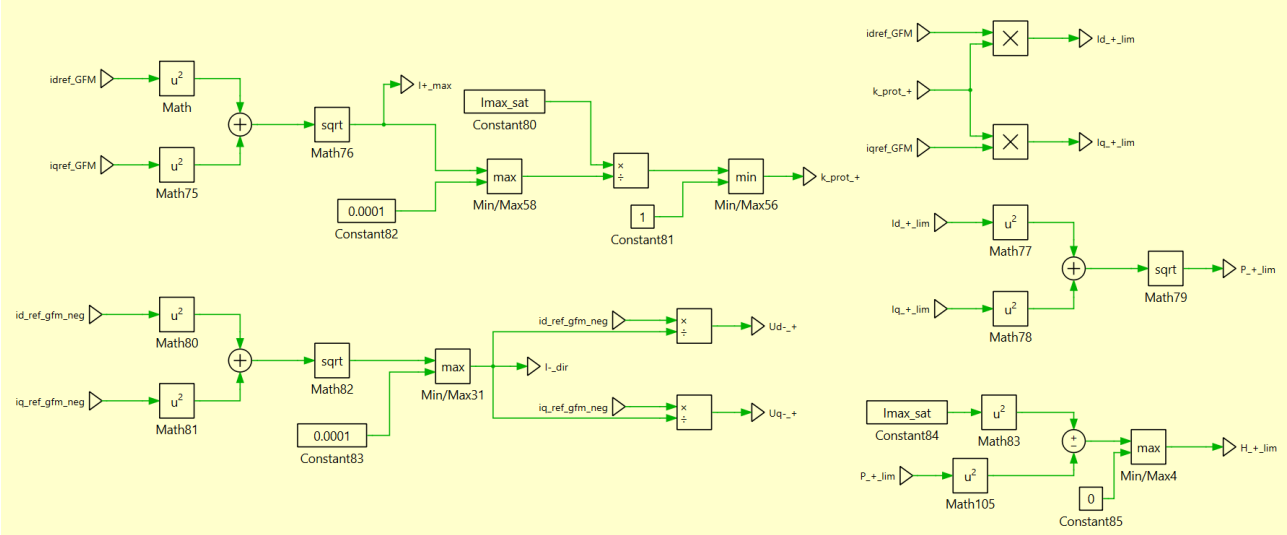


Figure 4.5.: PLECS implementation of the positive-sequence current allocation and calculation of the remaining current capability.

Negative-Sequence Current Direction

After the positive-sequence contribution has been established, the remaining converter current capability is assigned to the negative sequence. Rather than using only the magnitude of the original negative-sequence reference, its direction is first determined.

The magnitude of the unrestricted negative-sequence current reference is

$$I_{\mathrm{ref}}^{-} = \sqrt{\left(I_{d,\mathrm{ref}}^{-}\right)^2 + \left(I_{q,\mathrm{ref}}^{-}\right)^2}. \tag{4.17}$$

The corresponding unit vector components are defined as

$$u_d^{-} = \frac{I_{d,\mathrm{ref}}^{-}}{\max\left(I_{\mathrm{ref}}^{-}, C\right)}, \qquad u_q^{-} = \frac{I_{q,\mathrm{ref}}^{-}}{\max\left(I_{\mathrm{ref}}^{-}, C\right)}. \tag{4.18}$$

The negative-sequence current that can be assigned within the remaining converter capability can therefore be written as

$$\begin{bmatrix} I_{d,\mathrm{alloc}}^{-} \\ I_{q,\mathrm{alloc}}^{-} \end{bmatrix} = I_{\mathrm{alloc}}^{-} \begin{bmatrix} u_d^{-} \\ u_q^{-} \end{bmatrix}, \tag{4.19}$$

where I_{alloc}^{-} denotes the negative-sequence magnitude that can be accommodated without violating the phase current limit.

Allocation of the Remaining Current Capability

A simple calculation based on the difference $I_{\max} - I_{\text{ref}}^{+}$ would be unnecessarily conservative under asymmetrical operating conditions. Positive- and negative-sequence currents rotate in opposite directions and their contributions combine differently in each phase. Therefore, the available negative-sequence current is determined directly from the current constraint of each phase.

The interaction between the already allocated positive-sequence current and the negative-sequence direction is described by

$$A_{-} = I_{d,\text{lim}}^{+} u_{d}^{-} - I_{q,\text{lim}}^{+} u_{q}^{-}, \tag{4.20}$$

and

$$B_{-} = I_{d,\text{lim}}^{+} u_{q}^{-} + I_{q,\text{lim}}^{+} u_{d}^{-}. \tag{4.21}$$

For completeness, the mathematical derivation of the auxiliary quantities A_{-} and B_{-} used in the phase-aware current allocation is provided in Appendix C.

Taking into account the 120° displacement between the three phases, the corresponding phase coefficients are

$$L_{a,-} = 2A_{-}, \tag{4.22}$$
$$L_{b,-} = -A_{-} + \sqrt{3}B_{-}, \tag{4.23}$$
$$L_{c,-} = -A_{-} - \sqrt{3}B_{-}. \tag{4.24}$$

The PLECS implementation of these phase-dependent quantities is shown in Fig. 4.6.

For a generic phase $\phi \in \{a, b, c\}$, the squared phase current magnitude resulting from the fixed positive-sequence contribution and an allocated negative-sequence magnitude I_{alloc}^{-} is

$$I_{\phi}^{2} = P^{2} + \left(I_{\text{alloc}}^{-}\right)^{2} + L_{\phi,-} I_{\text{alloc}}^{-}. \tag{4.25}$$

The maximum admissible negative-sequence contribution for each phase is obtained by imposing

$$I_{\phi} = I_{\max}. \tag{4.26}$$

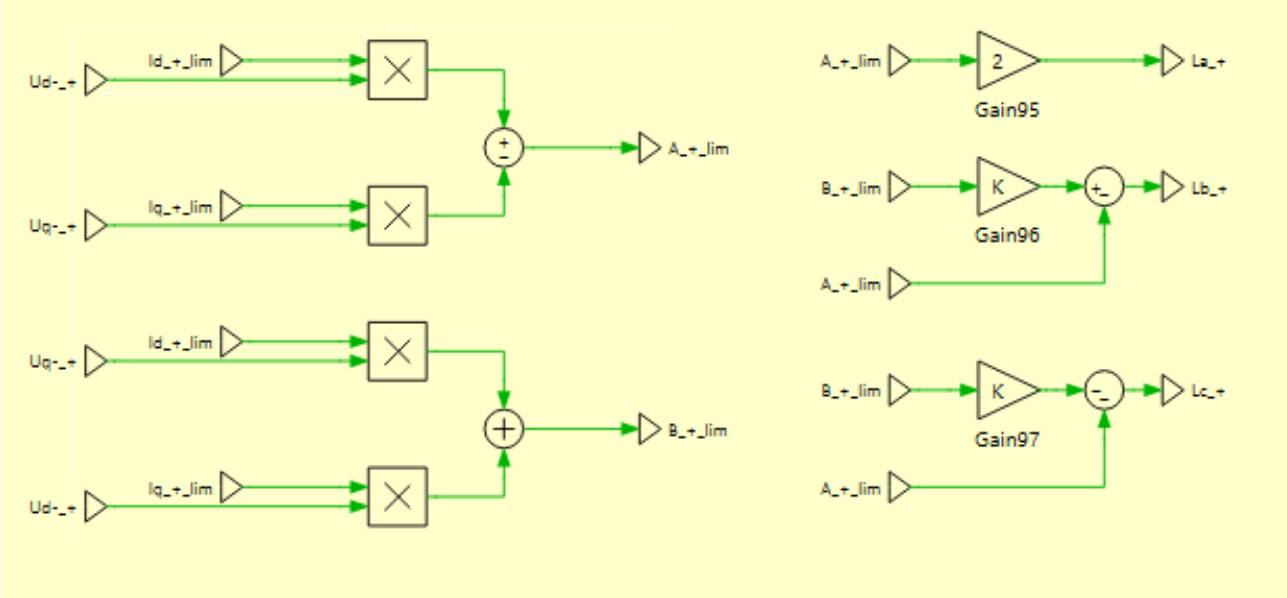


Figure 4.6.: PLECS implementation of the phase-dependent quantities used for the negative-sequence allocation under positive-sequence priority.

Using the definition of H_{-}, the corresponding equation becomes

$$\left(I^{-}_{\text{alloc}}\right)^2 + L_{\phi,-} I^{-}_{\text{alloc}} - H_{-} = 0. \tag{4.27}$$

The maximum admissible allocated negative-sequence magnitude imposed by phase ϕ is

$$I^{-}_{\text{alloc},\phi,\max} = \frac{-L_{\phi,-} + \sqrt{L^2_{\phi,-} + 4H_{-}}}{2}. \tag{4.28}$$

Accordingly, the three phase dependent limits are

$$I^{-}_{a,\max} = \frac{-L_{a,-} + \sqrt{L^2_{a,-} + 4H_{-}}}{2}, \tag{4.29}$$

$$I^{-}_{b,\max} = \frac{-L_{b,-} + \sqrt{L^2_{b,-} + 4H_{-}}}{2}, \tag{4.30}$$

$$I^{-}_{c,\max} = \frac{-L_{c,-} + \sqrt{L^2_{c,-} + 4H_{-}}}{2}. \tag{4.31}$$

The maximum negative-sequence magnitude that can be accommodated without exceeding the phase current limit is determined by the most restrictive phase:

$$I^{-}_{\text{alloc}} = \min\left(I^{-}_{a,\max}, I^{-}_{b,\max}, I^{-}_{c,\max}, I_{\max}\right). \tag{4.32}$$

The resulting negative-sequence dq current references are therefore

$$I_{d,\mathrm{lim}}^{-} = I_{\mathrm{alloc}}^{-} u_d^{-}, \tag{4.33}$$

$$I_{q,\mathrm{lim}}^{-} = I_{\mathrm{alloc}}^{-} u_q^{-}. \tag{4.34}$$

The phase-aware calculation of the available negative-sequence current and the resulting current allocation is presented in Fig. 4.7.

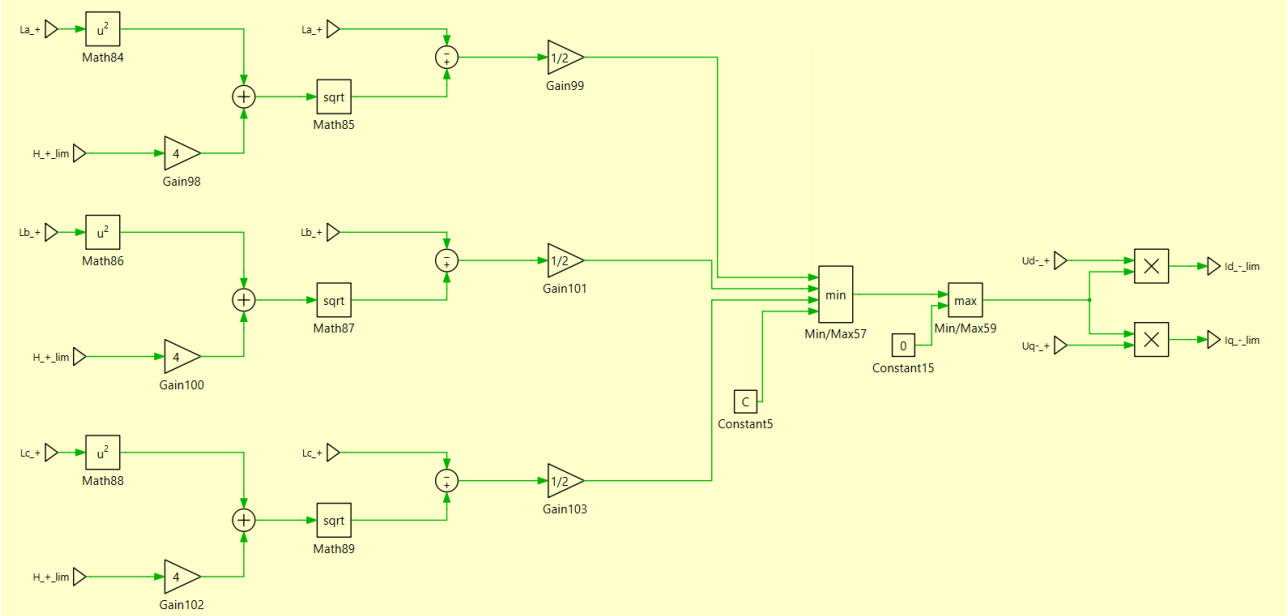


Figure 4.7.: PLECS implementation of the phase aware negative-sequence current allocation after the positive-sequence demand has been satisfied.

The resulting strategy therefore gives the positive-sequence controller first access to the available converter current capability. Once the positive-sequence demand has been satisfied, the remaining capability is allocated to the negative sequence.

This formulation differs from a conservative sequence magnitude constraint such as

$$I^{+} + I^{-} \leq I_{\mathrm{max}}. \tag{4.35}$$

Instead, the interaction between the two sequence current vectors is evaluated separately for phases a, b, and c. The negative-sequence contribution can therefore use the actual remaining phase current capability, improving utilization of the converter current rating during asymmetrical disturbances without intentionally exceeding I_{max}.

The activation of the priority-based current limiter is determined from the measured converter phase currents. The instantaneous magnitudes of the three phase currents are monitored and the largest value is calculated as

$$I_{\text{meas,max}} = \max\left(|i_a|, |i_b|, |i_c|\right). \tag{4.36}$$

When $I_{\text{meas,max}}$ exceeds the predefined current limit I_{max}, a limiting signal is generated. This signal controls a switch that selects the current references supplied to the inner current control loops. Under normal operating conditions, the unrestricted positive- and negative-sequence current references are used. Once the measured converter current exceeds the specified limit, the switch selects the references generated by the positive-sequence-priority allocation algorithm. In this way, the priority-based limiter is introduced only when the actual converter current requires limitation, rather than being continuously applied. The three phase currents are calculated as they did in the elliptical limiter, as shown in Fig. 4.1.

4.1.4. Negative-Sequence Priority Current Limiter

The negative-sequence priority current limiter reverses the allocation order considered in the positive-sequence priority strategy. Its main objective is to provide the negative-sequence controller with preferential access to the available converter current capability during asymmetrical grid disturbances. This allows a larger current contribution to be used for the compensation of the negative-sequence voltage and therefore for the reduction of voltage unbalance [2], [8].

However, completely suppressing the positive-sequence current may be undesirable for a grid-forming converter, since the positive-sequence control path is responsible for the main voltage forming and synchronization functions. For this reason, the implemented strategy allows a minimum positive-sequence current magnitude $I_{+,\text{min}}$ to be reserved before the remaining converter current capability is allocated. The negative-sequence current is then given priority, while any unused current capability is returned to the positive-sequence controller.

The allocation order can therefore be summarized as

$$I_{+,\text{min}} \quad \rightarrow \quad I^{-}_{\text{ref}} \quad \rightarrow \quad I^{+}_{\text{ref}}, \tag{4.37}$$

where the arrows indicate the order in which the available converter current capability is assigned. Setting $I_{+,\text{min}} = 0$ removes the reserved positive-sequence contribution and represents the limiting case of unrestricted negative-sequence priority.

Sequence Current Directions and Minimum Positive-Sequence Allocation

The magnitudes of the unrestricted positive- and negative-sequence current reference vectors are first calculated as

$$I^{+}_{\text{ref}} = \sqrt{\left(I^{+}_{d,\text{ref}}\right)^2 + \left(I^{+}_{q,\text{ref}}\right)^2}, \tag{4.38}$$

and

$$I_{\mathrm{ref}}^{-}=\sqrt{\left(I_{d,\mathrm{ref}}^{-}\right)^{2}+\left(I_{q,\mathrm{ref}}^{-}\right)^{2}}. \tag{4.39}$$

The direction of each sequence current vector is retained through the corresponding unit vector components:

$$u_{d}^{+}=\frac{I_{d,\mathrm{ref}}^{+}}{\max\left(I_{\mathrm{ref}}^{+},C\right)},\qquad u_{q}^{+}=\frac{I_{q,\mathrm{ref}}^{+}}{\max\left(I_{\mathrm{ref}}^{+},C\right)}, \tag{4.40}$$

$$u_{d}^{-}=\frac{I_{d,\mathrm{ref}}^{-}}{\max\left(I_{\mathrm{ref}}^{-},C\right)},\qquad u_{q}^{-}=\frac{I_{q,\mathrm{ref}}^{-}}{\max\left(I_{\mathrm{ref}}^{-},C\right)}, \tag{4.41}$$

where C is a small positive constant introduced to avoid numerical problems (division by zero) when the magnitude of a sequence current reference approaches zero.

Before assigning current capability to the negative sequence, a minimum positive-sequence contribution is reserved according to

$$I_{d,0}^{+}=I_{+,\min}u_{d}^{+},\qquad I_{q,0}^{+}=I_{+,\min}u_{q}^{+}. \tag{4.42}$$

The squared magnitude of this reserved contribution is

$$P_{0}^{2}=\left(I_{d,0}^{+}\right)^{2}+\left(I_{q,0}^{+}\right)^{2}. \tag{4.43}$$

The remaining squared current margin available before the negative-sequence allocation is therefore

$$H_{-}=\max\left(I_{\max}^{2}-P_{0}^{2},0\right). \tag{4.44}$$

The corresponding implementation, including the normalization of the two sequence current references, the minimum positive-sequence allocation, and the calculation of H_{-}, is shown in Fig. 4.8.

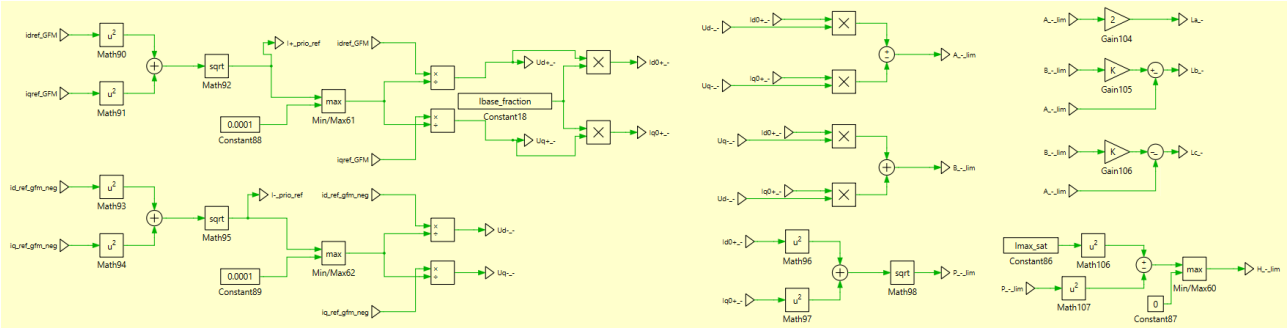


Figure 4.8.: PLECS implementation of the sequence current direction calculation, minimum positive-sequence allocation, and initial current margin for the negative-sequence priority limiter.

Negative-Sequence Current Allocation

The negative-sequence current is allocated after the minimum positive-sequence contribution has been reserved. As in the positive-sequence priority method, the available current is determined from the resulting current of each phase rather than from a conservative addition of the two sequence magnitudes.

The interaction between the reserved positive-sequence current and the negative-sequence direction is described by

$$A_- = I^+_{d,0} u^-_d - I^+_{q,0} u^-_q, \tag{4.45}$$

and

$$B_- = I^+_{d,0} u^-_q + I^+_{q,0} u^-_d. \tag{4.46}$$

The corresponding phase coefficients are

$$L_{a,-} = 2A_-, \tag{4.47}$$
$$L_{b,-} = -A_- + \sqrt{3}B_-, \tag{4.48}$$
$$L_{c,-} = -A_- - \sqrt{3}B_-. \tag{4.49}$$

For a generic phase ϕ, the squared phase-current magnitude can then be written as

$$I^2_\phi = P^2_0 + (I^-)^2 + L_{\phi,-} I^-, \tag{4.50}$$

where I^- represents the magnitude of the negative-sequence current to be allocated.

Imposing the phase current constraint

$$I_\phi = I_{\max} \tag{4.51}$$

results in

$$(I^-)^2 + L_{\phi,-} I^- - H_- = 0. \tag{4.52}$$

The maximum negative-sequence magnitude permitted by each individual phase is therefore

$$I^-_{\phi,\max} = \frac{-L_{\phi,-} + \sqrt{L^2_{\phi,-} + 4H_-}}{2}. \tag{4.53}$$

Consequently,

$$I^-_{a,\max} = \frac{-L_{a,-} + \sqrt{L^2_{a,-} + 4H_-}}{2}, \tag{4.54}$$

$$I^-_{b,\max} = \frac{-L_{b,-} + \sqrt{L^2_{b,-} + 4H_-}}{2}, \tag{4.55}$$

$$I^-_{c,\max} = \frac{-L_{c,-} + \sqrt{L^2_{c,-} + 4H_-}}{2}. \tag{4.56}$$

The maximum negative-sequence magnitude that can be accommodated by the converter is selected from the most restrictive phase:

$$I^-_{\text{allow}} = \min\left(I^-_{a,\max}, I^-_{b,\max}, I^-_{c,\max}, I_{\max}\right). \tag{4.57}$$

The applied negative-sequence magnitude is therefore

$$I^-_{\text{cmd}} = \min\left(I^-_{\text{ref}}, I^-_{\text{allow}}\right). \tag{4.58}$$

and the corresponding dq references become

$$I^-_{d,\text{lim}} = I^-_{\text{cmd}} u^-_d, \qquad I^-_{q,\text{lim}} = I^-_{\text{cmd}} u^-_q. \tag{4.59}$$

The PLECS implementation of the phase-dependent negative-sequence current limits and the final negative-sequence allocation is presented in Fig. 4.9.

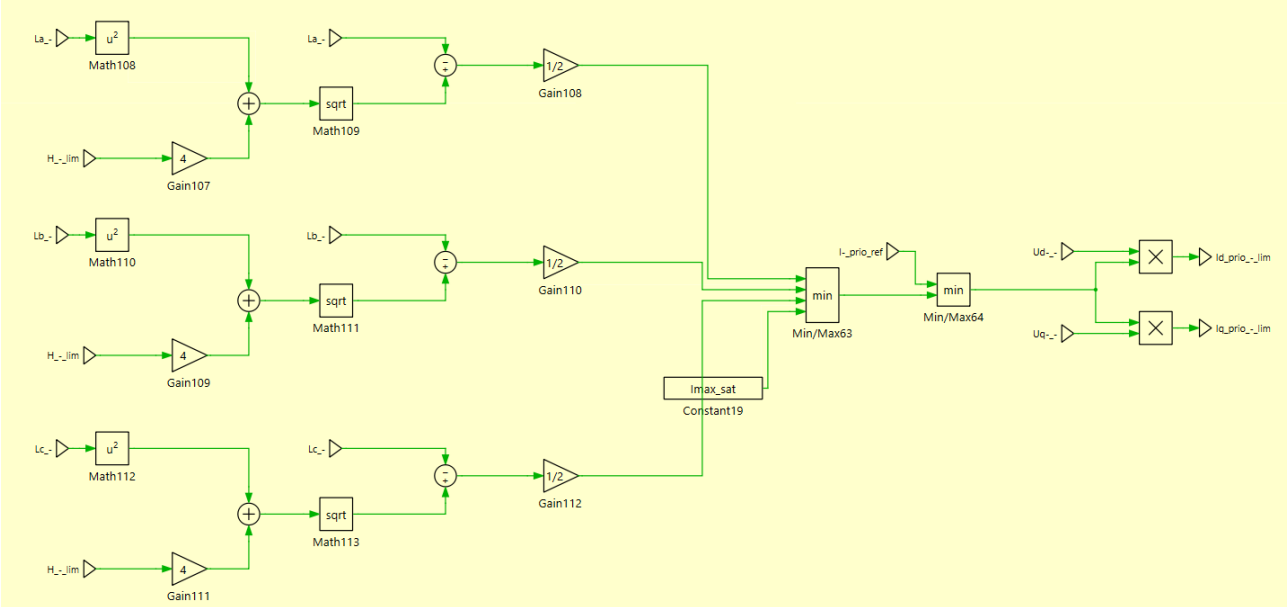


Figure 4.9.: PLECS implementation of the phase aware negative-sequence current allocation.

Allocation of the Remaining Current Capability

Once the negative-sequence current has been determined, the current capability that remains available to the positive-sequence is calculated. The squared magnitude of the allocated negative-sequence current is

$$N^2 = \left(I^-_{d,\mathrm{lim}}\right)^2 + \left(I^-_{q,\mathrm{lim}}\right)^2. \tag{4.60}$$

The remaining squared current margin becomes

$$H_+ = \max\left(I^2_{\mathrm{max}} - N^2, 0\right). \tag{4.61}$$

The interaction between the allocated negative-sequence current and the positive-sequence current direction is expressed as

$$A_+ = u^+_d I^-_{d,\mathrm{lim}} - u^+_q I^-_{q,\mathrm{lim}}, \tag{4.62}$$

$$B_+ = u^+_d I^-_{q,\mathrm{lim}} + u^+_q I^-_{d,\mathrm{lim}}. \tag{4.63}$$

The corresponding phase coefficients are

$$L_{a,+} = 2A_+, \tag{4.64}$$
$$L_{b,+} = -A_+ + \sqrt{3}B_+, \tag{4.65}$$
$$L_{c,+} = -A_+ - \sqrt{3}B_+. \tag{4.66}$$

The calculation of the remaining positive-sequence current capability is illustrated in Fig. 4.10.

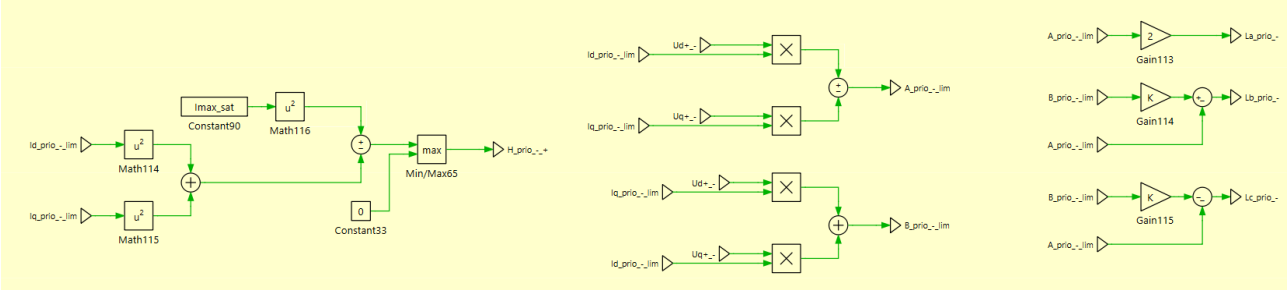


Figure 4.10.: PLECS calculation of the remaining phase current capability available to the positive-sequence controller after the negative-sequence allocation.

Following the same phase current constraint used for the negative-sequence allocation, the maximum positive-sequence magnitude allowed by each phase is

$$I^{+}_{\phi,\max} = \frac{-L_{\phi,+} + \sqrt{L^2_{\phi,+} + 4H_+}}{2}. \tag{4.67}$$

Thus,

$$I^{+}_{a,\max} = \frac{-L_{a,+} + \sqrt{L^2_{a,+} + 4H_+}}{2}, \tag{4.68}$$

$$I^{+}_{b,\max} = \frac{-L_{b,+} + \sqrt{L^2_{b,+} + 4H_+}}{2}, \tag{4.69}$$

$$I^{+}_{c,\max} = \frac{-L_{c,+} + \sqrt{L^2_{c,+} + 4H_+}}{2}. \tag{4.70}$$

The available positive-sequence magnitude is therefore

$$I^{+}_{\text{allow}} = \min\left(I^{+}_{a,\max}, I^{+}_{b,\max}, I^{+}_{c,\max}, I_{\max}\right). \tag{4.71}$$

The positive-sequence current demand is first restricted by this remaining capability:

$$\tilde{I}^{+}_{\text{cmd}} = \min\left(I^{+}_{\text{ref}}, I^{+}_{\text{allow}}\right). \tag{4.72}$$

The final magnitude is then constrained so that the selected minimum positive-sequence contribution is retained:

$$I^{+}_{\text{cmd}} = \max\left(I_{+,\min}, \tilde{I}^{+}_{\text{cmd}}\right). \tag{4.73}$$

Finally,

$$I^{+}_{d,\text{lim}} = I^{+}_{\text{cmd}} u^{+}_{d}, \qquad I^{+}_{q,\text{lim}} = I^{+}_{\text{cmd}} u^{+}_{q}. \tag{4.74}$$

The final positive-sequence allocation is shown in Fig. 4.11.

The resulting strategy therefore reserves the selected minimum positive-sequence contribution, gives the negative-sequence controller preferential access to the remaining converter current capability, and only afterwards assigns any unused capability back to the positive-sequence controller.

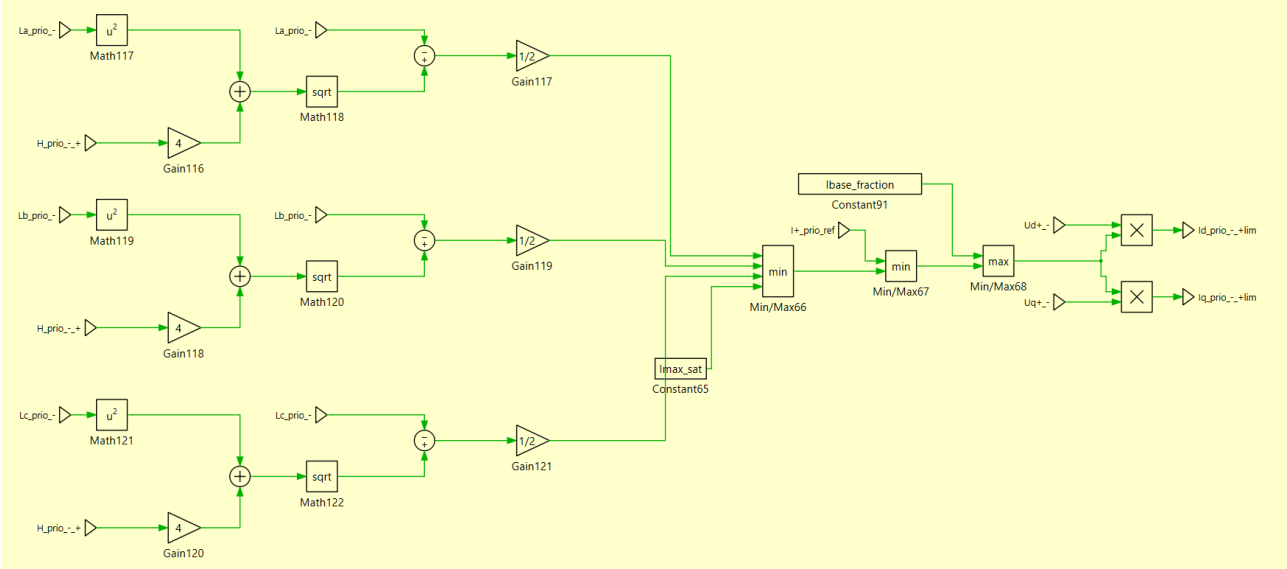


Figure 4.11.: PLECS implementation of the final positive-sequence allocation after the negative-sequence current demand has been satisfied.

An important feature of the implemented formulation is that the current capability is determined directly from the resulting three phase current constraints. Therefore, the conservative condition

$$I^+ + I^- \leq I_{\max} \tag{4.75}$$

is not used as the principal allocation rule. Instead, the positive- and negative-sequence interaction is considered separately for phases a, b, and c, and the most restrictive phase determines the allowable sequence current magnitude. This improves the utilization of the available converter current capability under asymmetrical conditions.

As in the positive-sequence priority limiter, the activation of the priority-based current limiter is determined from the measured converter phase currents. The instantaneous magnitudes of the three phase currents are monitored and the largest value is calculated as

$$I_{\text{meas,max}} = \max\left(|i_a|, |i_b|, |i_c|\right). \tag{4.76}$$

As already mentioned the calculation of the each phase current it is shown in Fig. 4.1.

When $I_{\text{meas,max}}$ exceeds the predefined current limit $I_{\max}$, a limiting signal is generated. This signal controls a switch that selects the current references supplied to the inner current control loops, exactly as it did in the positive-sequence priority limiter.

4.2. Grid Disturbance Implementation

The grid disturbances considered in the simulation studies are introduced directly in the three phase grid voltage generation stage. The implemented structure allows independent modification of the voltage magnitude and phase angle, as shown in Fig. 4.12. Step signals are used to determine the instants at which each disturbance is applied and removed.

Voltage magnitude disturbances are implemented through the switches located on the left hand side of Fig. 4.12. During normal operation, the nominal sinusoidal voltage is selected. When the corresponding switching command is activated, a scaled version of the same waveform is selected instead. In this way, the voltage magnitude of each phase can be reduced independently, allowing both symmetrical and asymmetrical voltage sags to be simulated.

Phase-angle disturbances are introduced through the switching stage located on the right hand side of the figure. In this case, the switches select between sinusoidal signals having different phase angles. The transition from one waveform to another produces an abrupt change in the grid voltage phase angle, thereby simulating a phase jump. The phase shift can be selected independently for each phase, making it possible to investigate both symmetrical and asymmetrical phase-angle disturbances.

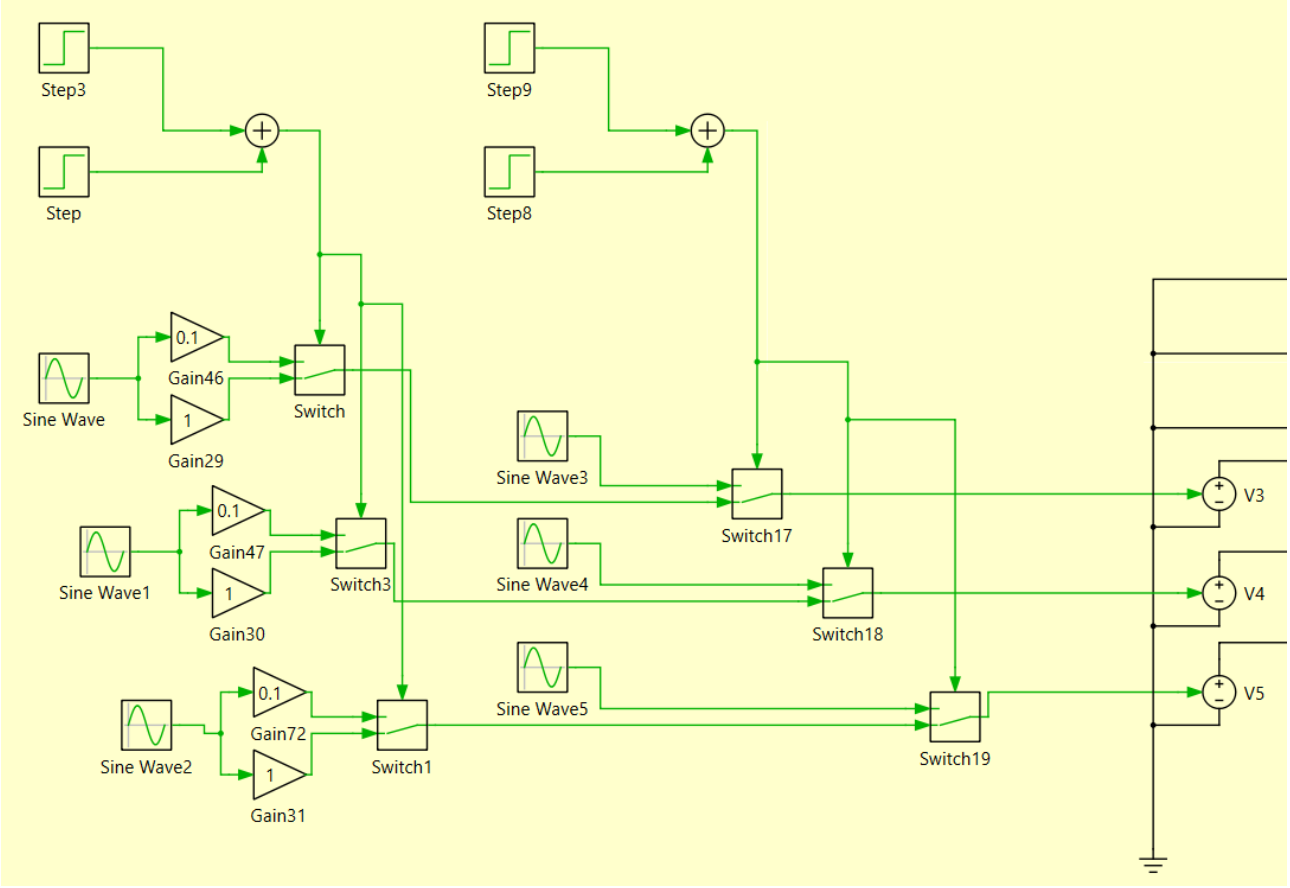


Figure 4.12.: PLECS implementation of grid voltage magnitude disturbances and phase-angle jumps.

Chapter 5.

Simulation Results

This section presents the time-domain simulation results of the current-limiting strategies on the grid-forming converter. The objective is to evaluate and compare the converter response under the grid disturbances, while examining the operation of the methods.

The results are organized according to the applied current-limiting strategy. First, the converter response without current-limiting action is presented as a reference. The elliptical current limiter and the virtual-admittance based current-limiting method with also the positive- and negative-sequence priority-based limiting strategies are then examined. For each method, the same four disturbances are applied.

The system configuration and operating conditions are maintained among the different strategies and faults, so when comparing the current-limiting approaches, the effect of only the limiting method can be identified directly from the simulation results. Interpretation and comparison of the observed responses are reserved for the following chapter.

5.1. Grid Disturbances

Four grid disturbances are considered in order to evaluate the dynamic response of the grid-forming converter under both symmetrical and asymmetrical operating conditions. The disturbances include three voltage magnitude sags and one phase angle disturbance, as illustrated in Fig. 5.1. In all cases, the disturbance is maintained for 0.3 s.

The first case is a **single-phase voltage sag**, in which the voltage magnitude of phase A is reduced from 1 p.u. to 0.1 p.u., while the remaining two phases retain their nominal voltage magnitude. This disturbance introduces a severe asymmetrical voltage condition and is used to examine the converter response in the presence of positive- and negative-sequence components.

The second case is a **two-phase voltage sag**. The voltage magnitudes of two phases are simultaneously reduced from 1 p.u. to 0.1 p.u., while the third phase remains at its nominal value. Compared with the single phase sag, this case represents a different and more severe asymmetrical voltage condition.

The third case is a **symmetrical three-phase voltage sag**, during which the voltage magnitudes of all three phases are reduced simultaneously from 1 p.u. to 0.1 p.u.. Since all three phases are affected equally, the voltage system remains balanced and the disturbance primarily affects the positive-sequence response of the converter.

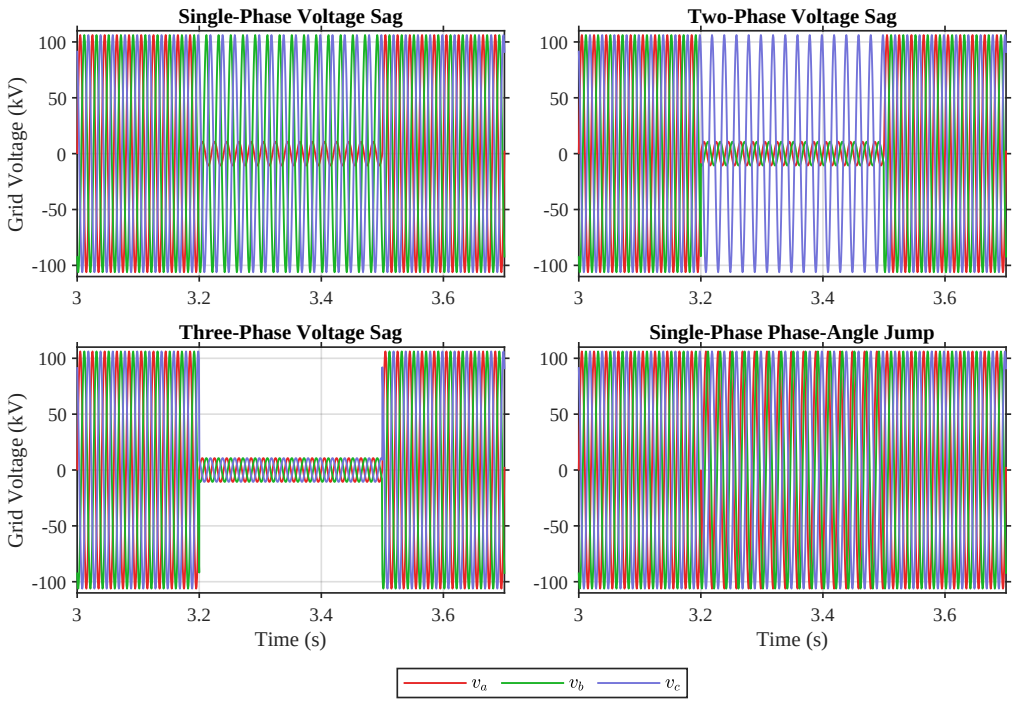


Figure 5.1.: Grid disturbances considered in the simulation study.

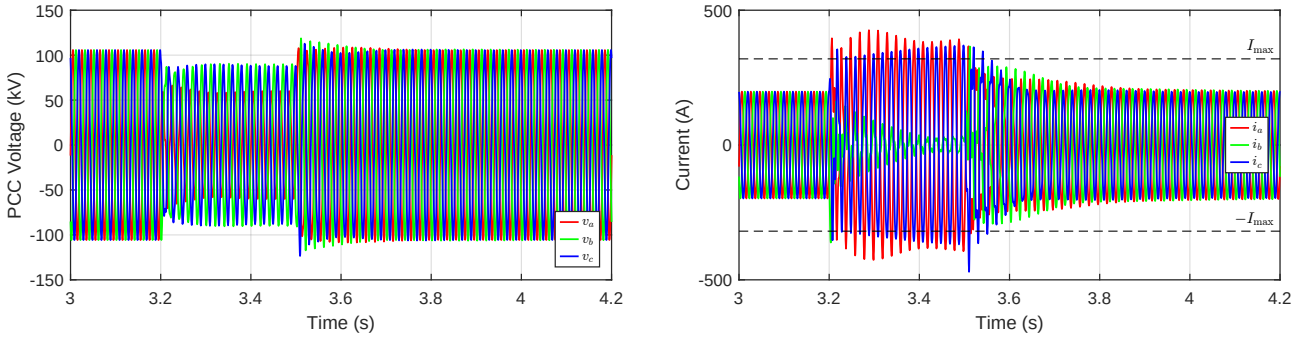
Finally, a **single-phase phase-angle disturbance** is considered. The phase angle of phase A is shifted by 60° for 0.3 s, while the voltage magnitudes and the phase angles of the remaining two phases remain unchanged. This produces an abrupt asymmetrical change in the angular relationship between the converter and the grid and allows the response to a phase angle disturbance to be evaluated separately from the voltage magnitude sags.

5.2. Operation Without Current Limitation

Before evaluating the implemented current-limiting methods, the converter is simulated with the current-limiting mechanisms disabled. This case provides a reference for the current levels and transient response that would occur if the references were supplied directly to the inner current controller without anything interfering. The same types of grid disturbances used in the following subsections are applied to this reference case.

5.2.1. Single-phase voltage sag

The first disturbance considered is a single-phase voltage sag. This produces a severe asymmetrical voltage condition at the PCC. The current references are applied directly to the inner current controller, since no current-limiting mechanism is active.

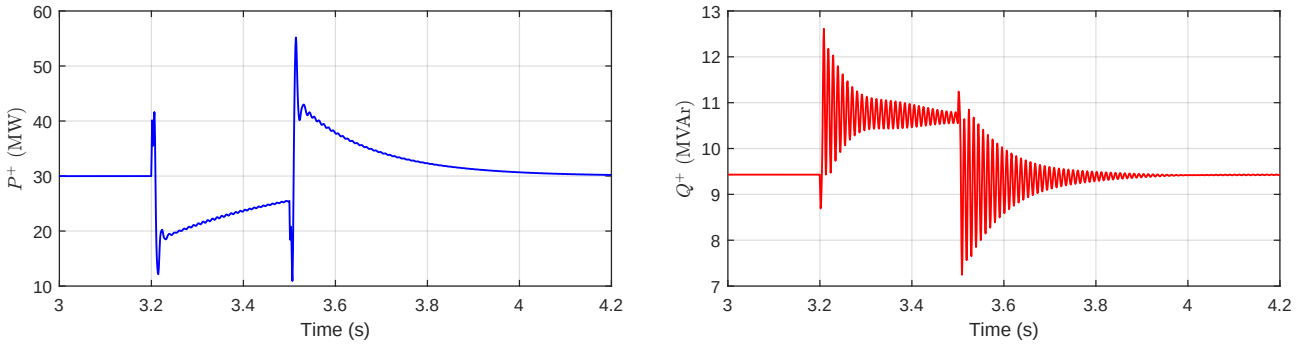


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.2.: PCC voltage and converter current during the single-phase voltage sag without current limitation.

Figure 5.2 shows the PCC voltage and the resulting three phase converter current. The applied single-phase voltage sag introduces a clear asymmetry in the PCC voltages and consequently produces an unbalanced converter current response. Since the current references are not limited, the converter is free to increase its current according to the voltage error generated during the disturbance. This case therefore illustrates the current stress that may occur when the grid-forming controller operates without a dedicated current-limiting mechanism.



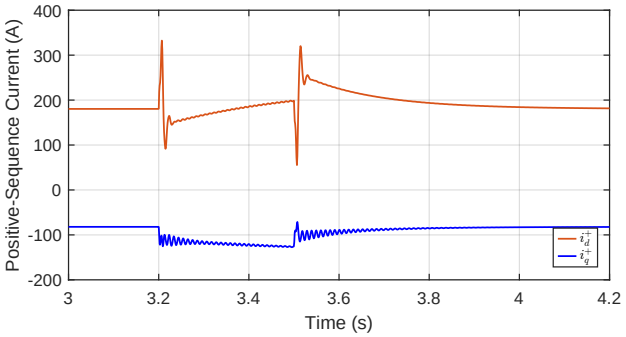
(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

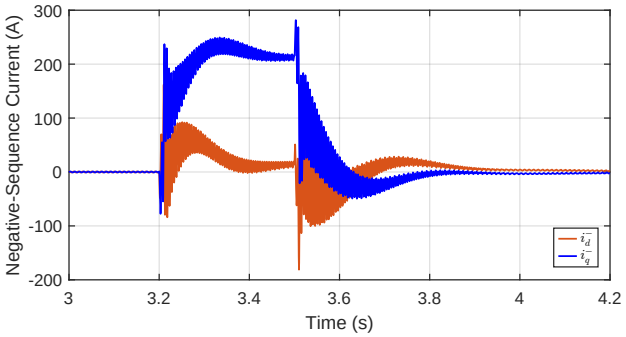
Figure 5.3.: Positive-sequence active and reactive power during the single-phase voltage sag without current limitation.

The corresponding positive-sequence active and reactive powers are presented in Fig. 5.3. These quantities represent the signals used by the outer grid-forming control loops and therefore provide an indication of how the disturbance affects the active power synchronization and reactive power voltage control mechanisms. Their transient response during the fault and their recovery after fault clearing are examined together with the converter current response.

The sequence current response is shown in Fig. 5.4. The asymmetrical disturbance results in the appearance of negative-sequence current components in addition to the positive-sequence current required for the normal grid-forming operation. The simultaneous presence of positive- and negative-sequence currents explains the unbalanced phase current waveforms observed in Fig. 5.2. These quantities are particularly important for the subsequent evaluation of the current-limiting methods, since the available converter current capability must be shared between the two sequence components during asymmetrical disturbances.



(a) Positive-sequence dq currents.

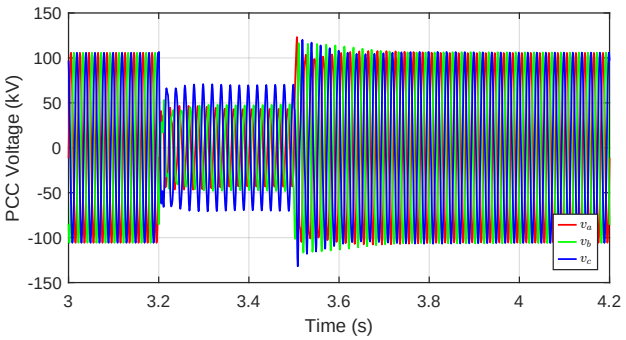


(b) Negative-sequence dq currents.

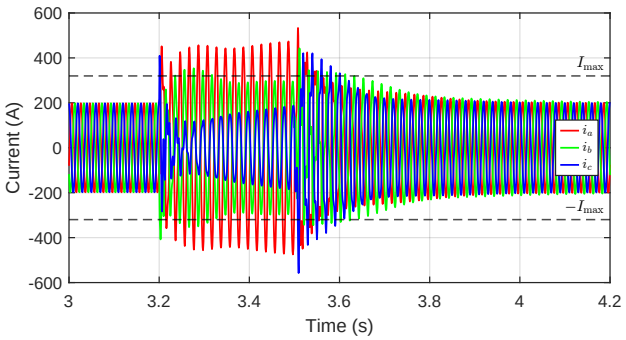
Figure 5.4.: Positive- and negative-sequence dq current components during the single-phase voltage sag without current limitation.

5.2.2. Two-phase voltage sag

The second asymmetrical disturbance considered is a two-phase voltage sag. As in the previous case, all current-limiting mechanisms are disabled, allowing the unrestricted response of the converter to be examined under a more severe asymmetrical voltage disturbance.



(a) Three phase PCC voltage.

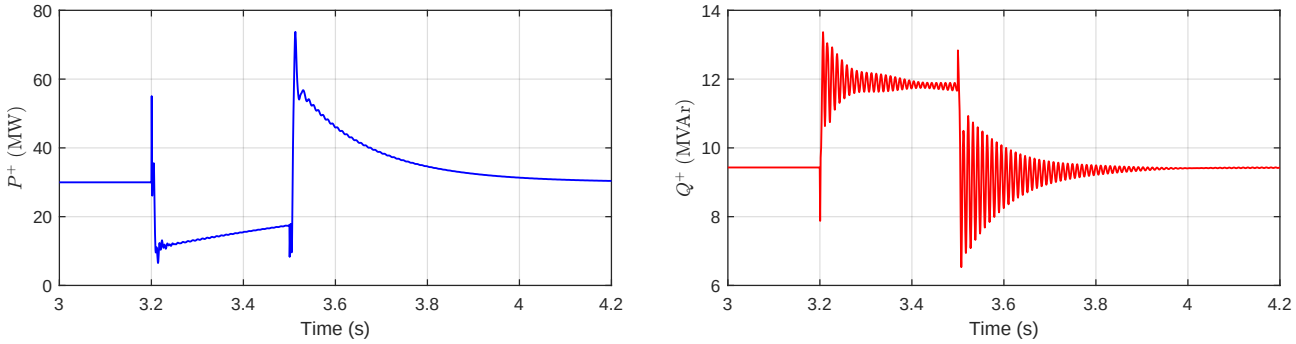


(b) Three phase converter current.

Figure 5.5.: PCC voltage and converter current during the two-phase voltage sag without current limitation.

Figure 5.5 shows the PCC voltage and the corresponding converter current response. The simultaneous reduction of two phase voltages produces a strongly unbalanced voltage condition at the PCC. Compared with the single phase disturbance, a larger portion of the three phase voltage system is affected, resulting in a more severe disturbance for the converter.

The positive-sequence active and reactive powers during the disturbance are shown in Fig. 5.6. The considerable reduction of the positive-sequence voltage during the two phase sag directly affects the power transferred between the converter and the grid. Consequently, both outer grid-forming control loops experience a transient deviation from their pre-disturbance operating conditions. The post-fault response also provides an indication of how effectively the converter returns to its original operating point once the grid voltages are restored.

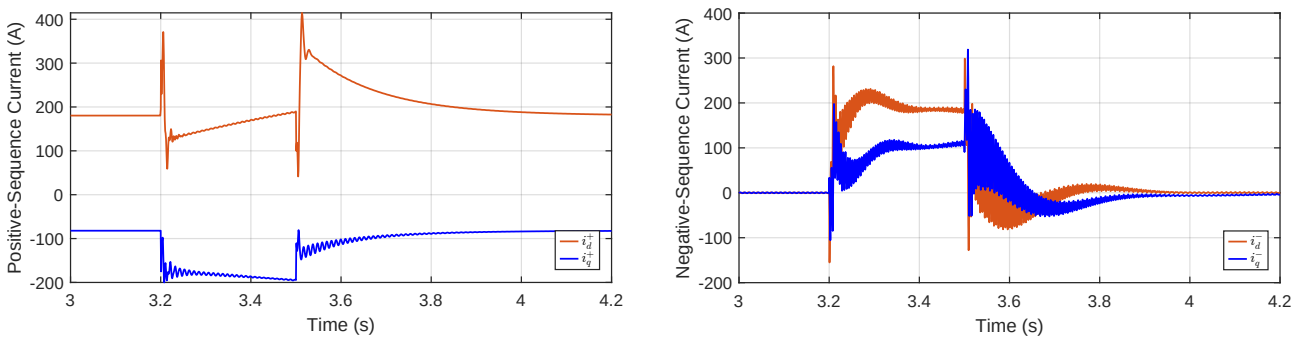


(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

Figure 5.6.: Positive-sequence active and reactive power during the two-phase voltage sag without current limitation.

The positive- and negative-sequence current components are presented in Fig. 5.7. Due to the asymmetrical nature of the two-phase voltage sag, a significant negative-sequence component is generated in addition to the positive-sequence current. The simultaneous contribution of the two sequences results in the unbalanced phase current waveforms observed in Fig. 5.5.



(a) Positive-sequence *dq* currents.

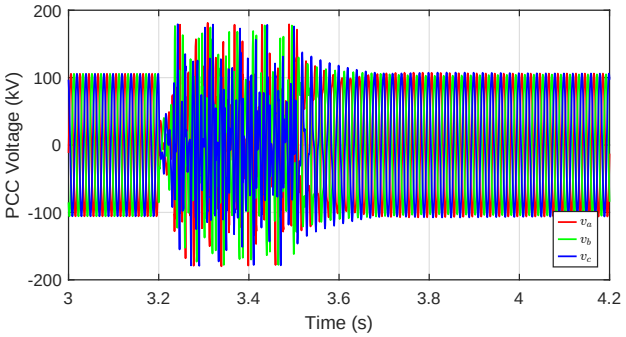
(b) Negative-sequence *dq* currents.

Figure 5.7.: Positive- and negative-sequence *dq* current components during the two-phase voltage sag without current limitation.

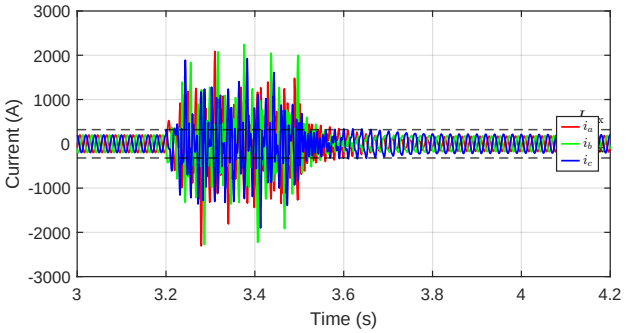
5.2.3. Three-Phase Voltage Sag

The final voltage magnitude disturbance considered in the unrestricted case is a symmetrical three-phase voltage sag. In contrast to the previous single and two phase disturbances, the three phase voltages remain balanced during the fault and therefore the disturbance mainly affects the positive-sequence component of the system.

Figure 5.8 presents the PCC voltage and the corresponding converter current response. Since the voltage magnitude of all three phases is reduced by the same amount, the PCC voltage would normally be expected to remain balanced during the disturbance. However, because the voltage sag is very severe, a noticeable voltage unbalance appears. Nevertheless, the large reduction in voltage produces a considerable voltage error within the converter control structure. With no current-limiting mechanism active, the resulting current references are applied directly to the inner current controller, allowing a substantial increase in the converter current during the sag.



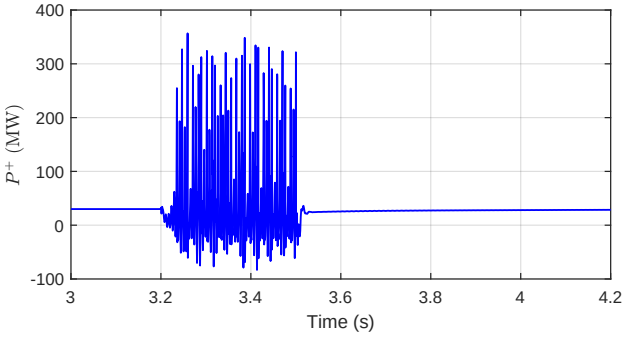
(a) Three phase PCC voltage.



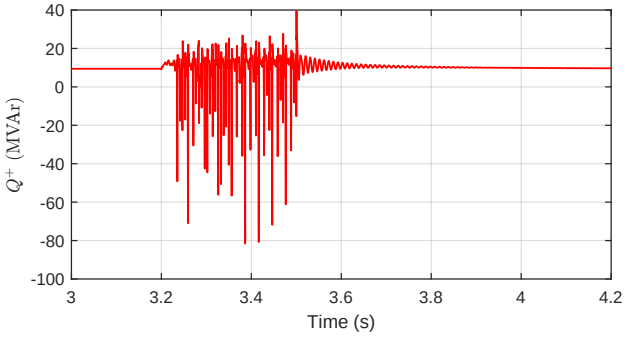
(b) Three phase converter current.

Figure 5.8.: PCC voltage and converter current during the three-phase voltage sag without current limitation.

Here it must be noted, that because of the voltage saturation before the converter, extreme currents like this can make the GFM fall out of synchronization, or even worse, destroy the sinusoidal nature of the voltage generated.



(a) Positive-sequence active power.

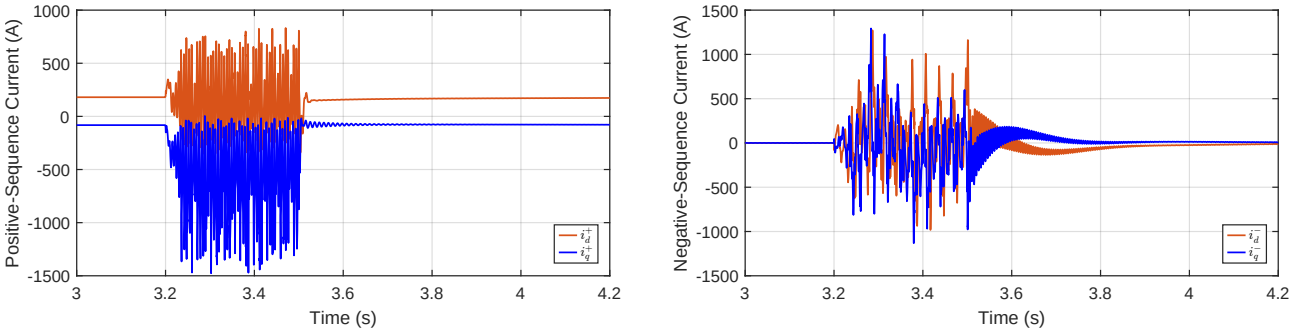


(b) Positive-sequence reactive power.

Figure 5.9.: Positive-sequence active and reactive power during the three-phase voltage sag without current limitation.

The positive-sequence active and reactive powers are shown in Fig. 5.9. The substantial reduction of the positive-sequence voltage directly influences the power exchanged between the converter and the grid and consequently produces a pronounced transient response in both outer grid-forming control loops. Following fault clearing, the power response is also used to assess the recovery of the converter toward its pre disturbance operating condition.

The sequence current response is presented in Fig. 5.10. Since the applied voltage sag is symmetrical, no negative-sequence voltage component is intentionally introduced by the disturbance. Consequently, the converter response is primarily governed by the positive-sequence control path. The negative sequence component observed in this case is only caused by the unbalance that develops in the system at very low voltage levels. This further highlights the importance of current limitation during severe disturbances.



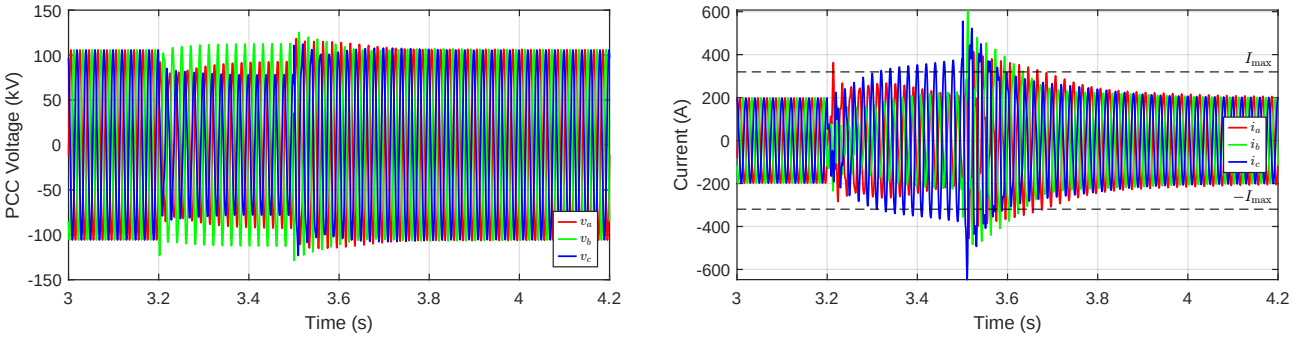
(a) Positive-sequence dq currents.

(b) Negative-sequence dq currents.

Figure 5.10.: Positive- and negative-sequence dq current components during the three-phase voltage sag without current limitation.

5.2.4. Single-Phase Phase-Angle Jump

The phase angle disturbance considered is an asymmetrical single-phase phase jump. This test allows the converter response to an abrupt change in the angular position of one grid phase to be examined, independently of voltage magnitude variations.



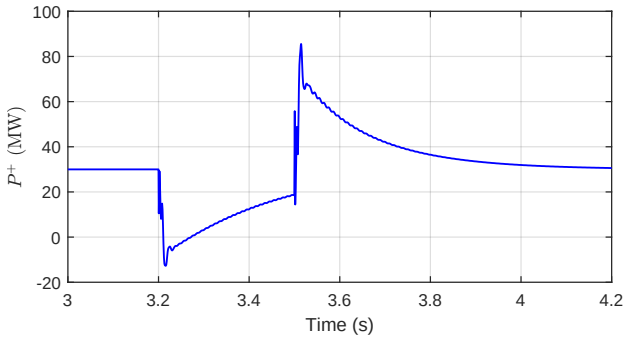
(a) Three phase PCC voltage.

(b) Three phase converter current.

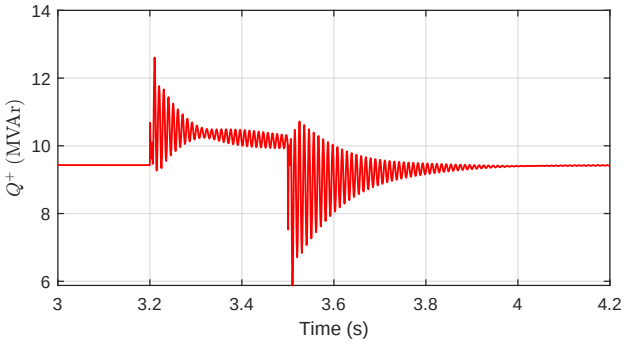
Figure 5.11.: PCC voltage and converter current during the single-phase 60° phase-angle jump without current limitation.

Figure 5.11 shows the three phase PCC voltage and the corresponding converter current response. Since the voltage magnitude remains unchanged, the disturbance appears as an abrupt displacement of the affected phase with respect to the other two phases. This immediately introduces an asymmetrical condition at the PCC and produces a transient response in the converter current.

The positive-sequence active and reactive powers are presented in Fig. 5.12. The sudden change in the angular relationship between the converter and the grid modifies the instantaneous power transfer conditions and consequently produces a transient response in both power components. Since these positive sequence quantities are used by the outer grid-forming loops, their response also indicates how the synchronization and voltage control mechanisms react to the imposed phase angle disturbance.

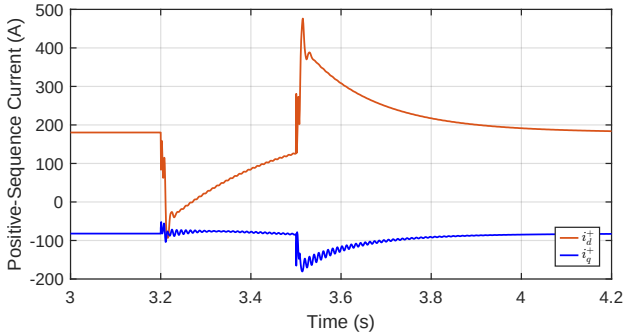


(a) Positive-sequence active power.

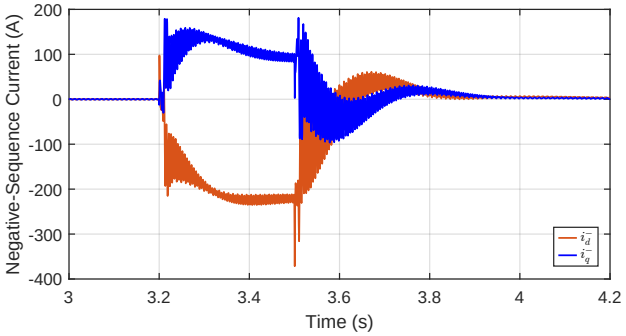


(b) Positive-sequence reactive power.

Figure 5.12.: Positive-sequence active and reactive power during the single-phase 60° phase-angle jump without current limitation.



(a) Positive-sequence *dq* currents.



(b) Negative-sequence *dq* currents.

Figure 5.13.: Positive- and negative-sequence *dq* current components during the single-phase 60° phase angle jump without current limitation.

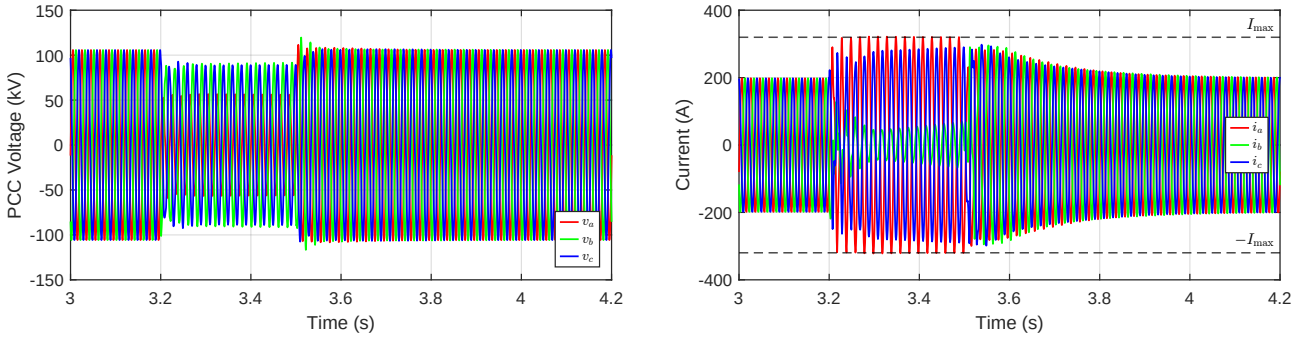
The corresponding sequence current components are shown in Fig. 5.13. Negative-sequence current components are generated together with the positive-sequence current response. The simultaneous presence of the two sequences contributes to the unbalanced phase currents observed in Fig. 5.11.

5.3. Elliptical Current Limiter

The performance of the elliptical current limiter is evaluated under the same grid disturbances considered in the unrestricted case. The limiter acts on the current references when the resulting converter current demand exceeds the specified current limit. The same disturbance conditions are maintained in order to allow a direct comparison with the results obtained without current limitation, as we already mentioned.

5.3.1. Single-phase voltage sag

In contrast to the unrestricted case, the elliptical current limiter is enabled in order to restrict the converter current demand generated during the asymmetrical disturbance.

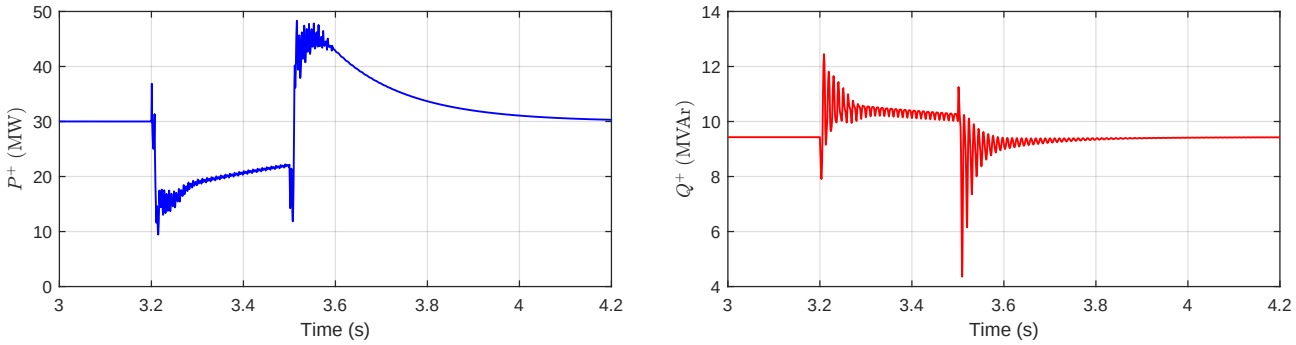


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.14.: PCC voltage and converter current during the single phase voltage sag with the elliptical current limiter.

Figure 5.14 shows the PCC voltage and the corresponding three phase converter current. In this case, the elliptical limiter modifies the current references when the imposed current boundary is reached. The resulting phase current response can therefore be directly compared with the unrestricted response presented previously in order to evaluate the effectiveness of the current-limiting action.



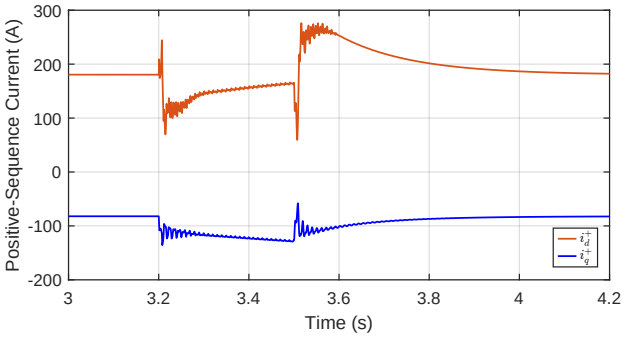
(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

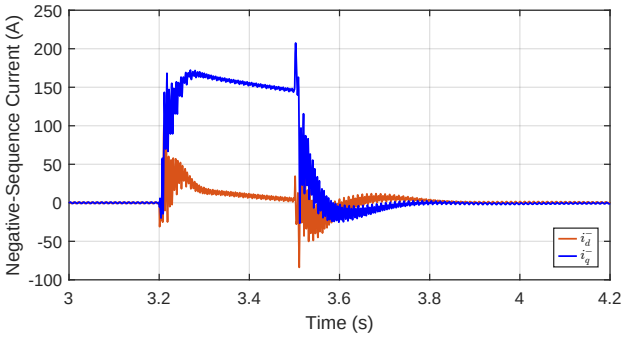
Figure 5.15.: Positive-sequence active and reactive power during the single-phase voltage sag with the elliptical current limiter.

The corresponding positive-sequence active and reactive powers are presented in Fig. 5.15. Current limitation modifies the current references available to the converter and can therefore influence the power transferred during the disturbance. These waveforms are used to evaluate the interaction between the current-limiting action and the outer grid-forming control loops, as well as the recovery of the converter following fault clearing.

The positive- and negative-sequence current components are shown in Fig. 5.16. As expected for an asymmetrical disturbance, both sequence components participate in the converter current response. The elliptical limiter acts on the overall current demand rather than treating the two sequences independently. The resulting sequence current response provides an important indication of how the limiting action affects the contribution of each sequence during the fault.



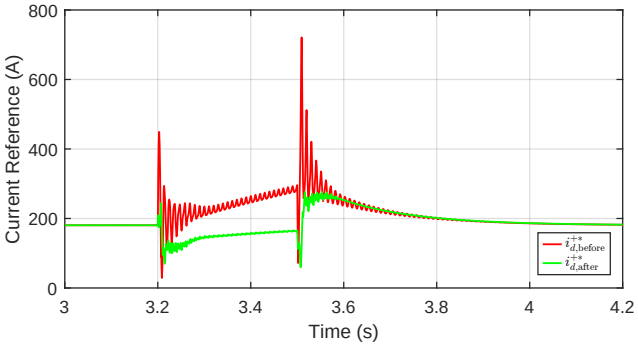
(a) Positive-sequence dq currents.



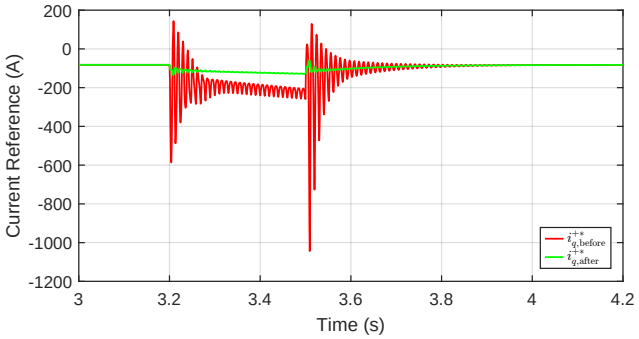
(b) Negative-sequence dq currents.

Figure 5.16.: Positive- and negative-sequence dq current components during the single-phase voltage sag with the elliptical current limiter.

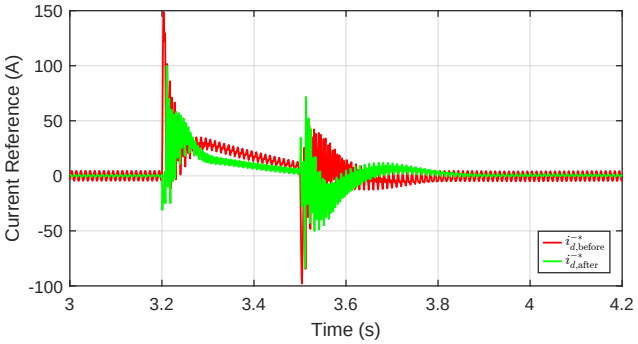
To further illustrate the action of the elliptical limiter, the current references before and after limitation are shown in Fig. 5.17. In the following plots, the reference before limitation is shown together with the corresponding limited reference, in order to illustrate the action of the elliptical current limiter on the sequence current commands.



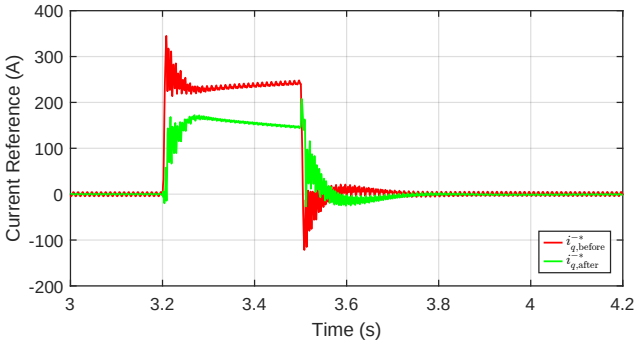
(a) Positive-sequence d-axis current reference.



(b) Positive-sequence q-axis current reference.



(c) Negative-sequence d-axis current reference.

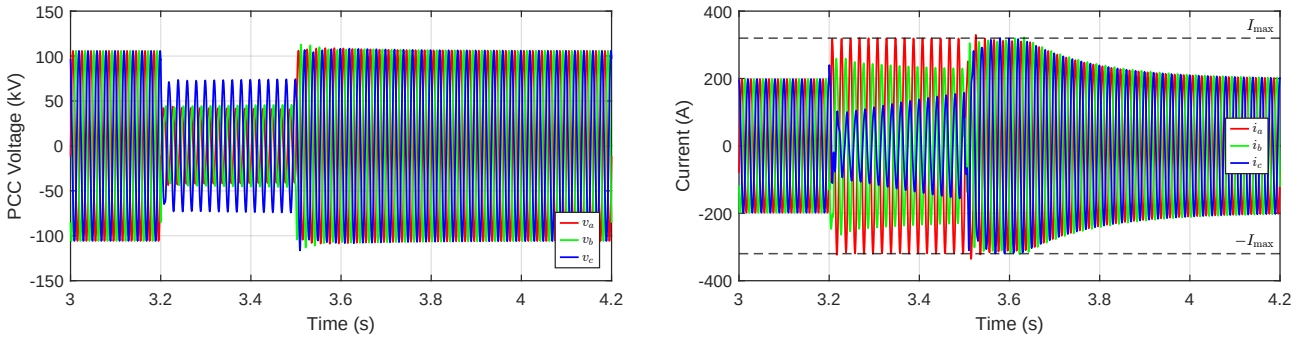


(d) Negative-sequence q-axis current reference.

Figure 5.17.: Positive- and negative-sequence d- and q-axis current references before and after limitation with the elliptical limiter during the single-phase voltage sag.

5.3.2. Two-phase voltage sag

The elliptical current limiter remains active throughout the disturbance in order to restrict the converter current demand under this more severe asymmetrical operating condition.

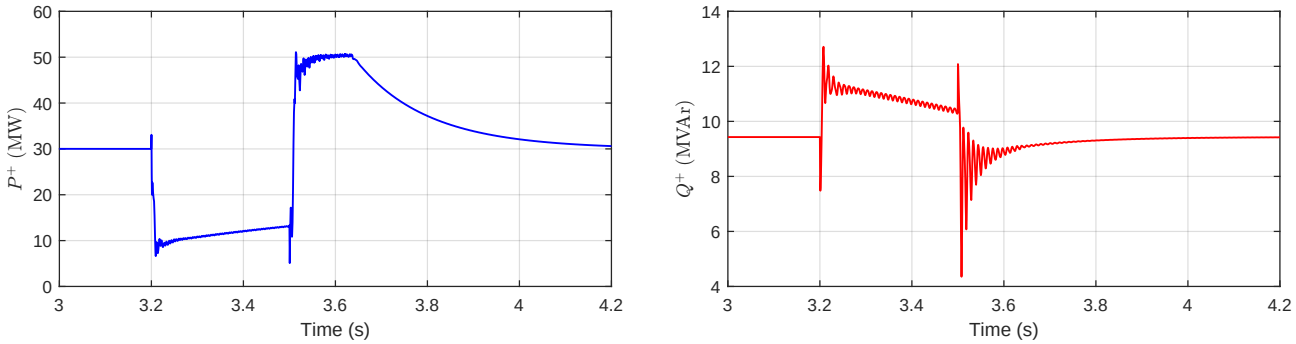


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.18.: PCC voltage and converter current during the two phase voltage sag with the elliptical current limiter.

Figure 5.18 presents the PCC voltage and the corresponding three phase converter current. The reduction of two phase voltages produces a strongly unbalanced condition and increases the current demand generated by the voltage control structure. Once the current boundary is reached, the elliptical limiter modifies the current references in order to restrict the resulting converter current.



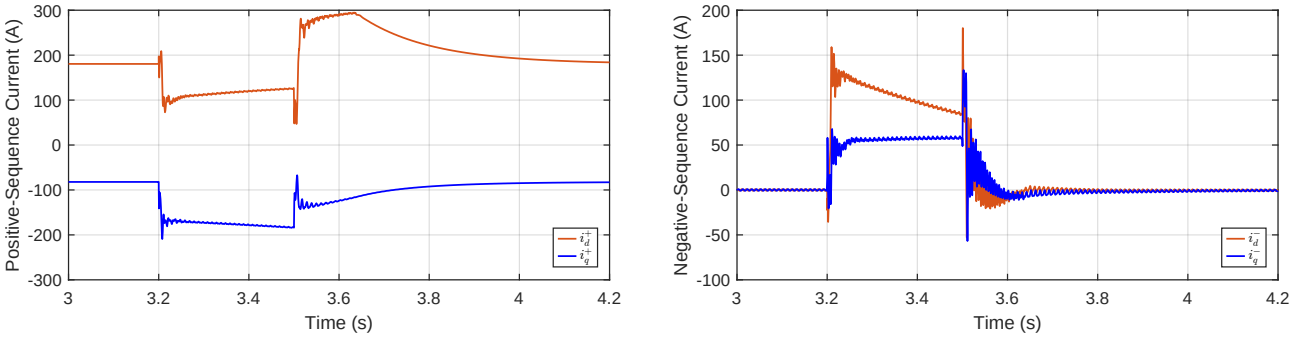
(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

Figure 5.19.: Positive-sequence active and reactive power during the two-phase voltage sag with the elliptical current limiter.

The corresponding positive-sequence active and reactive powers are shown in Fig. 5.19. The transient behavior of both power components and their recovery following fault clearing provide an indication of the interaction between the elliptical limiter and the outer grid-forming control loops.

The positive- and negative-sequence current components are presented in Fig. 5.20. Because of the asymmetrical nature of the two-phase voltage sag, both sequence components contribute to the total converter current demand.

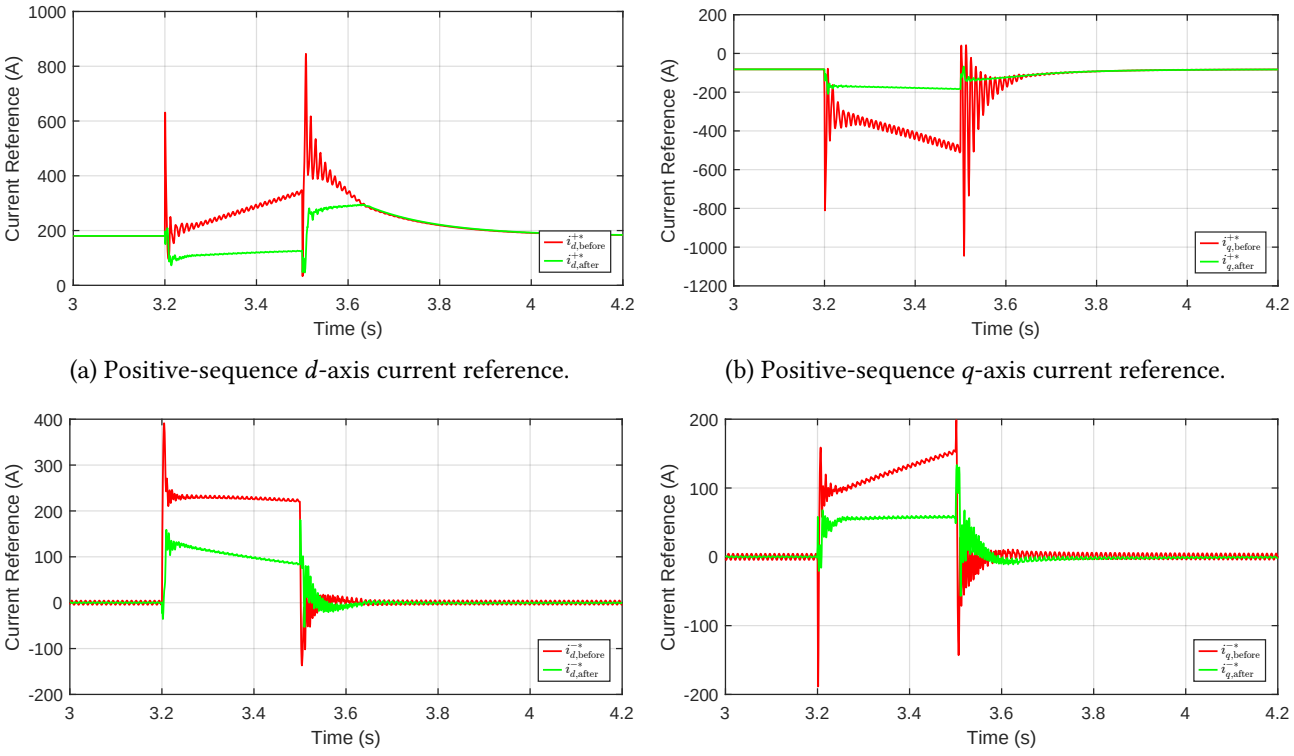


(a) Positive-sequence dq currents.

(b) Negative-sequence dq currents.

Figure 5.20.: Positive- and negative-sequence dq current components during the two-phase voltage sag with the elliptical current limiter.

The corresponding current references for the two-phase voltage sag are shown in Fig. 5.21. As in the single phase case, the asymmetrical disturbance gives rise to both positive- and negative-sequence reference components, whose amplitudes are modified by the elliptical limiter during the fault interval. In the following plots, the reference before limitation is shown together with the corresponding limited reference, in order to illustrate the action of the elliptical current limiter on the sequence current commands.



(a) Positive-sequence d-axis current reference.

(b) Positive-sequence q-axis current reference.

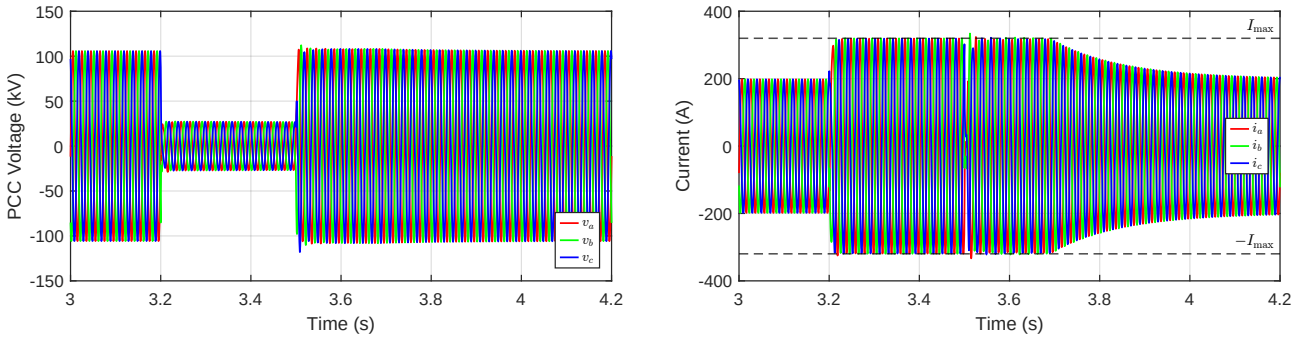
(c) Negative-sequence d-axis current reference.

(d) Negative-sequence q-axis current reference.

Figure 5.21.: Positive- and negative-sequence d- and q-axis current references before and after limitation with the elliptical limiter during the two-phase voltage sag.

5.3.3. Three-phase voltage sag

The next disturbance considered is a symmetrical three-phase voltage sag. The elliptical current limiter is enabled in order to restrict the large current demand produced by the severe reduction of the grid voltage.

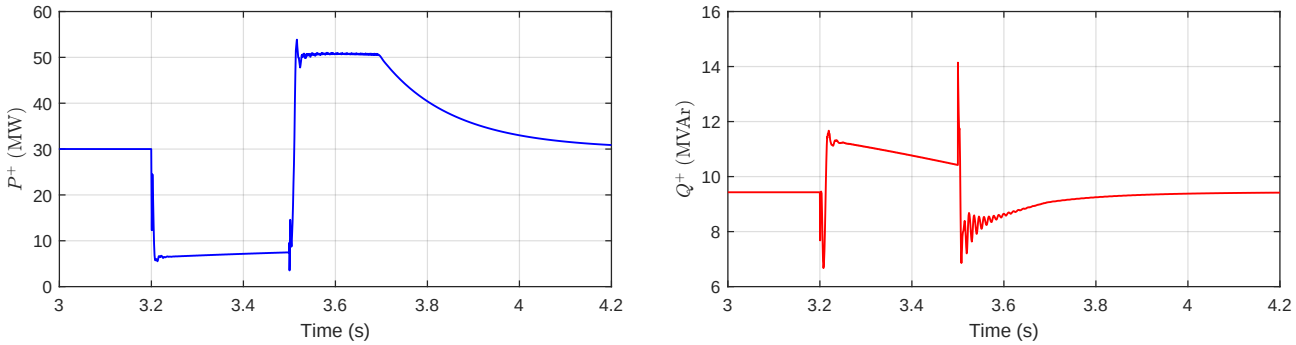


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.22.: PCC voltage and converter current during the symmetrical three-phase voltage sag with the elliptical current limiter.

Figure 5.22 shows the PCC voltage and the corresponding converter current response. The simultaneous voltage reduction in all three phases produces a substantial increase in the current demand of the converter. Once the specified current boundary is reached, the elliptical limiter reduces the current references in order to prevent the converter current from increasing without restriction. In contrast to the asymmetrical fault cases, the three phase currents remain balanced since the applied disturbance is symmetrical.

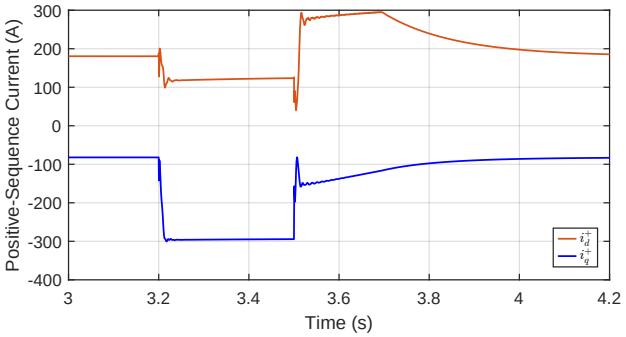


(a) Positive-sequence active power.

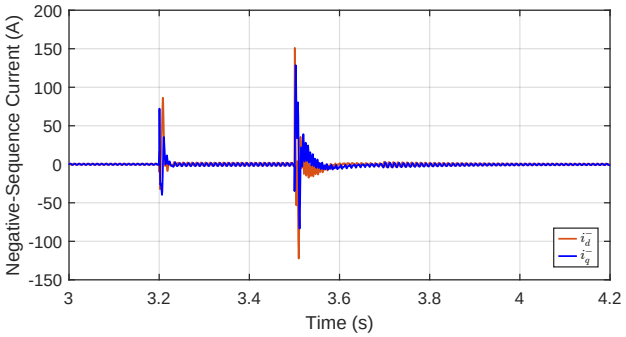
(b) Positive-sequence reactive power.

Figure 5.23.: Positive-sequence active and reactive power during the symmetrical three-phase voltage sag with the elliptical current limiter.

The positive-sequence active and reactive powers are presented in Fig. 5.23. The large reduction in the positive-sequence grid voltage significantly changes the power transfer conditions during the disturbance. At the same time, the activation of the current limiter restricts the current available to the converter and therefore also affects the active and reactive power response. Following fault clearing, the recovery of these quantities indicates how the outer grid-forming loops return toward its pre-disturbance operating condition.



(a) Positive-sequence dq currents.

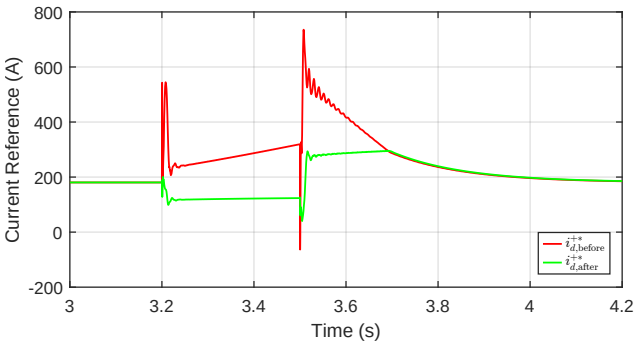


(b) Negative-sequence dq currents.

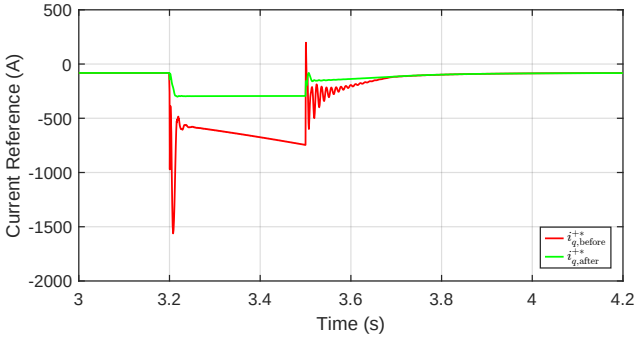
Figure 5.24.: Positive- and negative-sequence dq current components during the symmetrical three-phase voltage sag with the elliptical current limiter.

The sequence current response is shown in Fig. 5.24. Since the applied voltage sag is symmetrical, the converter current response is primarily determined by the positive-sequence components. The negative-sequence current is therefore expected to remain very small compared with the positive-sequence current.

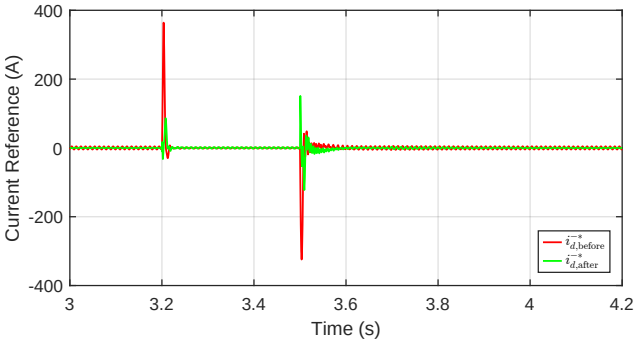
Figure 5.25 presents the current references during the three-phase symmetrical voltage sag. In the following plots, the reference before limitation is shown together with the corresponding limited reference, in order to illustrate the action of the elliptical current limiter on the sequence current commands.



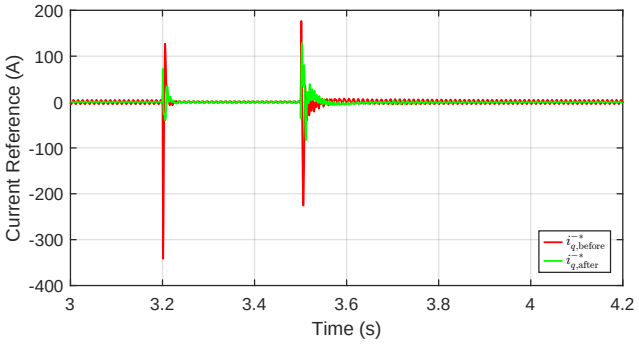
(a) Positive-sequence d-axis current reference.



(b) Positive-sequence q-axis current reference.



(c) Negative-sequence d-axis current reference.

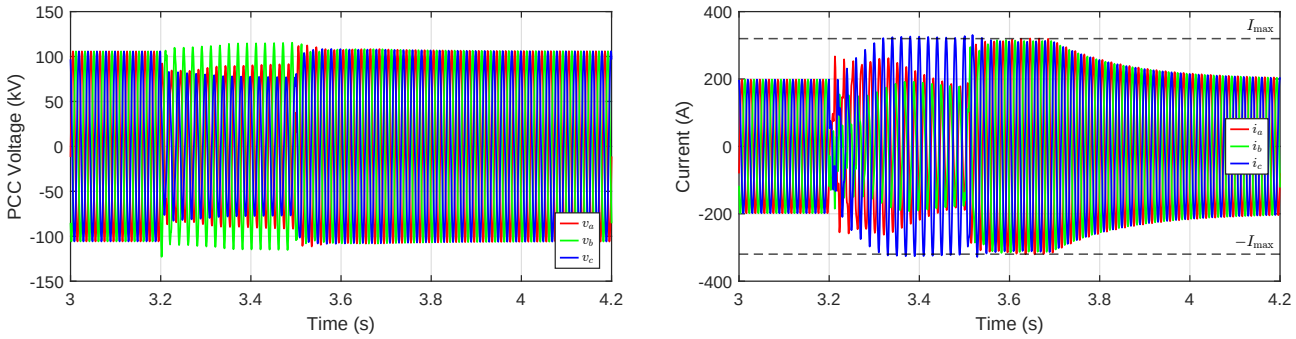


(d) Negative-sequence q-axis current reference.

Figure 5.25.: Positive- and negative-sequence d- and q-axis current references before and after limitation with the elliptical limiter during the three-phase symmetrical voltage sag.

5.3.4. Single-Phase Phase-Angle Jump

The final asymmetrical disturbance considered for the elliptical current limiter is a single-phase phase-angle jump. This produces an abrupt asymmetrical change in the grid voltage phase relationships. The elliptical current limiter is enabled in order to restrict the converter current demand resulting from the sudden angular displacement.

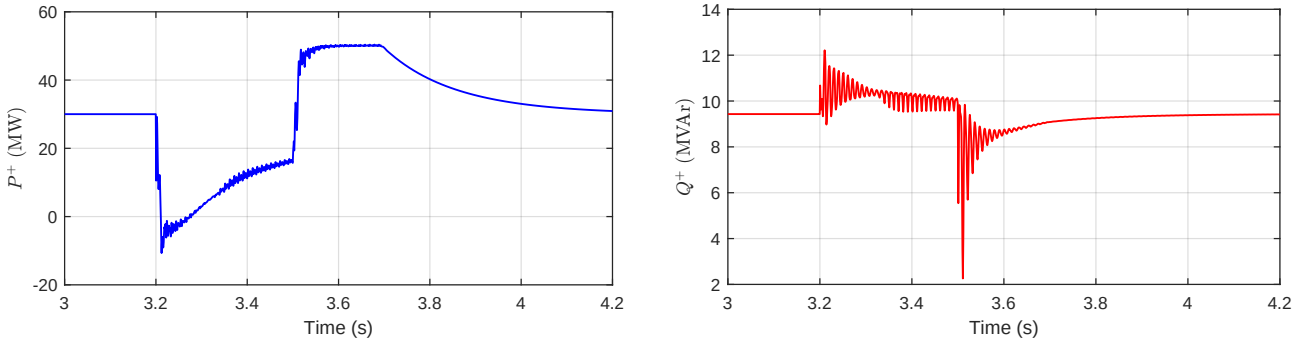


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.26.: PCC voltage and converter current during the single-phase 60° phase-angle jump with the elliptical current limiter.

Figure 5.26 presents the PCC voltage and the corresponding converter current response. Unlike the voltage sag cases, the grid voltage magnitude remains unchanged, while the sudden angular displacement of one phase introduces a severe asymmetry in the three phase voltage system. This change produces an increased current demand as the converter reacts to the phase mismatch.



(a) Positive-sequence active power.

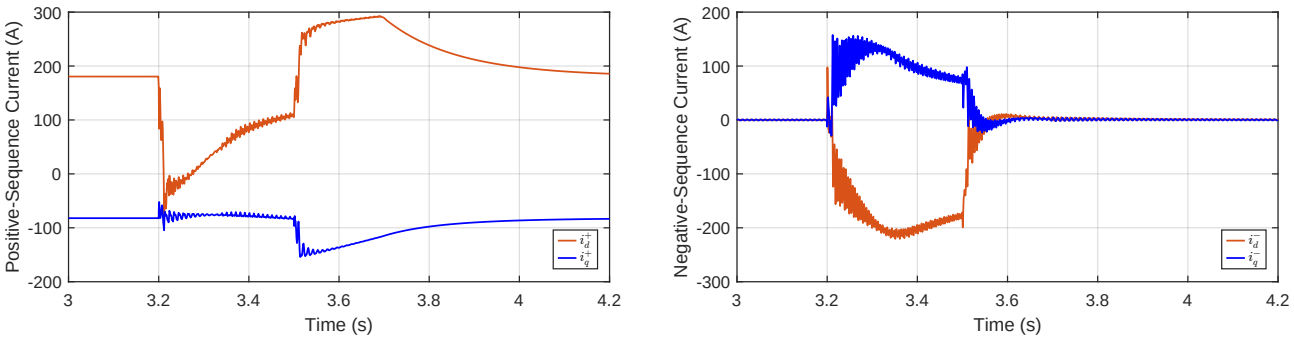
(b) Positive-sequence reactive power.

Figure 5.27.: Positive-sequence active and reactive power during the single-phase 60° phase-angle jump with the elliptical current limiter.

The corresponding positive-sequence active and reactive powers are shown in Fig. 5.27. The abrupt change in the angular relationship between the converter and the grid modifies the power transfer conditions and produces a transient response in both quantities. At the same time, the activation of the elliptical limiter restricts the available converter current and therefore influences the amount of active and reactive power that can be exchanged during the disturbance. The post-disturbance response is used to assess the recovery of the outer grid-forming control loops after the original phase angle is restored.

The positive- and negative-sequence current components are presented in Fig. 5.28. Since the phase angle

disturbance affects only one phase, both positive- and negative-sequence components are present in the converter current response. The elliptical limiter acts on the combined current demand and therefore constrains the current capability available to both sequence components.

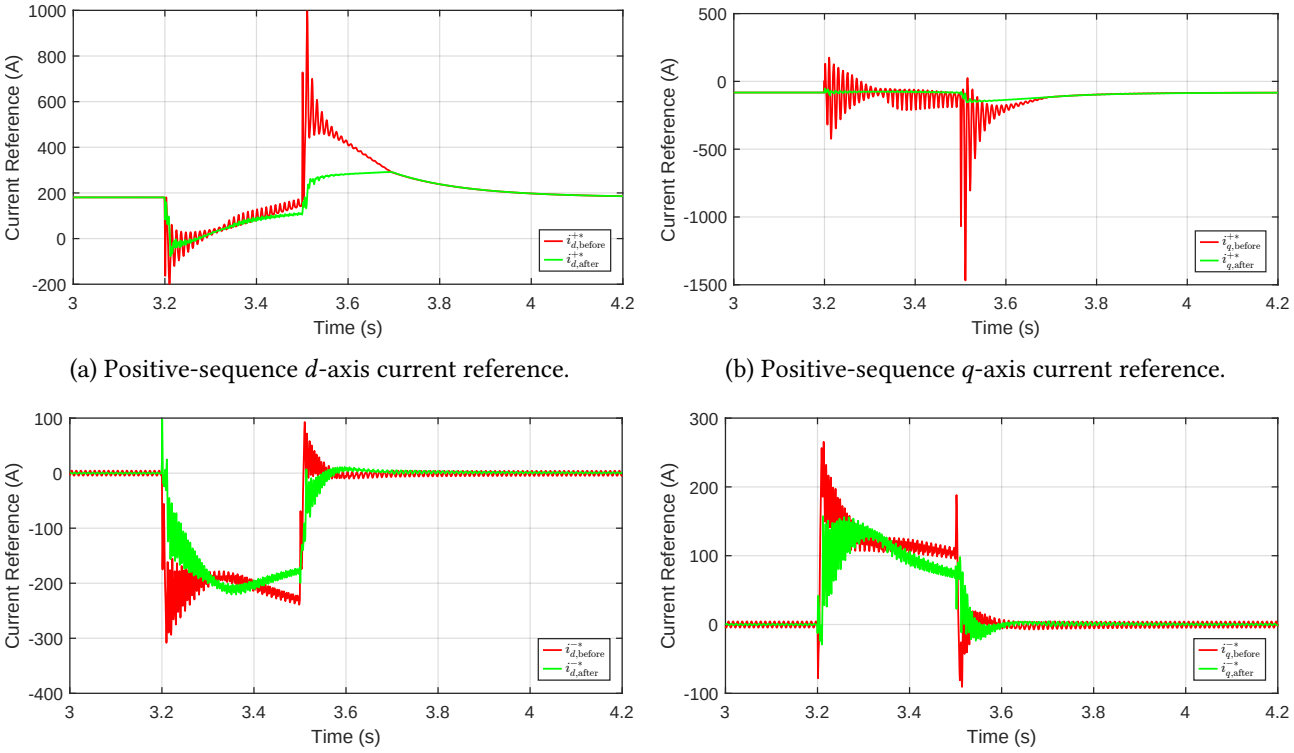


(a) Positive-sequence dq currents.

(b) Negative-sequence dq currents.

Figure 5.28.: Positive- and negative-sequence dq current components during the single-phase 60° phase-angle jump with the elliptical current limiter.

The current references obtained during the single-phase angle jump are shown in Fig. 5.29. Although the disturbance is caused by a phase angle variation, rather than a voltage magnitude reduction, it still creates an asymmetrical operating condition. In the following plots, the reference before limitation is shown together with the corresponding limited reference, in order to illustrate the action of the elliptical current limiter on the sequence current commands.



(a) Positive-sequence d-axis current reference.

(b) Positive-sequence q-axis current reference.

(c) Negative-sequence d-axis current reference.

(d) Negative-sequence q-axis current reference.

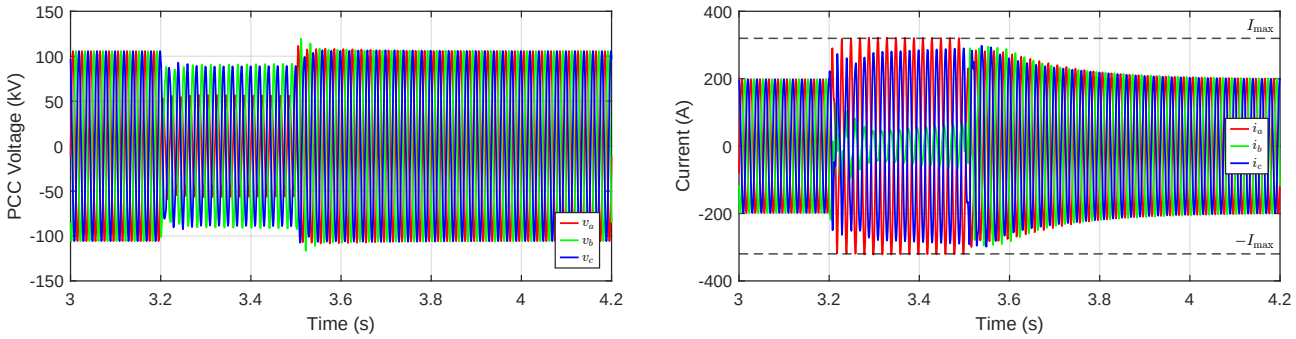
Figure 5.29.: Positive- and negative-sequence d- and q-axis current references before and after limitation with the elliptical limiter during the single-phase angle jump.

5.4. Virtual-Admittance-Based Current Limitation

The second current-limiting method is evaluated under the same four grid disturbances considered in the previous cases. Contrary to the elliptical current limiter, which acts directly on the current references, this method limits the current by modifying the parameters of the virtual admittance when excessive current demand is detected. The resulting increase in the effective virtual impedance reduces the current generated by the voltage difference between the converter internal voltage and the PCC voltage.

The same disturbance magnitudes and durations are maintained throughout the following simulations in order to allow a direct comparison in the chapter.

5.4.1. Single-phase voltage sag

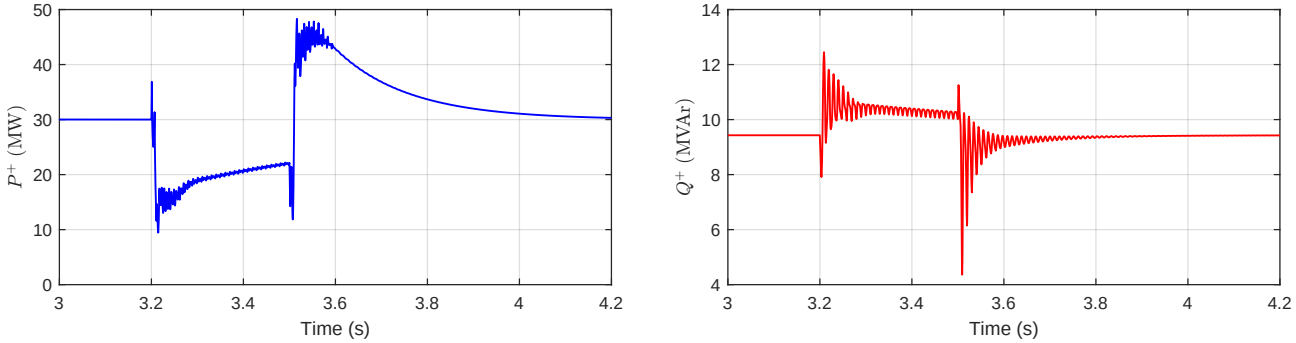


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.30.: PCC voltage and converter current during the single phase voltage sag with virtual-admittance based current limitation.

Figure 5.30 shows the PCC voltage and the corresponding three phase converter current. In contrast to the unrestricted case, the current response is now influenced by the variation of the virtual admittance, which reduces the current generated by the voltage difference between the converter internal voltage and the PCC voltage.

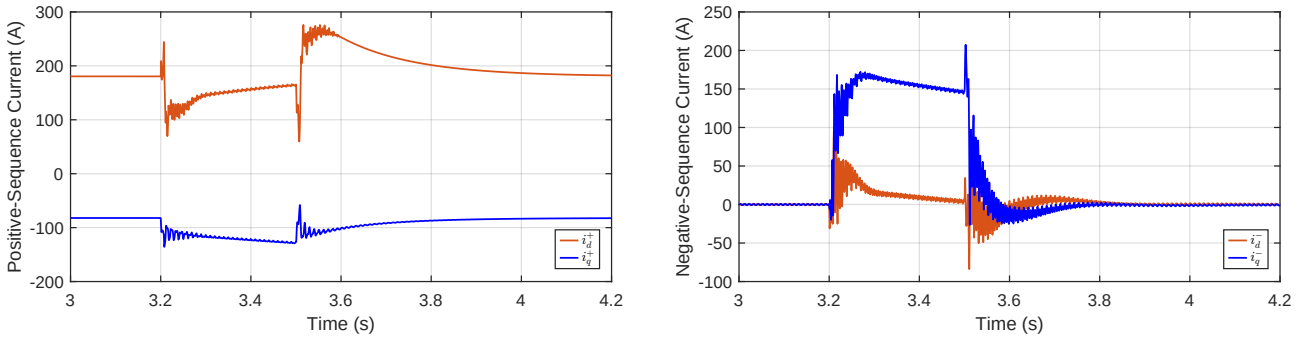


(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

Figure 5.31.: Positive-sequence active and reactive power during the single-phase voltage sag with virtual-admittance based current limitation.

The corresponding positive-sequence active and reactive powers are presented in Fig. 5.31.



(a) Positive-sequence dq currents.

(b) Negative-sequence dq currents.

Figure 5.32.: Positive- and negative-sequence dq current components during the single-phase voltage sag with virtual-admittance based current limitation.

The positive- and negative-sequence current components are shown in Fig. 5.32. As expected for an asymmetrical disturbance, both sequence components contribute to the converter current response. In this case, the current-limiting action does not directly saturate the current references, but instead changes the effective virtual admittance seen by the controller. For this reason, there is no distinct current reference before and after the limiter, since the limiting action is introduced directly in the control law that generates the current references.

The evolution of the virtual resistance and virtual reactance during the disturbance is shown in Fig. 5.33.

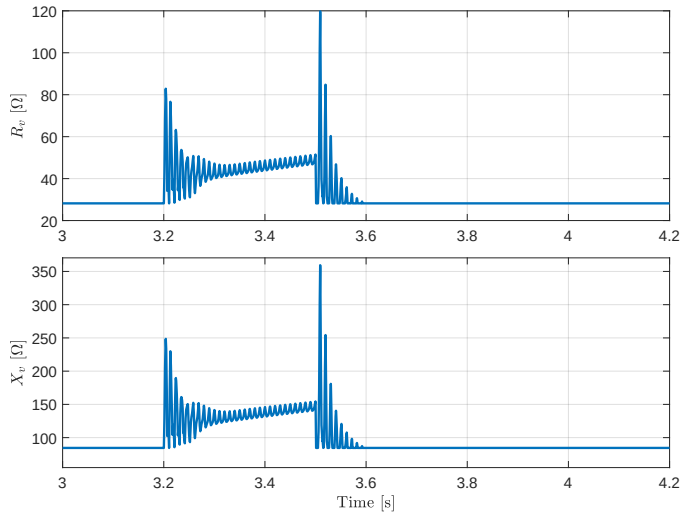
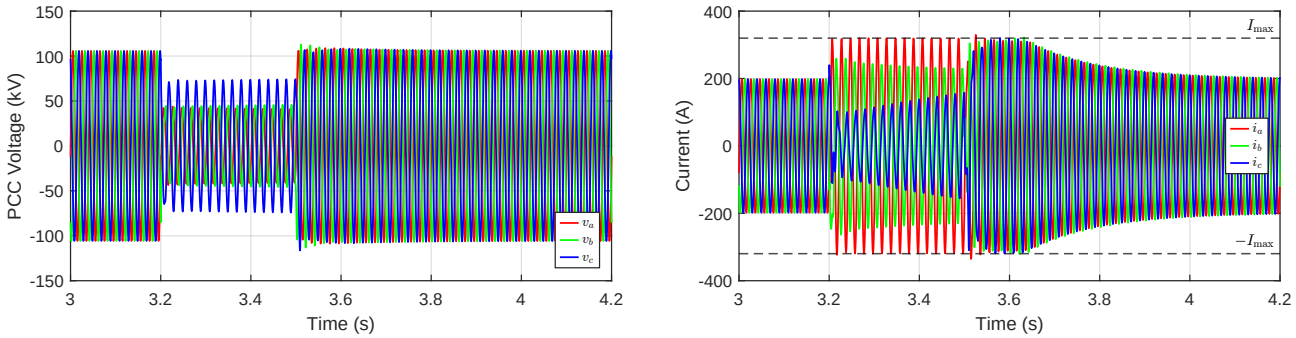


Figure 5.33.: Virtual resistance and virtual reactance during the single phase voltage sag with virtual-admittance based current limitation.

Figure 5.33 shows the variation of the virtual resistance and virtual reactance during the disturbance. Both parameters increase when current limiting is activated, limiting the converter current indirectly. After fault clearance, they return to their nominal values.

5.4.2. Two-phase voltage sag

The virtual-admittance-based current-limiting method remains active throughout the disturbance in order to reduce the converter current demand produced by the severe voltage unbalance.

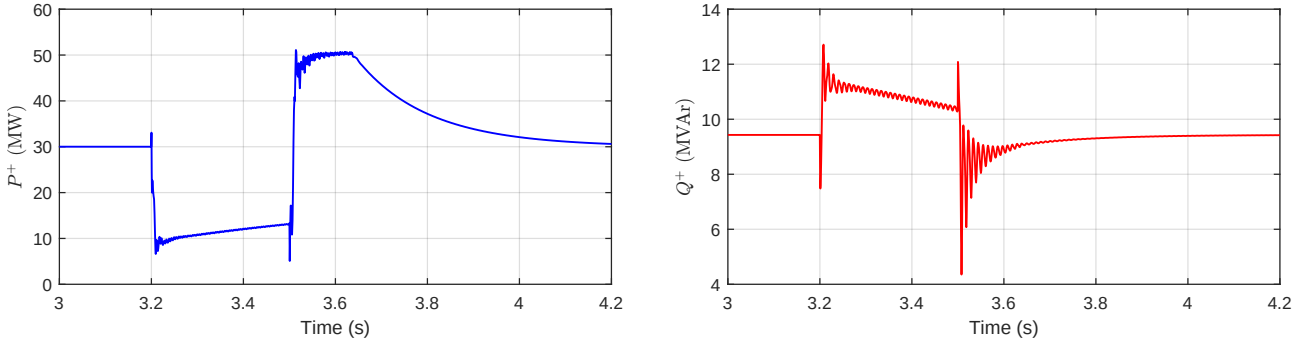


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.34.: PCC voltage and converter current during the two phase voltage sag with virtual-admittance-based current limitation.

Figure 5.34 presents the PCC voltage and the corresponding three phase converter current. The current-limiting mechanism responds by modifying the virtual admittance parameters, thereby increasing the effective virtual impedance and reducing the resulting converter current.



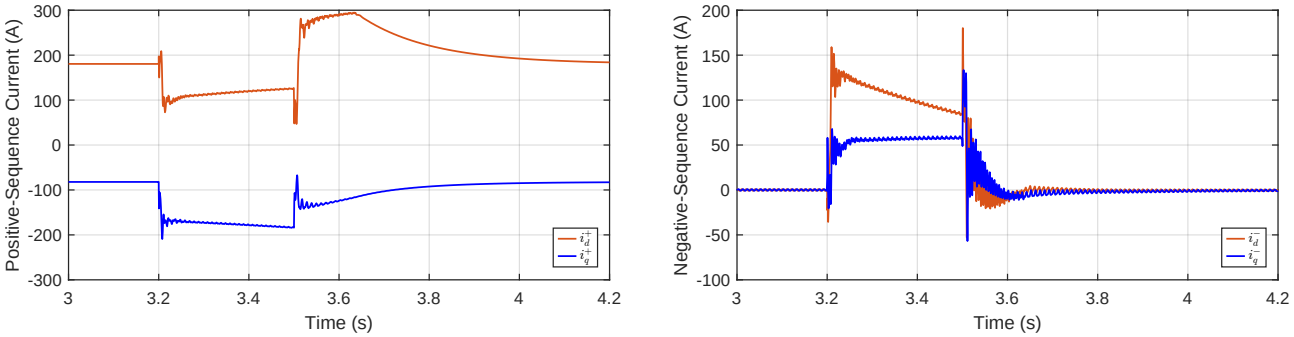
(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

Figure 5.35.: Positive-sequence active and reactive power during the two-phase voltage sag with virtual-admittance based current limitation.

The positive-sequence active and reactive powers are shown in Fig. 5.35. The virtual admittance limiter changes the current that can be supplied by the converter and therefore affects the active and reactive power response.

The corresponding positive- and negative-sequence current components are presented in Fig. 5.36. Because the disturbance is asymmetrical, both sequence components contribute to the total converter current.



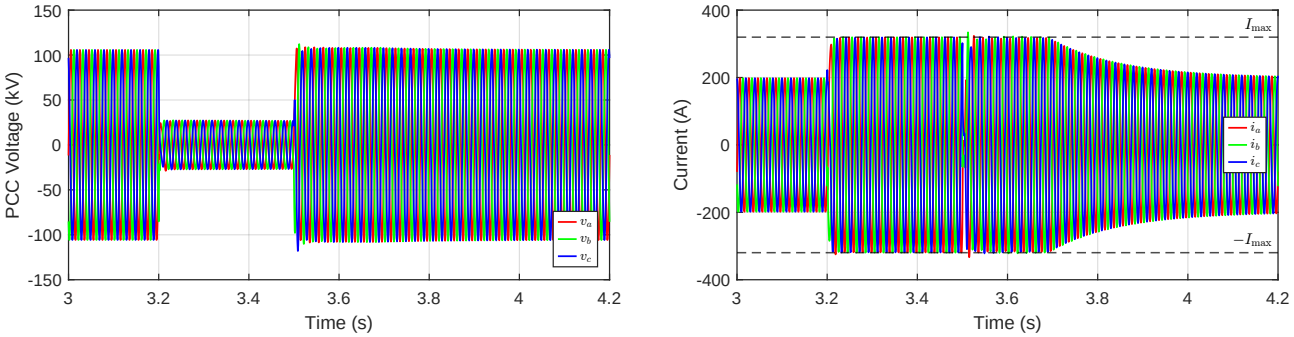
(a) Positive-sequence dq currents.

(b) Negative-sequence dq currents.

Figure 5.36.: Positive- and negative-sequence dq current components during the two-phase voltage sag with virtual-admittance based current limitation.

5.4.3. Three-phase voltage sag

The next disturbance considered is a symmetrical three-phase voltage sag. Since all three phases are affected equally, the voltage system remains balanced throughout the disturbance. The virtual-admittance-based current-limiting method is enabled in order to restrict the increased converter current demand caused by the severe reduction of the grid voltage.



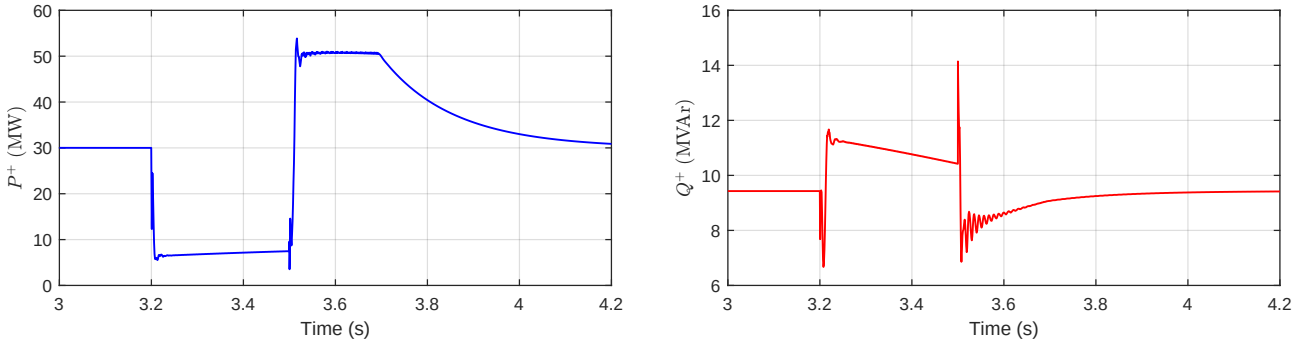
(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.37.: PCC voltage and converter current during the symmetrical three-phase voltage sag with virtual-admittance-based current limitation.

Figure 5.37 shows the PCC voltage and the corresponding converter current response. The simultaneous reduction of all three phase voltages increases the voltage difference between the converter internal voltage and the grid and therefore produces a significant increase in the current demand.

The positive-sequence active and reactive powers are presented in Fig. 5.38. During the voltage sag, the reduced PCC voltage changes the power transfer conditions of the converter. Simultaneously, the increase of the effective virtual impedance restricts the current available for power exchange with the grid. The resulting power response therefore reflects both the severity of the voltage disturbance and the action of the current-limiting mechanism. Following the fault clearing, the converter returns toward its pre-disturbance operating point as the virtual admittance parameters recover their normal values.

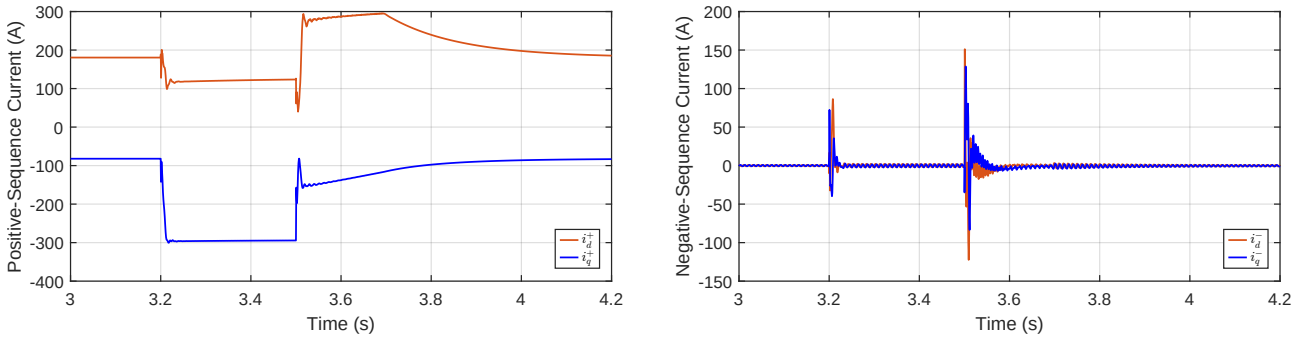


(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

Figure 5.38.: Positive-sequence active and reactive power during the symmetrical three-phase voltage sag with virtual-admittance based current limitation.

The sequence current response is shown in Fig. 5.39. Since the applied disturbance is symmetrical, the converter response is primarily associated with the positive-sequence components, while the negative-sequence current is expected to remain very small.

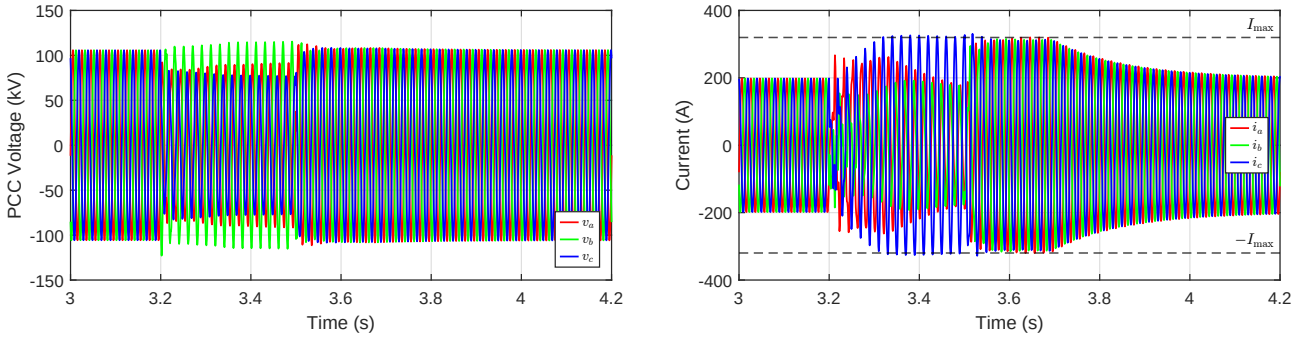


(a) Positive-sequence dq currents.

(b) Negative-sequence dq currents.

Figure 5.39.: Positive- and negative-sequence dq current components during the symmetrical three-phase voltage sag with virtual-admittance based current limitation.

5.4.4. Single-Phase Phase-Angle Jump



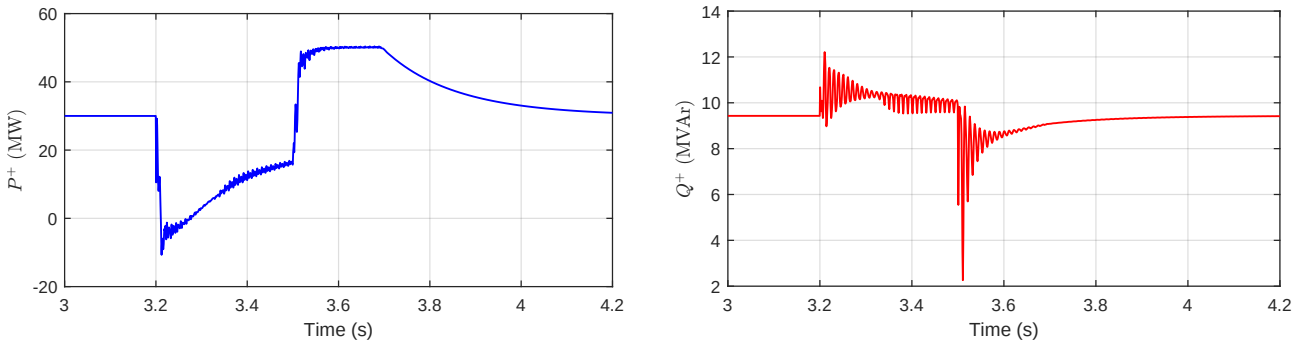
(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.40.: PCC voltage and converter current during the single-phase 60° phase-angle jump with virtual-admittance-based current limitation.

The final disturbance considered for the virtual-admittance-based current-limiting method is a single-phase phase-angle jump.

Figure 5.40 shows the PCC voltage and the corresponding converter current response. This results in an increased converter current demand.

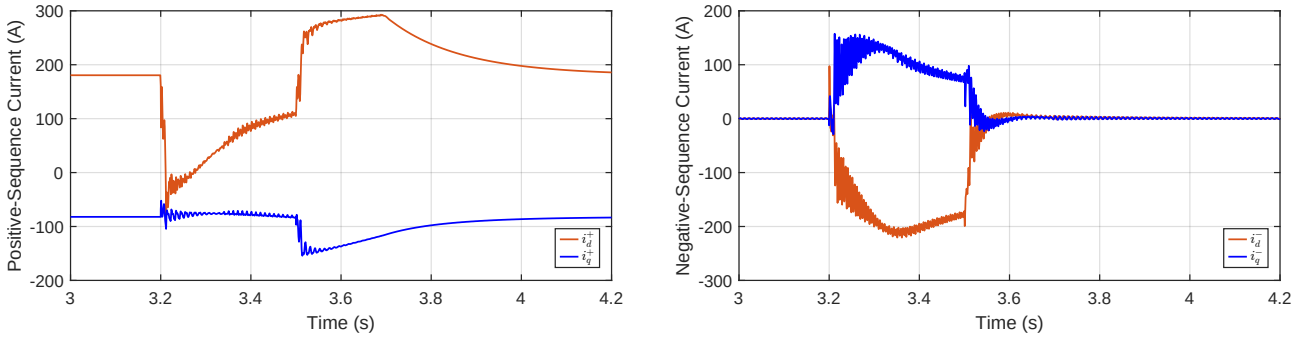


(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

Figure 5.41.: Positive-sequence active and reactive power during the single-phase 60° phase-angle jump with virtual-admittance-based current limitation.

The positive-sequence active and reactive powers are presented in Fig. 5.41. The sudden phase displacement changes the angular relationship between the converter and the grid and therefore modifies the instantaneous power transfer conditions. After the original phase angle is restored, the converter returns toward its pre-disturbance operating point.



(a) Positive-sequence *dq* currents.

(b) Negative-sequence *dq* currents.

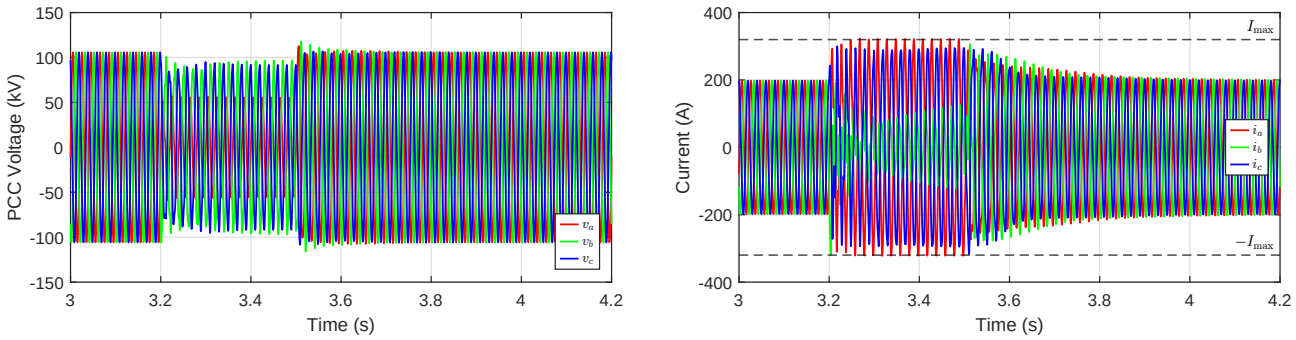
Figure 5.42.: Positive- and negative-sequence *dq* current components during the single-phase 60° phase-angle jump with virtual-admittance based current limitation.

The positive- and negative-sequence current components are shown in Fig. 5.42. Since the phase angle disturbance is applied to only one phase, both sequence components participate in the converter current response.The resulting sequence current response therefore illustrates how the positive- and negative-sequence components are affected by the change in effective virtual impedance during an asymmetrical angular disturbance.

5.5. Positive-Sequence Priority Current Limiter

5.5.1. Single-phase voltage sag

The first disturbance considered for the positive-sequence priority limiter is the single-phase voltage sag. The resulting asymmetrical operating condition produces both positive- and negative-sequence current demands. In this case, the current-limiting algorithm gives priority to the positive-sequence current and reduces the negative-sequence contribution when the total available converter current capability is exceeded.

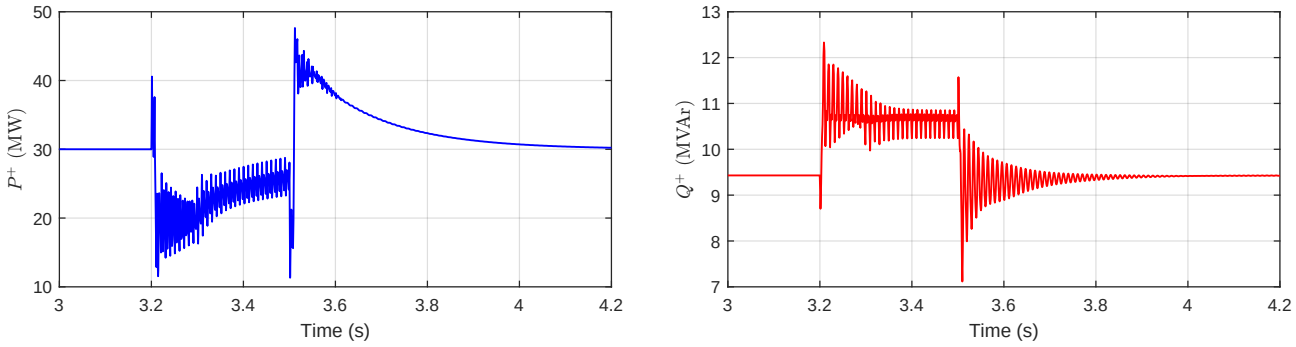


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.43.: PCC voltage and converter current during the single phase voltage sag with positive-sequence current priority.

Figure 5.43 shows the PCC voltage and the corresponding converter current. The asymmetrical voltage sag produces an increased and unbalanced current demand. Once the available converter current capability is reached, the priority-based limiter redistributes the available current between the two sequences, first to the positive one and then the negative.

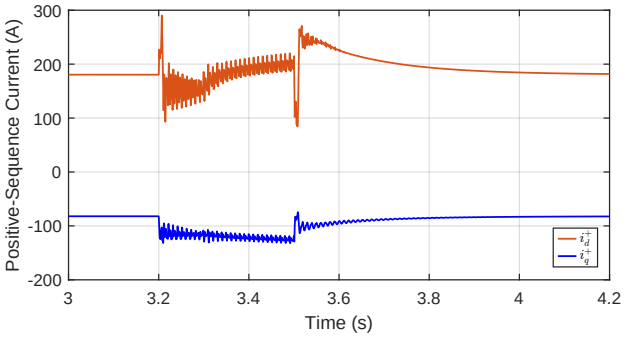


(a) Positive-sequence active power.

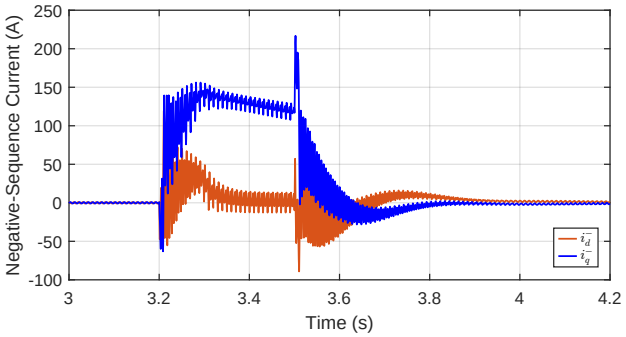
(b) Positive-sequence reactive power.

Figure 5.44.: Positive-sequence active and reactive power during the single-phase voltage sag with positive-sequence current priority.

The positive-sequence active and reactive powers are shown in Fig. 5.44. Since the limiting strategy prioritizes the positive-sequence current contribution, these quantities are particularly useful for examining how the available converter current capability is used to support the positive-sequence grid-forming response during the fault.



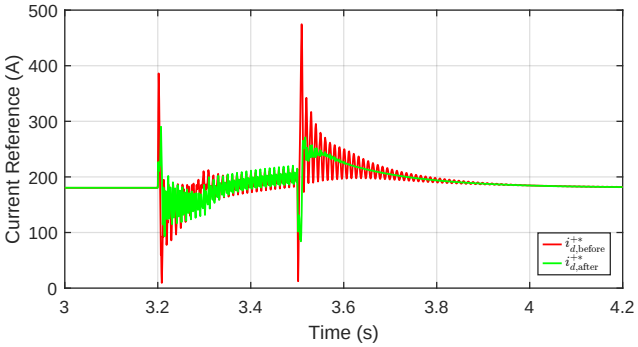
(a) Positive-sequence dq currents.



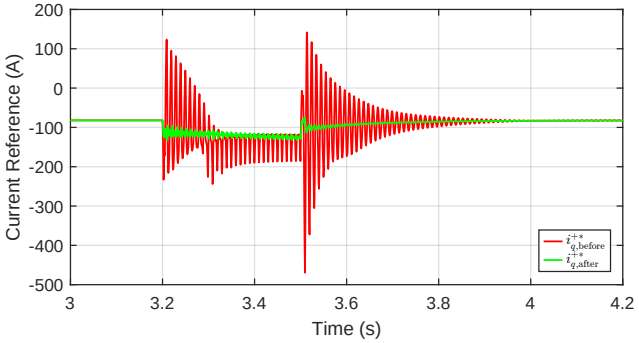
(b) Negative-sequence dq currents.

Figure 5.45.: Positive- and negative-sequence dq current components during the single-phase voltage sag with positive-sequence current priority.

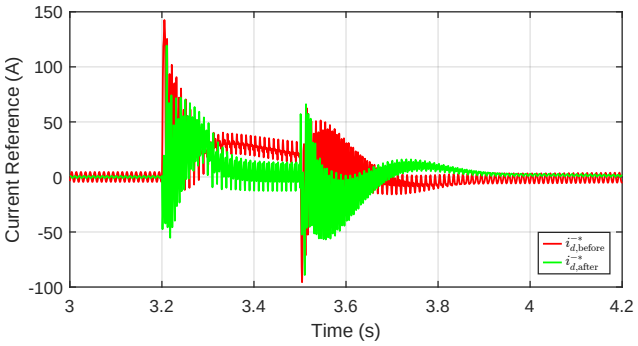
The sequence current components shown in Fig. 5.45 provide a direct view of the priority mechanism. Both sequences are present because of the asymmetrical disturbance.



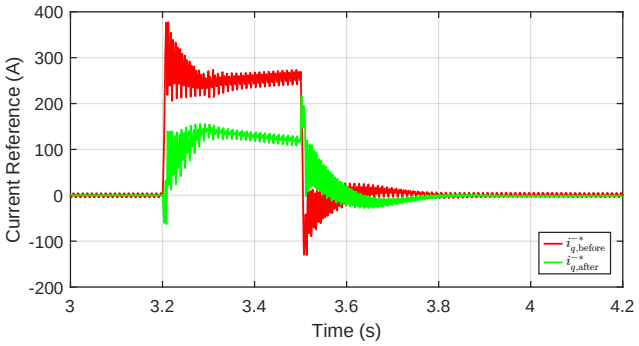
(a) Positive-sequence d-axis current reference.



(b) Positive-sequence q-axis current reference.



(c) Negative-sequence d-axis current reference.



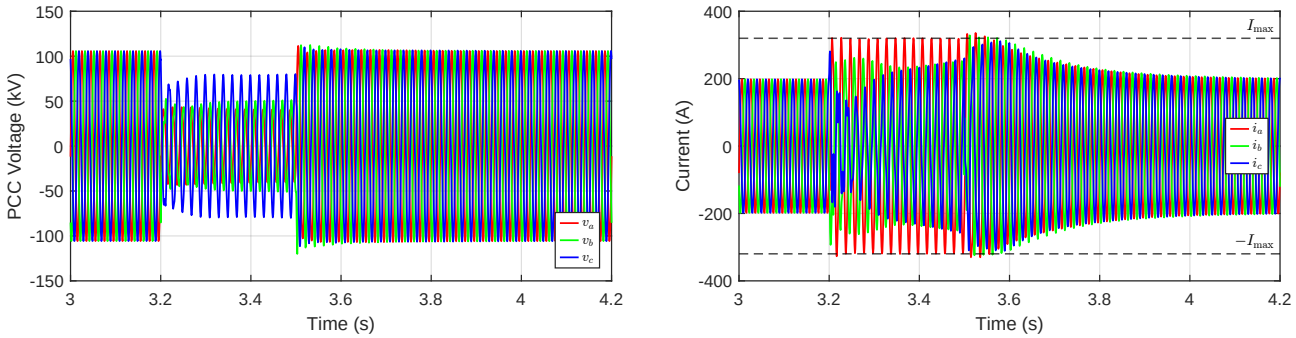
(d) Negative-sequence q-axis current reference.

Figure 5.46.: Positive- and negative-sequence d- and q-axis current references before and after limitation during the single phase voltage sag with positive-sequence current priority.

Figure 5.46 compares the current references before and after limitation. The reference trajectories illustrate directly how the priority allocation modifies the requested sequence currents. During the interval in which current limitation is required, the positive-sequence reference is prioritized, whereas the negative-sequence reference in total is reduced, to satisfy the available current capability.

5.5.2. Two-phase voltage sag

The second asymmetrical disturbance is a two-phase voltage sag. Compared with the single phase case, the disturbance produces a different distribution of the positive- and negative-sequence quantities and therefore provides a more severe test of the positive-sequence priority current-limiting strategy.

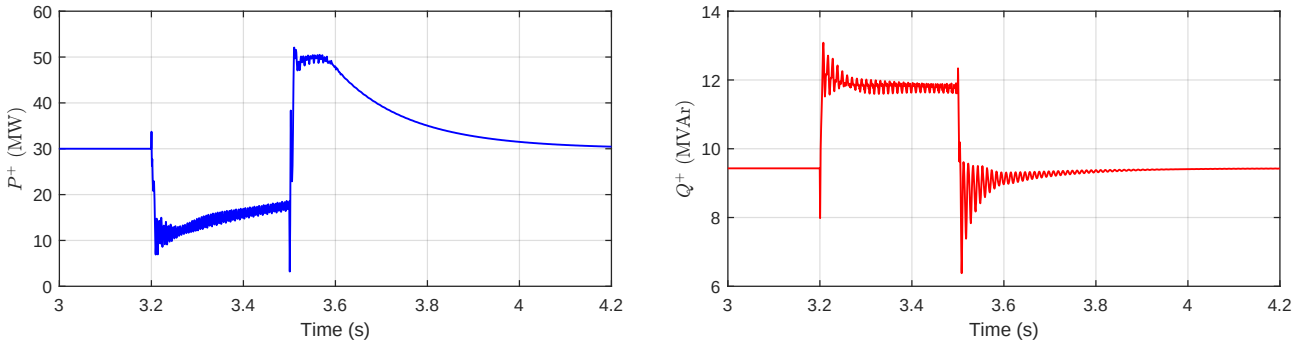


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.47.: PCC voltage and converter current during the two phase voltage sag with positive-sequence current priority.

Figure 5.47 shows the resulting PCC voltage and converter current response. The two phase voltage reduction produces a strongly unbalanced operating condition and a significant current demand.



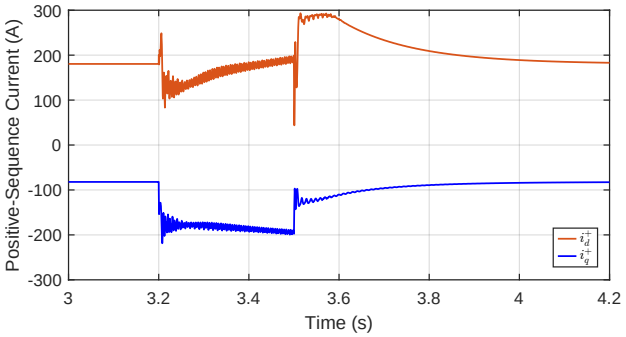
(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

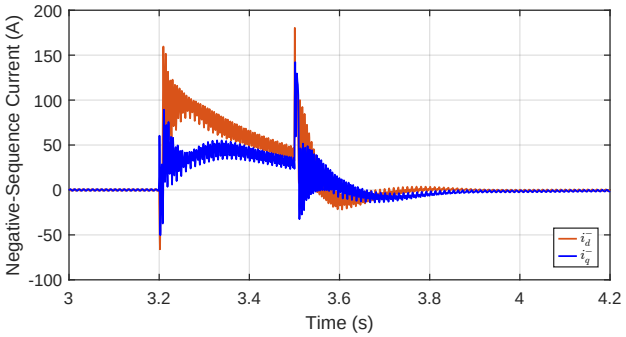
Figure 5.48.: Positive-sequence active and reactive power during the two-phase voltage sag with positive-sequence current priority.

The active and reactive power responses are presented in Fig. 5.48.

The sequence currents in Fig. 5.49 illustrate the allocation of the limited current capability between the two sequences. Because the fault is asymmetrical, a substantial negative-sequence current demand is generated. However, once the total demand exceeds the available limit, the negative-sequence contribution is reduced before the positive-sequence current is further restricted.

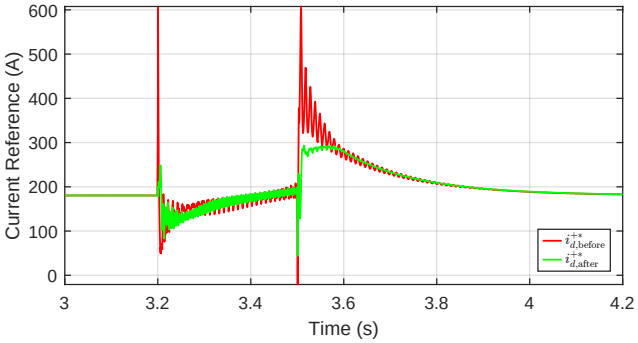


(a) Positive-sequence dq currents.

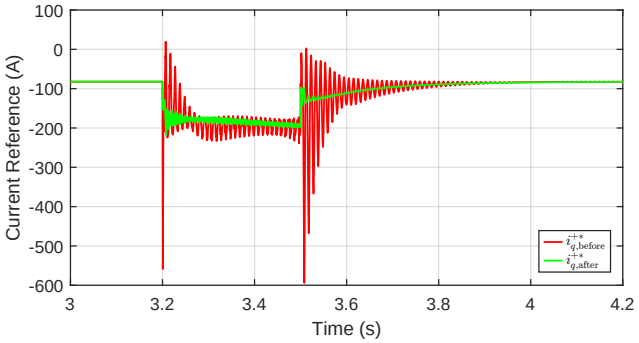


(b) Negative-sequence dq currents.

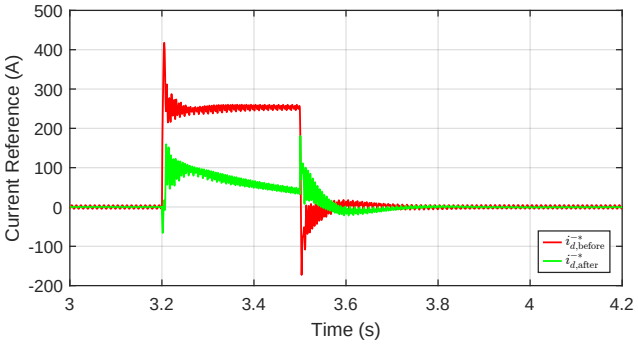
Figure 5.49.: Positive- and negative-sequence dq current components during the two-phase voltage sag with positive-sequence current priority.



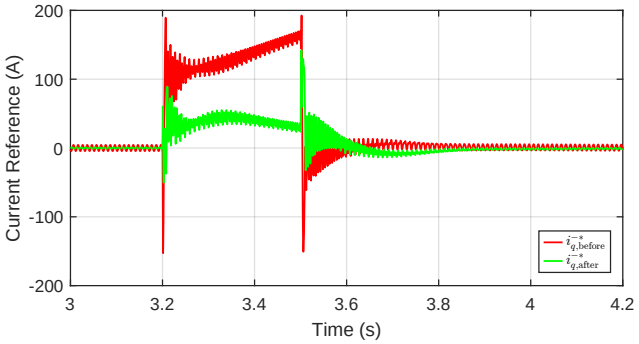
(a) Positive-sequence d-axis current reference.



(b) Positive-sequence q-axis current reference.



(c) Negative-sequence d-axis current reference.



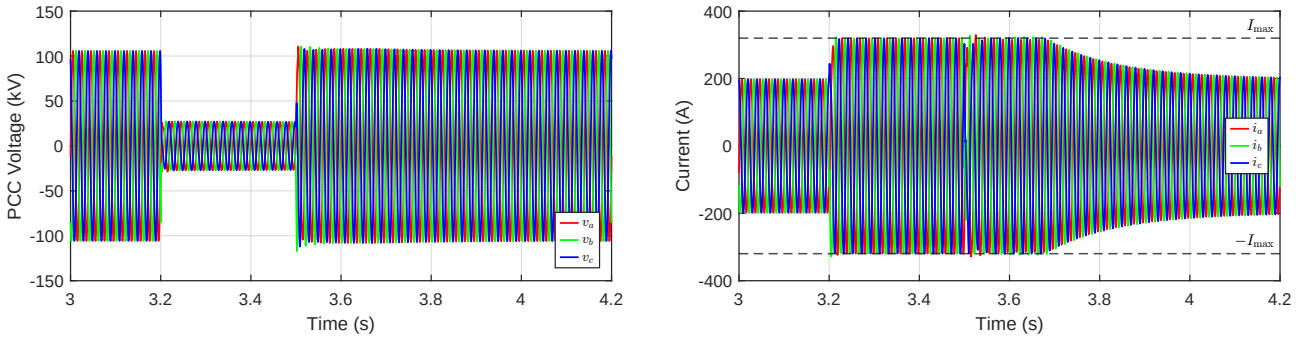
(d) Negative-sequence q-axis current reference.

Figure 5.50.: Positive- and negative-sequence current references before and after limitation during the two-phase voltage sag with positive-sequence current priority.

The reference quantities in Fig. 5.50 make the priority action directly visible. The difference between the unrestricted and limited negative-sequence references shows how this sequence is reduced to preserve the available current capability for the positive-sequence response.

5.5.3. Three-phase symmetrical voltage sag

Since the same voltage reduction is applied to each phase, the resulting disturbance is dominated by the positive sequence and does not generate a significant negative-sequence current demand.

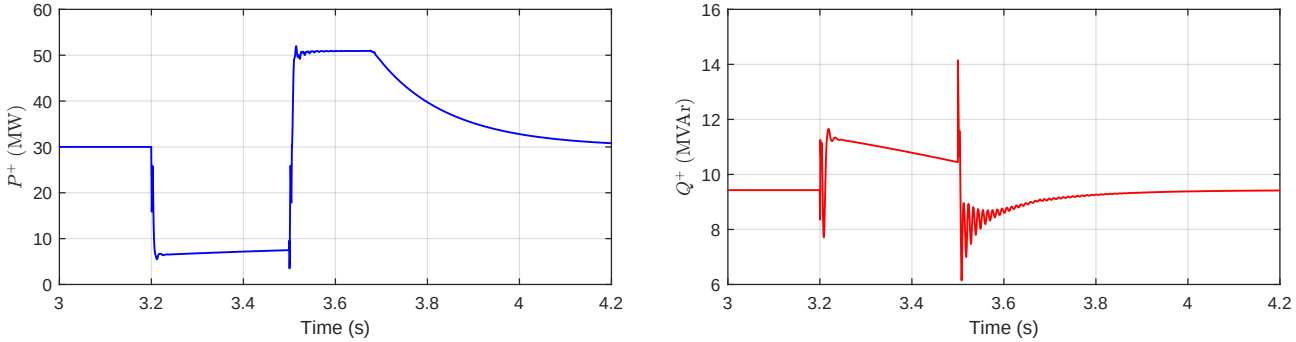


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.51.: PCC voltage and converter current during the three phase symmetrical voltage sag with positive-sequence current priority.

As shown in Fig. 5.51, the converter current response remains balanced because the imposed disturbance is symmetrical. In this operating condition, the distinction between positive- and negative-sequence priority becomes considerably less important, since practically the entire fault current demand belongs to the positive sequence. Consequently, the current-limiting action is mainly determined by the available positive-sequence current capability.

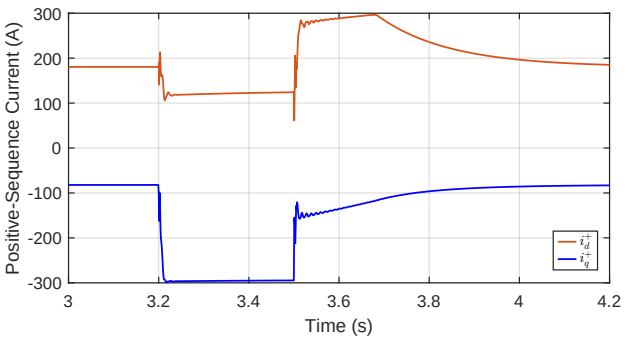


(a) Positive-sequence active power.

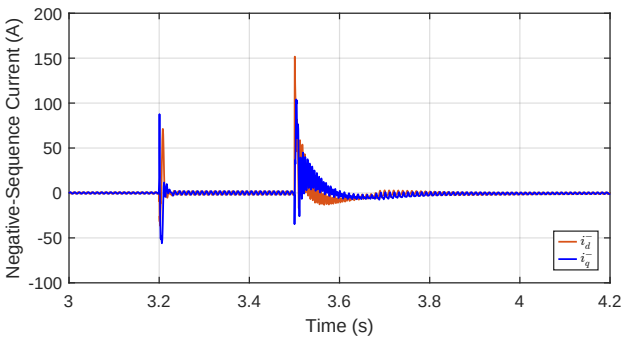
(b) Positive-sequence reactive power.

Figure 5.52.: Positive-sequence active and reactive power during the three-phase symmetrical voltage sag with positive-sequence current priority.

The positive-sequence active and reactive power responses are shown in Fig. 5.52. This case provides a useful reference for comparison with the asymmetrical disturbances, where the two sequences compete for the available current capability.



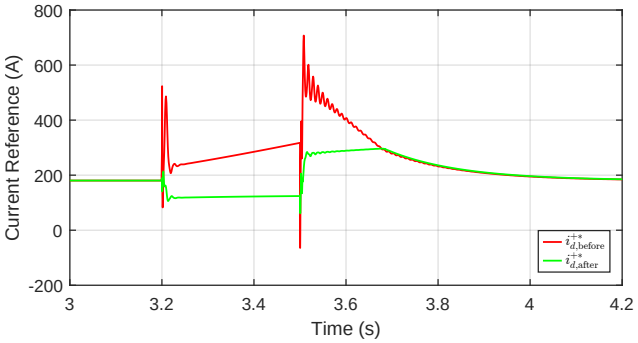
(a) Positive-sequence dq currents.



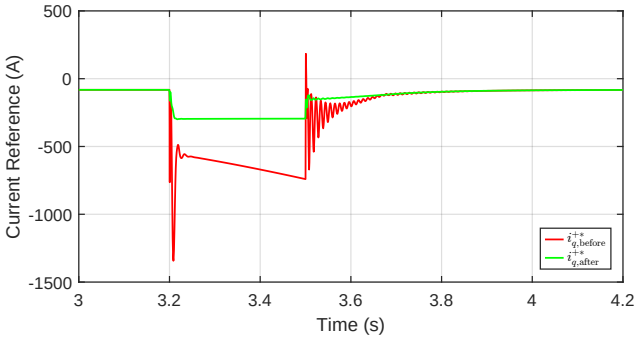
(b) Negative-sequence dq currents.

Figure 5.53.: Positive- and negative-sequence dq current components during the three-phase symmetrical voltage sag with positive-sequence current priority.

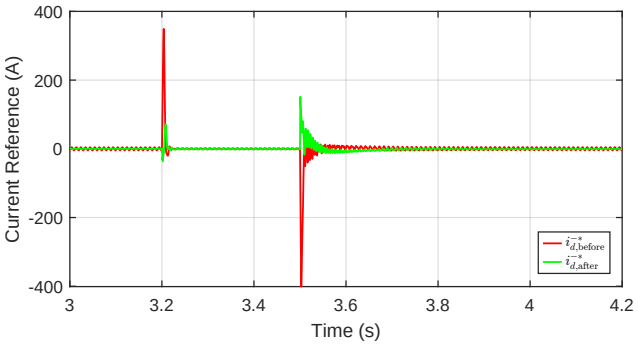
The sequence currents in Fig. 5.53 confirm that the response is dominated by the positive sequence.



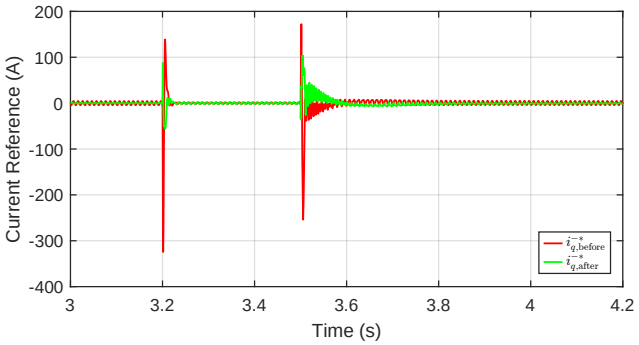
(a) Positive-sequence d-axis current reference.



(b) Positive-sequence q-axis current reference.



(c) Negative-sequence d-axis current reference.



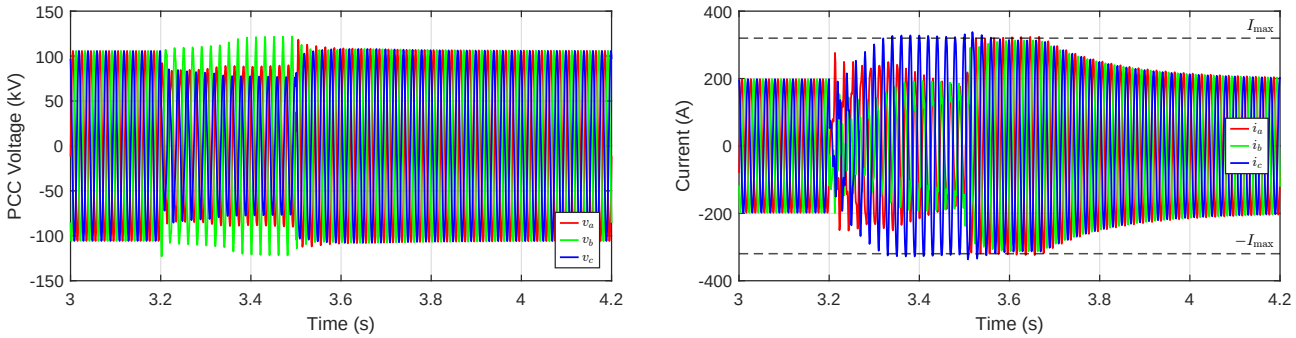
(d) Negative-sequence q-axis current reference.

Figure 5.54.: Positive- and negative-sequence current references before and after limitation during the three-phase symmetrical voltage sag with positive-sequence current priority.

The reference quantities in Fig. 5.70 make the limiting action visible. The difference between the unrestricted and limited positive-sequence references shows how the positive-sequence current is reduced to satisfy the converter current limit. The negative-sequence priority does not affect the current allocation in this case.

5.5.4. Single-Phase Angle Jump

The final disturbance considered is the single-phase angle jump. The resulting mismatch between the converter and grid voltage vectors generates an increased converter current demand and both positive- and negative-sequence current components.

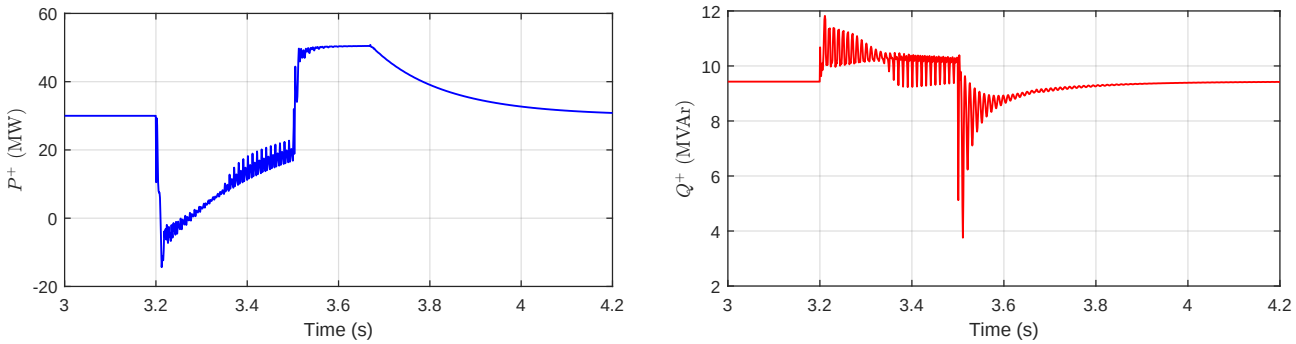


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.55.: PCC voltage and converter current during the single phase angle jump with positive-sequence current priority.

Figure 5.55 illustrates the converter response to the phase angle disturbance.

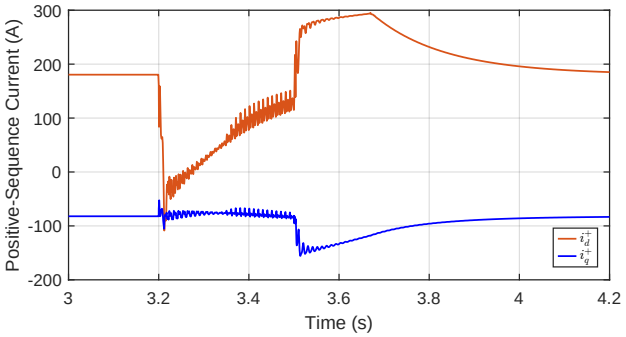


(a) Positive-sequence active power.

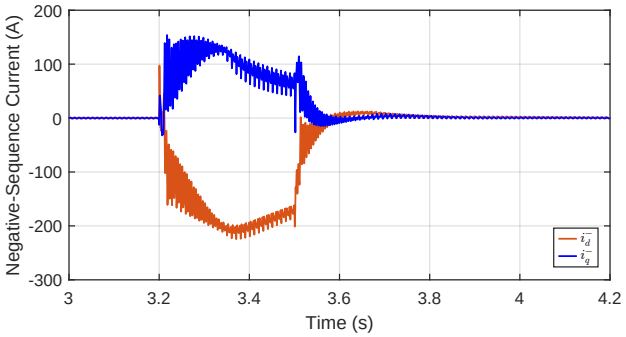
(b) Positive-sequence reactive power.

Figure 5.56.: Positive-sequence active and reactive power during the single-phase angle jump with positive-sequence current priority.

The power responses in Fig. 5.56 indicate the interaction between the current priority mechanism and the outer grid-forming loops during the angular disturbance. Since positive-sequence current is prioritized, the available current capability is preferentially retained for the sequence quantities directly associated with the grid-forming active and reactive power response.



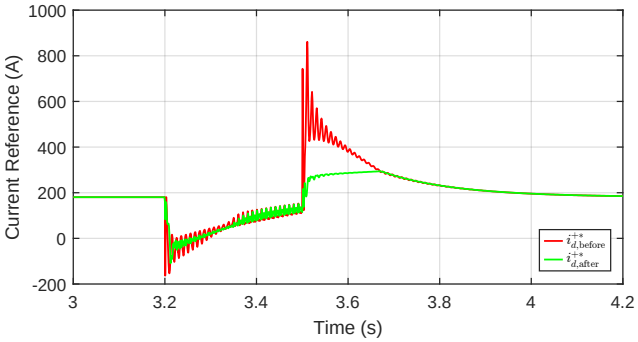
(a) Positive-sequence dq currents.



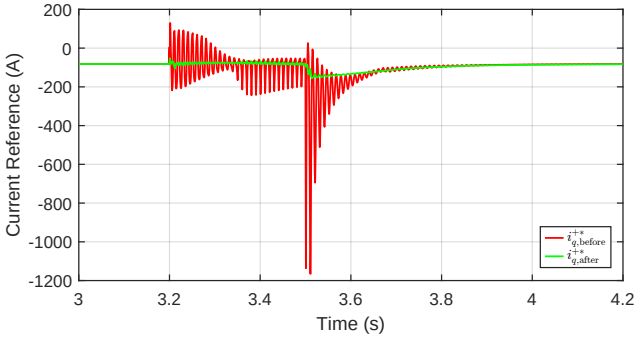
(b) Negative-sequence dq currents.

Figure 5.57.: Positive- and negative-sequence dq current components during the single-phase angle jump with positive-sequence current priority.

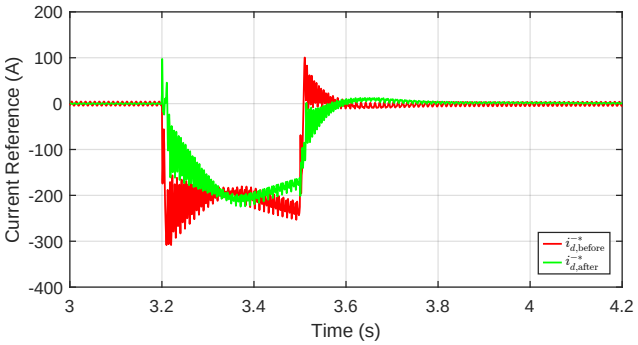
The sequence current components shown in Fig. 5.57 demonstrate that the single-phase angle jump generates both positive- and negative-sequence current components.



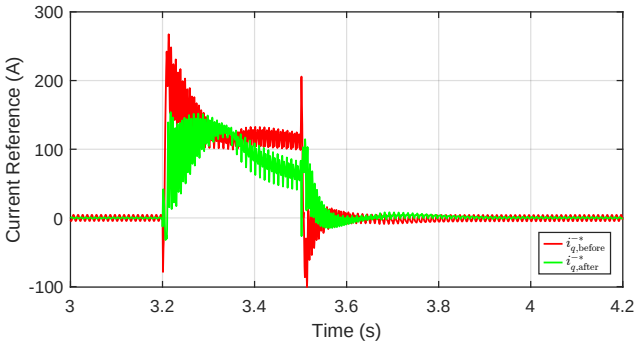
(a) Positive-sequence d-axis current reference.



(b) Positive-sequence q-axis current reference.



(c) Negative-sequence d-axis current reference.



(d) Negative-sequence q-axis current reference.

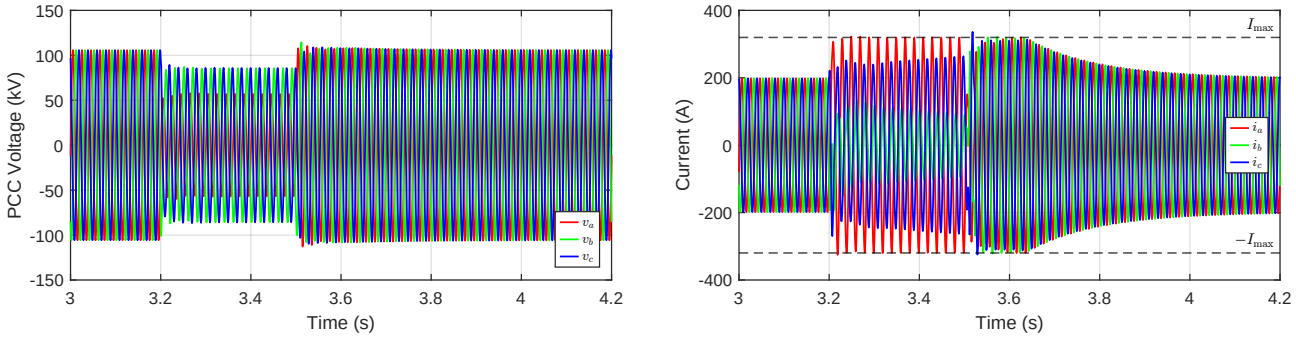
Figure 5.58.: Positive- and negative-sequence current references before and after limitation during the single-phase angle jump with positive-sequence current priority.

Figure 5.58 provides a direct representation of the priority action at the current reference level. The limited references show how the negative-sequence current demand is reduced when required, while the positive-sequence contribution is retained with priority.

5.6. Negative-Sequence Priority Current Limiter

5.6.1. Single-phase voltage sag

The first disturbance considered for the negative-sequence priority limiter is the single-phase voltage sag, as for every other limiter. The resulting asymmetrical operating condition produces both positive- and negative-sequence current demands. In this case, the current-limiting algorithm gives priority to the negative-sequence current, while the positive-sequence contribution is determined from the remaining converter current capability, subject to the imposed minimum positive-sequence current requirement.

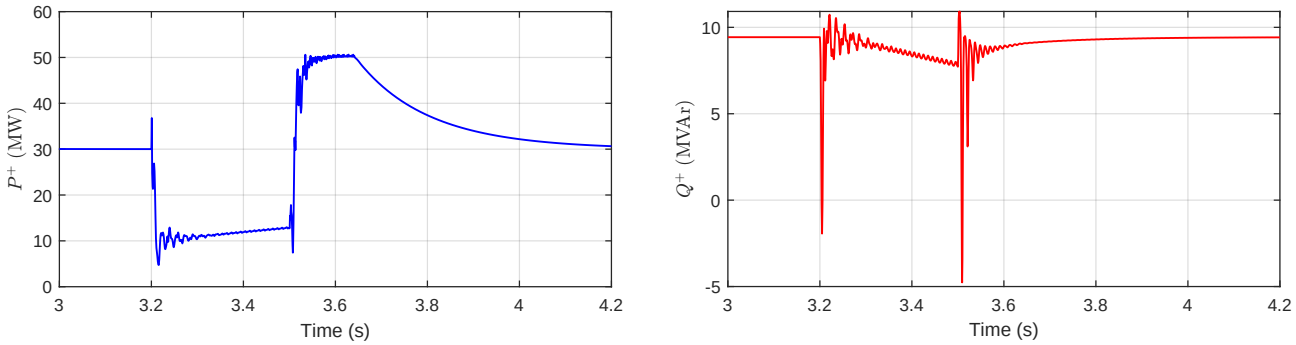


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.59.: PCC voltage and converter current during the single phase voltage sag with negative-sequence current priority.

Figure 5.59 shows the PCC voltage and the corresponding converter current. The single-phase voltage sag creates a strongly unbalanced operating condition and increases the demand for both sequence current components. Once the available converter current capability is reached, the priority-based limiter reallocates the available current between the two sequences. In contrast to the positive-sequence priority case, the negative-sequence demand is satisfied first, up to the phase current constraint. The remaining current capability is subsequently made available to the positive-sequence controller.



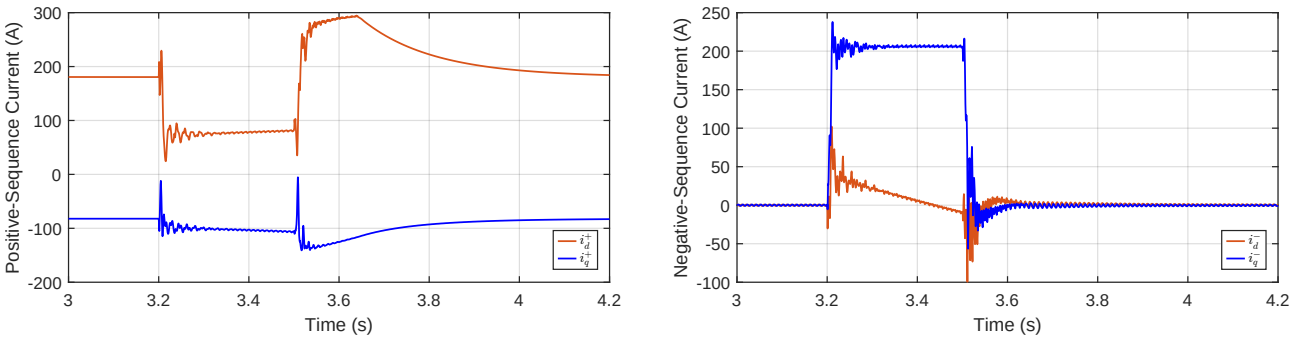
(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

Figure 5.60.: Positive-sequence active and reactive power during the single-phase voltage sag with negative-sequence current priority.

The positive-sequence active and reactive powers are shown in Fig. 5.60. Since part of the available converter current capability is preferentially assigned to the negative-sequence controller, less current may remain available

for the positive-sequence power transfer objectives during the disturbance. The resulting power response therefore reflects the trade off introduced by negative-sequence prioritization between voltage unbalance compensation and the preservation of the positive-sequence grid-forming contribution.



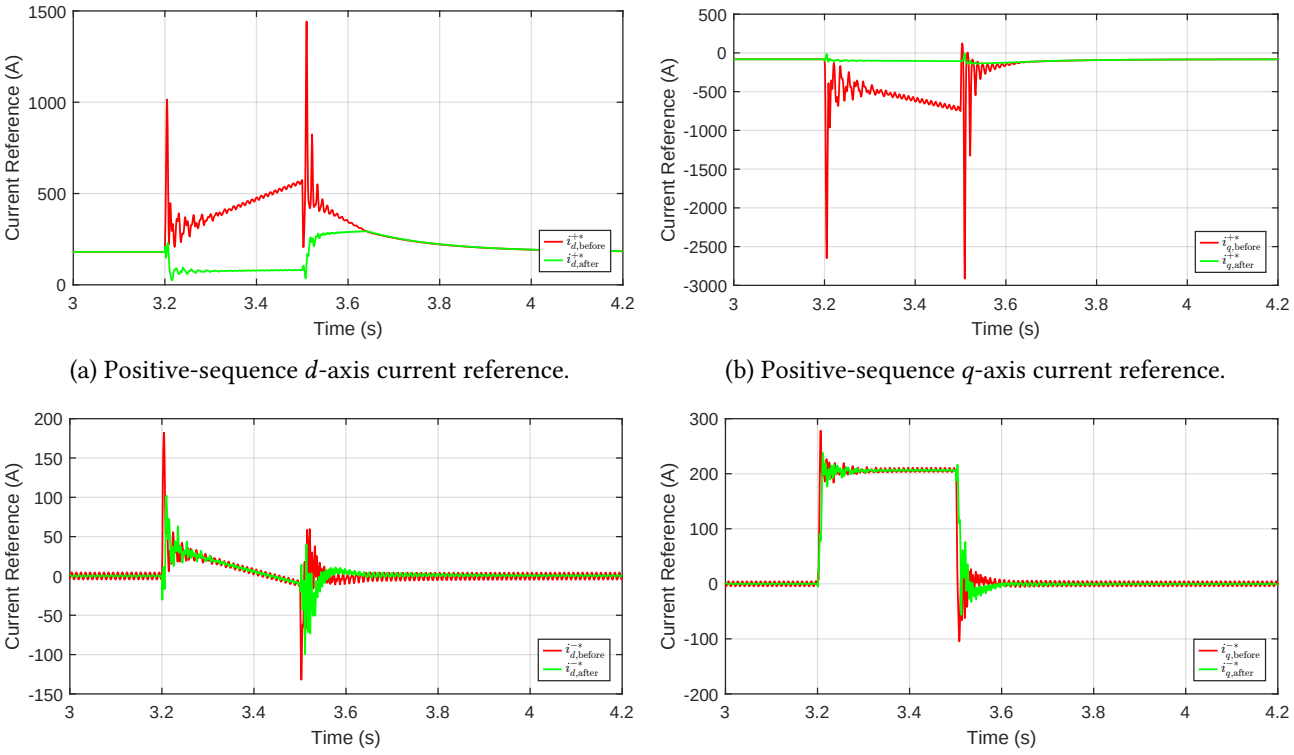
(a) Positive-sequence dq currents.

(b) Negative-sequence dq currents.

Figure 5.61.: Positive- and negative-sequence dq current components during the single-phase voltage sag with negative-sequence current priority.

The sequence current components shown in Fig. 5.61 provide a direct indication of the current allocation mechanism.

This behaviour differs from both the elliptical limiter and the positive-sequence priority strategy. In the present case, their relative contribution is deliberately modified so that a larger portion of the available current can be used for negative-sequence compensation.



(a) Positive-sequence d-axis current reference.

(b) Positive-sequence q-axis current reference.

(c) Negative-sequence d-axis current reference.

(d) Negative-sequence q-axis current reference.

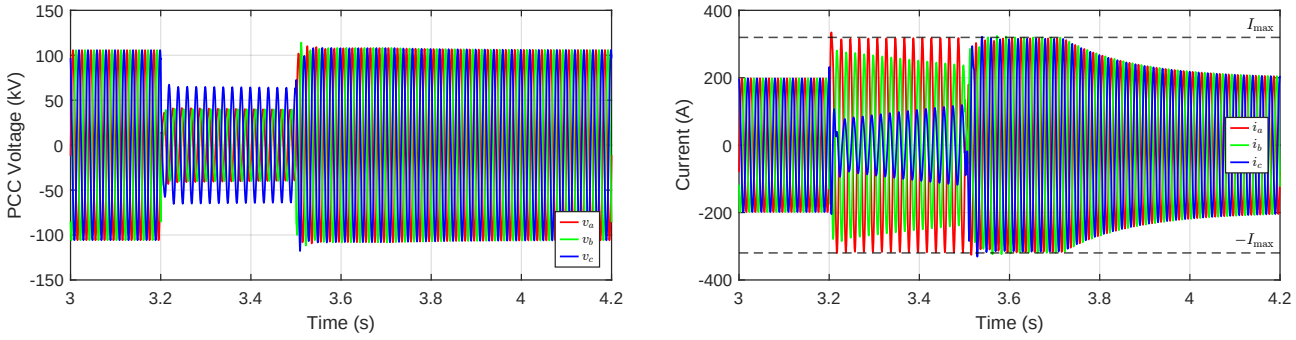
Figure 5.62.: Positive- and negative-sequence d- and q-axis current references before and after limitation during the single phase voltage sag with negative-sequence current priority.

Figure 5.62 compares the current references before and after the priority-based allocation. The reference trajectories therefore provide the clearest indication of how the negative-sequence priority method modifies the distribution of converter current during the asymmetrical fault.

5.6.2. Two-phase voltage sag

The next disturbance considered for the negative-sequence priority limiter is a two-phase voltage sag. Compared with the single phase case, the resulting disturbance produces a different distribution of the positive- and negative-sequence components and therefore modifies the current capability required by the two sequence controllers.

The negative-sequence priority strategy is again applied during the fault. After the selected minimum positive-sequence contribution has been reserved, the negative-sequence current demand is satisfied first, up to the phase dependent current capability. Any remaining converter current capability is then made available to the positive-sequence controller, along with the minimum current already specified.

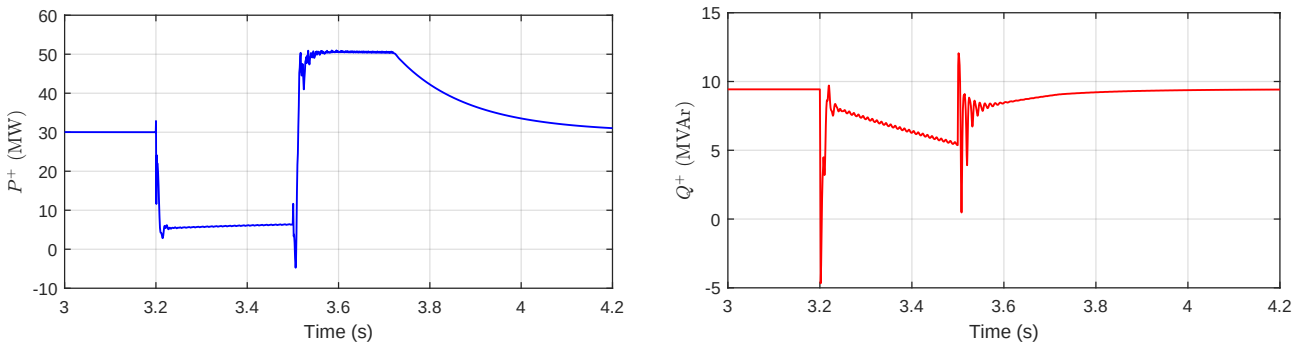


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.63.: PCC voltage and converter current during the two phase voltage sag with negative-sequence current priority.

Figure 5.63 shows the PCC voltage and the corresponding converter current during the two phase disturbance.

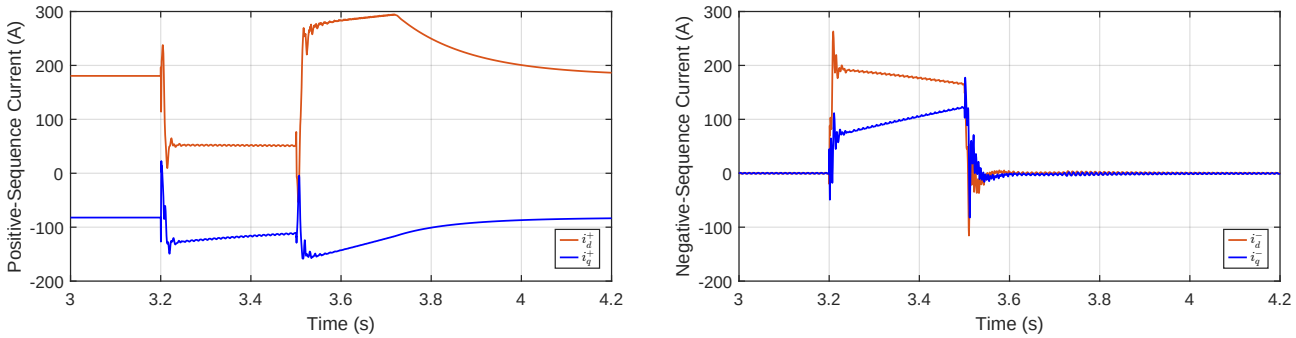


(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

Figure 5.64.: Positive-sequence active and reactive power during the two-phase voltage sag with negative-sequence current priority.

The positive-sequence active and reactive powers are presented in Fig. 5.64. The reduction of the current capability available to the positive-sequence controller affects the power exchange during the fault, since part of the converter capability is preferentially used by the negative-sequence controller. Following fault clearance, the power quantities recover as the converter returns to its normal current reference allocation.

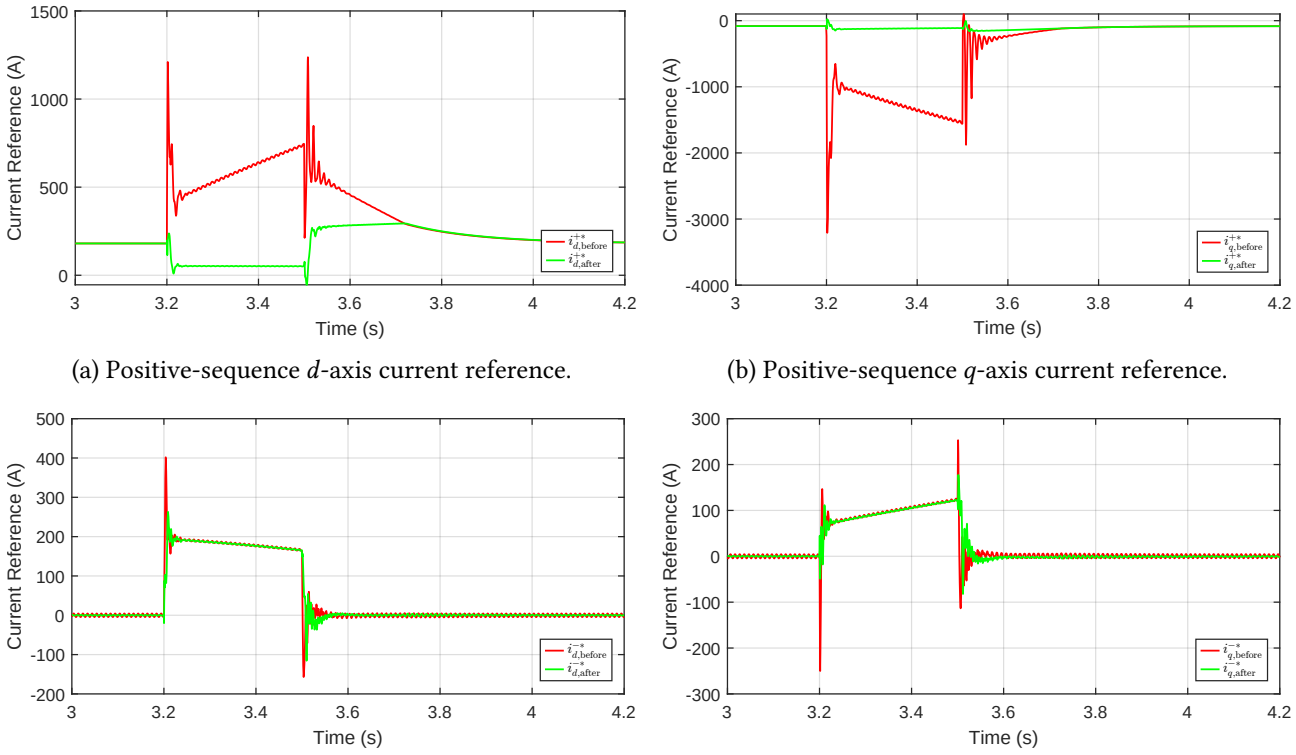


(a) Positive-sequence dq currents.

(b) Negative-sequence dq currents.

Figure 5.65.: Positive- and negative-sequence dq current components during the two-phase voltage sag with negative-sequence current priority.

The sequence current components shown in Fig. 5.65 illustrate the current distribution produced by the negative-sequence priority strategy. During the fault, the negative-sequence current is allowed to use the available converter current capability first. Consequently, the positive-sequence current can be reduced substantially when the negative-sequence demand becomes large.



(a) Positive-sequence d-axis current reference.

(b) Positive-sequence q-axis current reference.

(c) Negative-sequence d-axis current reference.

(d) Negative-sequence q-axis current reference.

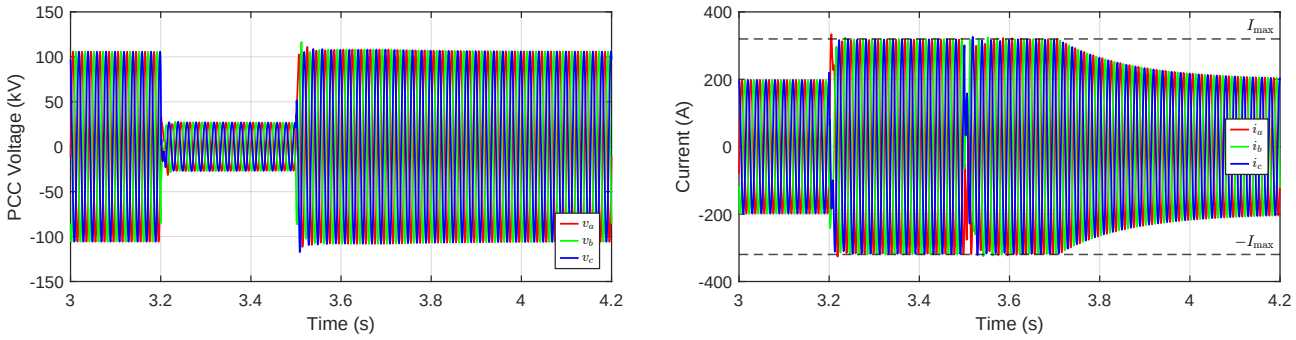
Figure 5.66.: Positive- and negative-sequence d- and q-axis current references before and after limitation during the two phase voltage sag with negative-sequence current priority.

Figure 5.66 compares the unrestricted and limited current references during the two-phase voltage sag. The reference trajectories provide a direct indication of the allocation process.

5.6.3. Three-phase voltage sag

The final voltage magnitude disturbance considered for the negative-sequence priority limiter is a symmetrical three phase voltage sag.

The negative-sequence current demand remains negligible apart from possible short transient components associated with the sequence extraction and control dynamics. Although the negative-sequence priority limiter remains enabled, there is effectively no negative-sequence current demand to prioritize. The available converter current capability can therefore be used predominantly by the positive-sequence controller.

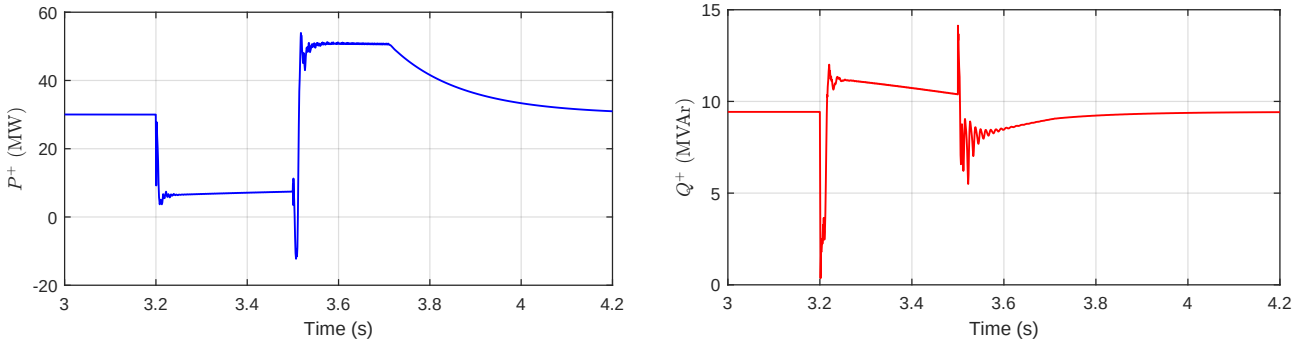


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.67.: PCC voltage and converter current during the symmetrical three-phase voltage sag with negative-sequence current priority.

Figure 5.67 shows the PCC voltage and the corresponding converter current during the severe three-phase voltage sag. Since all three phase voltages are reduced by the same amount, the disturbance remains balanced and the three converter currents retain a balanced three phase structure.



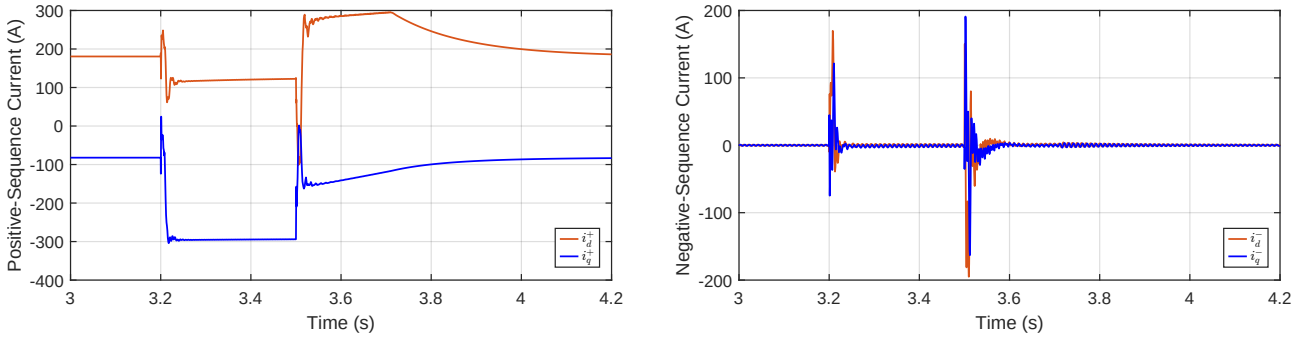
(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

Figure 5.68.: Positive-sequence active and reactive power during the symmetrical three-phase voltage sag with negative-sequence current priority.

The positive-sequence active and reactive powers are presented in Fig. 5.68.

Following fault clearance, the PCC voltage returns to its nominal value and the converter transitions back to its normal operating condition.

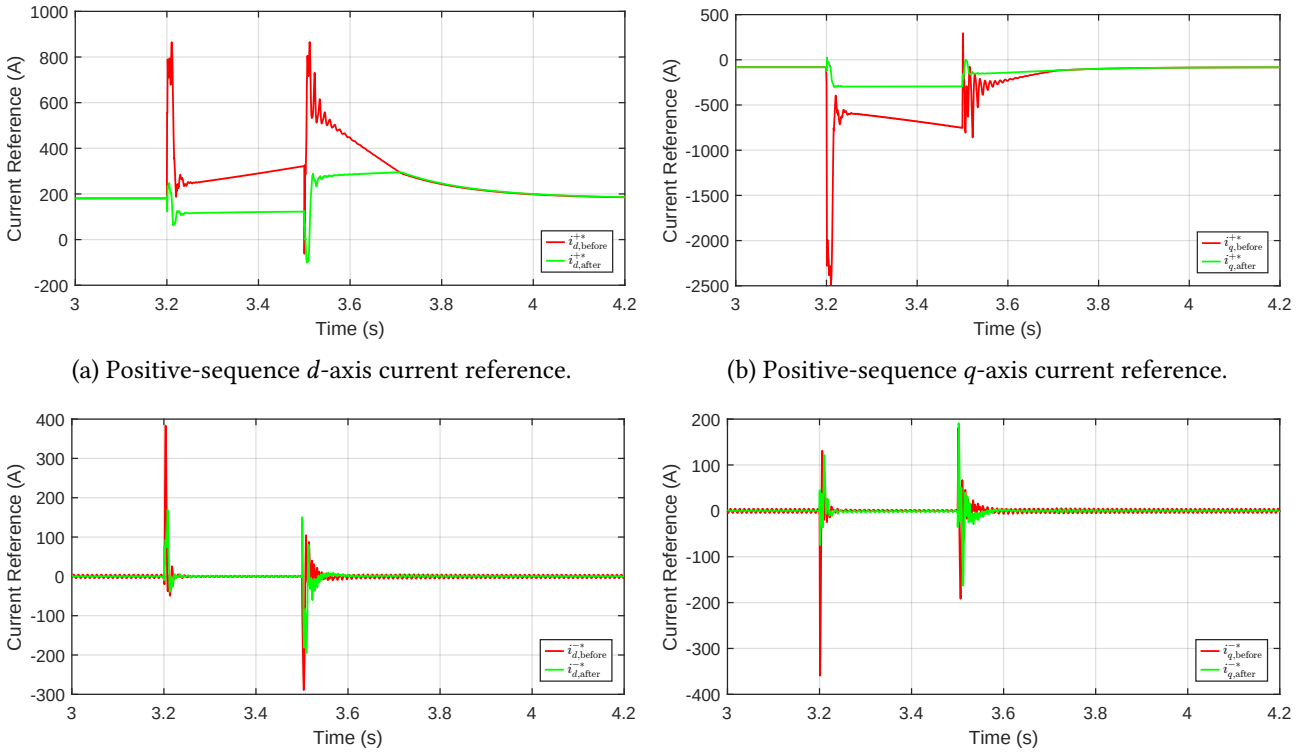


(a) Positive-sequence dq currents.

(b) Negative-sequence dq currents.

Figure 5.69.: Positive- and negative-sequence dq current components during the symmetrical three-phase voltage sag with negative-sequence current priority.

The sequence current components in Fig. 5.69 clearly distinguish the symmetrical three phase case from the preceding asymmetrical disturbances. The converter current is dominated by the positive-sequence component, whereas the negative-sequence current remains close to zero during the fault, except for possible transient components around the disturbance transitions.



(a) Positive-sequence d-axis current reference.

(b) Positive-sequence q-axis current reference.

(c) Negative-sequence d-axis current reference.

(d) Negative-sequence q-axis current reference.

Figure 5.70.: Positive- and negative-sequence d- and q-axis current references before and after limitation during the symmetrical three-phase voltage sag with negative-sequence current priority.

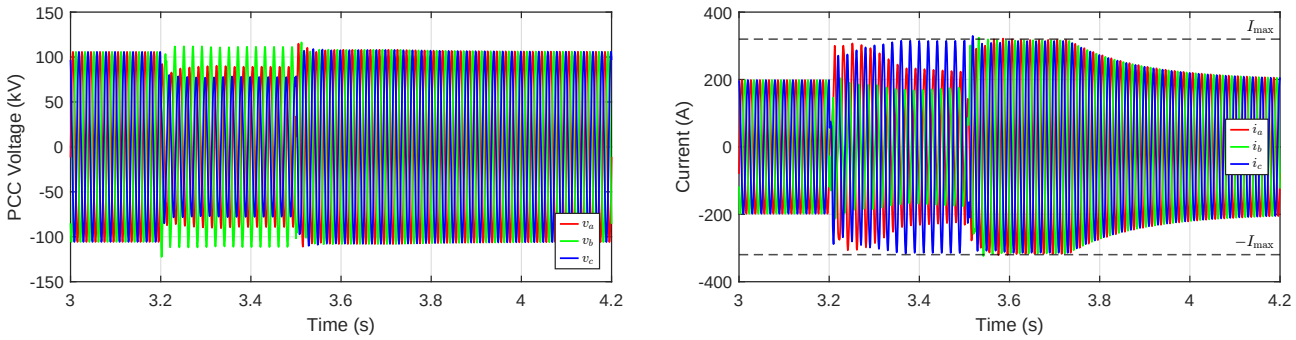
Figure 5.70 compares the unrestricted and limited sequence current references. The positive-sequence refer-

ences increase significantly in response to the deep voltage sag and are restricted when the requested current exceeds the available converter capability. In contrast, the negative-sequence references remain small because the disturbance does not introduce a sustained negative-sequence voltage component.

As a result, the selected negative-sequence priority has little effect on the current distribution during the three phase sag. Since no significant negative-sequence current is requested, the positive-sequence controller can use the available converter current capability. Therefore, this case demonstrates that the priority mechanism becomes relevant primarily when positive- and negative-sequence current demands are simultaneously present.

5.6.4. Single-Phase Angle Jump

Finally, the last disturbance considered for the negative-sequence priority limiter is a single-phase angle jump.

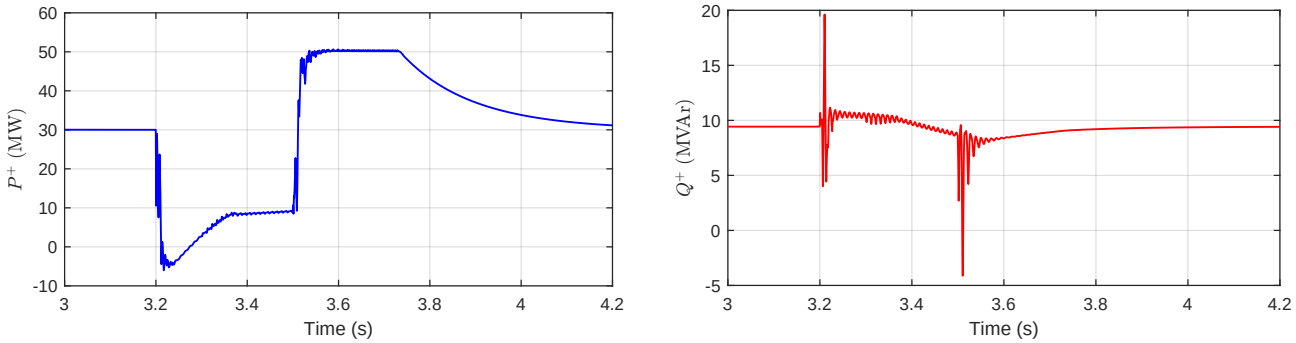


(a) Three phase PCC voltage.

(b) Three phase converter current.

Figure 5.71.: PCC voltage and converter current during the single phase angle jump with negative-sequence current priority.

Figure 5.71 shows the PCC voltage and the corresponding converter current. The phase angle disturbance does not reduce the voltage magnitude directly, but it creates a sudden change in the relative phase position between the converter and the grid. As a result, the converter experiences a significant transient current demand together with an unbalanced three phase response.

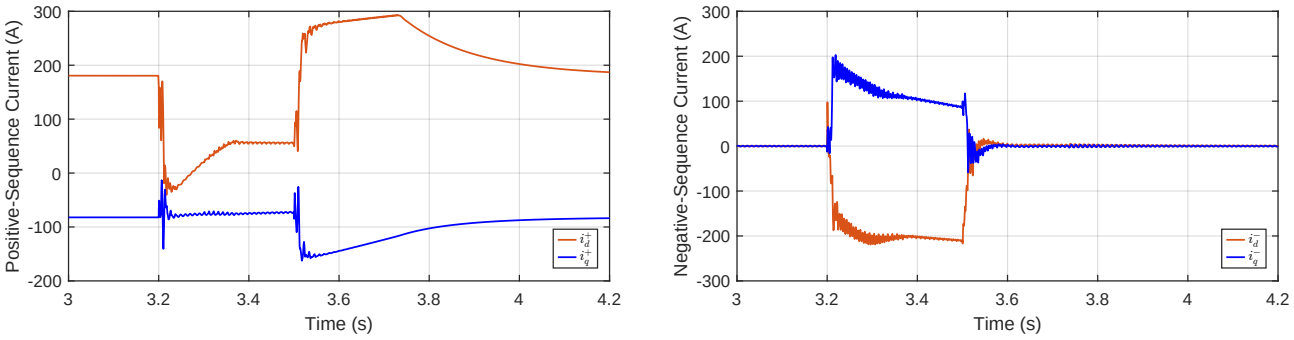


(a) Positive-sequence active power.

(b) Positive-sequence reactive power.

Figure 5.72.: Positive-sequence active and reactive power during the single-phase angle jump with negative-sequence current priority.

The positive-sequence active and reactive powers are shown in Fig. 5.72. The sudden phase displacement produces an abrupt change in the power exchange between the converter and the grid.

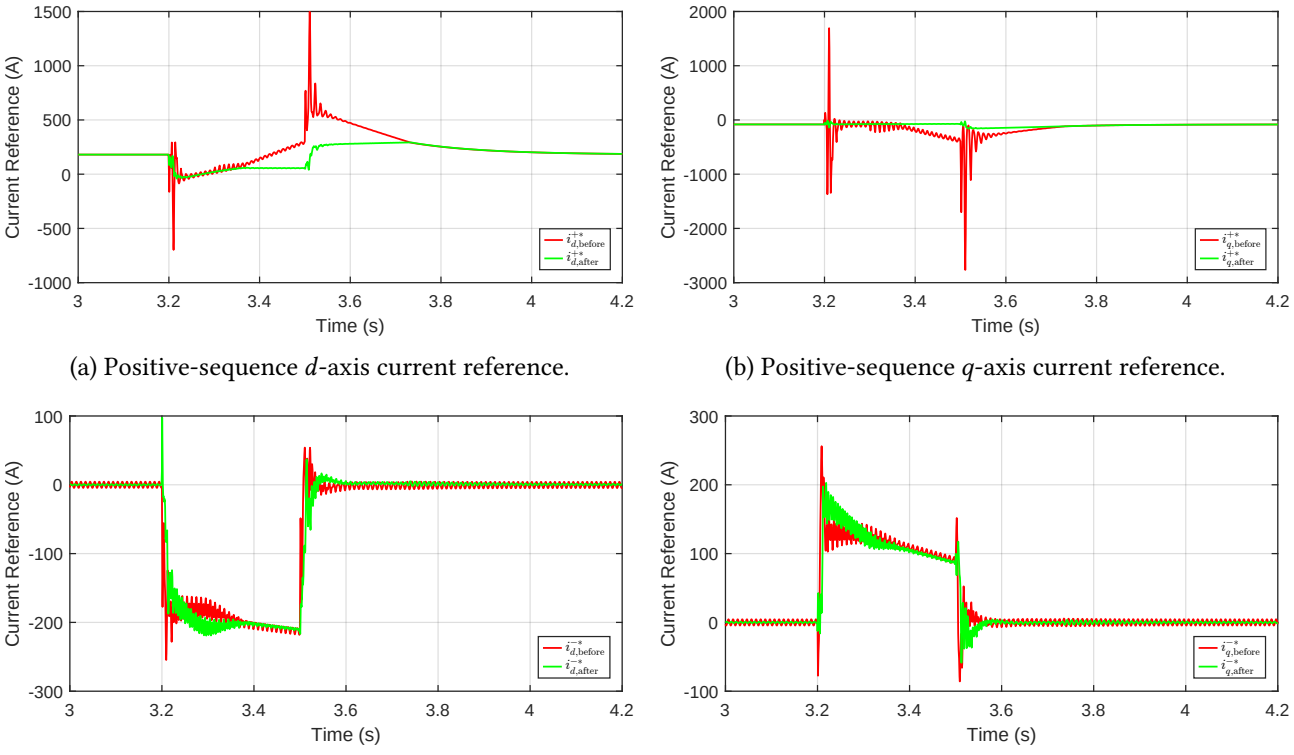


(a) Positive-sequence dq currents.

(b) Negative-sequence dq currents.

Figure 5.73.: Positive- and negative-sequence dq current components during the single-phase angle jump with negative-sequence current priority.

The sequence current components in Fig. 5.73 show the effect of the asymmetrical phase displacement on the two sequence controllers.



(a) Positive-sequence d-axis current reference.

(b) Positive-sequence q-axis current reference.

(c) Negative-sequence d-axis current reference.

(d) Negative-sequence q-axis current reference.

Figure 5.74.: Positive- and negative-sequence d- and q-axis current references before and after limitation during the single phase angle jump with negative-sequence current priority.

Figure 5.74 compares the unrestricted and limited current references during the phase angle disturbance. The reference trajectories illustrate directly how the sequence priority allocation is activated following the sudden phase displacement.

The single-phase angle jump case therefore confirms that the proposed priority mechanism is not restricted to voltage magnitude disturbances. It also operates during asymmetrical phase-angle disturbances, where the negative-sequence component arises from the unequal phase displacement rather than from unequal phase voltage magnitudes.

Chapter 6.

Discussion of Simulation Results

This chapter discusses the simulation results presented in the previous chapter and provides a comparative assessment of the four current-limiting methods considered in this thesis. Particular attention is given to their ability to restrict the converter current during grid disturbances, their transient response, their effect on the positive- and negative-sequence quantities, and the recovery of the converter after fault clearance.

6.1. General Comparison of the Current-Limiting Methods

The simulation results show that all methods are capable of reducing the converter current during severe grid disturbances compared with the unlimited operation. Nevertheless, their behavior differs because of the different variables on which the limiting action is applied.

The elliptical limiter provides a direct limitation of the current references. Once the calculated current demand becomes excessive, the reference components are scaled and the current demand seen by the inner control system is reduced.

The virtual-admittance-based method follows a different approach. Instead of directly modifying the current references through saturation, it increases the effective impedance seen between the internal converter voltage and the PCC voltage. The current reduction is therefore obtained indirectly through the electrical relation imposed by the virtual admittance.

The positive sequence priority-based limiter gives priority to the positive sequence current components when the total current demand exceeds the available converter capability. The positive sequence reference is therefore preserved as much as possible, while the negative sequence contribution is reduced according to the remaining current margin.

The negative sequence priority-based limiter follows the opposite allocation principle. When the total current demand exceeds the converter limit, the negative sequence current reference is prioritized, while the positive sequence contribution is reduced according to the remaining available current capability. Here it must be noted again, that a minimum positive sequence current is retained in order to preserve stable grid-forming operation.

6.2. Comparison of the Converter Current and the PCC voltage response

The first aspect considered in the comparison is the converter current response. The main objective the four current-limiting methods is to prevent excessive current during severe grid disturbances while maintaining a stable converter response. Although all methods achieve a significant reduction of the fault current compared with operation without current limitation, their behavior differs because of the different mechanisms through which the limiting action is introduced.

The comparison therefore considers not only the maximum current reached during the disturbance, but also the transient current excursions, the waveform quality during the fault, and the recovery following fault clearance.

Current limitation also affects the voltage behavior of the grid-forming converter. This is particularly important because limiting the converter current reduces the available control action that can be used to support the PCC voltage during a grid disturbance. Consequently, the assessment of a current-limiting method should not be based exclusively on the achieved current magnitude.

The PCC voltage responses obtained with the different limiters are therefore compared under the considered grid disturbances. Particular attention is given to the voltage magnitude during the fault, the behavior of the individual phases under asymmetrical conditions, and the recovery of the PCC voltage after fault clearance.

6.2.1. Asymmetrical Voltage Sags

Single-phase voltage sag

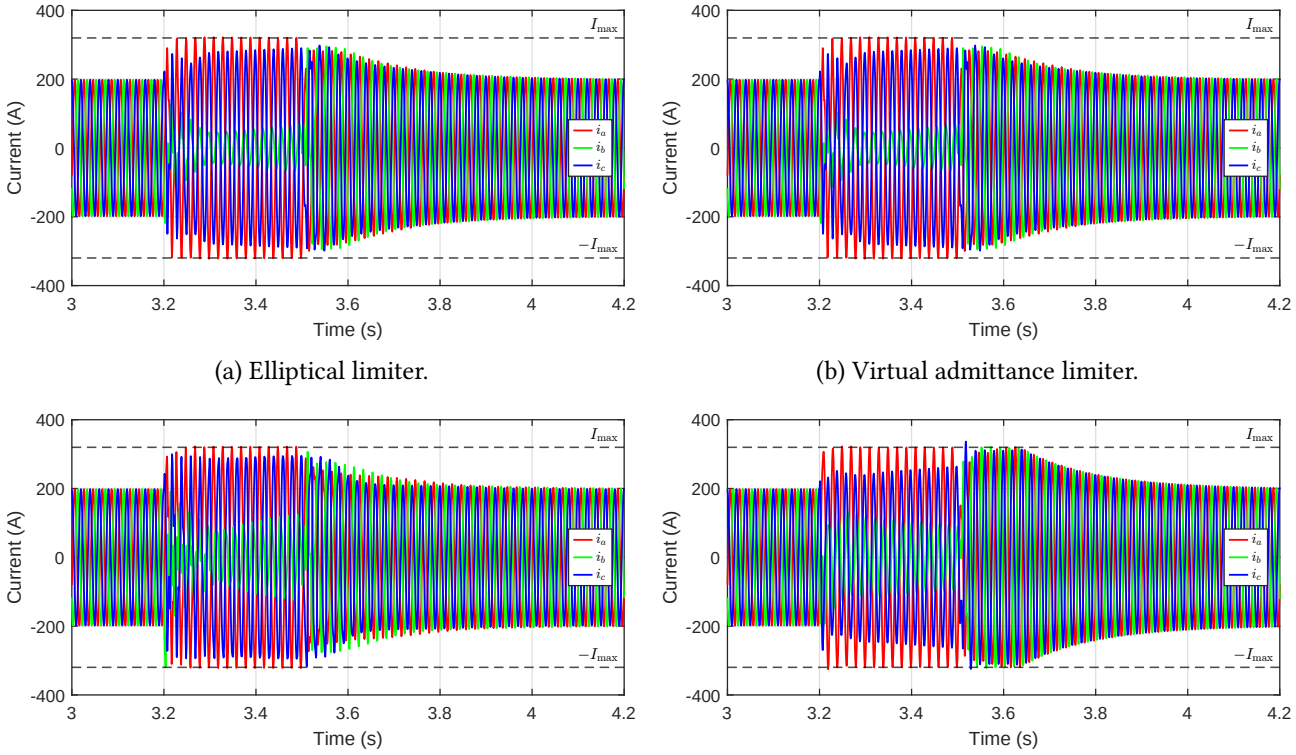
The current and PCC voltage responses obtained during the single phase voltage sag are compared for the four implemented current-limiting strategies.

Figure 6.1 compares the three phase converter currents. The elliptical limiter, shown in Fig. 6.1a, rapidly restricts the current following the application of the voltage sag, with the most heavily loaded phase remaining close to the imposed current limit. The phase currents are unbalanced, as expected during the asymmetrical disturbance, and gradually decrease as the converter approaches a new operating condition.

The virtual admittance limiter in Fig. 6.1b produces an almost identical current response. Both the initial rise in current and its subsequent evolution during the fault closely follow those obtained with the elliptical limiter. This is particularly interesting because the two methods introduce current limitation through different mechanisms. For the examined operating condition, their three phase current responses are practically indistinguishable.

The positive-sequence priority response in Fig. 6.1c exhibits a different distribution among the three phase currents. The maximum permissible current is still respected, but the phase current envelopes differ from those of the first two methods.

A further redistribution is observed with negative-sequence priority in Fig. 6.1d. The resulting phase current distribution is clearly different from that obtained with positive-sequence priority, although the same overall phase current constraint is applied.



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

(d) Negative-sequence priority.

Figure 6.1.: Comparison of the converter current responses obtained with the four current-limiting strategies during the single-phase voltage sag.

The current comparison consequently shows that all four strategies fulfill the main objective of restricting the converter current. The principal difference is not the value of the imposed current limit, but the manner in which the available capability is distributed.

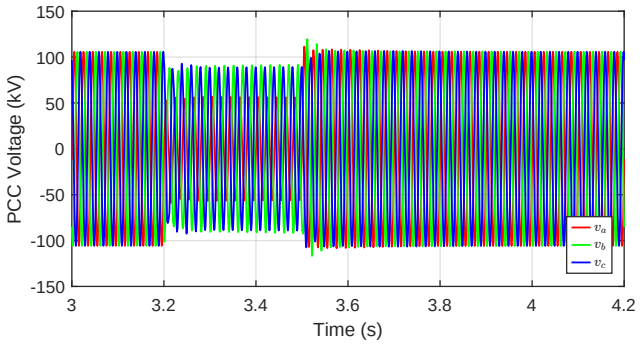
The corresponding PCC voltages are compared in Fig. 6.2. The elliptical response in Fig. 6.2a shows the voltage behavior resulting from the simultaneous proportional reduction of the positive- and negative-sequence current references. The virtual admittance result in Fig. 6.2b is again very close to the elliptical case, in agreement with the similarity already observed in their three phase currents.

The effect of explicitly assigning sequence priority can be examined from Figs. 6.2c and 6.2d. The two strategies influence the PCC voltage through different sequence current distributions, even though they operate under the same grid disturbance and the same converter current limit.

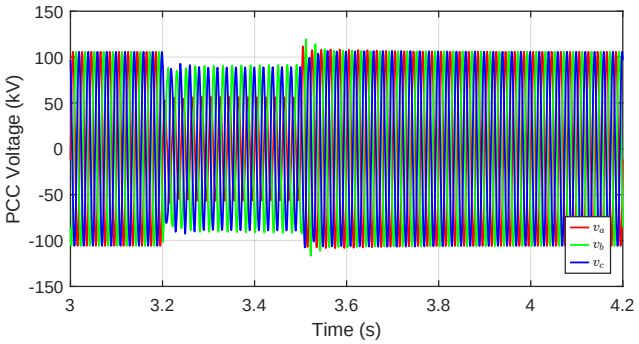
The voltage comparison therefore complements the phase current results. While the elliptical and virtual admittance approaches lead to very similar external responses, the sequence priority methods provide an additional degree of freedom by changing how the limited current capability contributes to the positive- and negative-sequence PCC voltage response.

The positive-sequence *dq* currents in Fig. 6.3 provide a clearer explanation for the differences observed in the phase current waveforms. With the elliptical limiter, shown in Fig. 6.3a, the positive-sequence components are limited together with the negative sequence components through the common scaling action. The virtual admittance response in Fig. 6.3b follows a similar evolution.

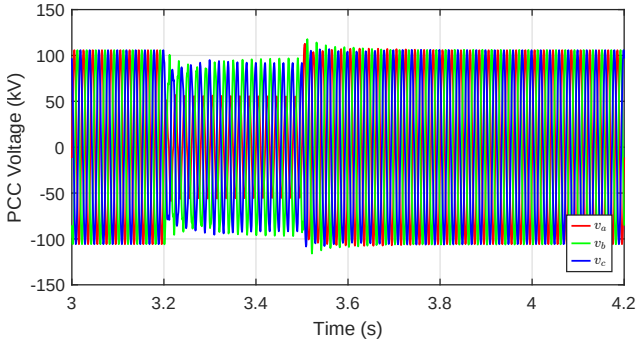
The positive-sequence priority case in Fig. 6.3c differs fundamentally in that the positive-sequence demand



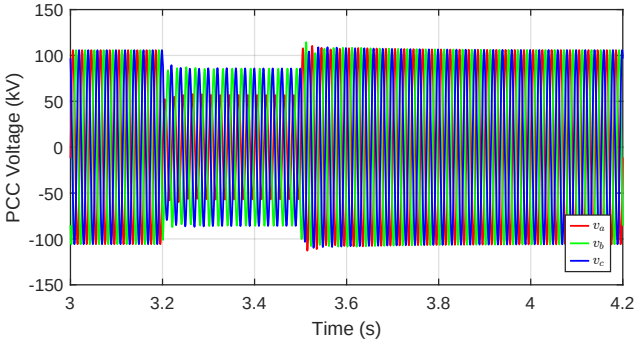
(a) Elliptical limiter.



(b) Virtual admittance limiter.

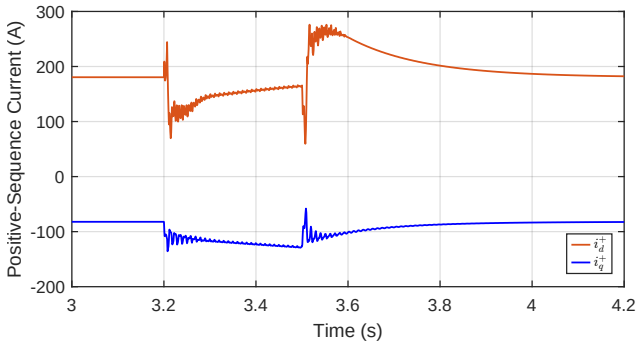


(c) Positive-sequence priority.

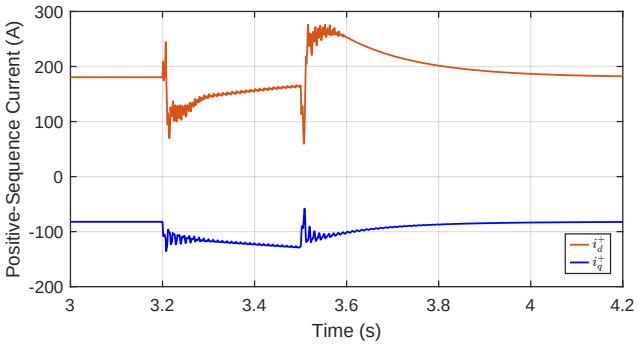


(d) Negative-sequence priority.

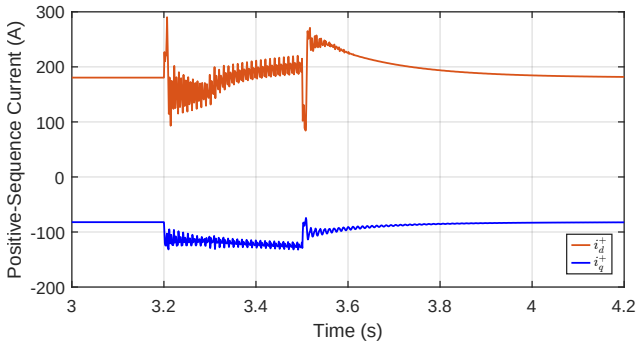
Figure 6.2.: Comparison of the PCC voltage responses obtained with the four current-limiting strategies during the single-phase voltage sag.



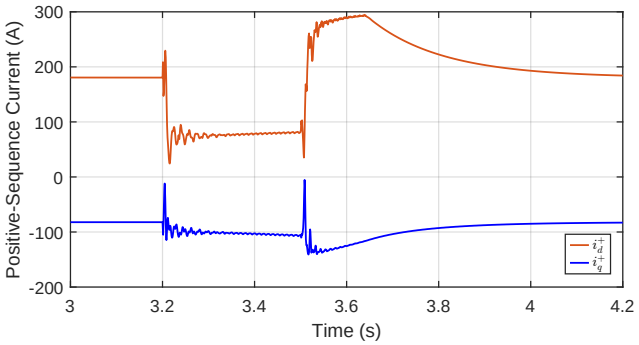
(a) Elliptical limiter.



(b) Virtual admittance limiter.



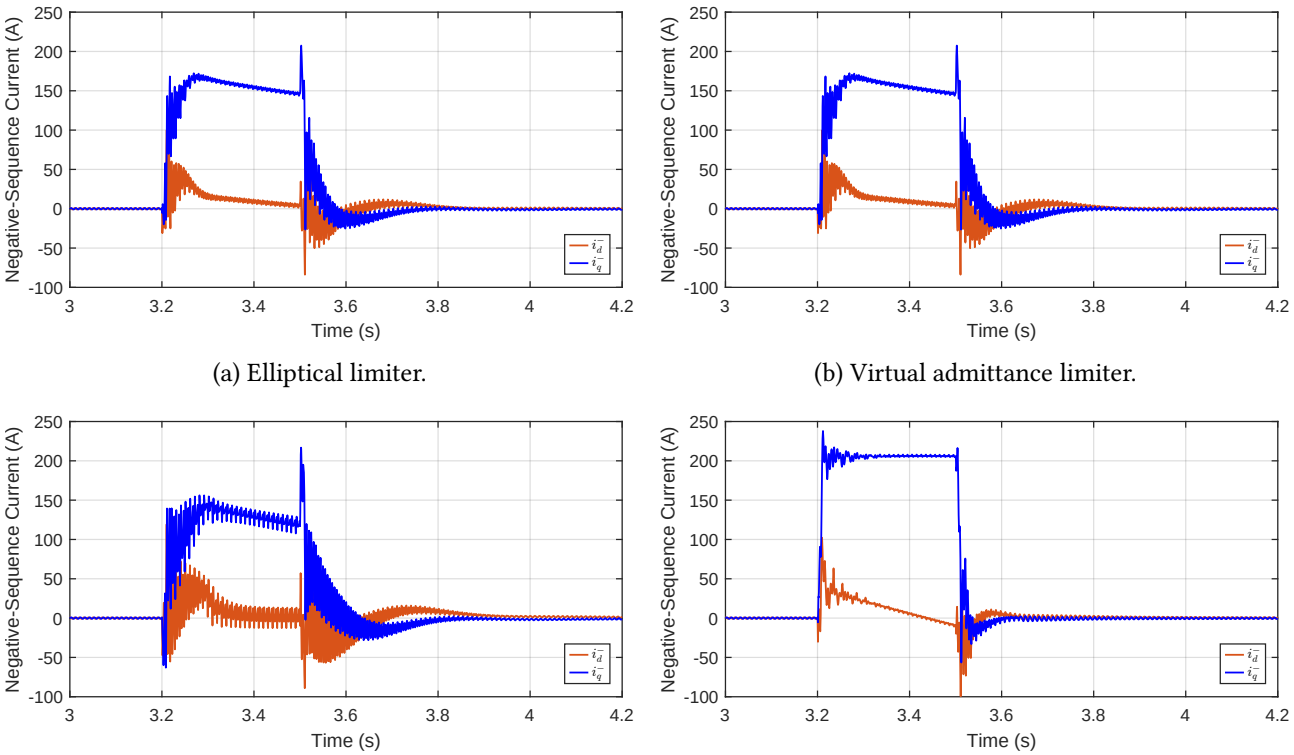
(c) Positive-sequence priority.



(d) Negative-sequence priority.

Figure 6.3.: Comparison of the positive-sequence *dq* current components obtained with the four current-limiting strategies during the single-phase voltage sag.

receives first access to the available converter current capability. In contrast, under negative-sequence priority, Fig. 6.3d, the positive-sequence current is the component that can be reduced after the minimum required contribution has been reserved.



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

(d) Negative-sequence priority.

Figure 6.4.: Comparison of the negative-sequence *dq* current components obtained with the four current-limiting strategies during the single-phase voltage sag.

The negative-sequence current components in Fig. 6.4 show the complementary effect of the current allocation strategies. In the elliptical case, Fig. 6.4a, the negative-sequence current is reduced by the same scaling mechanism applied to the positive sequence. The virtual admittance limiter, Fig. 6.4b, again produces a comparable response, which explains the close similarity between the phase current waveforms of the two methods.

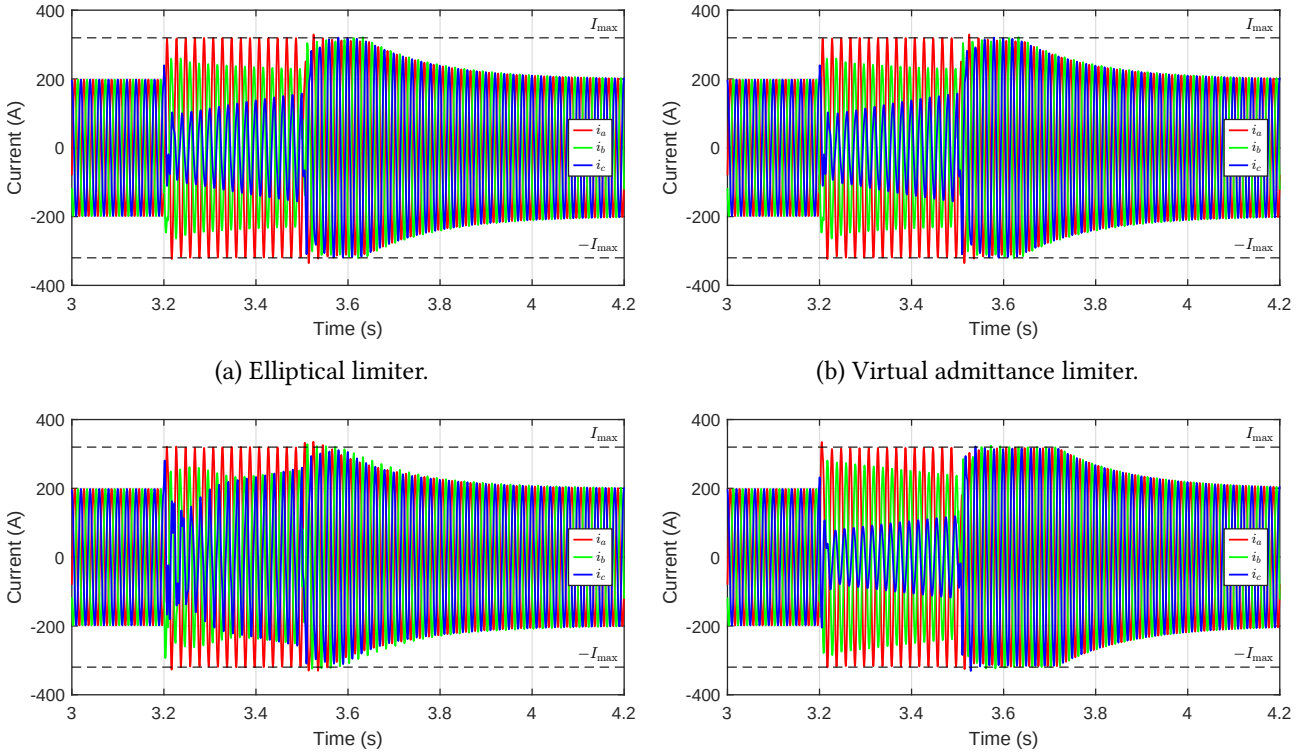
With positive-sequence priority, Fig. 6.4c, the negative-sequence current receives only the phase current capability that remains after the positive-sequence demand has been satisfied. The opposite behavior is obtained with negative-sequence priority, Fig. 6.4d, where the negative-sequence demand is given preferential access to the available capability.

The most suitable current-limiting strategy depends on the main control objective during this asymmetrical disturbance. In the examined case, the positive-sequence priority limiter provides the strongest support of the PCC voltage level, since the available converter current capability is assigned first to the positive-sequence component. Conversely, when the main objective is to provide stronger negative-sequence compensation and a more balanced voltage, the negative-sequence priority strategy may be preferable. The elliptical limiter represents a more neutral solution, since it reduces the positive- and negative-sequence current references proportionally without explicitly favoring either sequence. Similarly, the virtual admittance limiter does not impose a direct sequence priority, but limits the current indirectly by modifying the converter's effective virtual impedance. Since the elliptical and virtual admittance approaches produced very similar current and PCC voltage responses in the examined case, their main distinction lies in the internal limiting mechanism rather

than in the resulting external behavior. The preferred strategy therefore depends not only on the ability to respect the converter current limit, but also on whether the desired response favors positive-sequence voltage support, negative-sequence compensation, or a more balanced treatment of the two sequence components during asymmetrical faults.

Two-phase voltage sag

The four current-limiting strategies are now compared under a two phase voltage sag. As in the single phase case, the disturbance is asymmetrical and therefore produces simultaneous positive- and negative-sequence components. This allows the different current allocation mechanisms to be compared under a more severe unbalanced operating condition.



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

(d) Negative-sequence priority.

Figure 6.5.: Comparison of the converter current responses obtained with the four current-limiting strategies during the two-phase voltage sag.

Figure 6.5 compares the converter current responses obtained with the four limiting strategies during the two phase voltage sag. Before the disturbance, all four methods exhibit the same balanced three phase current. Following the application of the fault, the current becomes strongly unbalanced and the current-limiting mechanisms are activated. In all cases, the largest phase current magnitude remains close to the imposed limit I_{max}, confirming that all four strategies successfully restrict the converter current.

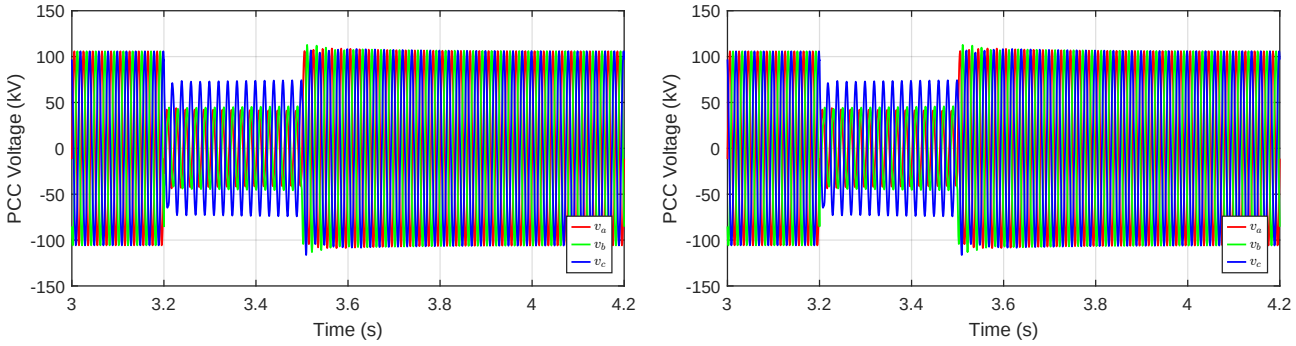
The elliptical limiter, Fig. 6.5a, produces a clearly unbalanced current distribution during the fault.

A very similar response is obtained with the virtual admittance limiter, shown in Fig. 6.5b. Both the magnitude and the distribution of the three phase currents closely follow those obtained with the elliptical limiter throughout the fault.

The positive-sequence priority limiter, Fig. 6.5c, produces a noticeably different three phase current distribution, with only a slight and brief exceedance of the limit over approximately two cycles.

The negative-sequence limiter exhibits the opposite tendency than the positive-sequence priority limiter in Fig. 6.5d. The maximum phase current continues to satisfy the imposed converter current constraint.

Following fault clearance, the responses of the four strategies become considerably more similar. The phase currents temporarily remain at elevated values and subsequently decrease towards their common pre fault operating level, with the negative-sequence priority limiter requiring slightly more time to return to its pre-fault operating condition after fault clearance, when compared to the others.



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

(d) Negative-sequence priority.

Figure 6.6.: Comparison of the PCC voltage responses obtained with the four current-limiting strategies during the two-phase voltage sag.

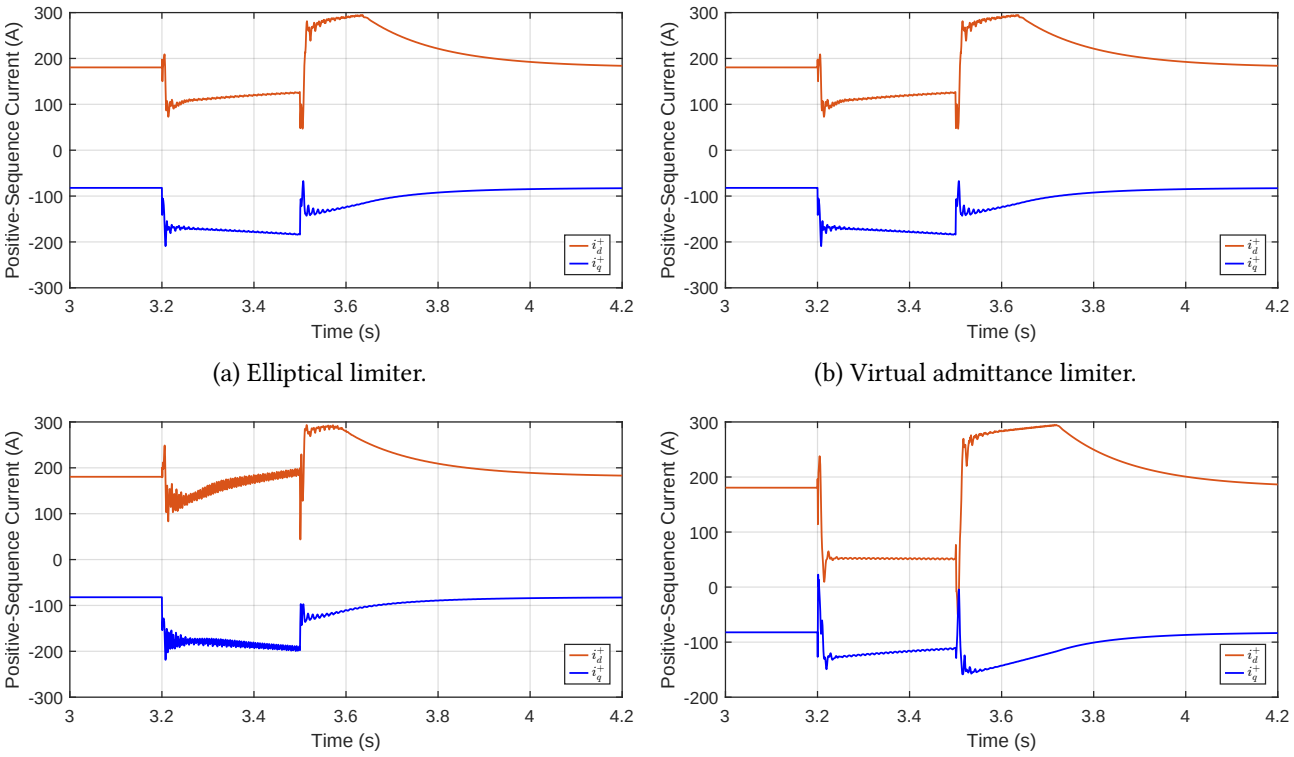
Figure 6.6 compares the PCC voltage responses obtained with the four current-limiting strategies during the two phase voltage sag. As expected, the disturbance produces a strongly unbalanced voltage profile during the fault interval, while all four methods allow the PCC voltage to recover following fault clearance.

The elliptical limiter, shown in Fig. 6.6a, results in a considerable reduction of the affected phase voltage magnitudes during the fault. The voltage remains bounded and recovers rapidly after the disturbance is removed. A nearly identical response is obtained with the virtual admittance limiter in Fig. 6.6b. The voltage magnitudes and their transient behavior closely follow those of the elliptical limiter, confirming once again the strong similarity between the responses of these two methods.

The positive-sequence priority limiter, Fig. 6.6c, provides the strongest preservation of the PCC voltage level among the four examined strategies. The voltage of the affected phases remains at a higher level compared with the other limiting methods, while the PCC voltage returns rapidly to its pre-fault condition following

fault clearance.

In contrast, the negative-sequence priority limiter, Fig. 6.6d, results in a different voltage distribution during the fault and a lower overall positive-sequence voltage support contribution compared with the positive-sequence priority case. The voltage nevertheless remains controlled and successfully recovers after the disturbance.



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

(d) Negative-sequence priority.

Figure 6.7.: Comparison of the positive-sequence *dq* current components obtained with the four current-limiting strategies during the two-phase voltage sag.

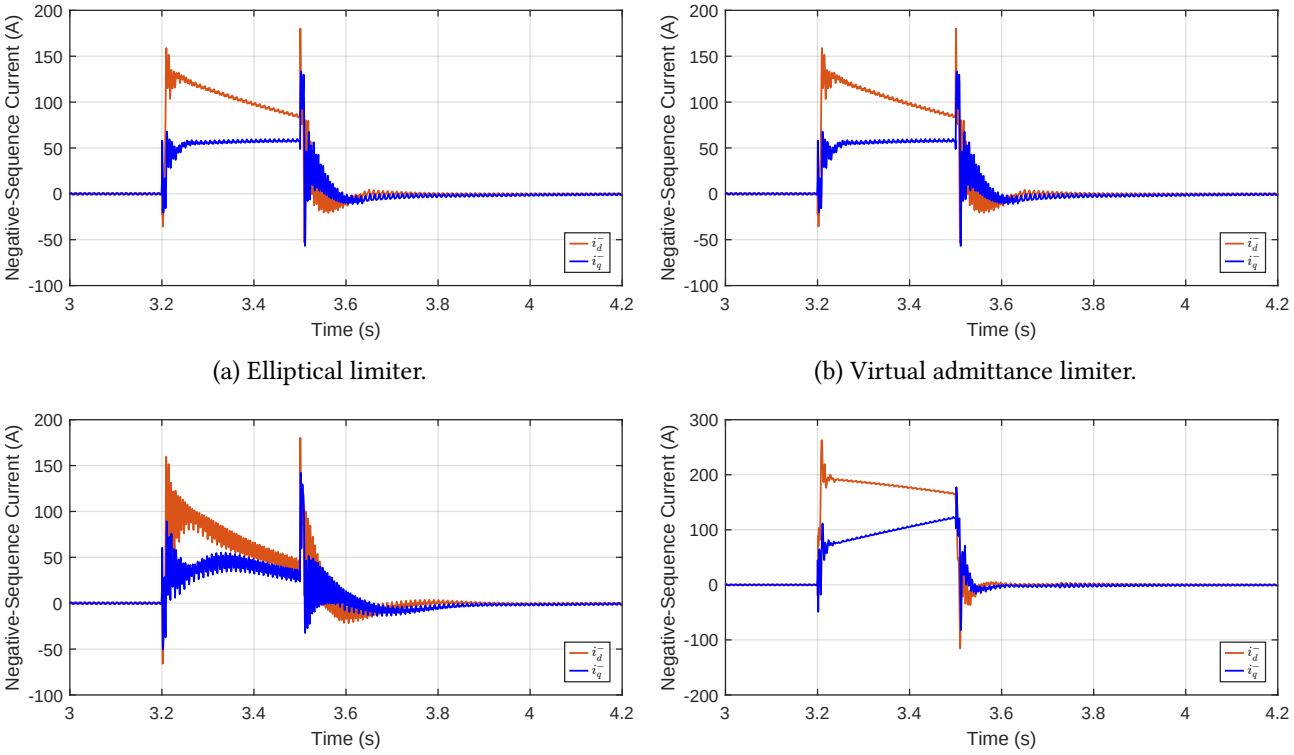
Figure 6.7 compares the positive-sequence *dq* current components. These waveforms provide a more direct indication of how each strategy uses the available current capability for the positive-sequence response.

The negative-sequence *dq* current components are compared in Fig. 6.8. Together with the positive-sequence results, these waveforms show how the converter current capability is divided between the two sequence controllers during the two phase disturbance. They therefore provide the basis for explaining the differences observed in both the three phase converter currents and the PCC voltages.

Overall, the two-phase voltage sag demonstrates that all four strategies successfully achieve the primary objective of limiting the maximum converter phase current, although they distribute the available current capability in different ways.

The elliptical and virtual admittance limiters again exhibit almost identical current and PCC voltage responses, despite relying on different limiting mechanisms. In both cases, no explicit preference is given to either sequence component, and the resulting behavior remains very similar.

The two priority-based strategies, on the other hand, modify the converter response according to the se-



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

(d) Negative-sequence priority.

Figure 6.8.: Comparison of the negative-sequence *dq* current components obtained with the four current-limiting strategies during the two-phase voltage sag.

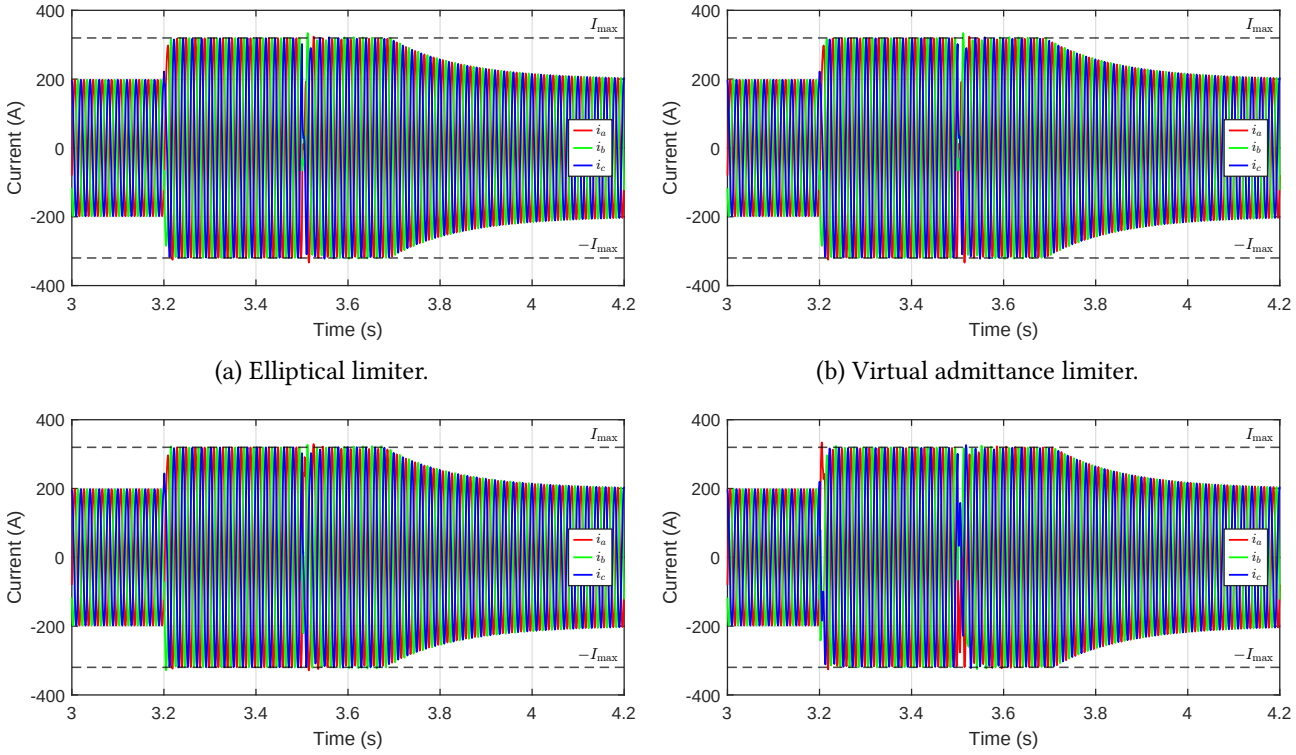
lected sequence objective. Positive-sequence priority maintains a stronger balanced current contribution and provides the best preservation of the PCC voltage level during the examined disturbance. Negative-sequence priority allows a larger phase current imbalance because a greater part of the available converter capability is assigned to the negative-sequence response, resulting in a different PCC voltage distribution.

The comparison therefore shows that the choice of current-limiting strategy should not be based only on whether the converter current remains within its specified limit. The preferred method also depends on the desired behavior of the positive- and negative-sequence components during the fault. If maintaining positive-sequence voltage support is the main objective, the positive-sequence priority limiter is the most suitable of the examined strategies. If stronger negative-sequence compensation is required, the negative-sequence priority method becomes more appropriate. The elliptical and virtual admittance limiters provide more neutral alternatives, since they limit the converter current without explicitly favoring either sequence.

6.2.2. Symmetrical Voltage Sag

Three-phase voltage sag

The four current-limiting strategies are finally compared under a severe symmetrical three-phase voltage sag. Unlike the single and two phase disturbances, this fault remains balanced and therefore does not introduce a sustained negative-sequence component.



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

(d) Negative-sequence priority.

Figure 6.9.: Comparison of the converter current responses obtained with the four current-limiting strategies during the symmetrical three phase voltage sag.

Figure 6.9 compares the converter current responses obtained with the four limiting strategies during the symmetrical three-phase voltage sag. Following the inception of the disturbance, the current magnitude increases rapidly and is restricted close to the predefined limit I_{max} in all four cases.

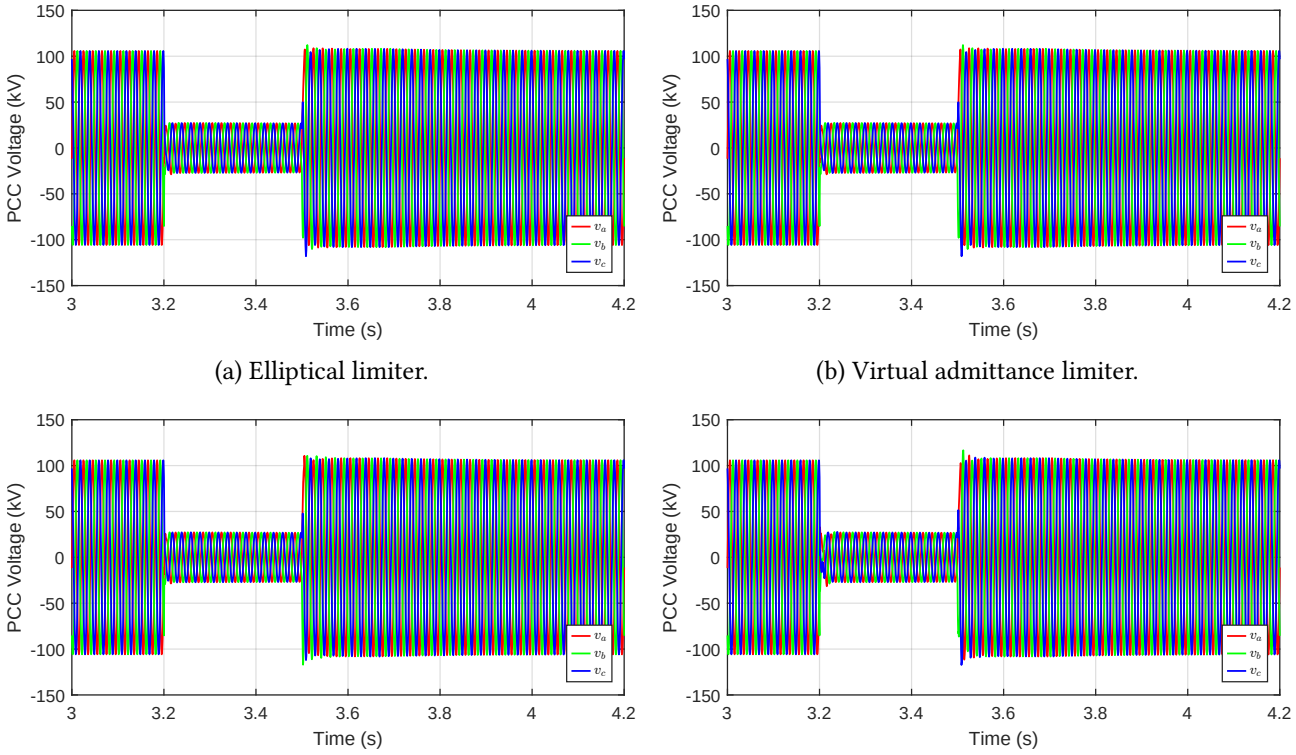
The elliptical limiter, Fig. 6.9a, produces a balanced current response with all three phase current magnitudes reaching approximately the same maximum value. The current remains close to the imposed limit during the voltage sag and returns towards its pre-fault value after the fault clearance.

A practically identical behavior again is obtained with the virtual admittance limiter, shown in Fig. 6.9b. The current magnitude, the balanced three phase structure, and the post-fault recovery closely follow those observed with the elliptical limiter.

The positive-sequence priority limiter, Fig. 6.9c, also results in a very similar balanced current waveform. Practically the entire available converter current capability is assigned to the positive-sequence component.

The negative-sequence priority response, Fig. 6.9d, is also very close to the other three cases during the steady part of the fault. Although the negative sequence is nominally given priority, there is no negative-sequence demand under the symmetrical disturbance, so the unused current capability is available to the positive-sequence component. A small transient deviation can be observed at the start of the fault.

The corresponding PCC voltage responses are presented in Fig. 6.10. Since all three phases are affected equally, the PCC voltage remains balanced during the fault.



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

(d) Negative-sequence priority.

Figure 6.10.: Comparison of the PCC voltage responses obtained with the four current-limiting strategies during the symmetrical three phase voltage sag.

The elliptical limiter, Fig. 6.10a, maintains a balanced PCC voltage response during the disturbance, with the three phase voltages reduced to approximately the same level. Once the fault is cleared, the PCC voltage recovers to its pre-fault value.

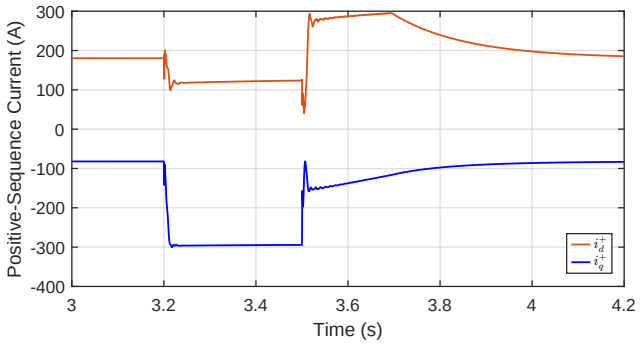
The virtual admittance limiter, Fig. 6.10b, produces an almost identical voltage response. Both the voltage level during the fault and the post-fault recovery closely follow those obtained with the elliptical limiter, confirming once again that the two methods provide identical converter behavior.

A very similar response is also obtained with the positive-sequence priority limiter, Fig. 6.10c. There is no competition between the two sequence controllers for the available converter current capability. The positive-sequence component dominates the response and the PCC voltage level remains practically the same as in the elliptical and virtual admittance cases.

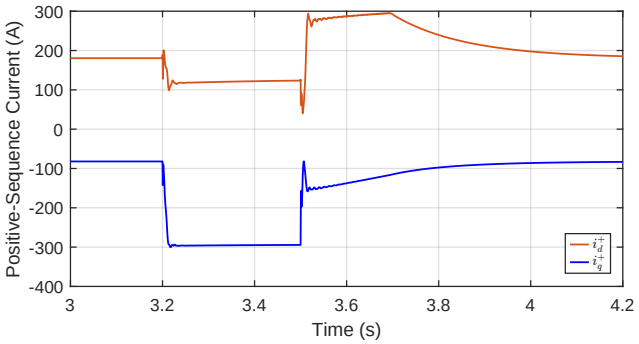
The negative-sequence priority limiter, Fig. 6.10d, also results in a nearly the same steady state voltage level during the fault. Although differences are observed during the transient intervals.

The positive-sequence *dq* currents in Fig. 6.11 represent the dominant (positive-) sequence current contribution during the symmetrical fault.

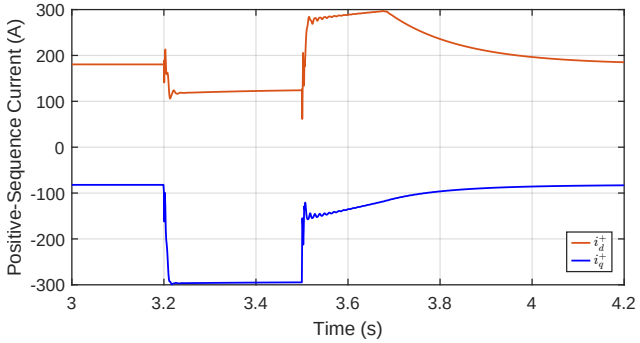
The negative-sequence current components are shown in Fig. 6.12. As expected for a symmetrical disturbance, these components remain close to zero during the steady part of the fault. Small transient values appear during the application and clearance of the fault because of the dynamic response of the sequence extraction and control loops, but no sustained negative-sequence current is required.



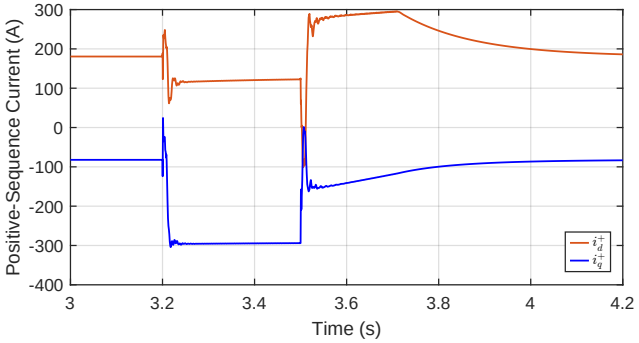
(a) Elliptical limiter.



(b) Virtual admittance limiter.

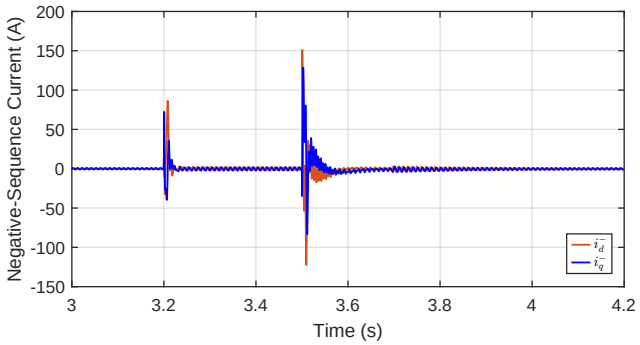


(c) Positive-sequence priority.

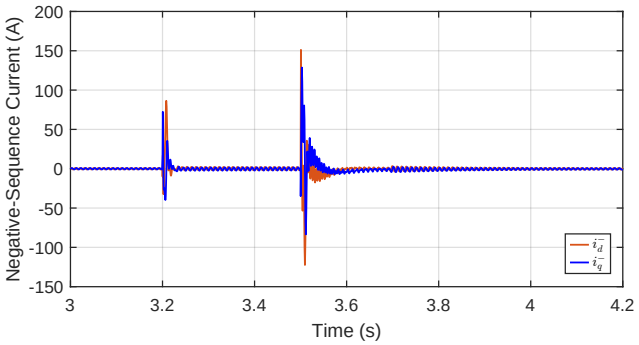


(d) Negative-sequence priority.

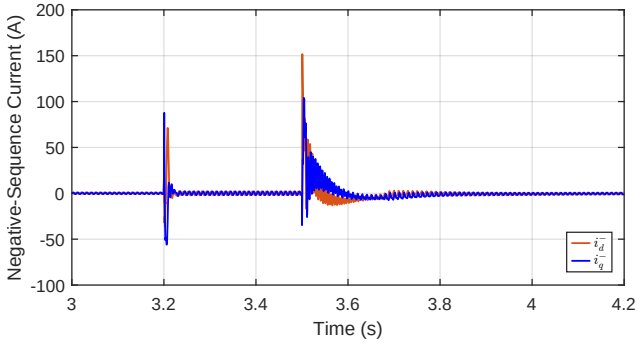
Figure 6.11.: Comparison of the positive-sequence *dq* current components obtained with the four current-limiting strategies during the symmetrical three-phase voltage sag.



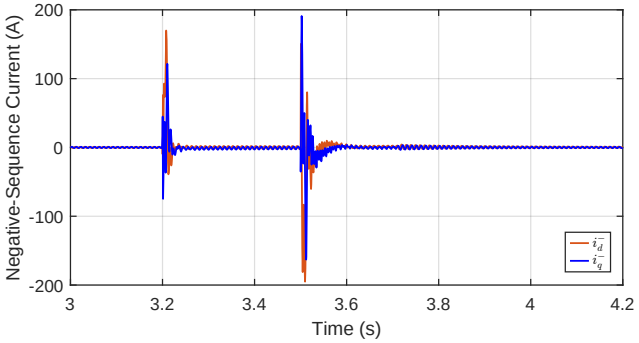
(a) Elliptical limiter.



(b) Virtual admittance limiter.



(c) Positive-sequence priority.



(d) Negative-sequence priority.

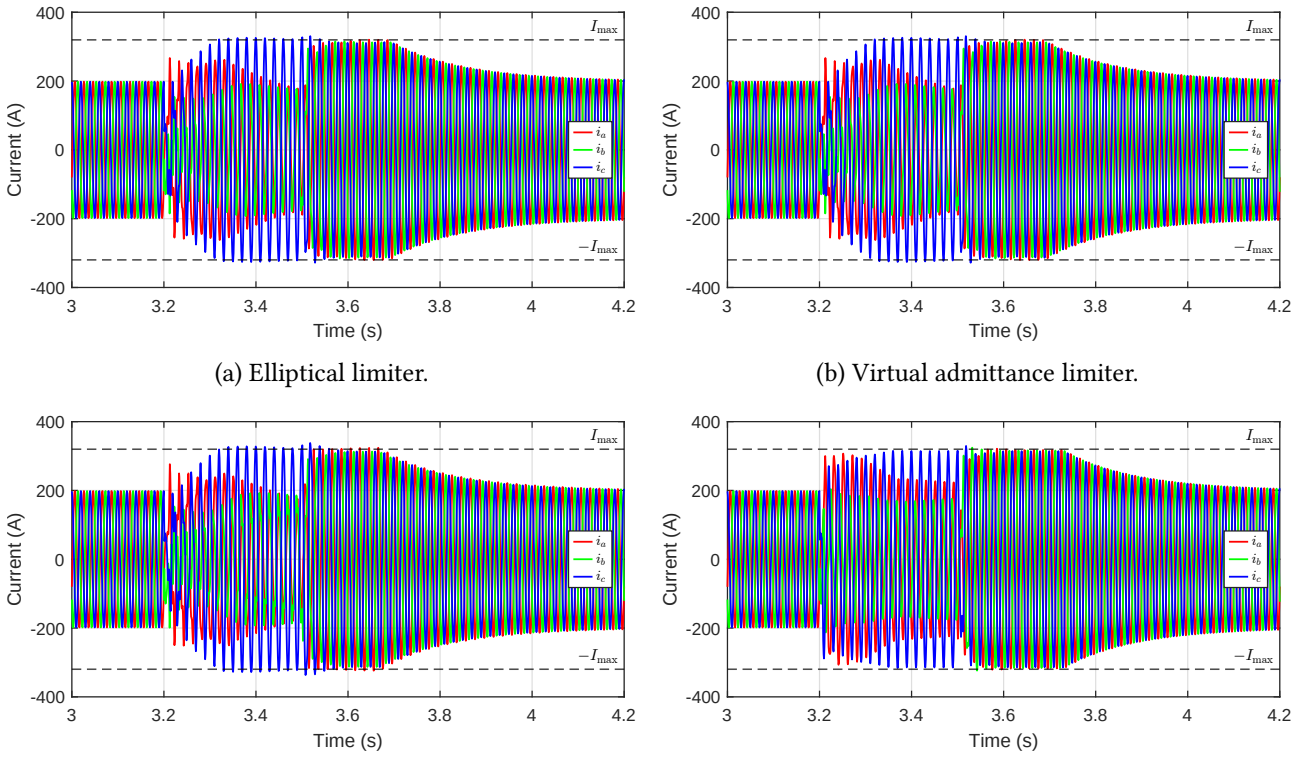
Figure 6.12.: Comparison of the negative-sequence *dq* current components obtained with the four current-limiting strategies during the symmetrical three-phase voltage sag.

Overall, the symmetrical three-phase voltage sag differs fundamentally from the preceding asymmetrical cases and produces only minor differences among the four current-limiting strategies in both the converter current and PCC voltage responses.

6.2.3. Phase-Angle Jump

Single-Phase Angle Jump

The final disturbance considered in the comparison of the four current-limiting strategies is a single-phase angle jump. Although the voltage magnitudes are not reduced, the unequal phase displacement creates an asymmetrical operating condition and introduces both positive- and negative-sequence components. This case therefore provides an additional comparison of the four limiting strategies under an unbalanced disturbance produced by a phase angle variation rather than by a voltage magnitude sag.



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

(d) Negative-sequence priority.

Figure 6.13.: Comparison of the converter current responses obtained with the four current-limiting strategies during the single-phase angle jump.

Figure 6.13 compares the converter current responses obtained with the four limiting strategies during the single-phase angle jump. Before the disturbance, all four methods exhibit the same balanced three-phase current, and during the fault they constrain the converter current under to the imposed limit I_{max}.

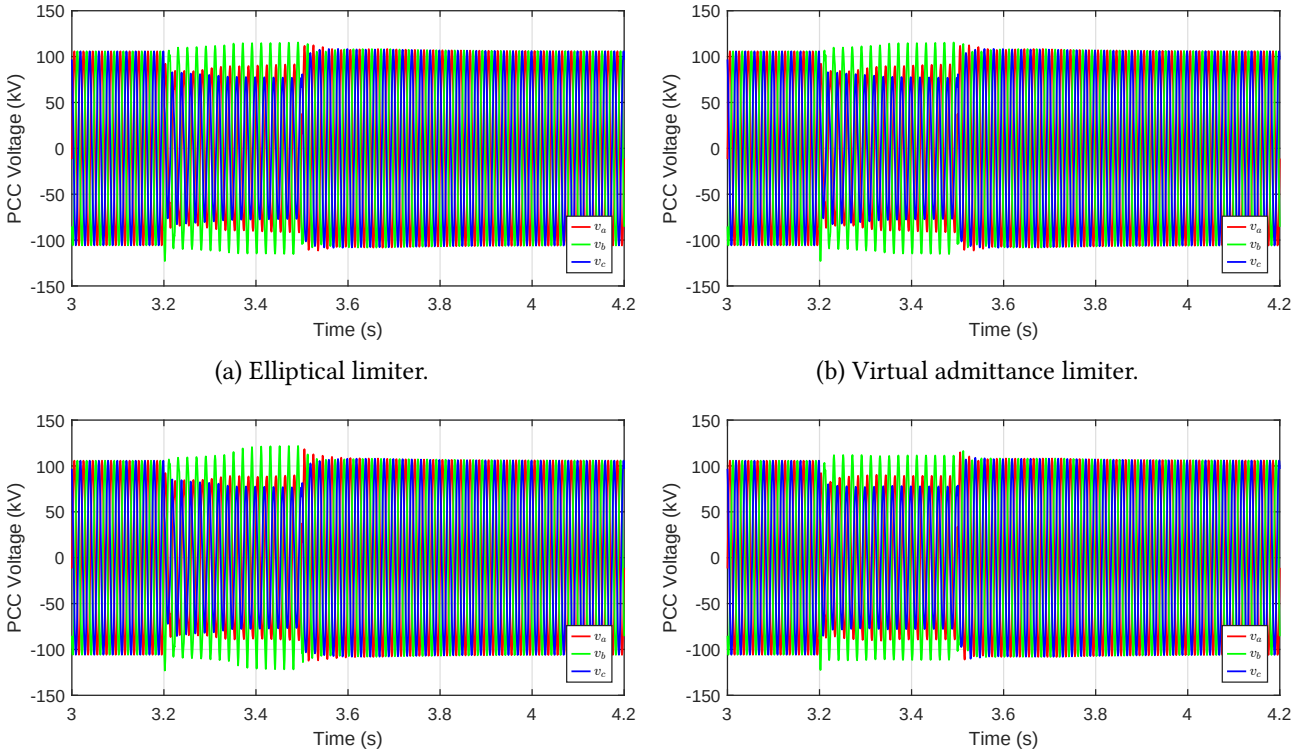
The elliptical limiter, Fig. 6.13a, produces an unbalanced current distribution during the fault interval, with the most heavily loaded phase approaching the current limit while the remaining two phases stay at lower magnitudes.

The virtual admittance limiter, Fig. 6.13b, exhibits practically the same behavior. The current amplitudes during

the fault, the degree of phase imbalance, and the post-fault recovery are all very close to those obtained with the elliptical limiter.

A different behavior is obtained with the positive-sequence priority limiter, Fig. 6.13c. During the disturbance, the three phase currents become much more evenly distributed and remain very close to the imposed current limit. The post-fault recovery is slower than in the elliptical and virtual admittance cases.

The negative-sequence priority limiter, Fig. 6.13d, retains a more pronounced phase imbalance during the disturbance. Compared with the other methods, the return to the pre-fault operating condition is slightly slower, indicating a more persistent post-fault transient.



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

(d) Negative-sequence priority.

Figure 6.14.: Comparison of the PCC voltage responses obtained with the four current-limiting strategies during the single-phase angle jump.

Figure 6.14 compares the PCC voltage responses obtained with the four current-limiting strategies during the single-phase angle jump. The phase displacement introduces a clear asymmetry in the PCC voltage during the disturbance, even though the voltage magnitudes of the grid sources are not directly reduced.

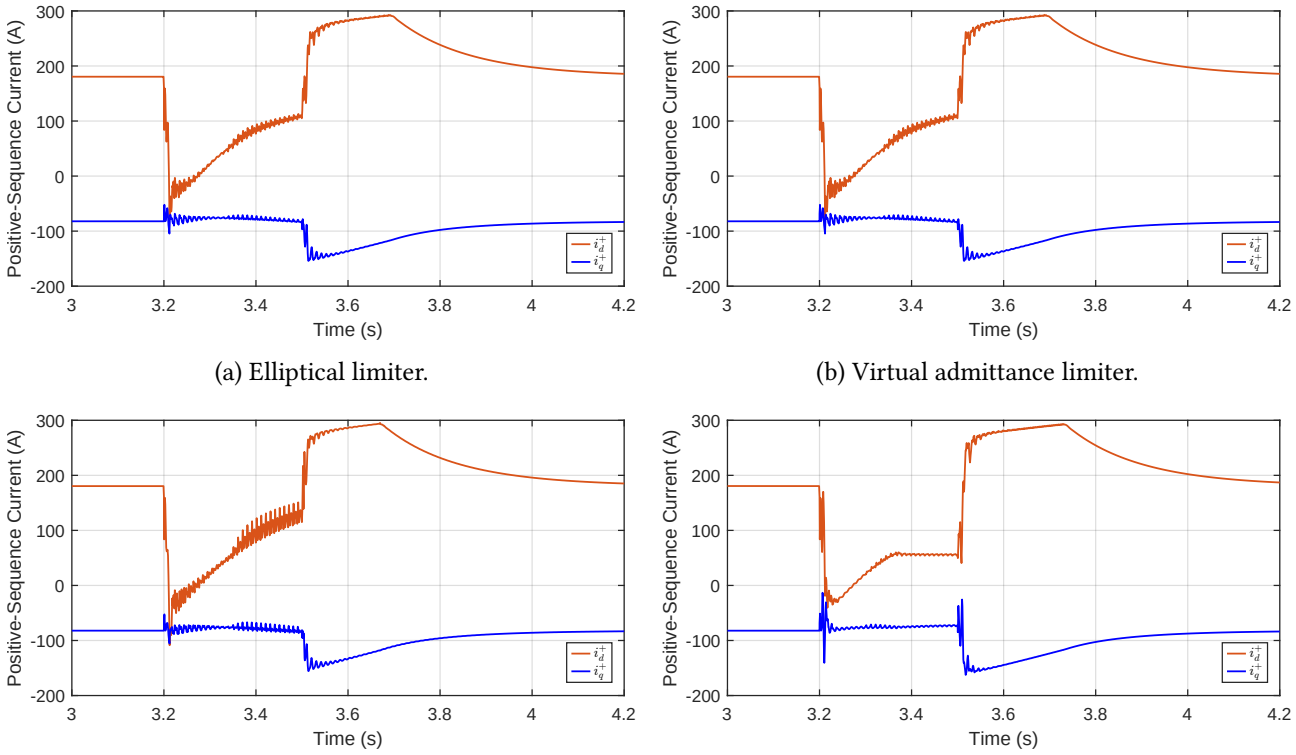
The elliptical limiter, Fig. 6.14a, produces an unbalanced voltage response during the disturbance. The magnitude of one phase increases, while the remaining phase voltages are maintained at lower levels. The voltage returns rapidly to its pre-fault condition after the phase angle disturbance is removed.

The virtual admittance limiter, Fig. 6.14b, produces an almost identical response, despite the different internal mechanisms through which the two methods introduce current limitation.

The positive-sequence priority limiter, Fig. 6.14c, produces the strongest positive-sequence voltage support response among the examined strategies, resulting in a stronger voltage contribution during the disturbance.

At the same time, the difference between the individual phase voltage magnitudes remains visible, since less current capability is available for negative-sequence compensation.

A different behavior is observed with the negative-sequence priority limiter in Fig. 6.14d. Since a larger part of the available converter current capability is assigned to the negative-sequence response, the difference between the phase voltage magnitudes is reduced compared to the other cases. This improved balancing is obtained at the expense of the stronger positive-sequence voltage support provided by the positive-sequence priority strategy. Following fault clearance, the PCC voltage returns to its pre-fault condition with only a short transient.



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

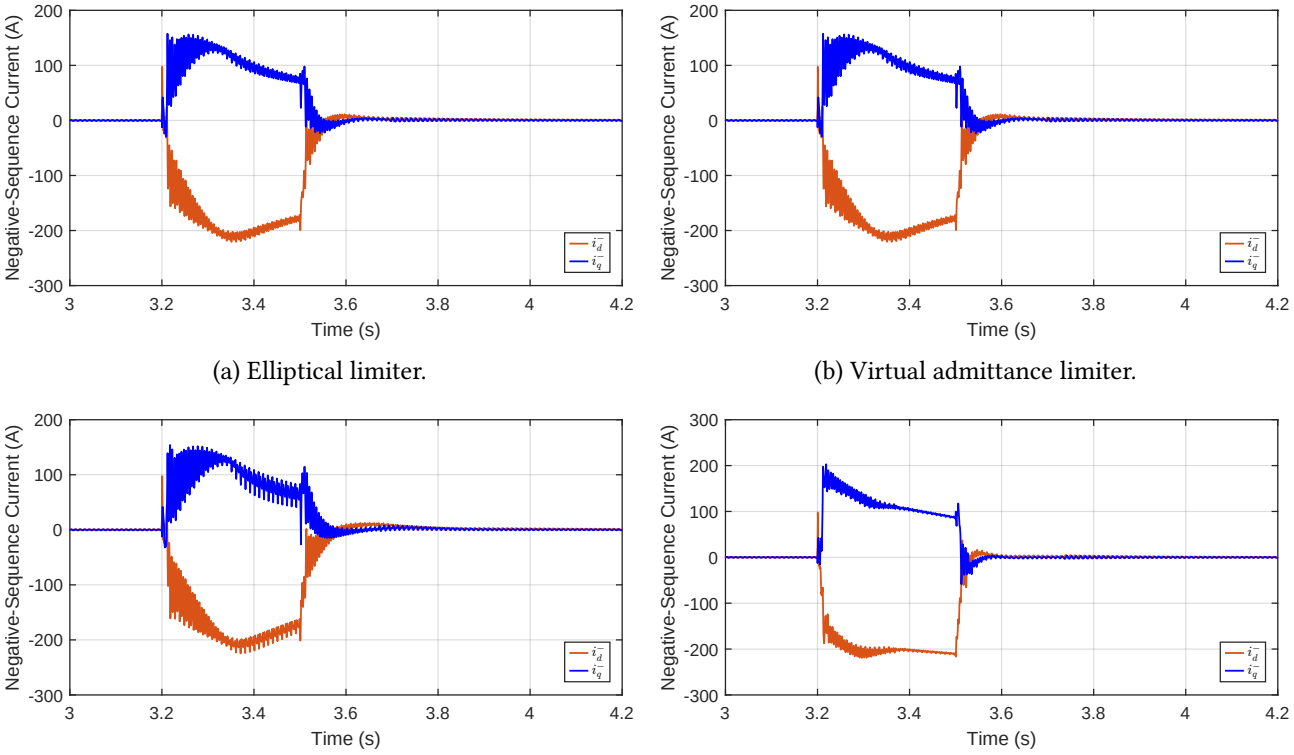
(d) Negative-sequence priority.

Figure 6.15.: Comparison of the positive-sequence *dq* current components obtained with the four current-limiting strategies during the single-phase angle jump.

The positive-sequence *dq* current components are compared in Fig. 6.15. These waveforms provide a direct indication of the current capability retained for the positive-sequence response during the disturbance.

The negative-sequence *dq* currents are presented in Fig. 6.16. Since the phase angle jump introduces a significant asymmetrical component, the negative-sequence response plays an important role in this case. Together with the positive-sequence results, these waveforms show how each strategy distributes the limited converter current capability between the two sequence controllers.

Overall, the single-phase angle jump highlights the different behavior of the four current-limiting strategies in both the converter current and the PCC voltage. The elliptical and virtual admittance limiters once again produce almost identical external responses. The two priority-based strategies, however, modify the sequence composition of the converter current differently.



(a) Elliptical limiter.

(b) Virtual admittance limiter.

(c) Positive-sequence priority.

(d) Negative-sequence priority.

Figure 6.16.: Comparison of the negative-sequence *dq* current components obtained with the four current-limiting strategies during the single-phase angle jump.

6.2.4. Comparison results

The comparison of the four implemented current-limiting strategies shows that no single method can be considered universally superior under all fault conditions. All approaches successfully fulfill the primary requirement of limiting the converter current, but they differ in the way the available current capability is used and allocated, consequently, in their influence on the PCC voltage and on the distribution of the phase currents. The selection of the most suitable method therefore depends on the desired behavior of the converter during the disturbance.

The elliptical and virtual admittance limiters produced remarkably similar external responses throughout the examined cases. These two methods are considered comparatively neutral with respect to sequence prioritization. Neither of them explicitly assigns the available converter current capability to the positive or negative sequence according to a predefined priority. The elliptical limiter preserves the relative contribution of the sequence current references through common scaling, while the virtual admittance limiter achieves current limitation indirectly by modifying the effective virtual impedance without imposing an explicit sequence priority. Their similar performance indicates that either approach can provide effective current limitation when no explicit sequence priority objective is required.

Significant differences were observed when the positive- and negative-sequence priority strategies were introduced. During asymmetrical faults, the positive-sequence priority limiter consistently produced a stronger positive-sequence contribution and generally maintained a more balanced three phase current response. More importantly, it provided the strongest preservation of the PCC voltage level in the investigated single- and two-phase voltage sags. A similar tendency was observed during the single-phase angle jump, where the method maintained a stronger positive-sequence voltage support contribution.

These characteristics are particularly relevant for grid-forming operation. Since the main voltage forming and synchronization functions are associated with the positive-sequence control path, preserving the positive-sequence current capability during a severe asymmetrical disturbance provides a direct advantage when maintaining the grid-forming response is considered the main objective.

The negative-sequence priority limiter offers a different advantage. By allocating a larger portion of the available converter current capability to the negative-sequence response, it can more strongly influence the voltage unbalance produced by an asymmetrical disturbance. This was particularly visible during the single-phase angle jump, where negative-sequence priority reduced the difference between the phase voltage magnitudes compared with positive-sequence priority. The corresponding phase currents, however, became more unbalanced, since the increased negative-sequence contribution naturally produces a larger difference between their magnitudes. In addition, the examined cases showed a slightly slower return towards the pre-fault operating condition with negative-sequence priority.

The negative-sequence priority strategy should therefore not be interpreted as inferior to positive-sequence priority. Instead, the two methods represent different objectives. If the primary objective is the reduction of voltage unbalance, a stronger negative-sequence contribution may be desirable. Conversely, if maintaining the positive-sequence voltage and the main grid-forming response is considered more important, positive-sequence priority provides the more suitable current allocation.

In symmetrical three-phase voltage sag, since the disturbance contains essentially no sustained negative-sequence component, the distinction between every current-limitation strategy becomes negligible. All four methods produced very similar balanced current and PCC voltage responses. The advantage of selecting a particular sequence priority therefore becomes significant mainly during asymmetrical disturbances, when the positive- and negative-sequence controllers simultaneously require part of the limited converter current capability.

Table 6.1.: Qualitative comparison of the investigated current-limiting strategies.

Method	**Current allocation**	**Main advantage**	**Main consideration**
Elliptical limiter	Common limitation of the sequence current references	Effective current limitation without explicit sequence priority	Does not independently control the allocation between positive and negative sequence
Virtual admittance limiter	Indirect limitation through modification of the virtual impedance	Effective limitation while retaining the virtual admittance control structure	External response was very similar to the elliptical limiter in the examined cases
Positive-sequence priority	Positive-sequence demand is satisfied before the remaining capability is assigned to the negative sequence	Strongest positive-sequence and PCC voltage support in the examined asymmetrical disturbances	Less current capability remains available for negative-sequence compensation
Negative-sequence priority	Negative-sequence demand is prioritized after reserving the minimum positive-sequence contribution	Stronger negative-sequence compensation and improved control of voltage unbalance	Reduced positive-sequence capability and more unbalanced phase current distribution

The main characteristics observed for the four strategies can be summarized above.

6.2.5. Similarity of the Responses of the Elliptical and Virtual-Admittance Methods

An important observation resulting from the comparison is the close similarity between the responses of the elliptical current limiter and the virtual-admittance-based current-limiting method. For the examined disturbances, the three phase converter currents obtained with the two methods are practically identical. A similar behavior is observed in the PCC voltages, as well as in the positive- and negative-sequence current components. Numerical comparison of the exported simulation data confirmed that the small differences that are not visible in the plotted waveforms are negligible with respect to the corresponding signal magnitudes.

At first, this result may appear unexpected, since the two methods introduce current limitation at different locations of the control structure. The elliptical limiter operates directly on the current reference and reduces its magnitude when the demanded current exceeds the predefined limit. In contrast, the virtual-admittance-based method limits the current indirectly by modifying the virtual impedance that relates the converter internal voltage to the PCC voltage. Therefore, the two methods represent different current-limiting concepts despite the similar responses observed in the present simulations.

For the elliptical limiter, the limited current reference can be expressed in a simplified form as

$$\mathbf{i}^*_{\mathrm{ell}} = \kappa \mathbf{i}^*_0, \qquad 0 < \kappa \leq 1, \tag{6.1}$$

where $\mathbf{i}^*_0$ denotes the original current reference and κ is the scaling factor determined by the current-limiting algorithm. Consequently, the magnitude of the current reference is reduced while its relative dq composition is preserved.

In the virtual admittance control structure, the current reference is generated from the difference between the internal converter voltage reference and the measured PCC voltage. In complex vector form, this relation can be written as

$$\mathbf{i}^*_0 = \mathbf{Y}_{v0} \Delta \mathbf{v}, \tag{6.2}$$

where

$$\Delta \mathbf{v} = \mathbf{e}^* - \mathbf{v}_{\mathrm{PCC}} \tag{6.3}$$

and $\mathbf{Y}_{v0}$ is the virtual admittance under normal operating conditions.

When current limitation is achieved mainly by changing the magnitude of the virtual admittance while approximately preserving its impedance angle, the limited admittance may be represented as

$$\mathbf{Y}_{v,\mathrm{lim}} = \kappa_v \mathbf{Y}_{v0}. \tag{6.4}$$

The corresponding current reference is then

$$\mathbf{i}^{*}_{\mathrm{VA}} = \mathbf{Y}_{v,\mathrm{lim}}\Delta\mathbf{v} = \kappa_v \mathbf{Y}_{v0}\Delta\mathbf{v}, \tag{6.5}$$

and therefore

$$\mathbf{i}^{*}_{\mathrm{VA}} = \kappa_v \mathbf{i}^{*}_{0}. \tag{6.6}$$

Consequently, when the scaling introduced by the virtual admittance adjustment is close to that imposed by the elliptical limiter, i.e.,

$$\kappa_v \approx \kappa, \tag{6.7}$$

the two methods generate approximately the same limited current reference,

$$\mathbf{i}^{*}_{\mathrm{VA}} \approx \mathbf{i}^{*}_{\mathrm{ell}}. \tag{6.8}$$

Since both current references are subsequently processed by the same inner current controller and the same converter and network model, very similar converter current trajectories are obtained. The similar injected currents also lead to almost identical PCC voltage responses, which explains why the similarity is not restricted to the current waveforms but is also observed in the remaining electrical quantities.

This result does not imply that the two current-limiting concepts are generally equivalent. Their similarity is a consequence of the particular control structure, parameter selection, and operating conditions considered in this study. A virtual admittance method can also modify the impedance angle, for example by changing the virtual resistance and reactance in different proportions. In such a case, both the magnitude and the phase of the resulting current reference may be modified, and a different response from a direct current reference limiter can be expected. The present results therefore indicate that, for the implemented configuration, the different limiting mechanisms lead to almost the same external converter response.

Chapter 7.

Conclusions and Future Work

7.1. Conclusions

This thesis investigated the control and current limitation of a grid-forming converter under symmetrical and asymmetrical grid disturbances, with particular emphasis on the interaction between the positive- and negative-sequence components. A grid-forming converter model with separate positive- and negative-sequence control paths was developed in PLECS, and four current-limiting strategies were implemented and studied: an elliptical current limiter, a virtual admittance-based limiter, a positive-sequence priority strategy, and a negative-sequence priority strategy.

The simulation results showed that all methods can restrict the converter current during the examined disturbances. Their differences become most significant under asymmetrical conditions, where both positive- and negative-sequence current components are present and compete for the available converter current capability. Under symmetrical disturbances, where the negative-sequence component remains small, the differences between the investigated methods are considerably reduced.

Overall, the results indicate that the elliptical and virtual admittance current-limiting methods are the safest choices for the converter considering that the main objective is to limit the converter current while preserving, as much as possible, the current's vector behavior requested by the control system. Given that all investigated methods satisfy the imposed converter current limits and exhibit relatively similar post-fault recovery, the main differences arise from the way in which the available current is distributed during asymmetrical disturbances.

The elliptical limiter preserves the direction and relative contribution of the positive- and negative-sequence current references through a common scaling action, while the virtual admittance method achieves current limitation without explicitly imposing a priority between the two sequences. For the operating conditions examined in this thesis, these two methods also produce very similar external converter responses. In contrast, the sequence priority strategies intentionally modify the relative contribution of the positive- and negative-sequence currents.

Consequently, the preferred current-limiting strategy should be selected according to the required fault support objective rather than only according to its ability to maintain the current below $I_{\max}$. When no explicit sequence prioritization is required, the results favor the elliptical and virtual admittance approaches, since they provide effective current limitation, without altering the relative current vector distribution requested by the converter control. Sequence priority strategies remain valuable alternatives when a positive-sequence voltage support or greater negative-sequence compensation is required.

7.2. Future Work

This study was based primarily on an averaged converter model, which allows the fundamental frequency control response and the interaction between the positive- and negative-sequence controllers to be examined without other complications. An extension of this work would therefore be the validation of the implemented current-limiting strategies using a detailed switching model. That way, the influence of PWM, switching ripple, converter voltage saturation, and discrete control effects would be investigated. In addition, the use of a detailed switching model would allow the investigation of current-limiting techniques acting directly at the modulation or semiconductor switching level. Such approaches may include the modification or blocking of switching pulses, modulation saturation, or other gate level protection mechanisms that can restrict the converter current independently of, or in coordination with, the higher level current-limiting controller. Experimental validation would also provide a further step towards assessing the current-limiting methods implemented in this work.

Further work could also investigate the performance of the current-limiting methods over a wider range of conditions and disturbances, such as different grid strengths, network impedance ratios, operating points, fault depths, and fault durations. The interaction of the proposed control system with multiple grid-forming and or grid-following converters would also be of interest, since the behavior of a larger converter-dominated power system may differ from the one examined in this thesis.

Finally, the current allocation strategy could be further developed by introducing adaptive or hybrid limiting techniques. Instead of applying a fixed sequence priority, the allocation between the positive- and negative-sequence components could be adjusted according to the type and severity of the disturbance or the required grid support objective.

Appendix A.

Systems and Controller Parameters

Table A.1.: System and controller parameters used in the simulations.

Quantity	Symbol	Value
Base apparent power	S_b	60 MVA
Base line-to-line voltage	V_b	130 kV
Nominal frequency	f_b	50 Hz
Nominal angular frequency	ω_b	314.159 rad s^{-1}
Base impedance	Z_b	281.667 Ω
Base inductance	L_b	0.896 57 H
Maximum phase current	I_{max}	320 A
Grid resistance	R_{src}	16.8 Ω
Grid inductance	L_{src}	0.179 31 H
Converter-side inductance	L_{cnv}	0.134 49 H
Converter side resistance	R_{cnv}	4.225 Ω
DC-link voltage	V_d	233.52 kV
Simulation time step	T_s	10 µs
Active power droop gain	k_p	4.1888×10^{-8}
Reactive power droop gain	k_q	1.9×10^{-2}
Current control bandwidth	α_c	2200 rad s^{-1}

Appendix B.

Inner Current Controller C-Script

This appendix presents the complete C-Script implementation of the inner current controller used in the PLECS model. The implementation is divided into three parts, corresponding to the structure of the PLECS C-Script block: the code declarations, the update function code, and the output function code.

B.1. Code Declarations

```
1
2     #include <stdio.h>
3     #include <stdlib.h>
4     #include <float.h>
5     #include <math.h>
6
7     // Static variable
8     //int sizeA, numRowsB;
9     //int outIdxC, outIdxD;
10    float kp,ki,errord,errorq,Rest,Lest;
11    float Ts, ac, wb, Lb, dblk;
12    float idref, iqref, id, iq, ud, uq, ww;
13
```

Listing B.1: Code declarations of the inner current controller.

B.2. Update Function

```
1
2     //DiscState(0) = InputSignal(0, 0);
3     id = InputSignal(0, 0);
4     iq = InputSignal(0, 1);
5     idref = InputSignal(0, 2);
6     iqref = InputSignal(0, 3);
7     dblk = InputSignal(0, 4);
8
9     Ts = ParamRealData(0, 0);
10    ac = ParamRealData(1, 0);
11
12    Rest = 4.225;
13    ki = ac*Rest;
14    errord = idref-id;
15    errorq = iqref-iq;
16
```

```
17      if (dblk == 0 ) {
18          DiscState(0) += 0;
19          DiscState(1) += 0;
20      } else {
21          DiscState(0) += ki*errord*Ts;
22          DiscState(1) += ki*errorq*Ts;
23      }
24
```

Listing B.2: Update function code of the inner current controller.

B.3. Output Function

```
1
2       id = InputSignal(0, 0);
3       iq = InputSignal(0, 1);
4       idref = InputSignal(0, 2);
5       iqref = InputSignal(0, 3);
6       dblk = InputSignal(0, 4);
7       ud = InputSignal(0, 5);
8       uq = InputSignal(0, 6);
9       ww = InputSignal(0, 7);
10
11      Ts = ParamRealData(0, 0);
12      ac = ParamRealData(1, 0);
13      wb = ParamRealData(2, 0);
14      Lb = ParamRealData(3, 0);
15
16      Lest = 0.15*Lb;
17      kp = ac*Lest;
18      errord = idref-id;
19      errorq = iqref-iq;
20
21      if (dblk == 0 ) {
22          OutputSignal(0, 0) = ud;
23          OutputSignal(0, 1) = uq;
24      } else {
25          OutputSignal(0, 0) = ud + kp*errord + DiscState(0) - ww*Lest*iqref;
26          OutputSignal(0, 1) = uq + kp*errorq + DiscState(1) + ww*Lest*idref;
27      }
28
```

Listing B.3: Output function code of the inner current controller.

Appendix C.

Derivation of the Sequence-Interaction Terms

The phase-aware positive-sequence priority limiter requires the interaction between the positive- and negative-sequence current components to be considered explicitly. This interaction is represented in the implemented algorithm through the auxiliary quantities A_- and B_-. The purpose of this section is to derive these quantities and explain their physical meaning. Here, it must be noted that also A_+ and B_+ of the negative-sequence priority limiter, are also formed following the same principle.

After the positive-sequence current has been allocated, its dq components are denoted by

$$I^+_{d,\mathrm{lim}}, \qquad I^+_{q,\mathrm{lim}}. \tag{C.1}$$

The negative-sequence current reference is first expressed in terms of its direction and an unknown magnitude. Its normalized direction is given by

$$u^-_d = \frac{I^-_{d,\mathrm{ref}}}{I^-_{\mathrm{ref}}}, \qquad u^-_q = \frac{I^-_{q,\mathrm{ref}}}{I^-_{\mathrm{ref}}}, \tag{C.2}$$

where

$$I^-_{\mathrm{ref}} = \sqrt{\left(I^-_{d,\mathrm{ref}}\right)^2 + \left(I^-_{q,\mathrm{ref}}\right)^2}. \tag{C.3}$$

Hence,

$$\left(u^-_d\right)^2 + \left(u^-_q\right)^2 = 1. \tag{C.4}$$

If I^-_{alloc} denotes the negative-sequence current magnitude that is eventually allocated by the limiter, its dq components can be written as

$$I^-_{d,\mathrm{alloc}} = I^-_{\mathrm{alloc}} u^-_d, \qquad I^-_{q,\mathrm{alloc}} = I^-_{\mathrm{alloc}} u^-_q. \tag{C.5}$$

Therefore, the direction of the original negative-sequence current reference is preserved, while only its magnitude is adjusted by the limiter.

C.1. Interaction in Phase A

The origin of the auxiliary quantity A_{-} can first be seen by considering phase A. Using the reference frame convention adopted in the implemented model, the positive-sequence contribution to the phase A current can be expressed as

$$i_a^+(t) = I_{d,\mathrm{lim}}^+ \cos\theta - I_{q,\mathrm{lim}}^+ \sin\theta. \tag{C.6}$$

Because the negative sequence rotates in the opposite direction, its phase-A contribution can be written as

$$i_a^-(t) = I_{\mathrm{alloc}}^- \left(u_d^- \cos\theta + u_q^- \sin\theta \right). \tag{C.7}$$

The total phase A current is therefore

$$\begin{aligned} i_a(t) &= i_a^+(t) + i_a^-(t) && \text{(C.8)} \\ &= \left(I_{d,\mathrm{lim}}^+ + I_{\mathrm{alloc}}^- u_d^- \right) \cos\theta \\ &\quad + \left(-I_{q,\mathrm{lim}}^+ + I_{\mathrm{alloc}}^- u_q^- \right) \sin\theta. && \text{(C.9)} \end{aligned}$$

A sinusoidal quantity of the general form

$$x(t) = X \cos\theta + Y \sin\theta \tag{C.10}$$

has a squared amplitude equal to

$$X^2 + Y^2. \tag{C.11}$$

Consequently, the squared amplitude of the phase A current is

$$\begin{aligned} I_a^2 &= \left(I_{d,\mathrm{lim}}^+ + I_{\mathrm{alloc}}^- u_d^- \right)^2 \\ &\quad + \left(-I_{q,\mathrm{lim}}^+ + I_{\mathrm{alloc}}^- u_q^- \right)^2. \end{aligned} \tag{C.12}$$

Expanding the two terms gives

$$
\begin{aligned}
I_a^2 = &\left(I_{d,\mathrm{lim}}^{+}\right)^2 + \left(I_{q,\mathrm{lim}}^{+}\right)^2 \\
&+ \left(I_{\mathrm{alloc}}^{-}\right)^2 \left[\left(u_d^{-}\right)^2 + \left(u_q^{-}\right)^2\right] \\
&+ 2I_{\mathrm{alloc}}^{-} \left(I_{d,\mathrm{lim}}^{+} u_d^{-} - I_{q,\mathrm{lim}}^{+} u_q^{-}\right).
\end{aligned} \tag{C.13}
$$

Since the negative-sequence direction vector has unit magnitude,

$$
\left(u_d^{-}\right)^2 + \left(u_q^{-}\right)^2 = 1, \tag{C.14}
$$

and defining the squared positive-sequence current magnitude as

$$
P^2 = \left(I_{d,\mathrm{lim}}^{+}\right)^2 + \left(I_{q,\mathrm{lim}}^{+}\right)^2, \tag{C.15}
$$

the previous expression becomes

$$
I_a^2 = P^2 + \left(I_{\mathrm{alloc}}^{-}\right)^2 + 2I_{\mathrm{alloc}}^{-} \left(I_{d,\mathrm{lim}}^{+} u_d^{-} - I_{q,\mathrm{lim}}^{+} u_q^{-}\right). \tag{C.16}
$$

The term multiplying I_{alloc}^{-} represents the interaction between the already allocated positive-sequence current and the direction of the negative-sequence current. For convenience, this term is defined as

$$
\boxed{A_{-} = I_{d,\mathrm{lim}}^{+} u_d^{-} - I_{q,\mathrm{lim}}^{+} u_q^{-}}. \tag{C.17}
$$

The phase A current magnitude can therefore be written in the compact form

$$
I_a^2 = P^2 + \left(I_{\mathrm{alloc}}^{-}\right)^2 + 2A_{-} I_{\mathrm{alloc}}^{-}. \tag{C.18}
$$

This shows that A_{-} is not an independently selected control parameter. It appears naturally as a cross term when the positive- and negative-sequence current contributions are combined in phase A.

C.2. Origin of the Second Interaction Term

For phases B and C, the positive- and negative-sequence quantities are displaced by 120°. Furthermore, because the two sequences rotate in opposite directions, their phase displacements have opposite signs. Using the phase ordering and inverse dq transformation convention adopted in the model, the phase B components are

$$i_b^+(t) = I_{d,\lim}^+ \cos\left(\theta + \frac{2\pi}{3}\right) - I_{q,\lim}^+ \sin\left(\theta + \frac{2\pi}{3}\right), \tag{C.19}$$

and

$$i_b^-(t) = I_{\text{alloc}}^- \left[u_d^- \cos\left(\theta - \frac{2\pi}{3}\right) + u_q^- \sin\left(\theta - \frac{2\pi}{3}\right)\right]. \tag{C.20}$$

Similarly, the phase C components are obtained by reversing the signs of the 120° displacement.

When these expressions are expanded using

$$\cos\left(\theta \pm \frac{2\pi}{3}\right) = -\frac{1}{2}\cos\theta \mp \frac{\sqrt{3}}{2}\sin\theta \tag{C.21}$$

and

$$\sin\left(\theta \pm \frac{2\pi}{3}\right) = -\frac{1}{2}\sin\theta \pm \frac{\sqrt{3}}{2}\cos\theta, \tag{C.22}$$

a second combination of the positive-sequence current components and the negative-sequence direction appears. This term is

$$\boxed{B_- = I_{d,\lim}^+ u_q^- + I_{q,\lim}^+ u_d^-}. \tag{C.23}$$

The two quantities A_- and B_- therefore contain the information required to describe the relative orientation of the positive-sequence current vector and the negative-sequence current direction.

Their origin can also be seen directly using a complex vector representation. Define

$$\underline{I}^+ = I_{d,\lim}^+ + jI_{q,\lim}^+ \tag{C.24}$$

and

$$\underline{u}^- = u_d^- + ju_q^-. \tag{C.25}$$

Their product is

$$\begin{aligned} \underline{I}^+\underline{u}^- &= \left(I_{d,\lim}^+ + jI_{q,\lim}^+\right)\left(u_d^- + ju_q^-\right) && \text{(C.26)} \\ &= \left(I_{d,\lim}^+ u_d^- - I_{q,\lim}^+ u_q^-\right) \\ &\quad + j\left(I_{d,\lim}^+ u_q^- + I_{q,\lim}^+ u_d^-\right). && \text{(C.27)} \end{aligned}$$

Hence,

$$\boxed{\underline{I}^{+}\underline{u}^{-} = A_{-} + jB_{-}}. \tag{C.28}$$

Thus, A_{-} and B_{-} may be interpreted as the real and imaginary parts of a complex quantity describing the relative orientation between the already allocated positive-sequence current and the requested negative-sequence current direction.

C.3. Phase-Dependent Interaction Coefficients

Applying the 120° phase displacements to this interaction gives the three phase-dependent coefficients used by the current-limiting algorithm:

$$L_{a,-} = 2A_{-}, \tag{C.29}$$

$$L_{b,-} = -A_{-} + \sqrt{3}B_{-}, \tag{C.30}$$

$$L_{c,-} = -A_{-} - \sqrt{3}B_{-}. \tag{C.31}$$

The squared current magnitude in each phase can therefore be written in the common form

$$I_{\phi}^{2} = P^{2} + \left(I_{\text{alloc}}^{-}\right)^{2} + L_{\phi,-}I_{\text{alloc}}^{-}, \qquad \phi \in \{a, b, c\}. \tag{C.32}$$

The quantities A_{-} and B_{-} therefore provide a compact description of how the positive- and negative-sequence current vectors interact, while $L_{a,-}$, $L_{b,-}$, and $L_{c,-}$ translate this interaction into the corresponding effect on each physical phase. This phase-dependent representation is required because the same negative-sequence current can increase the current magnitude in one phase while partially reducing it in another. Consequently, the remaining converter current capability cannot be determined accurately from the sequence magnitudes alone.

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